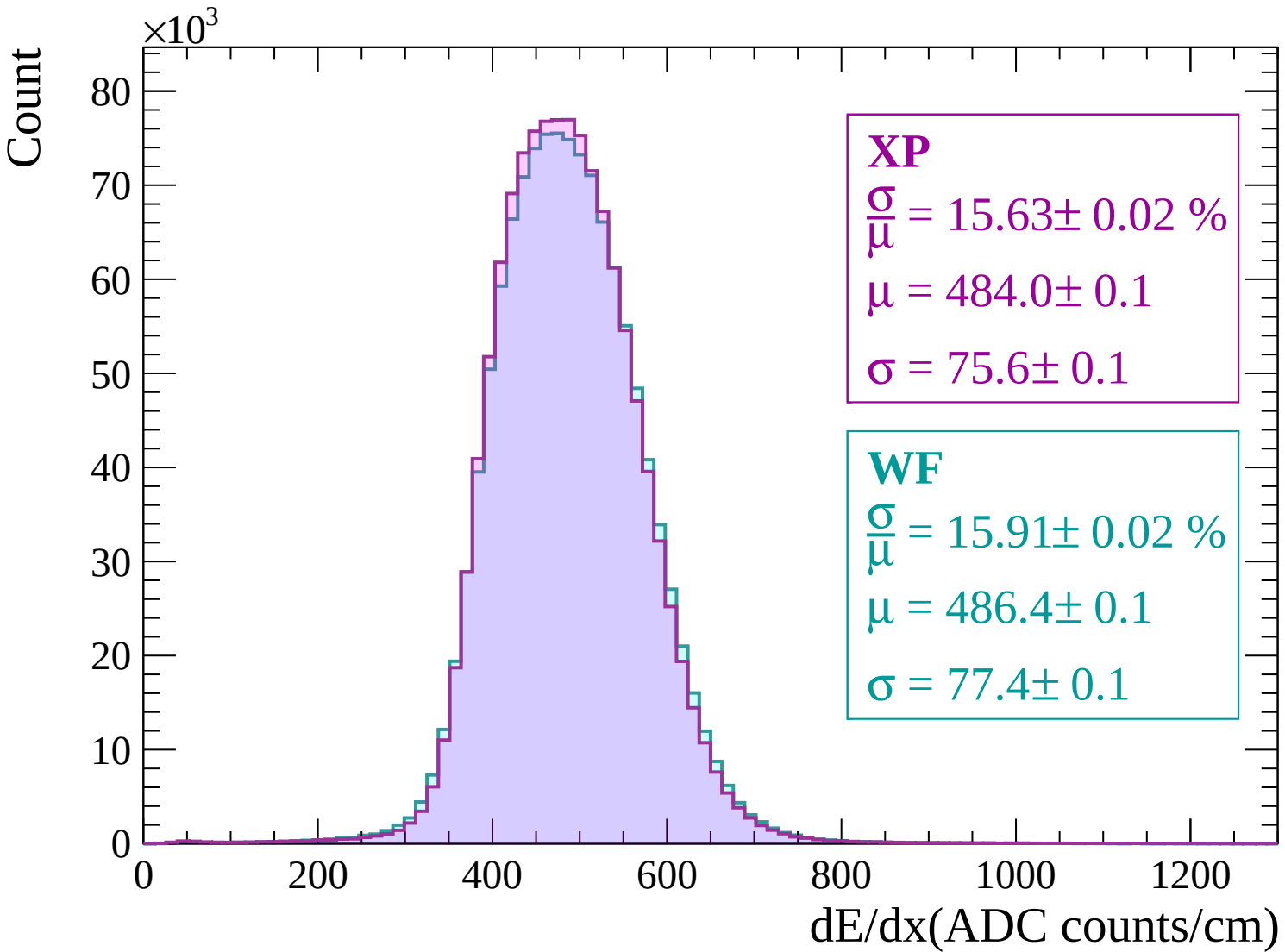
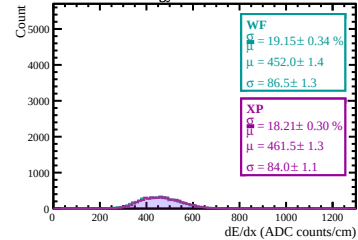
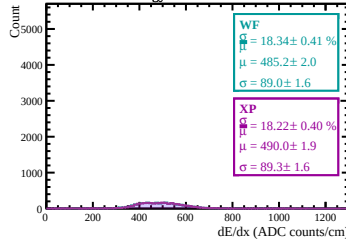




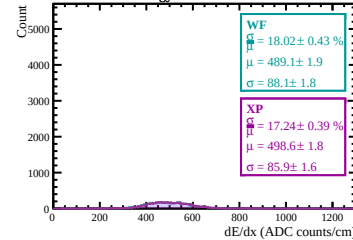
Energy loss in ERAM 24



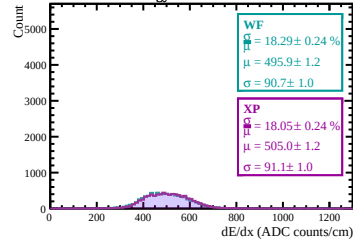
Energy loss in ERAM 30



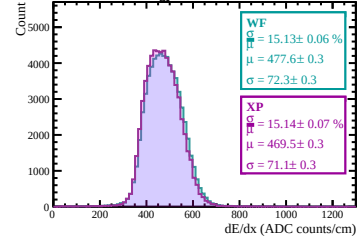
Energy loss in ERAM 68



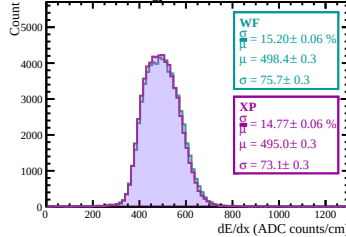
Energy loss in ERAM 19



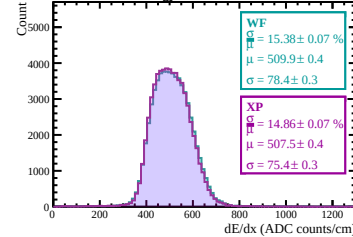
Energy loss in ERAM 21



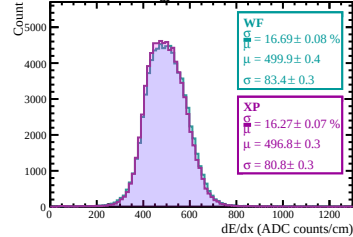
Energy loss in ERAM 13



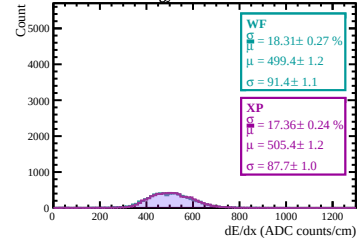
Energy loss in ERAM 9



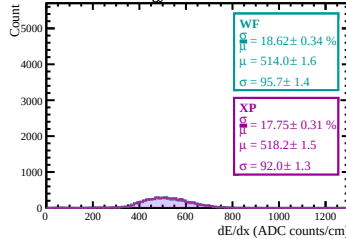
Energy loss in ERAM 2



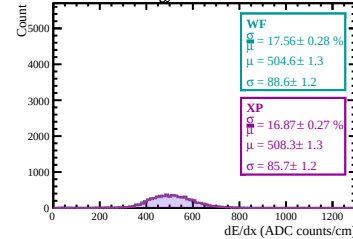
Energy loss in ERAM 26



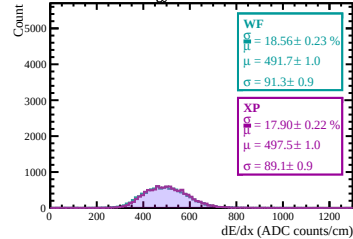
Energy loss in ERAM 17



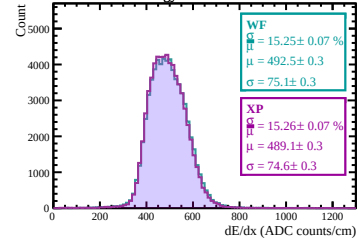
Energy loss in ERAM 23



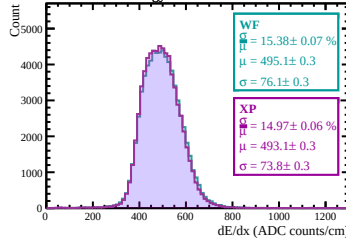
Energy loss in ERAM 29



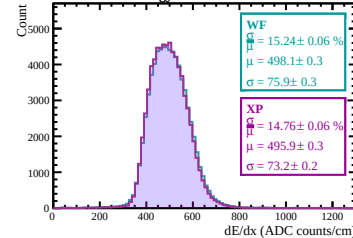
Energy loss in ERAM 1



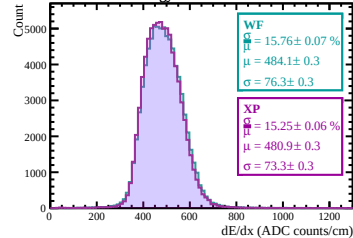
Energy loss in ERAM 10

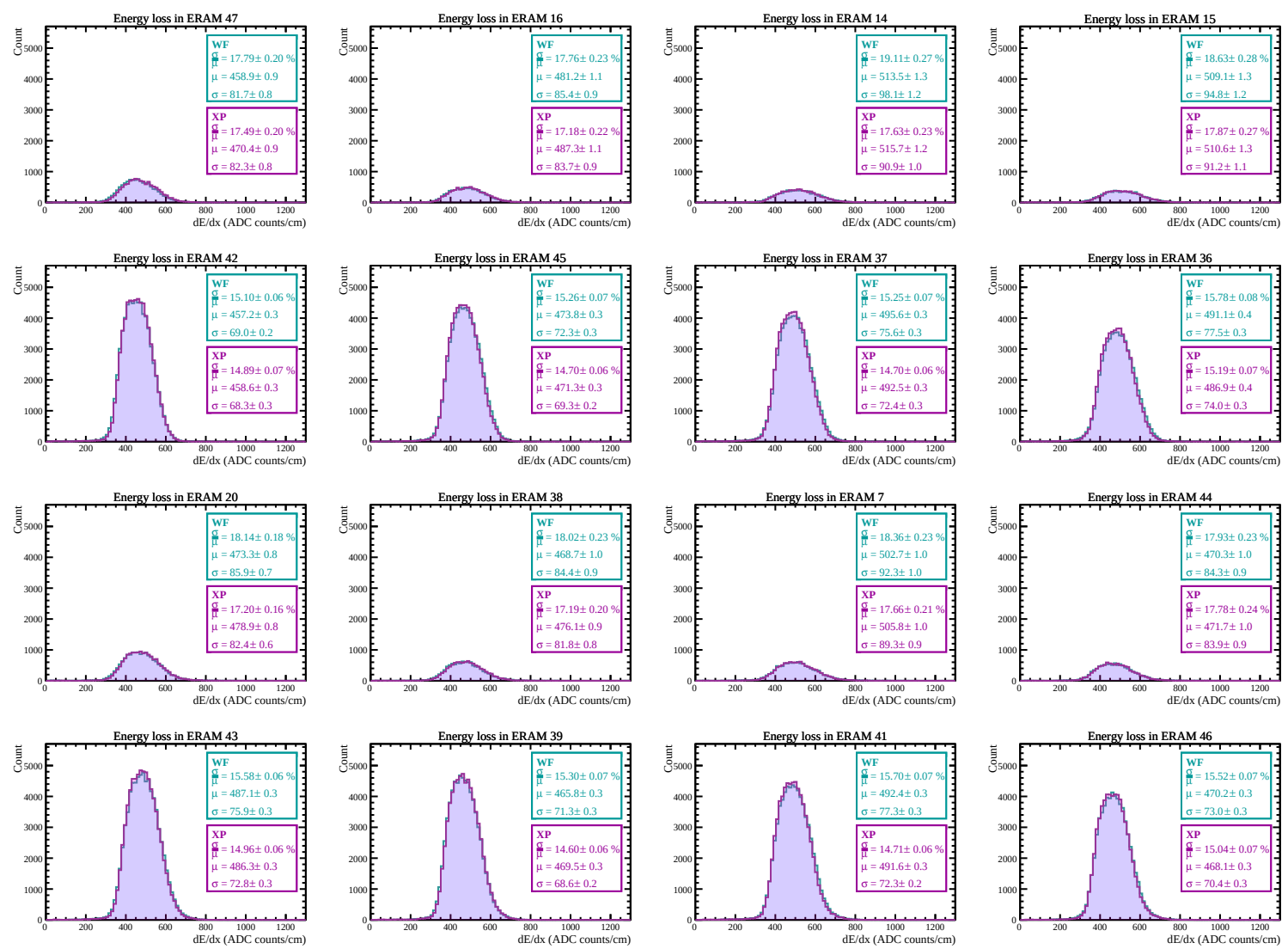


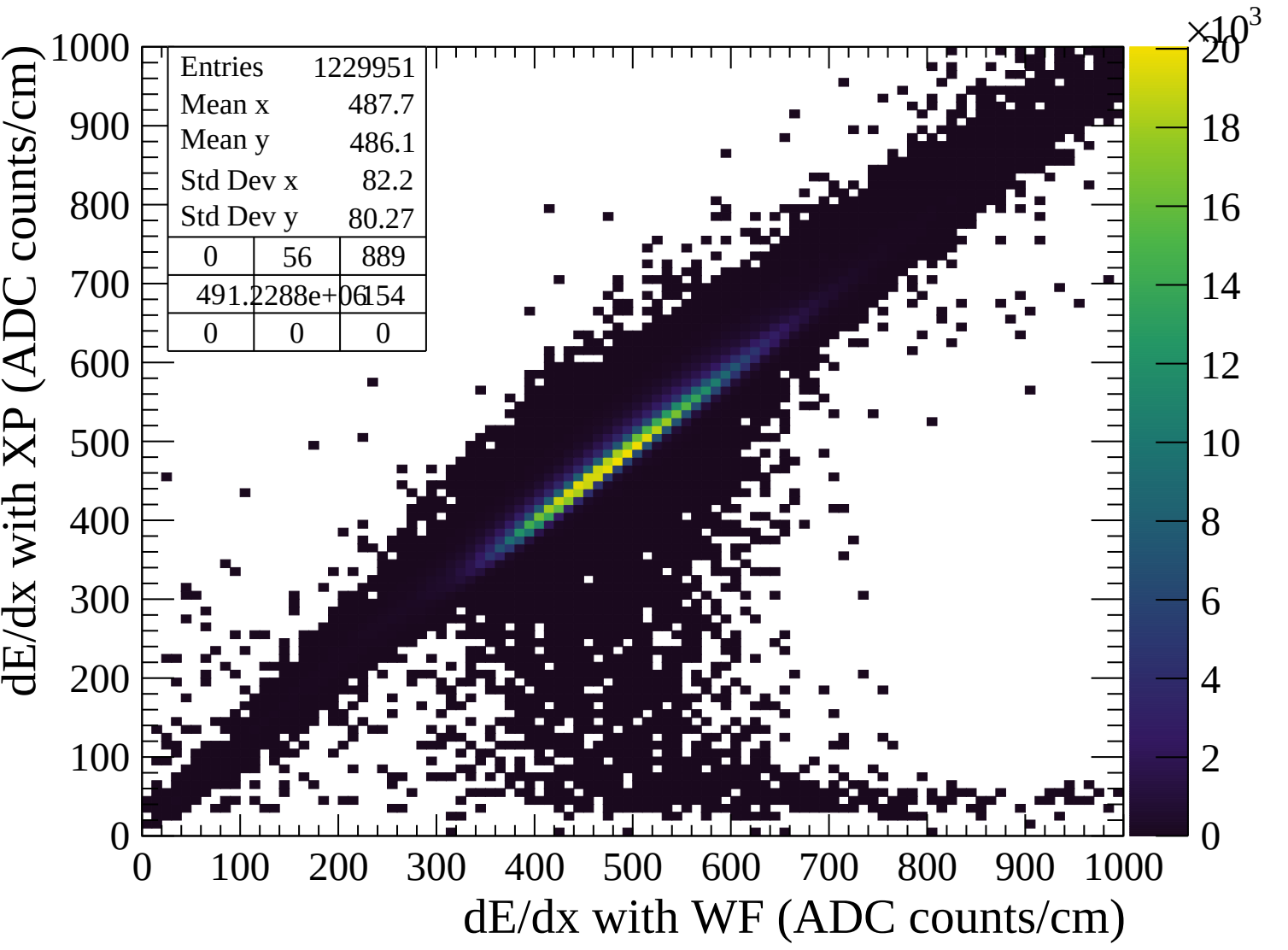
Energy loss in ERAM 11



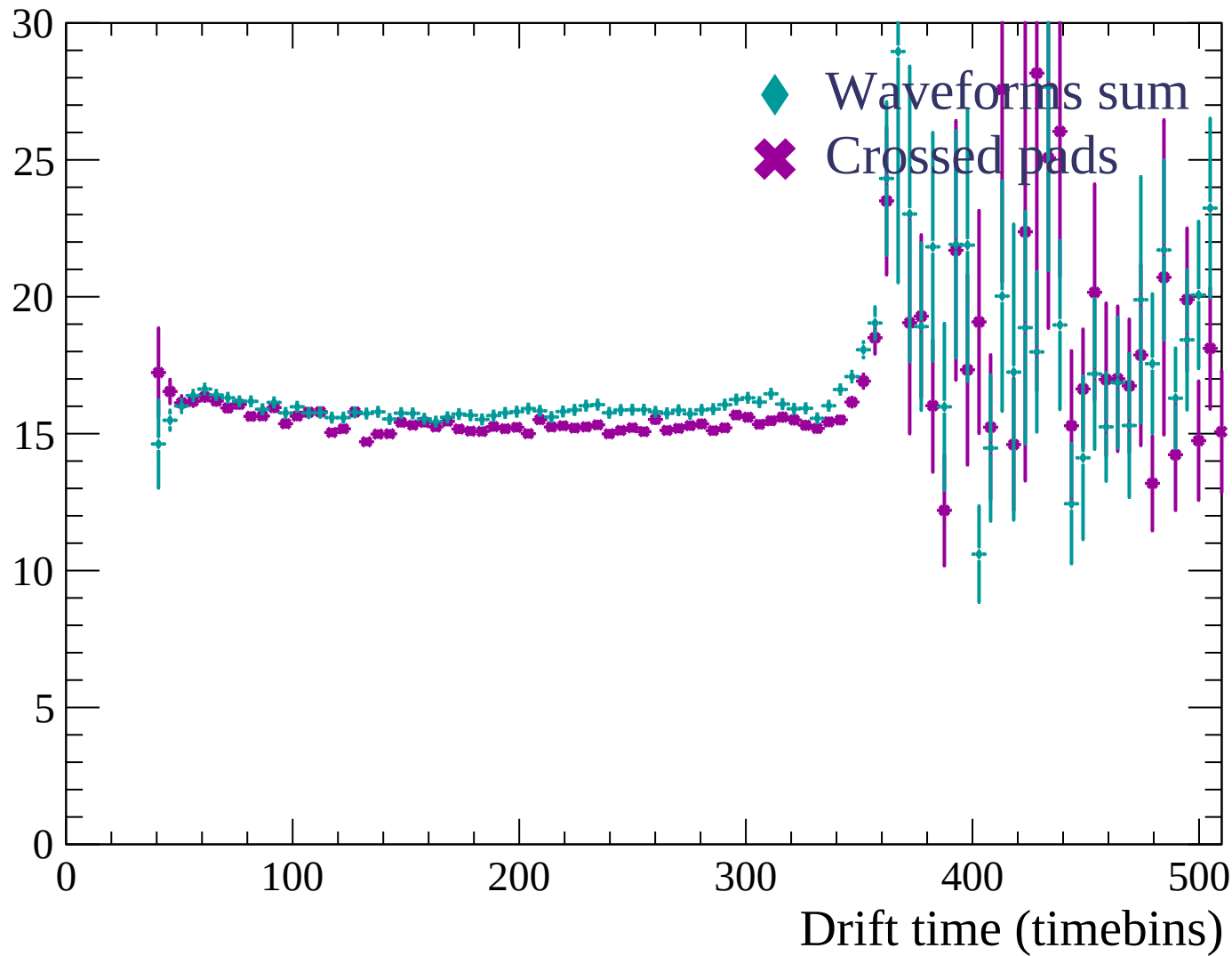
Energy loss in ERAM 3

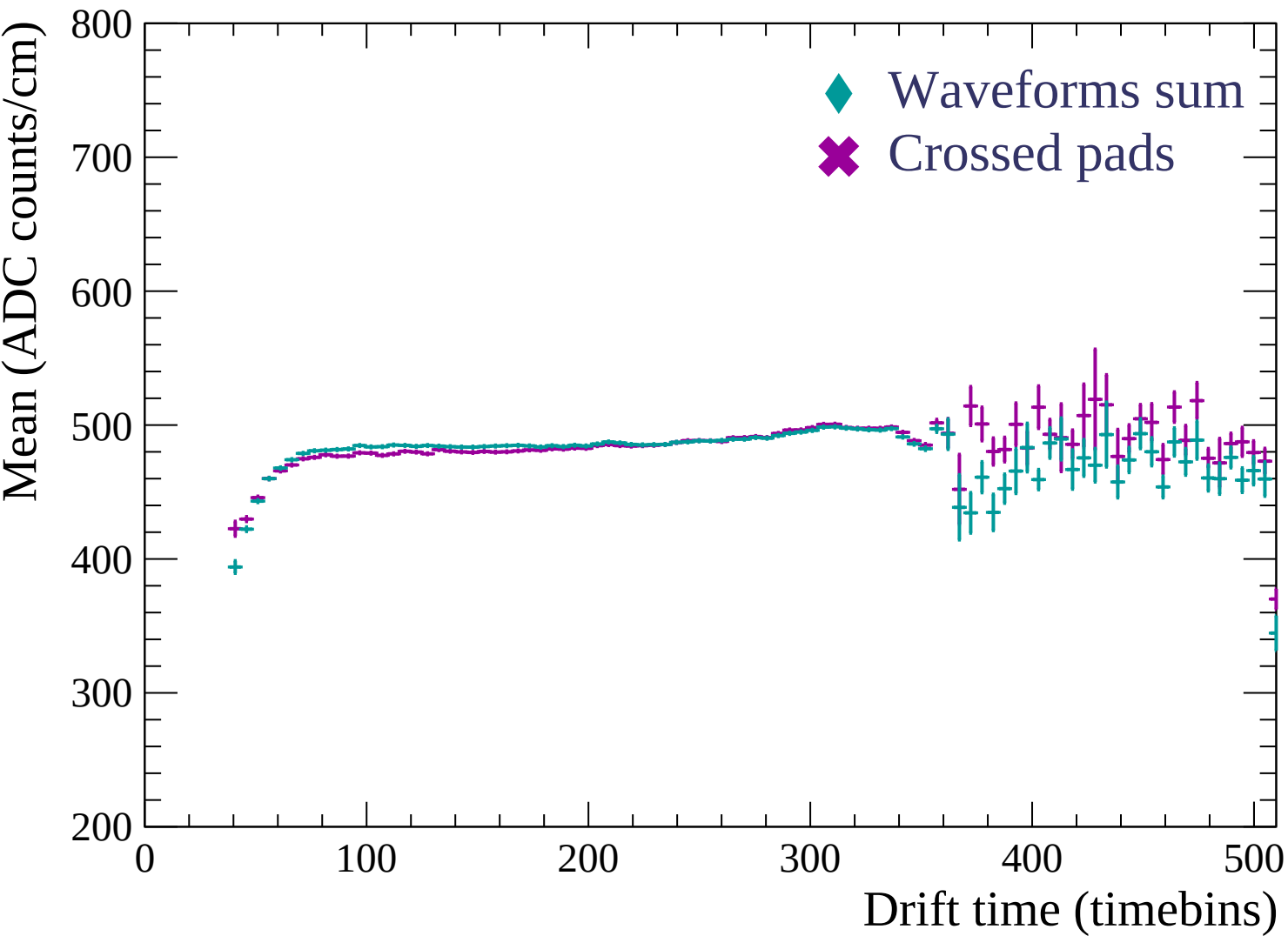


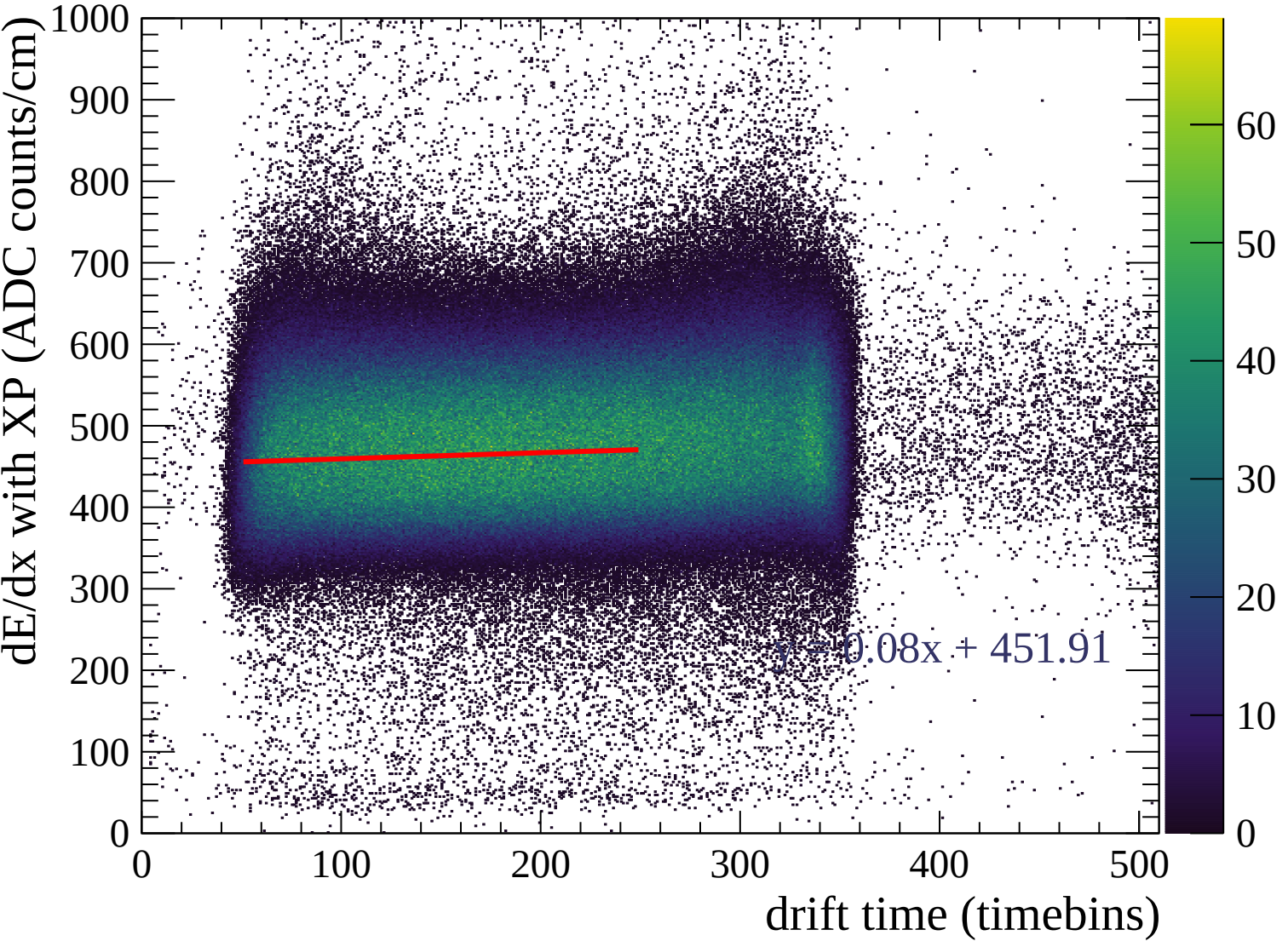


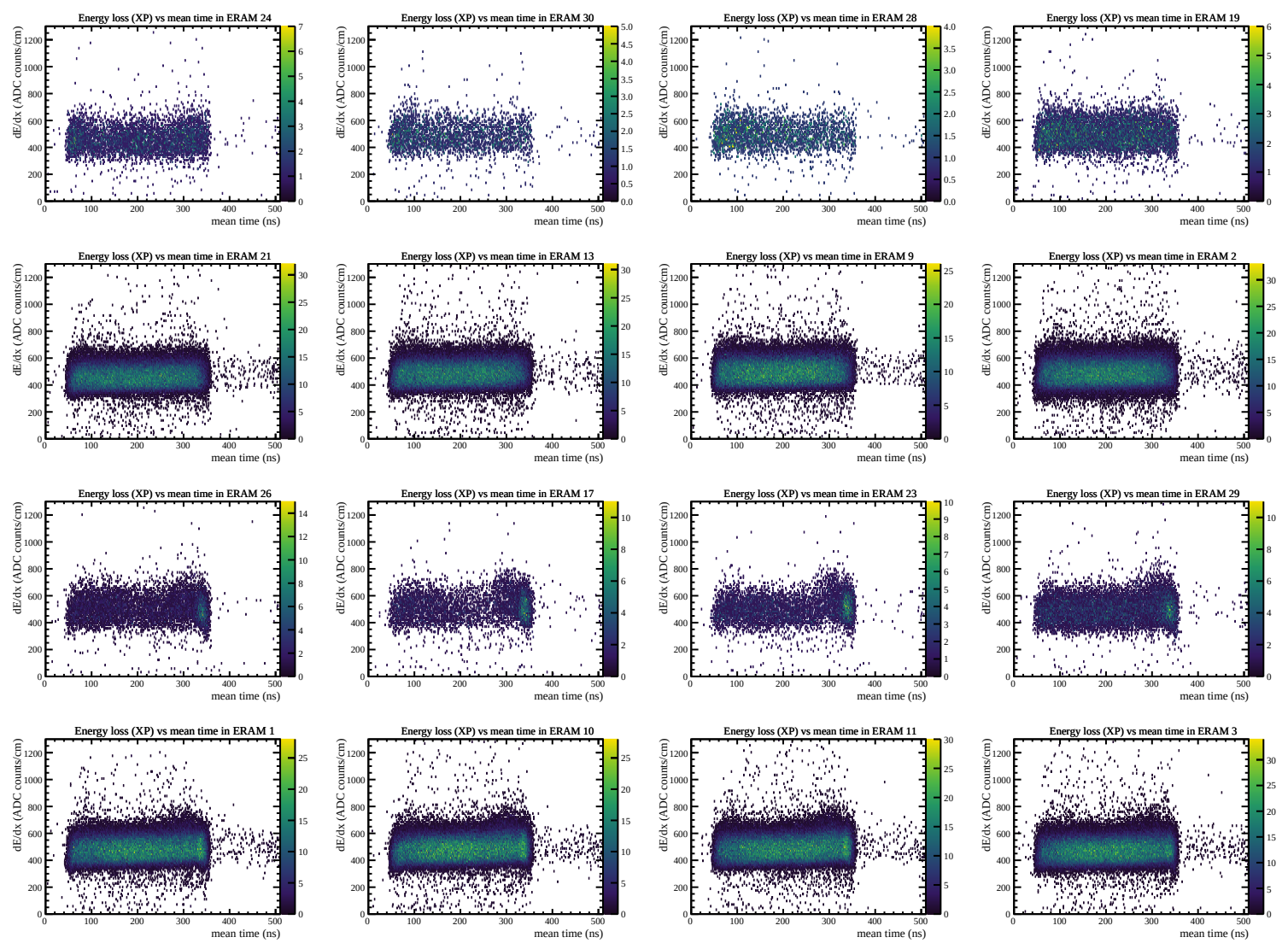


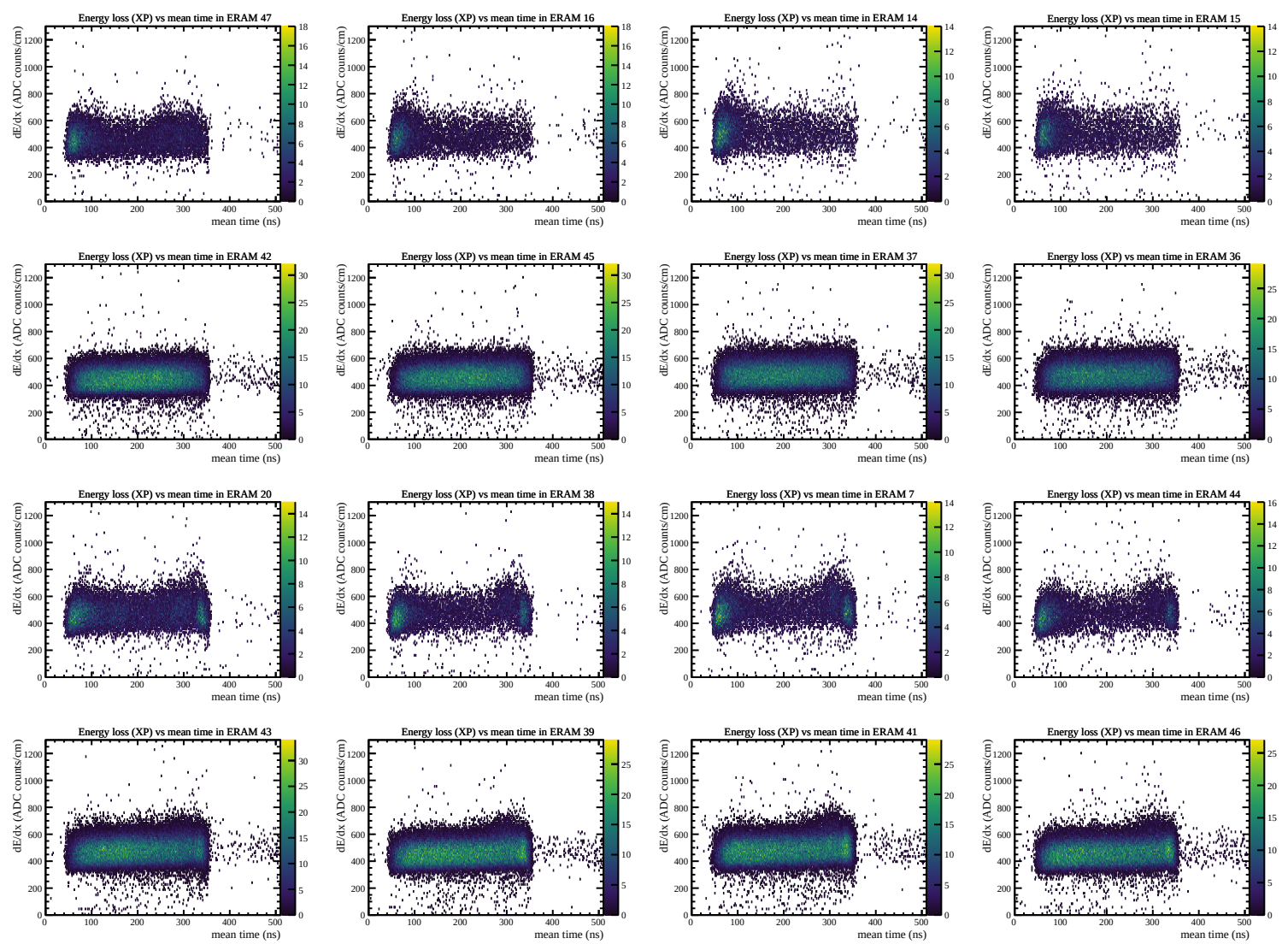
Resolution (%)

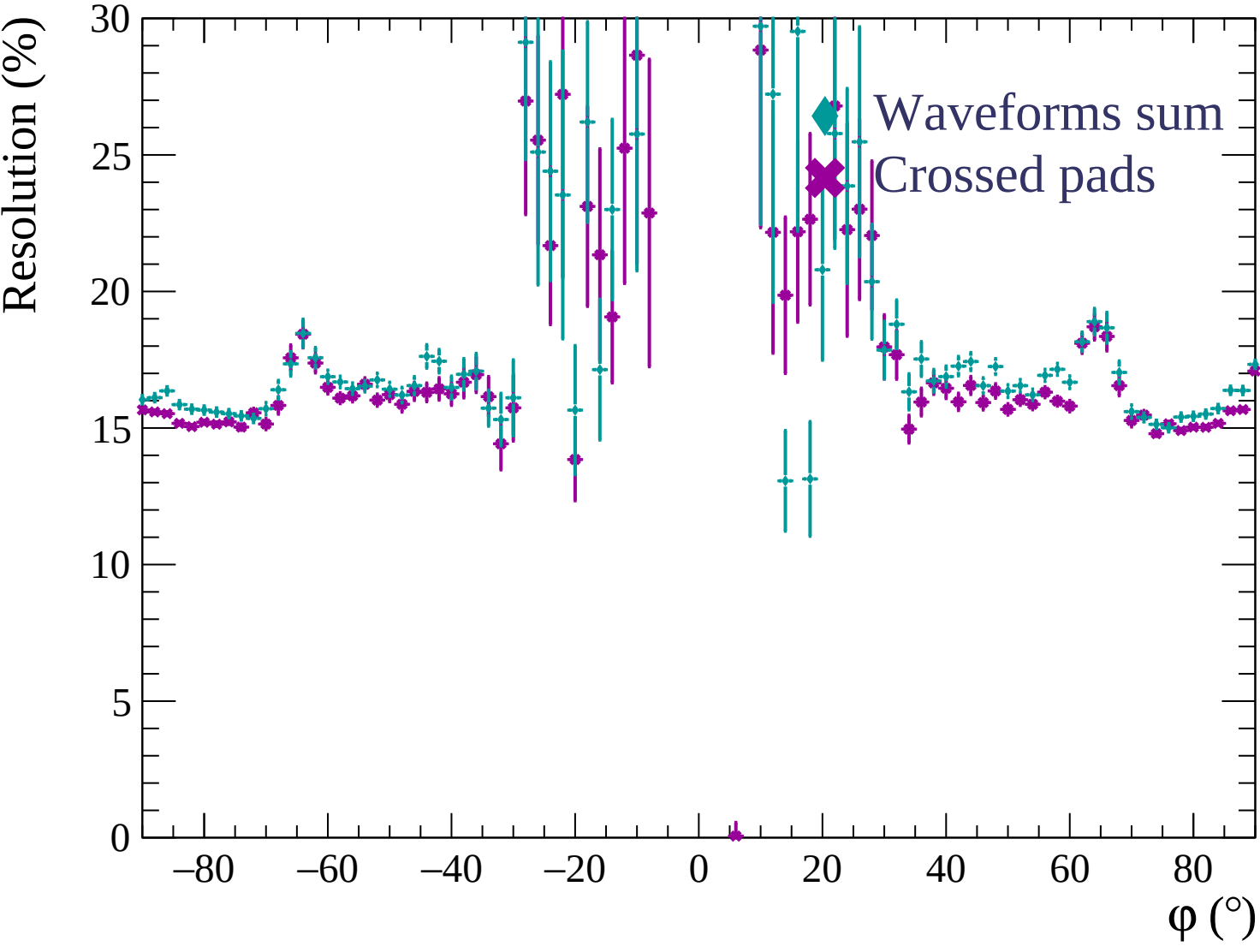


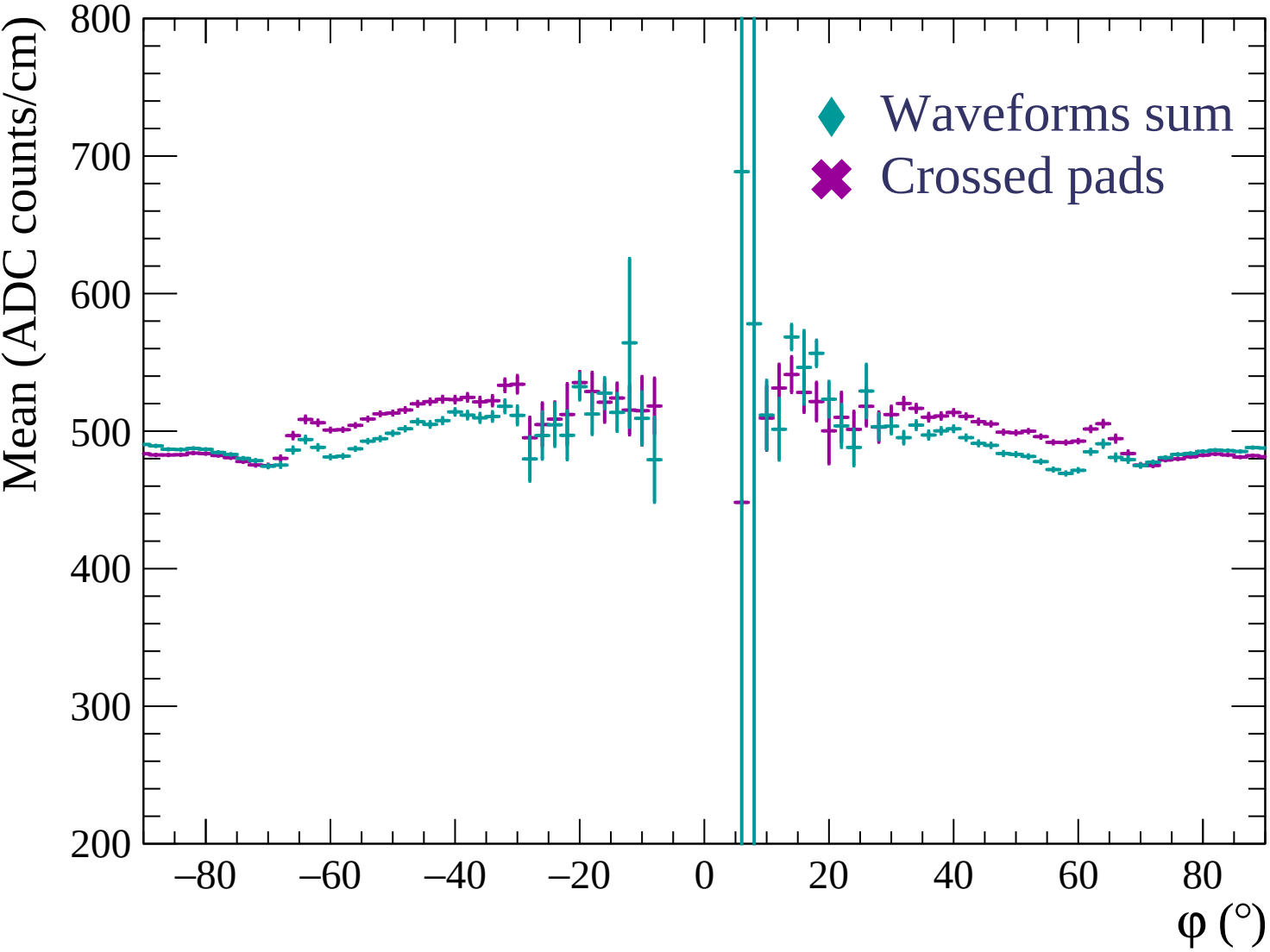


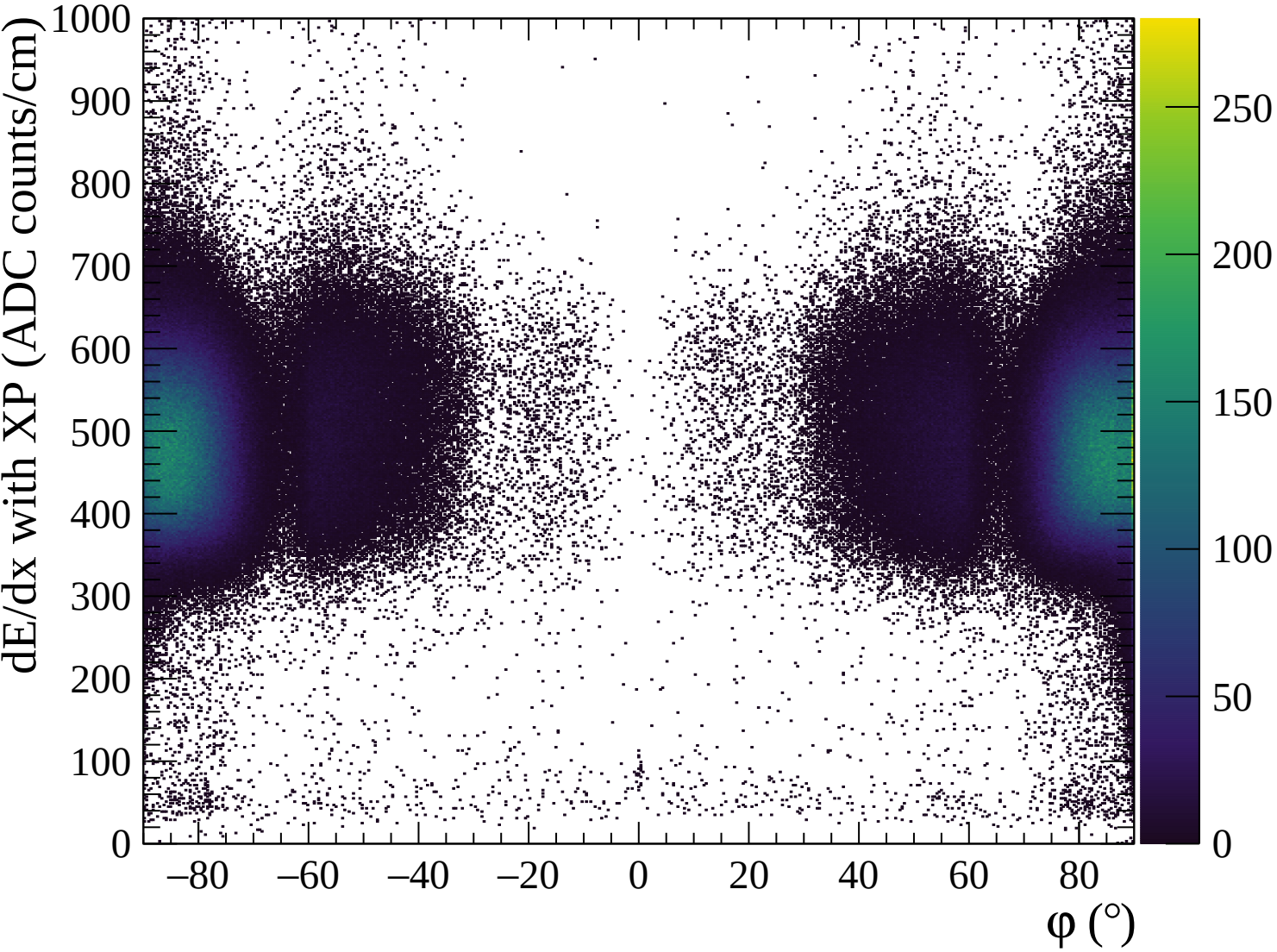


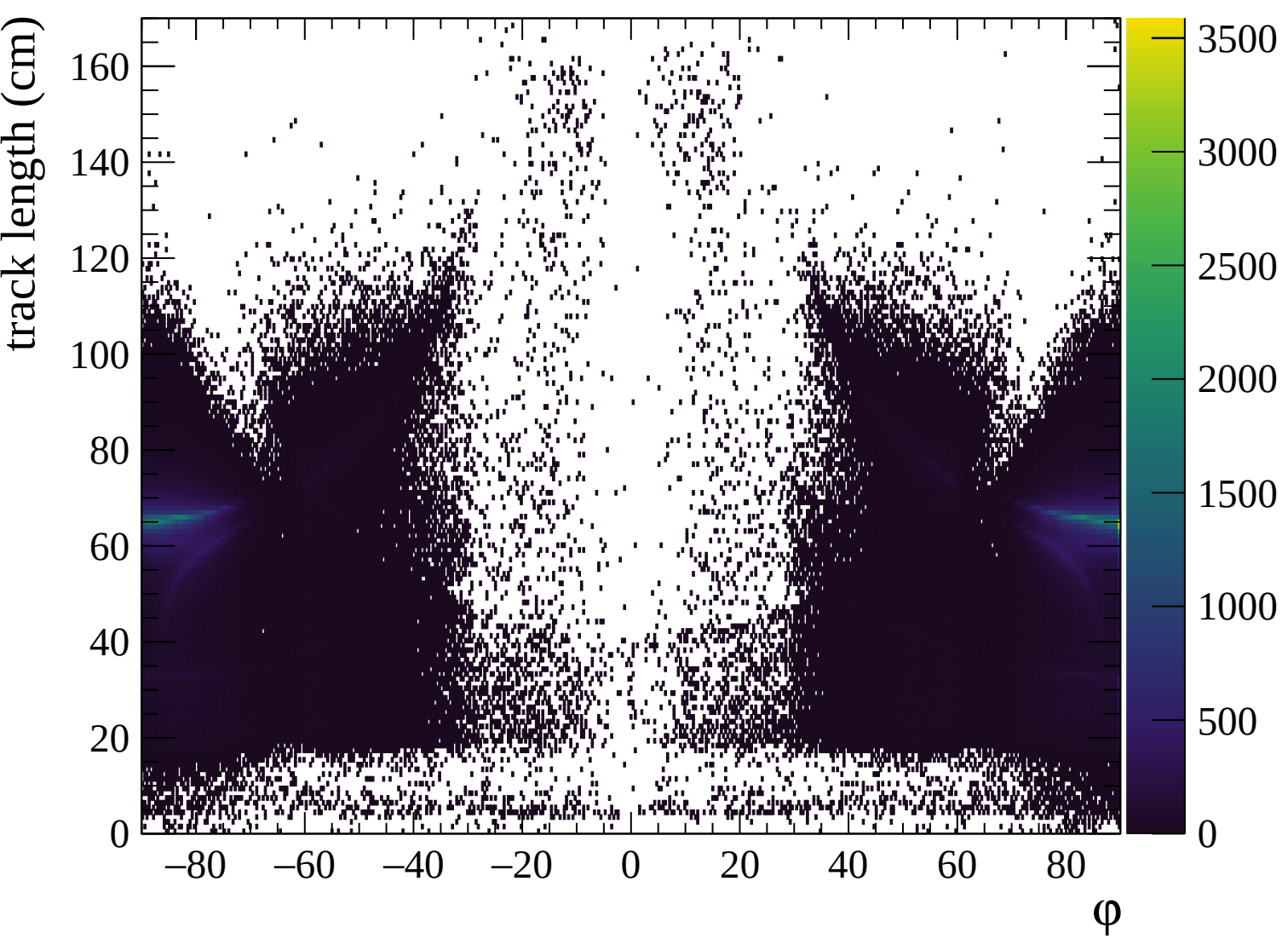


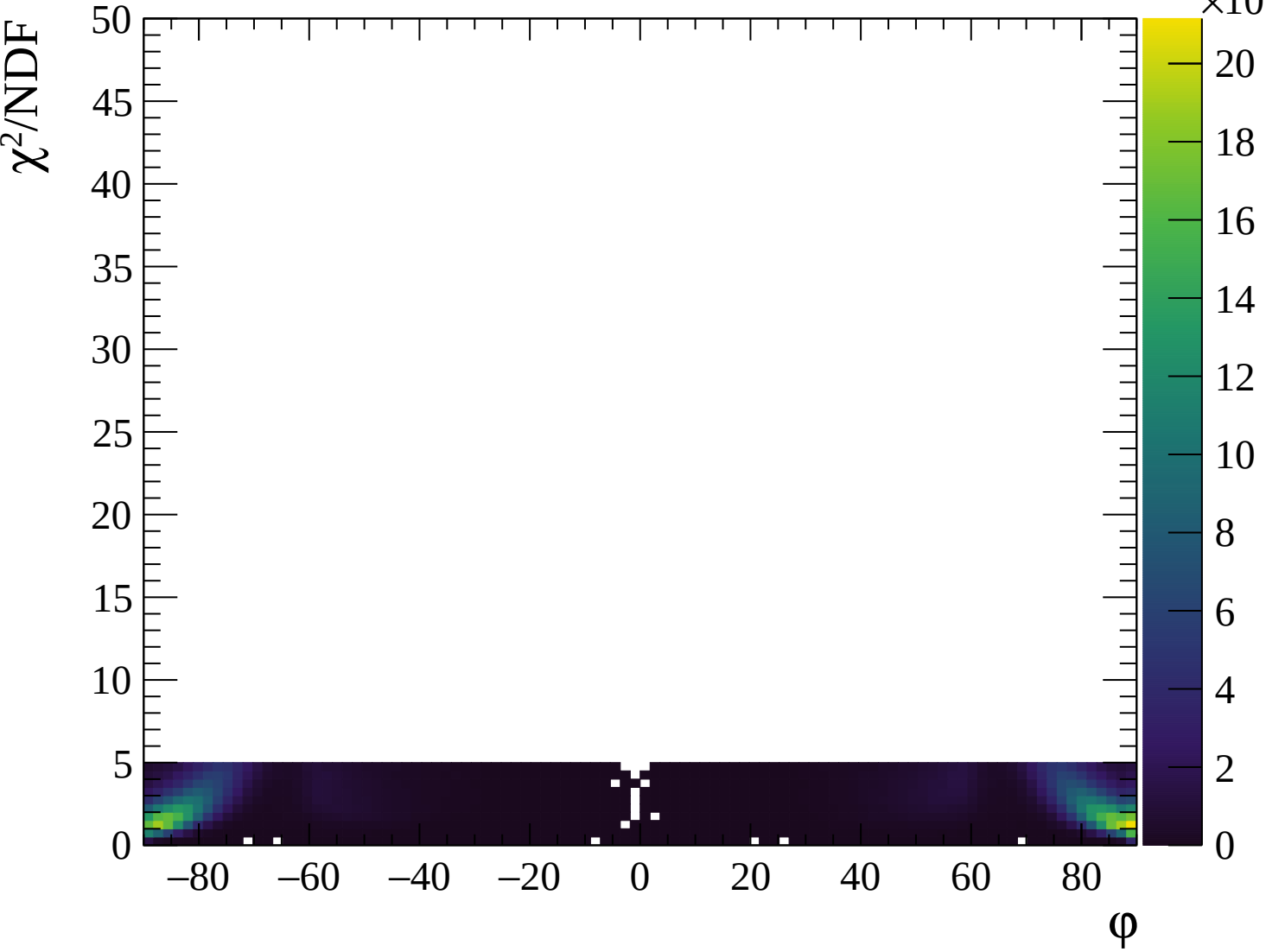


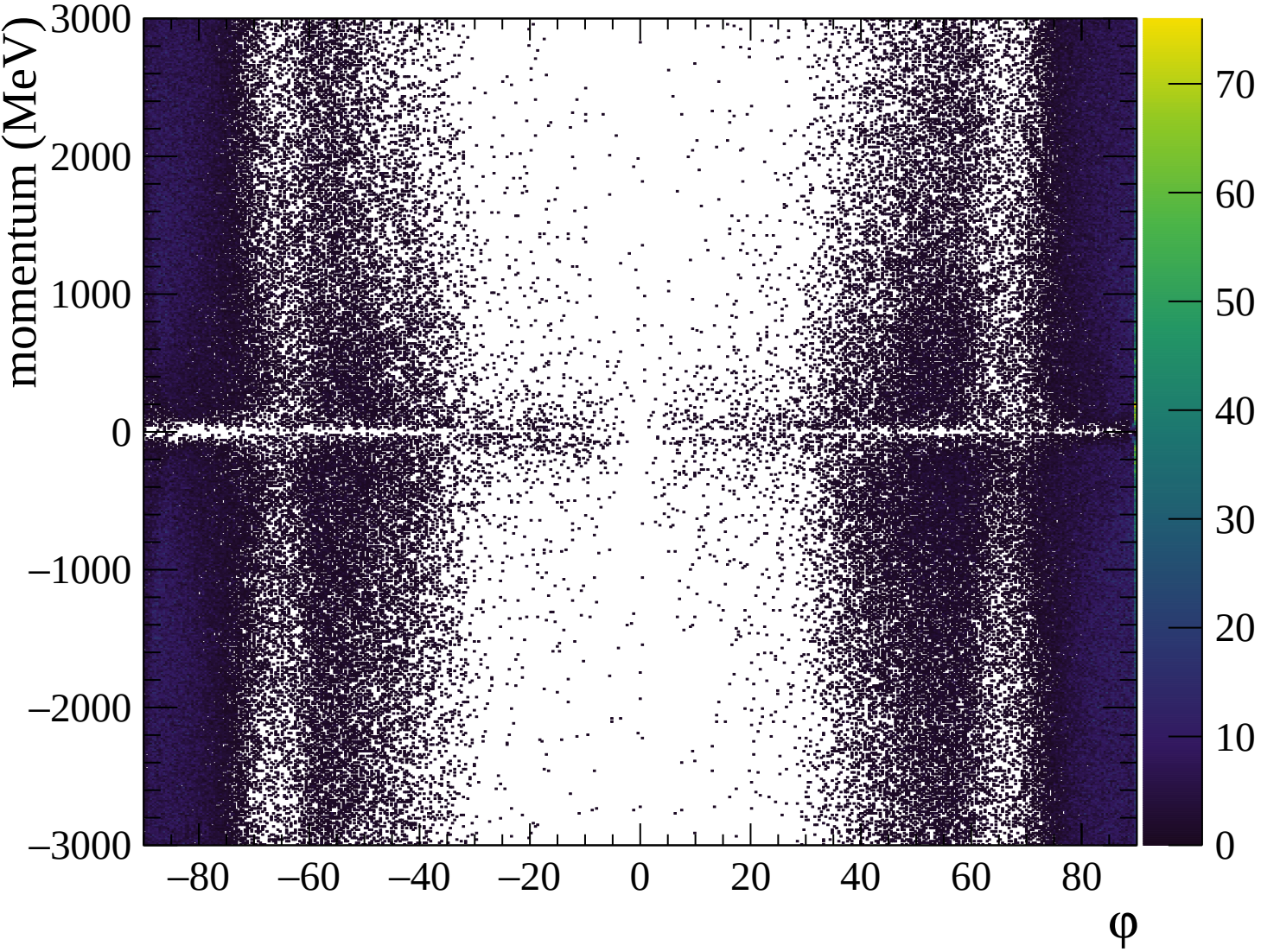


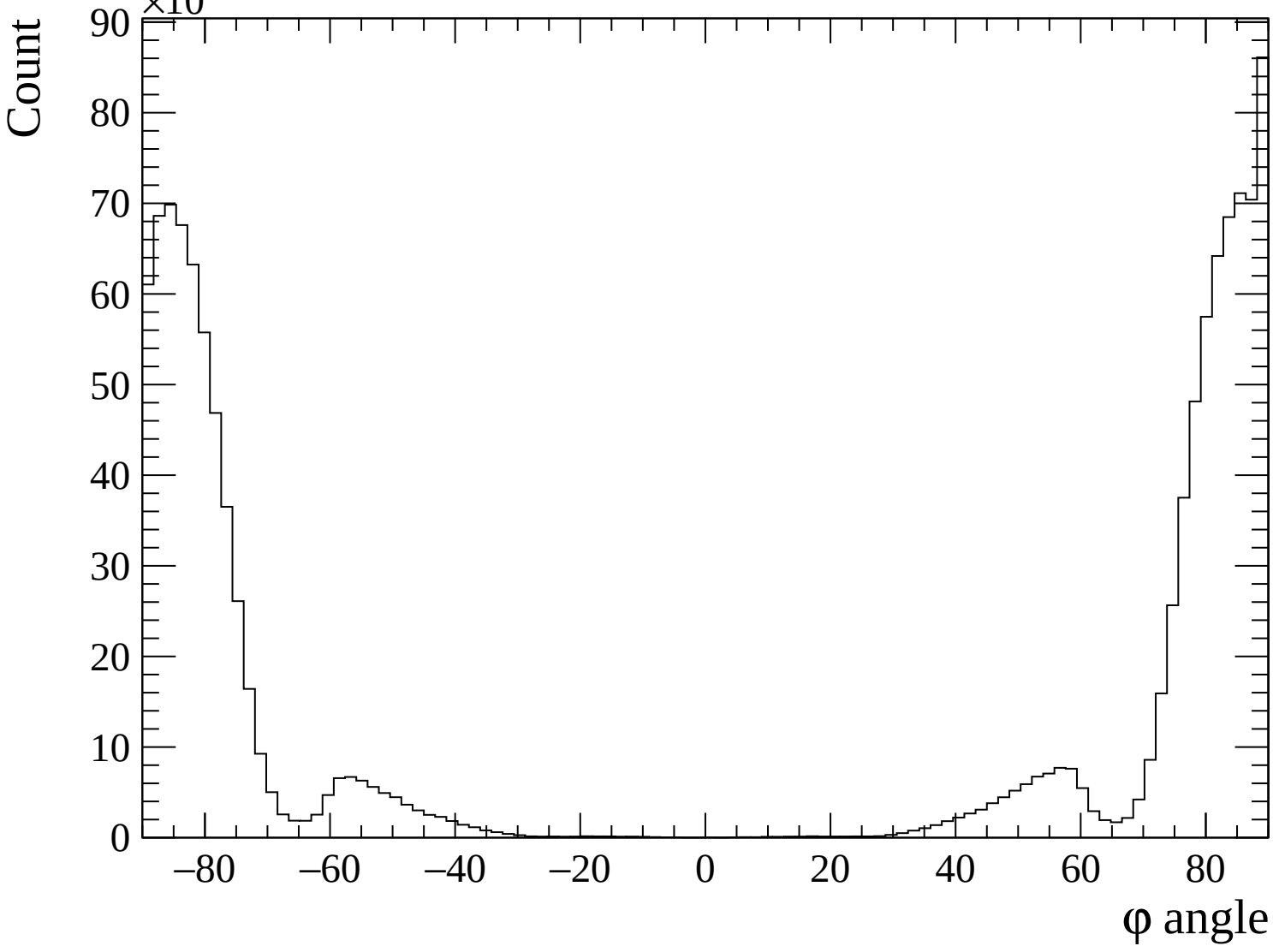


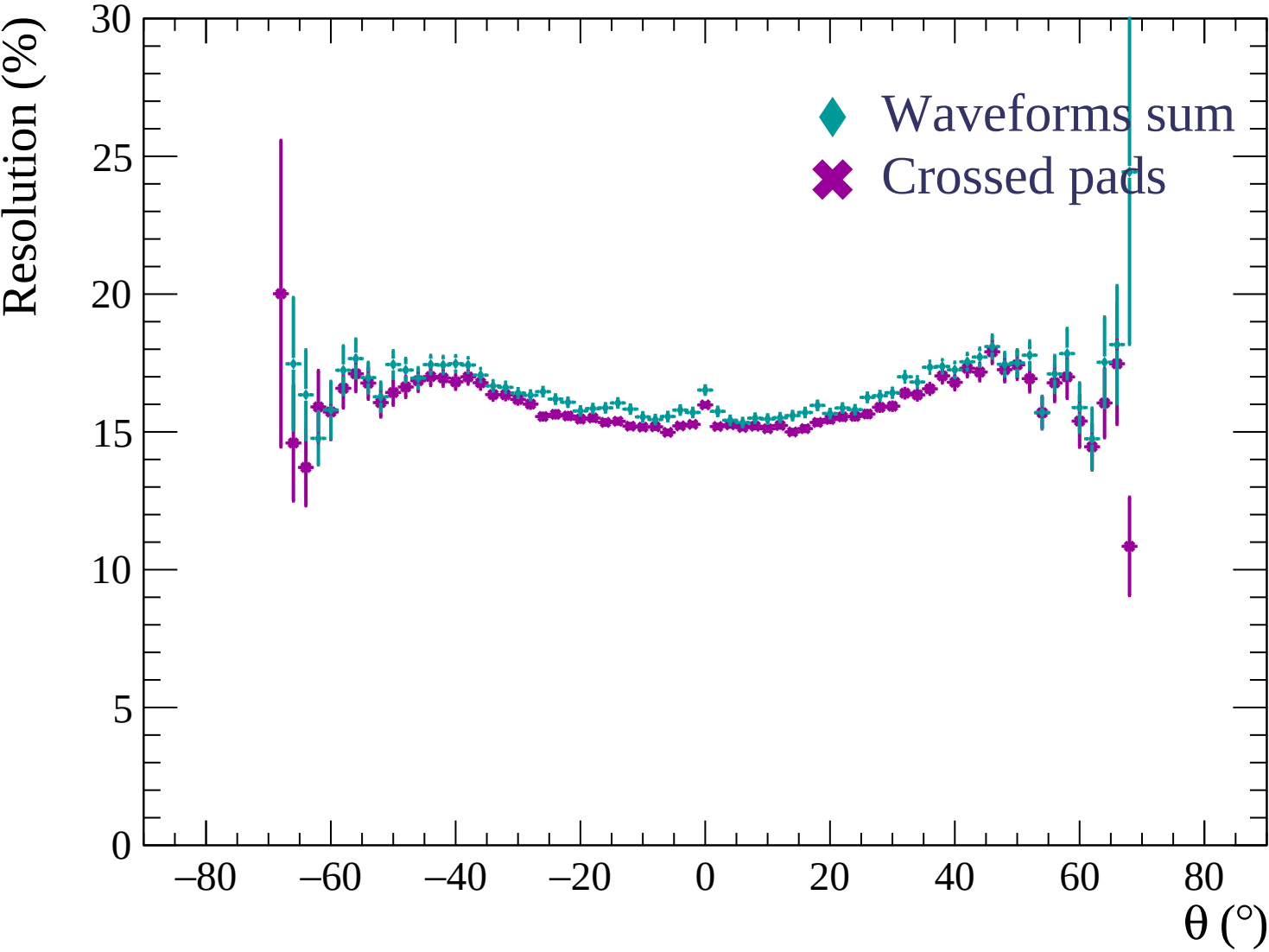


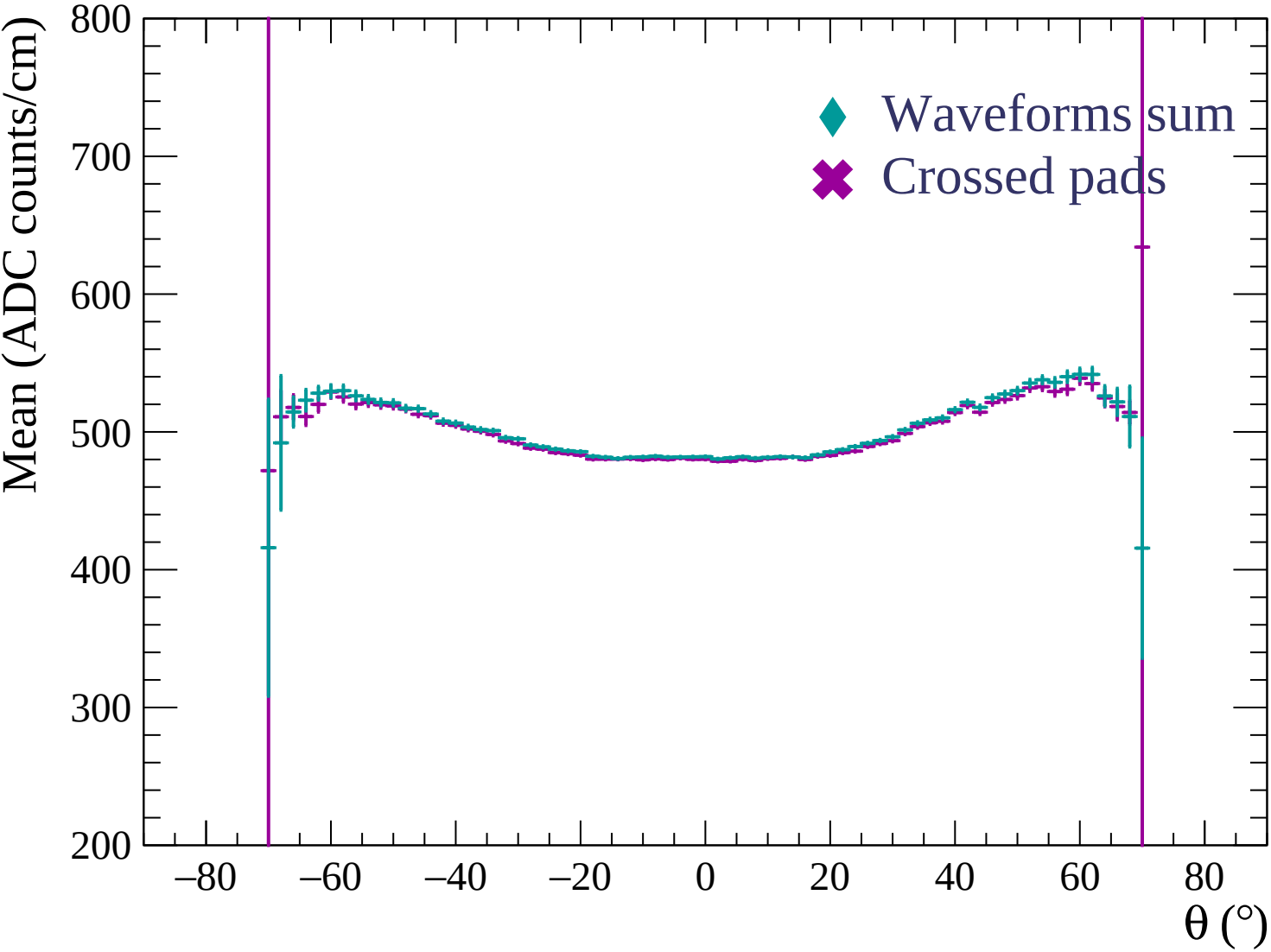


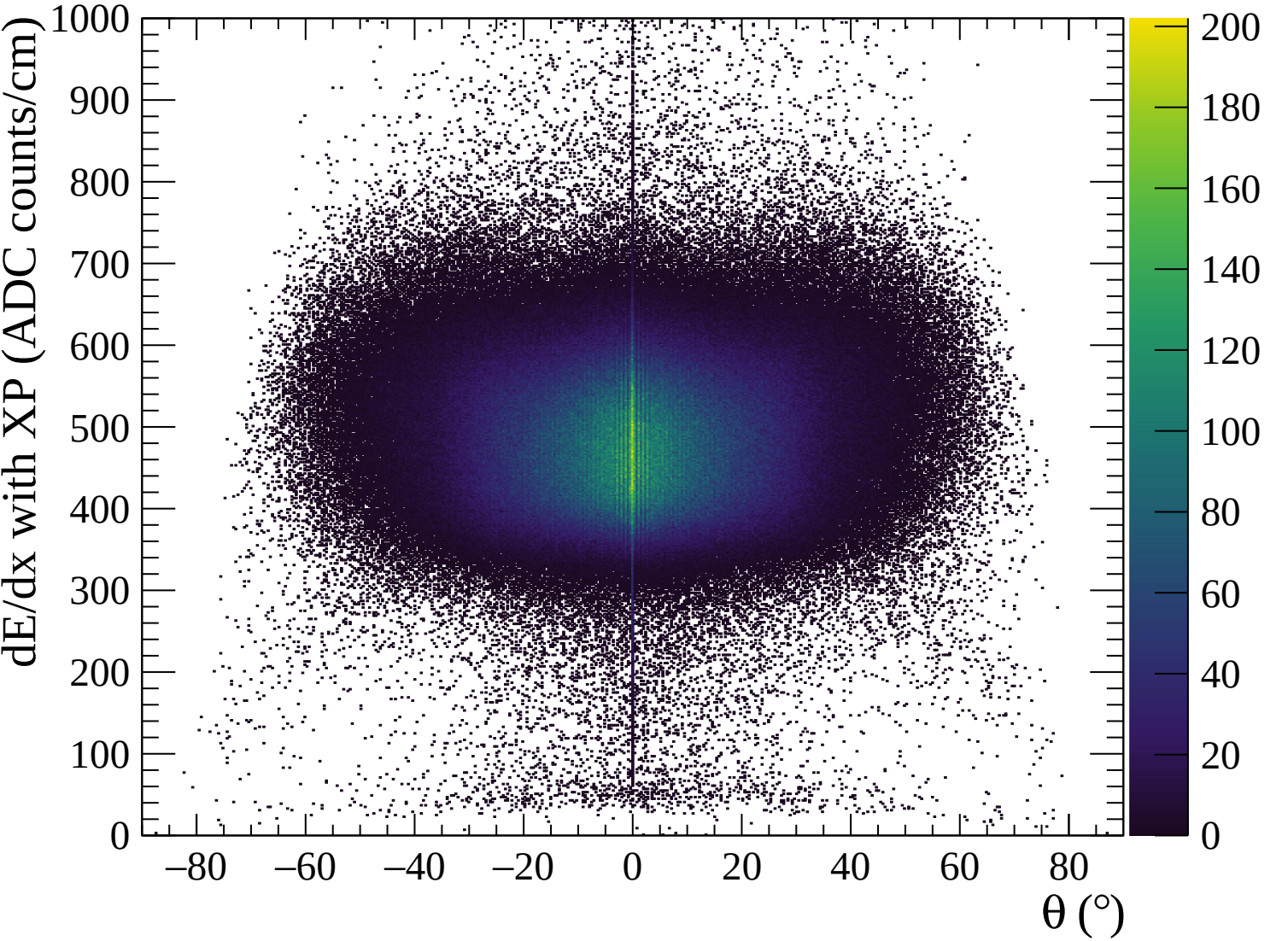


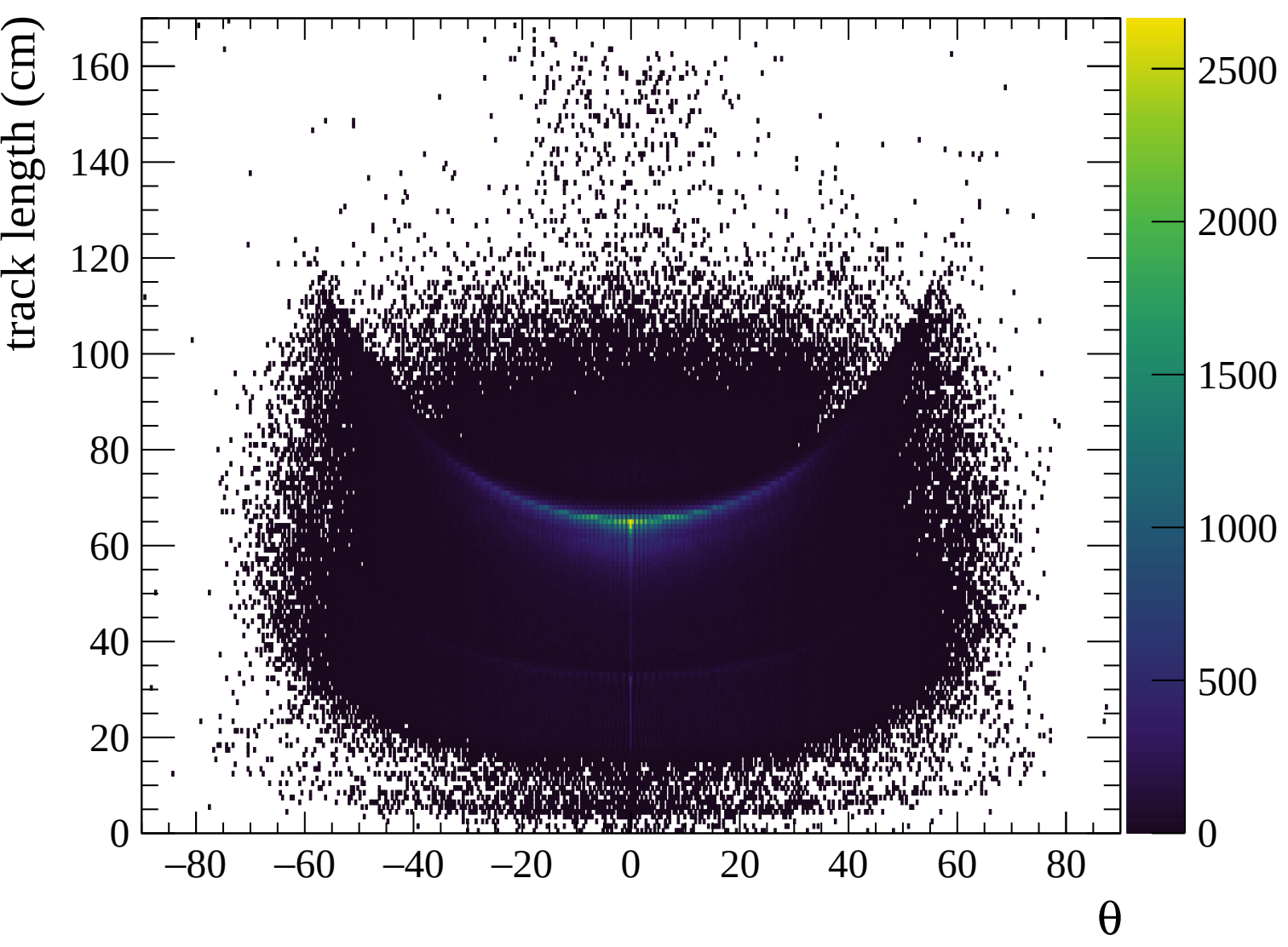


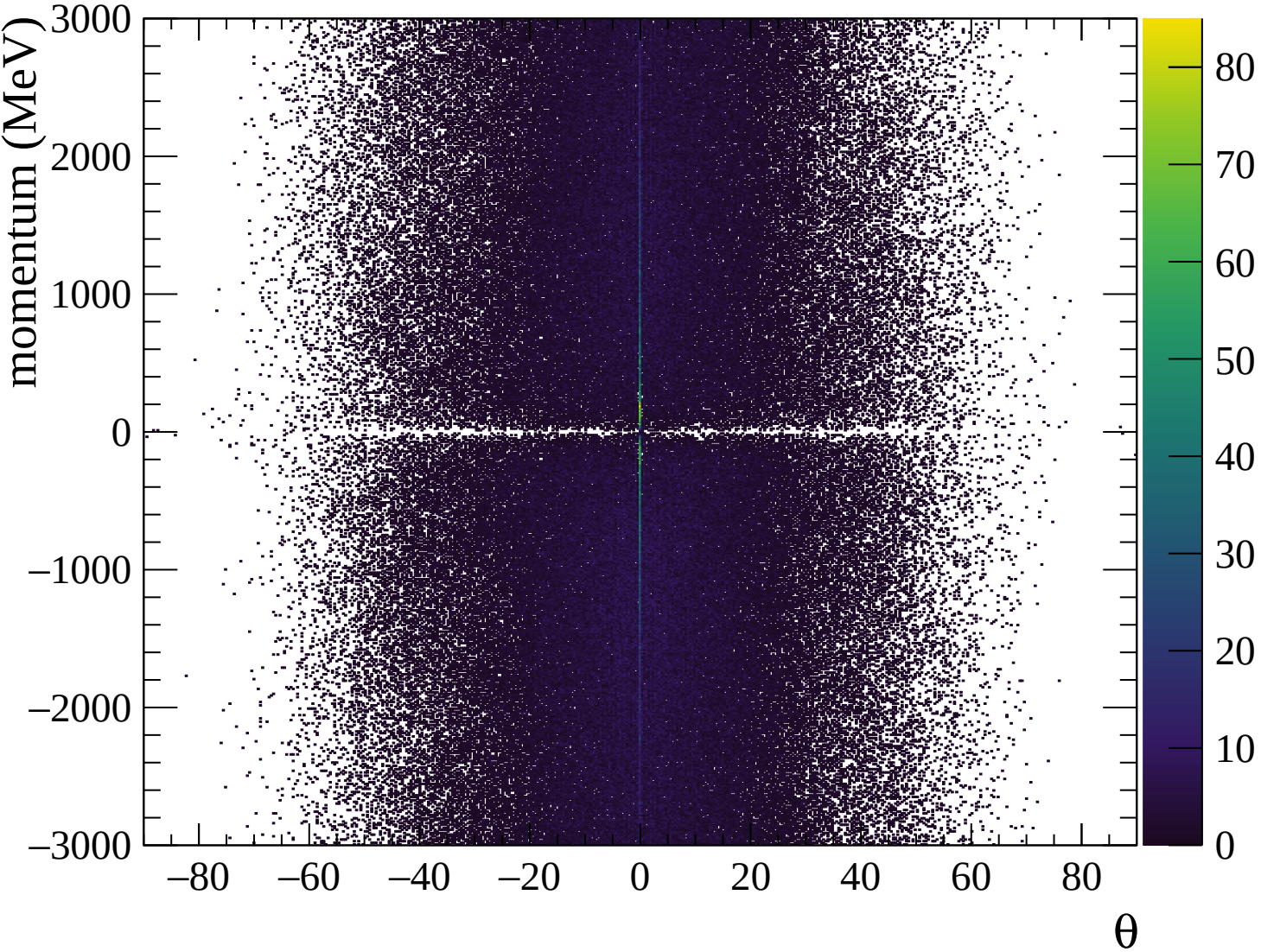


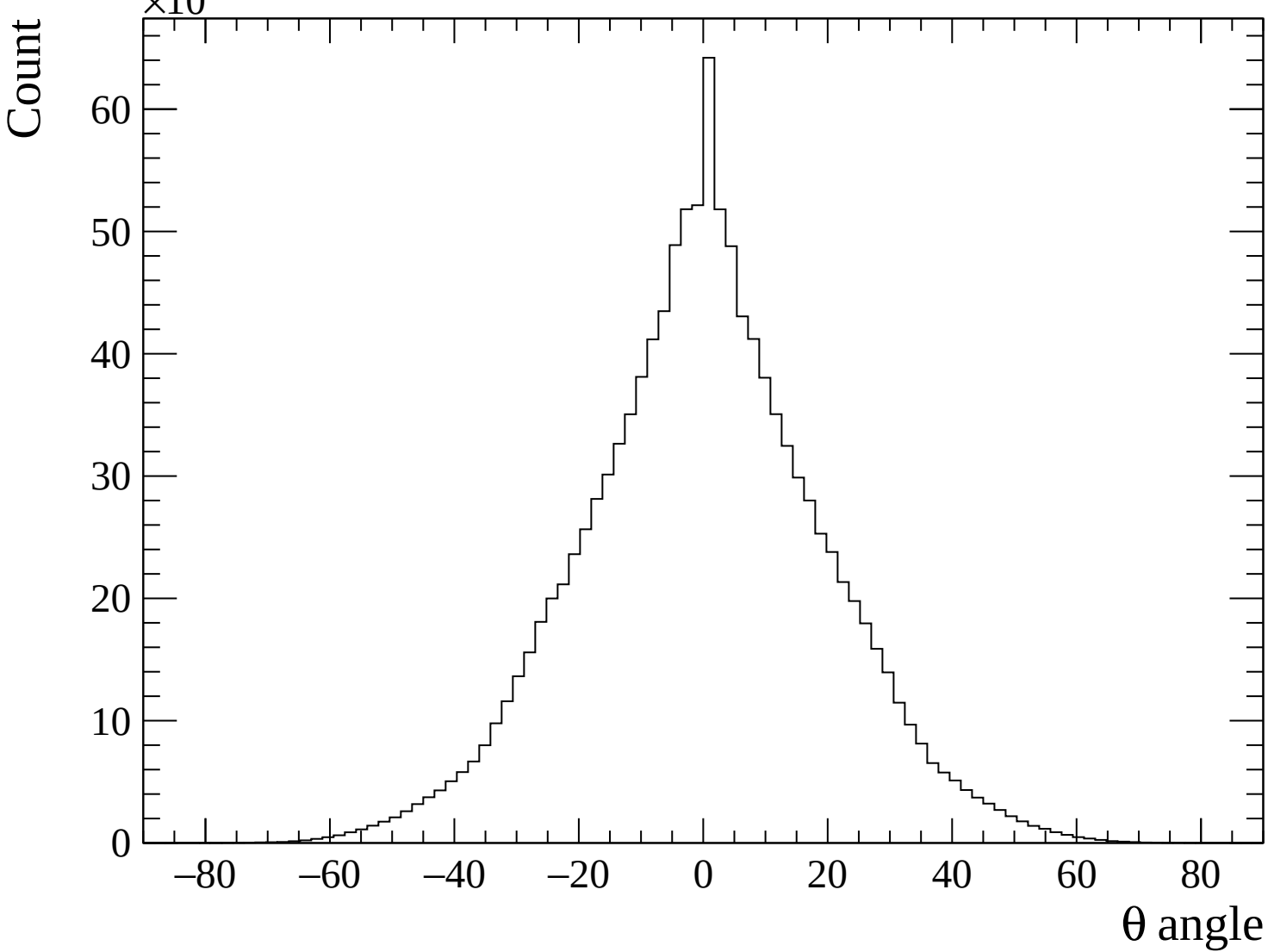


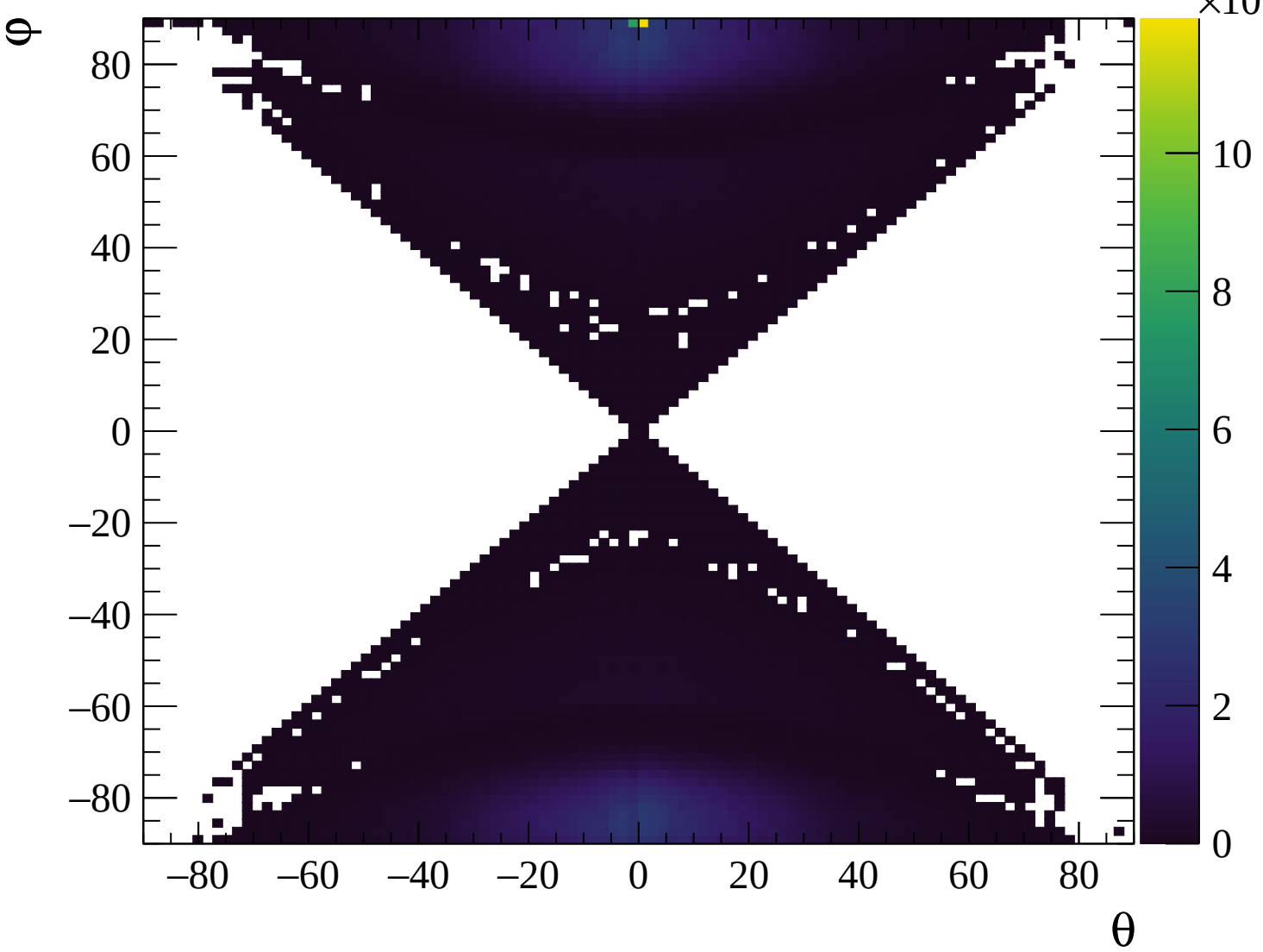


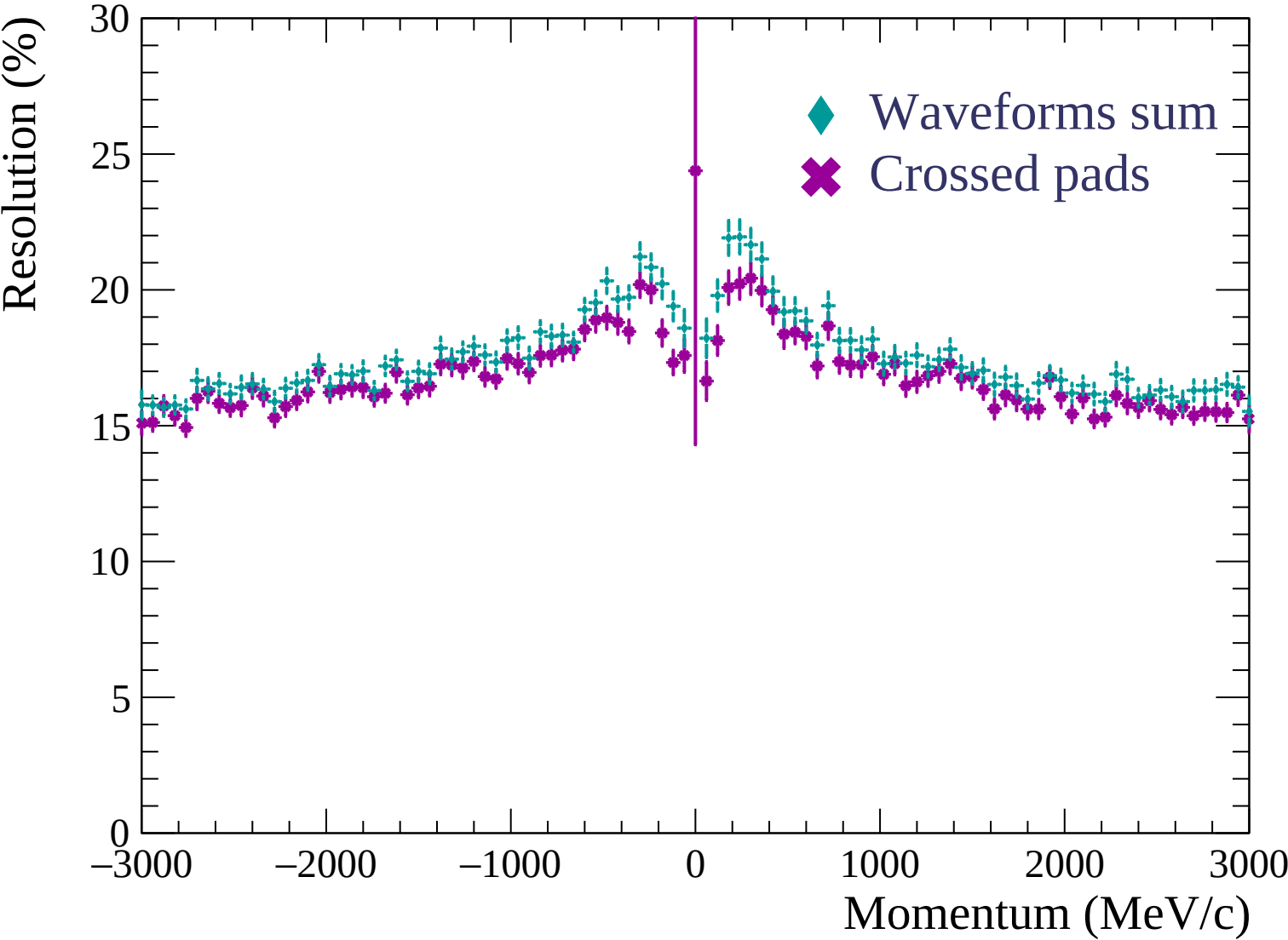


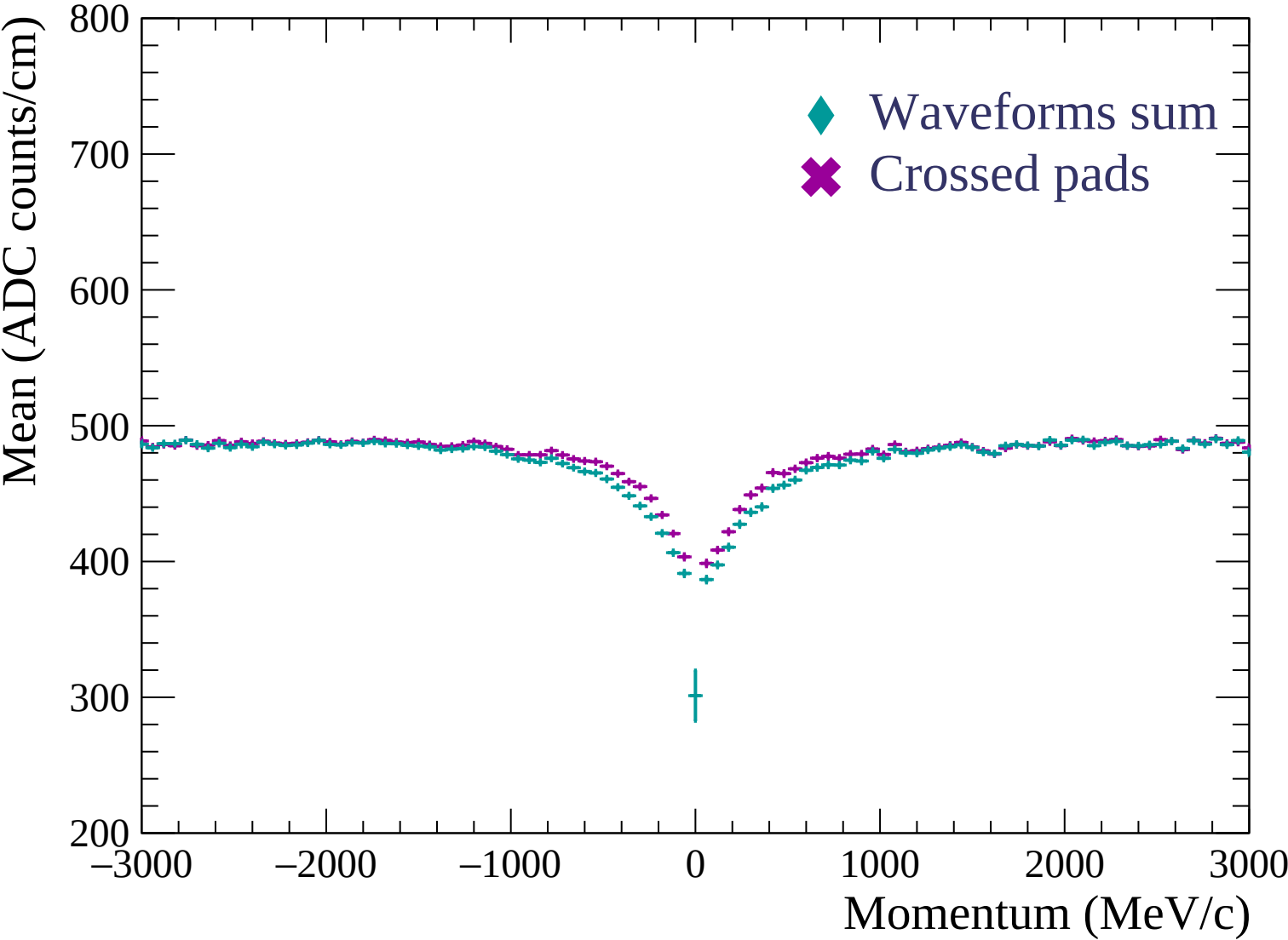


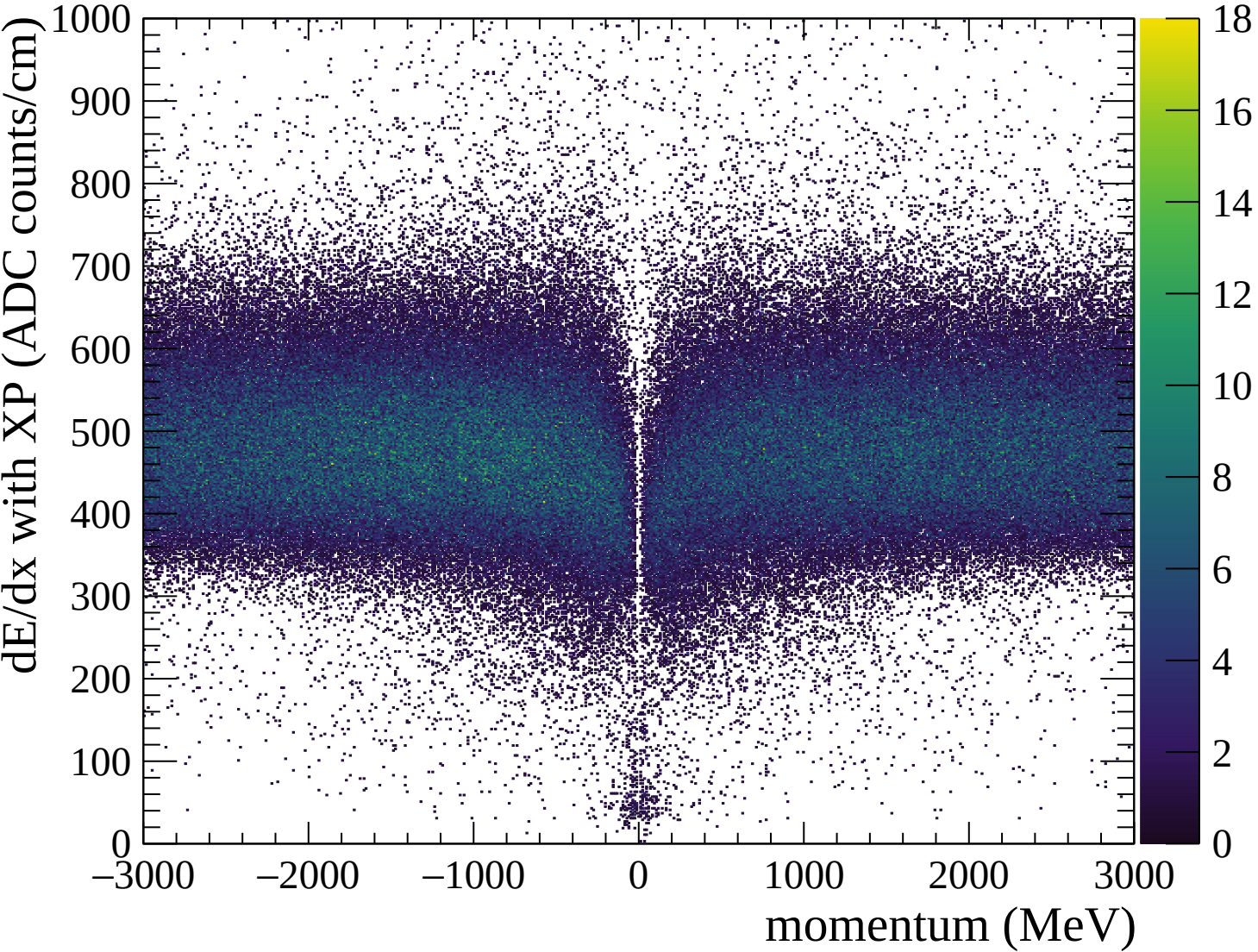


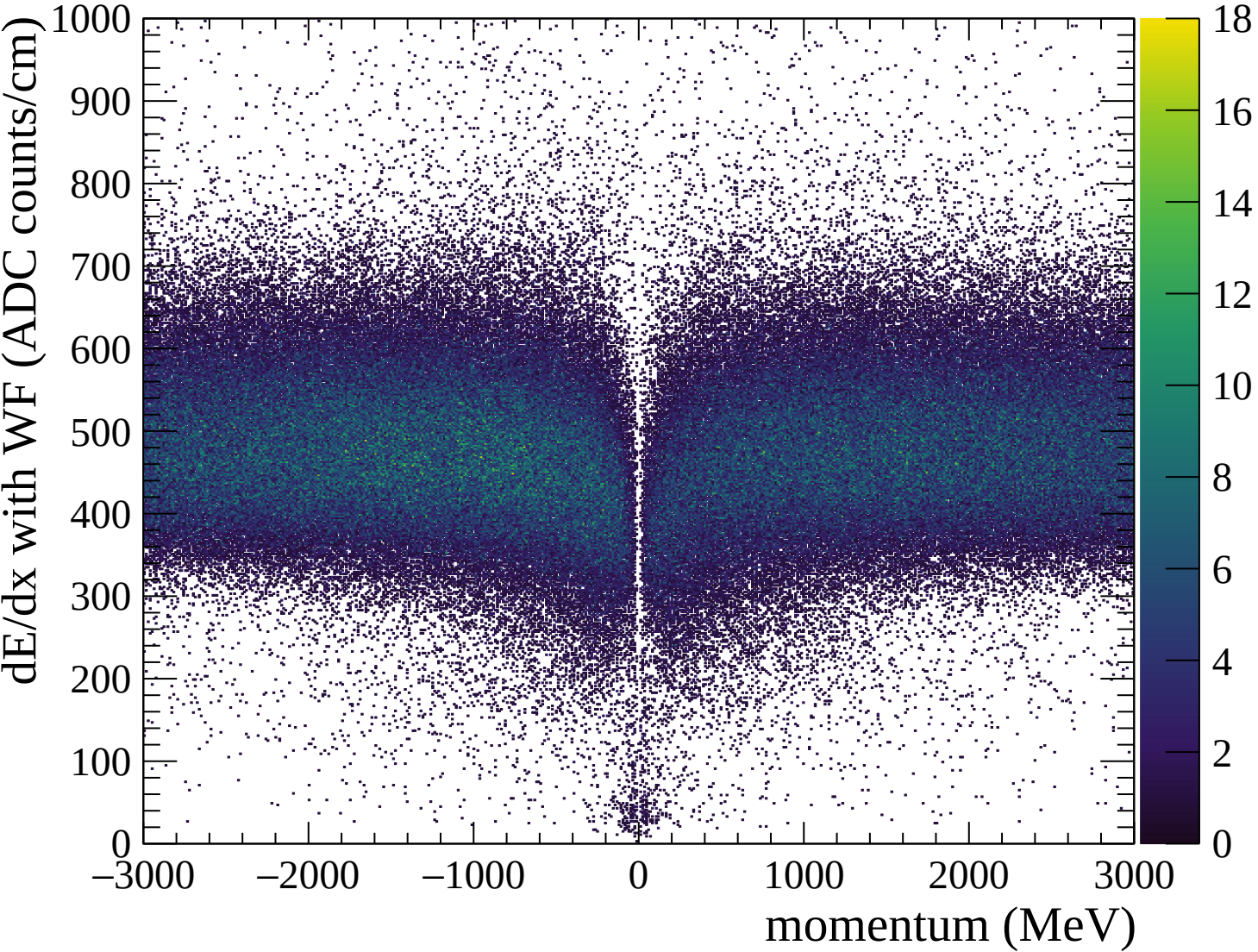


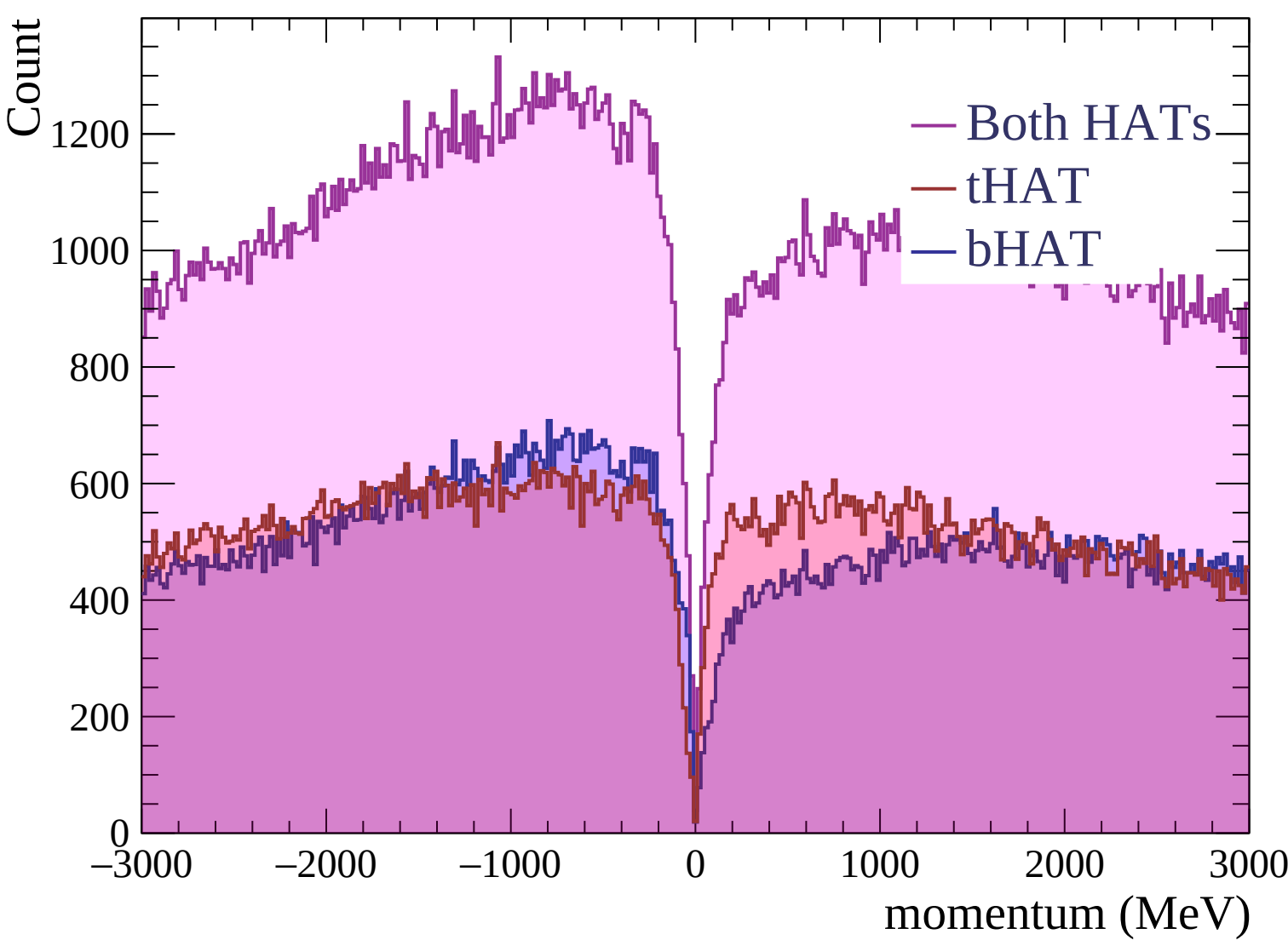


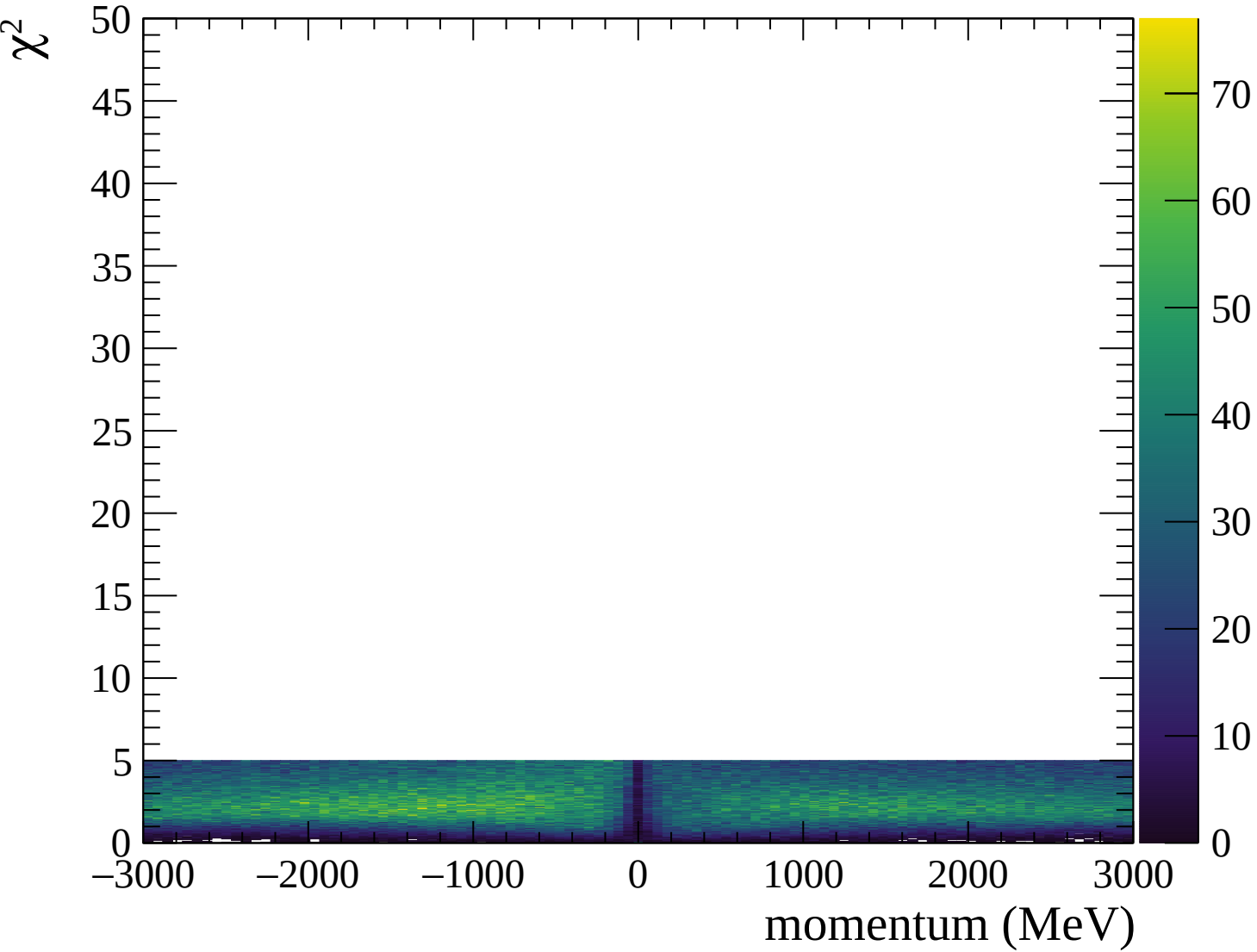


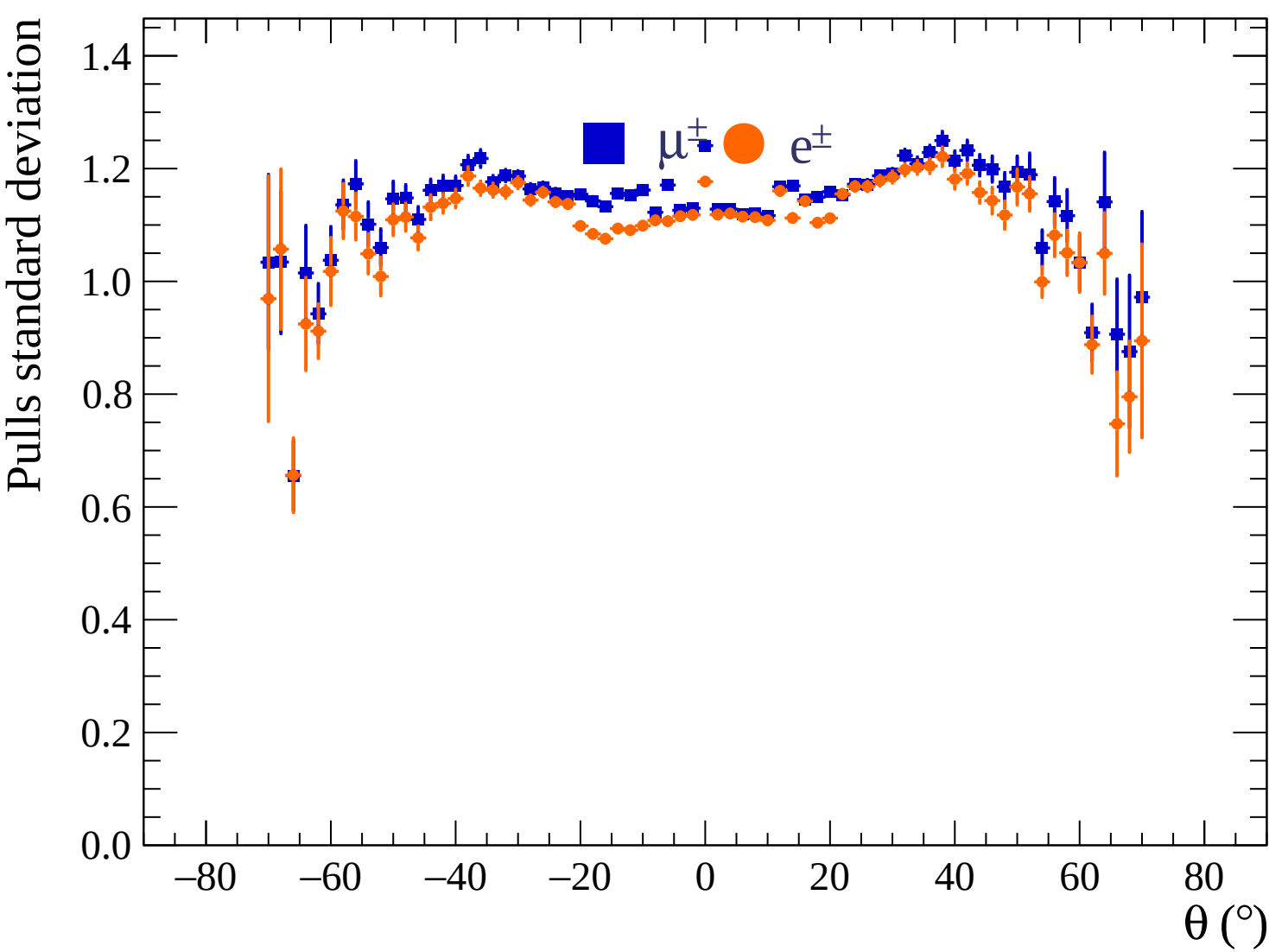


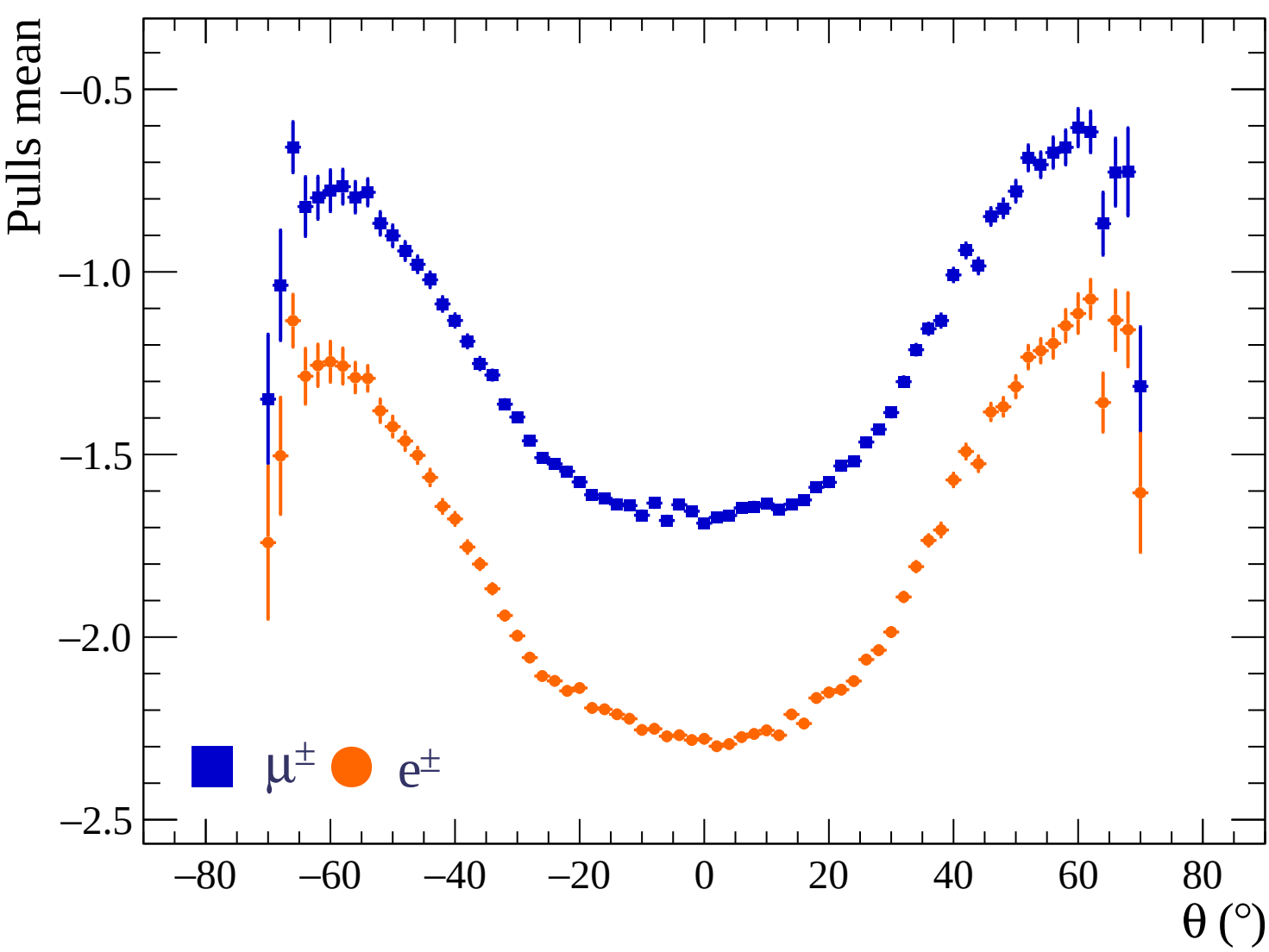


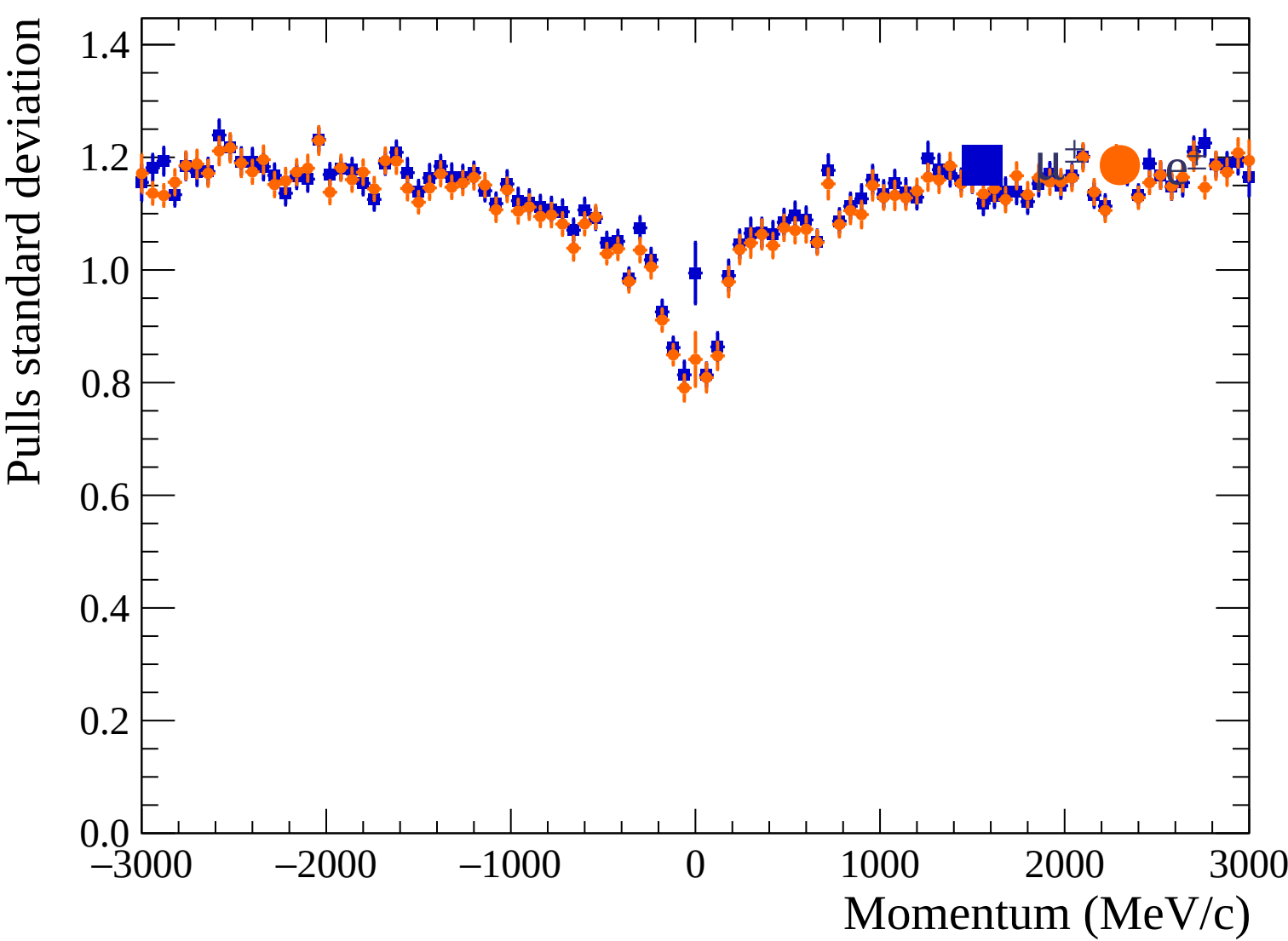


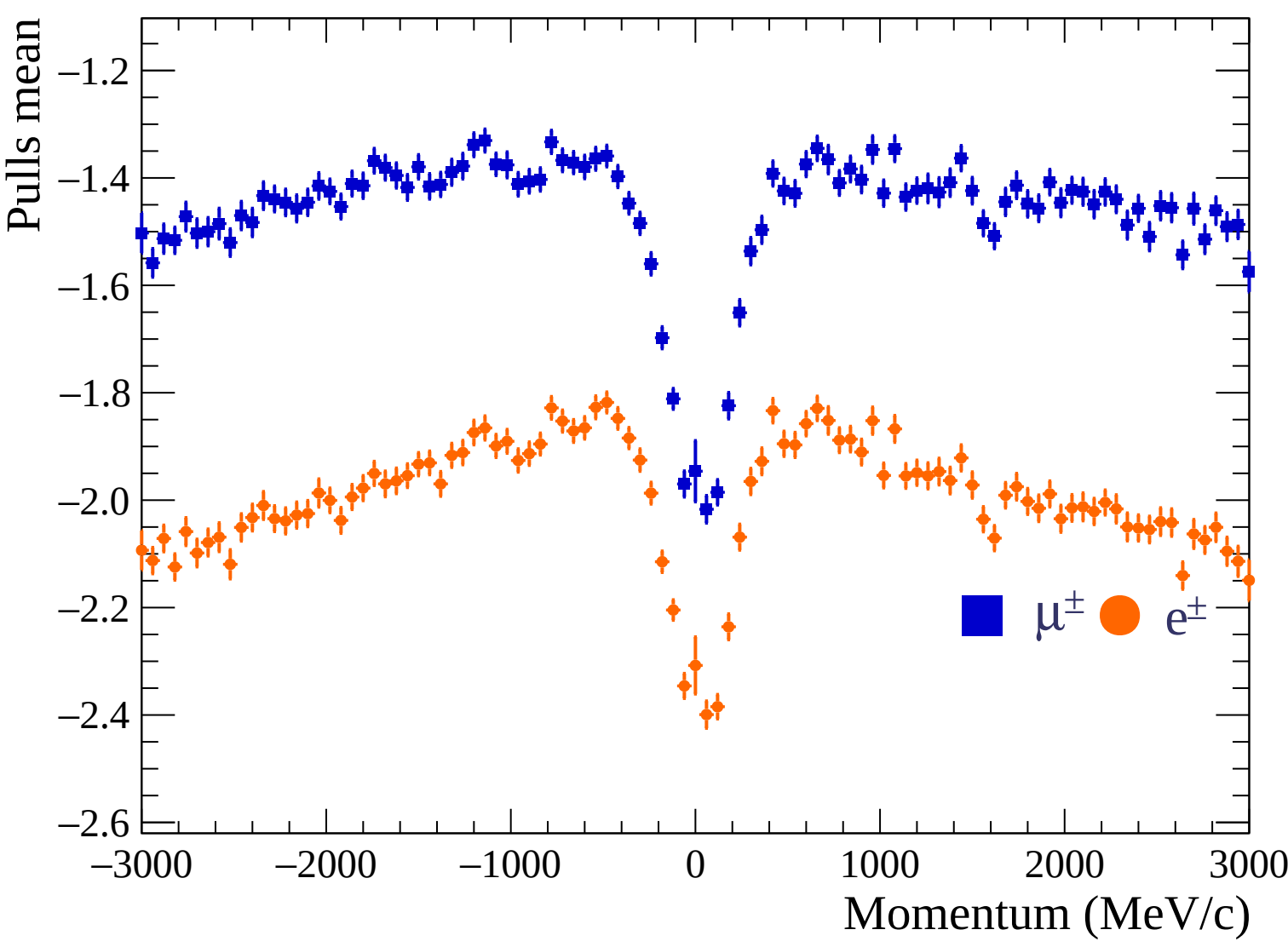


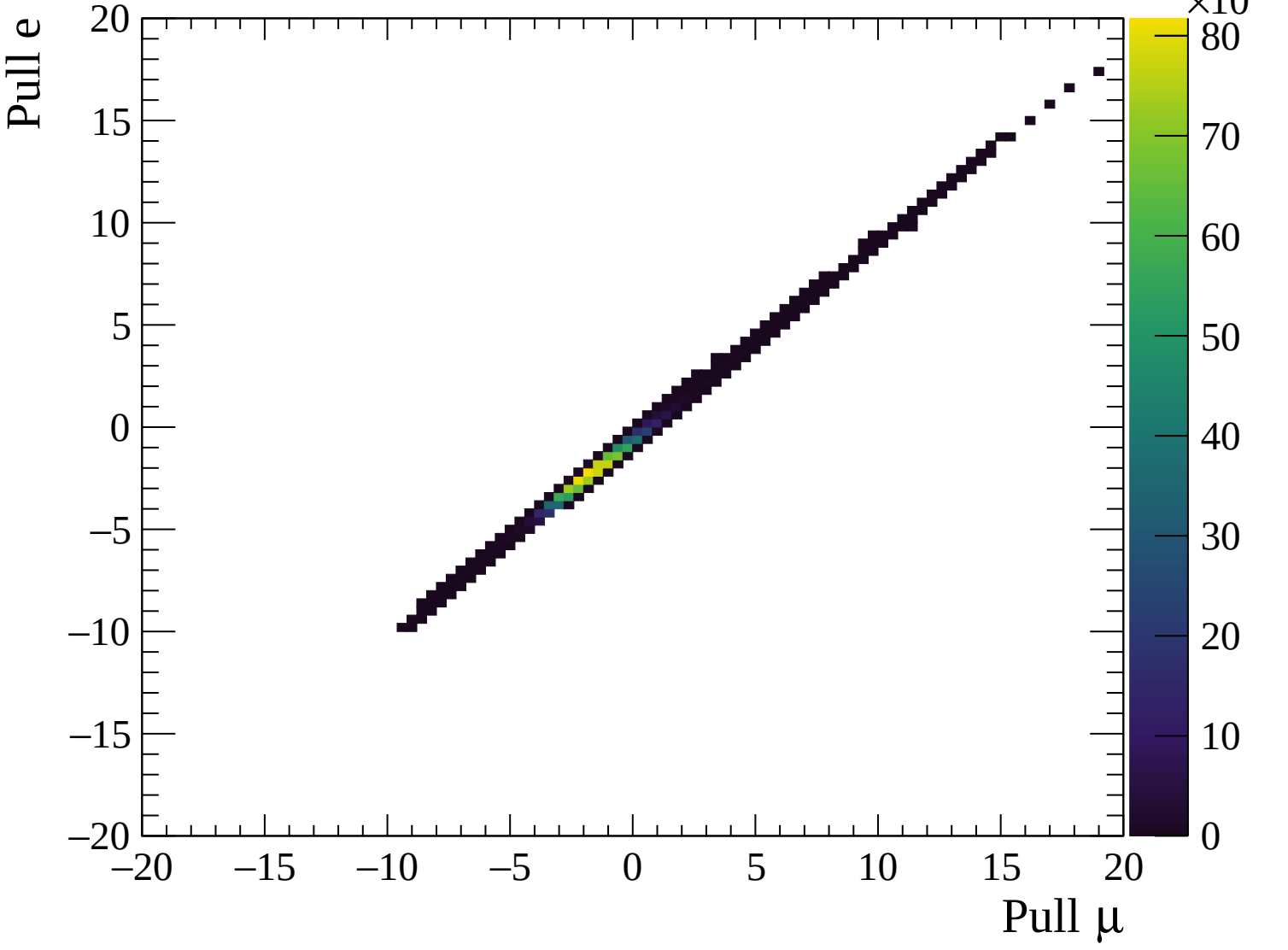


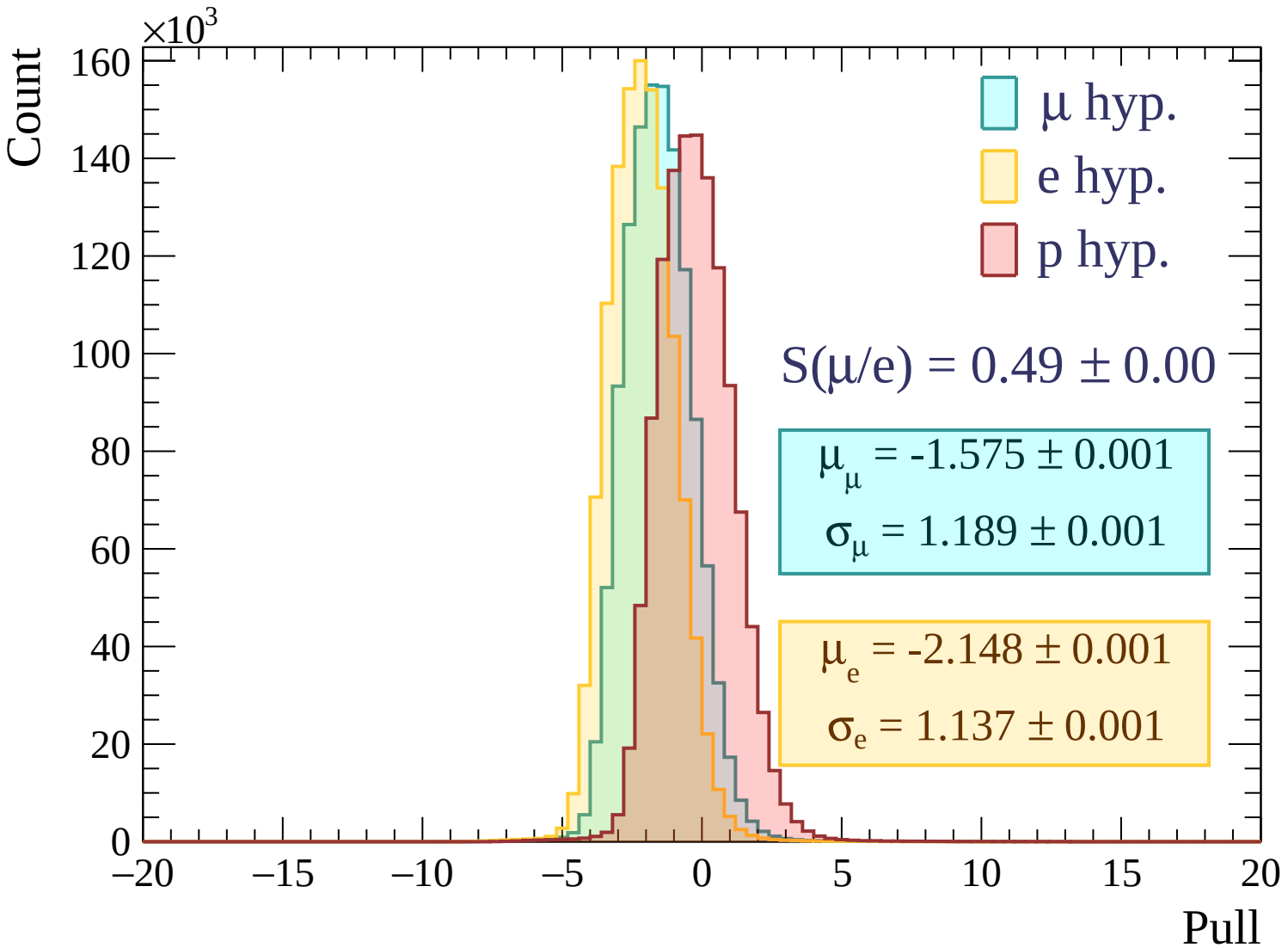


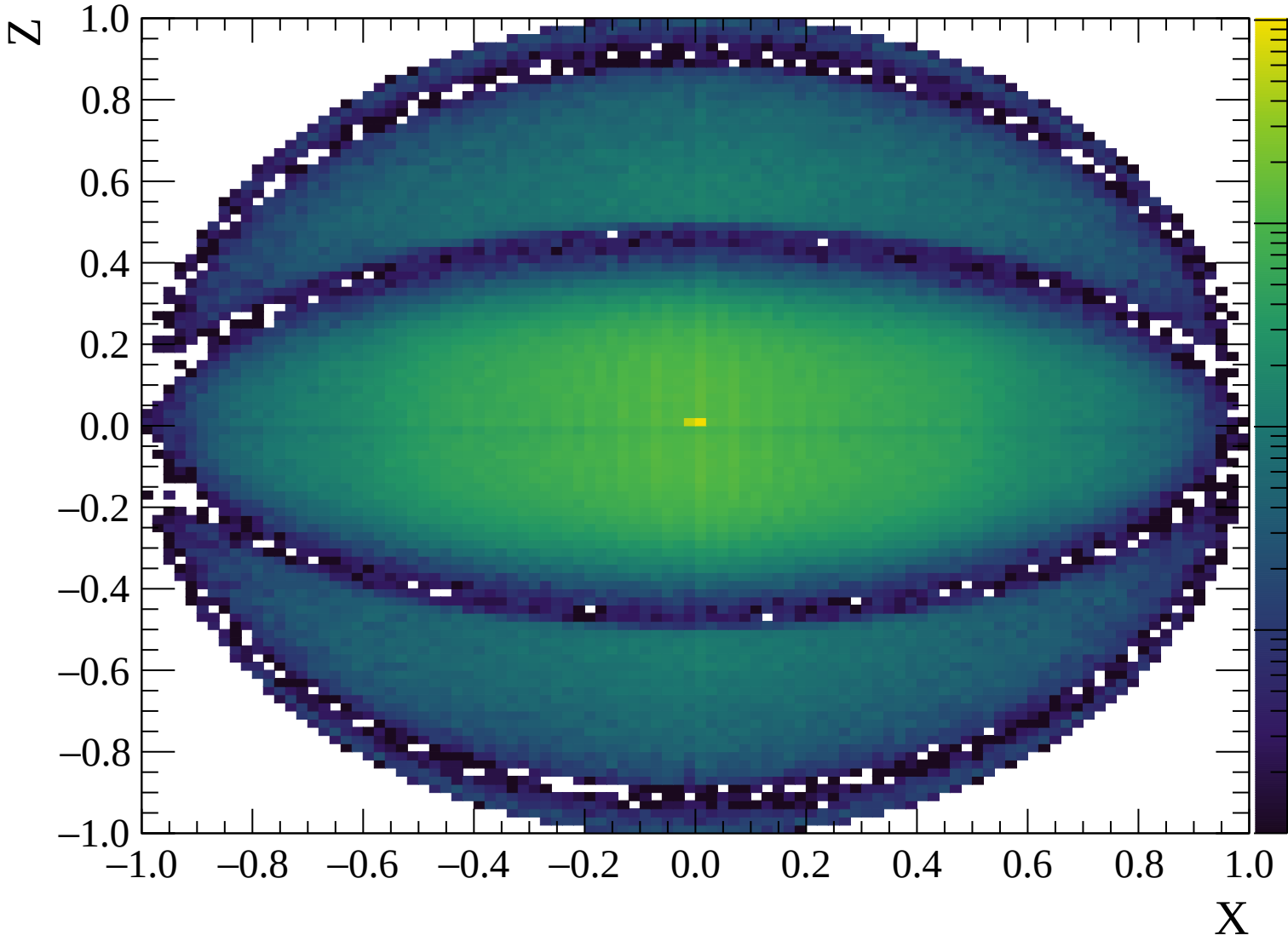


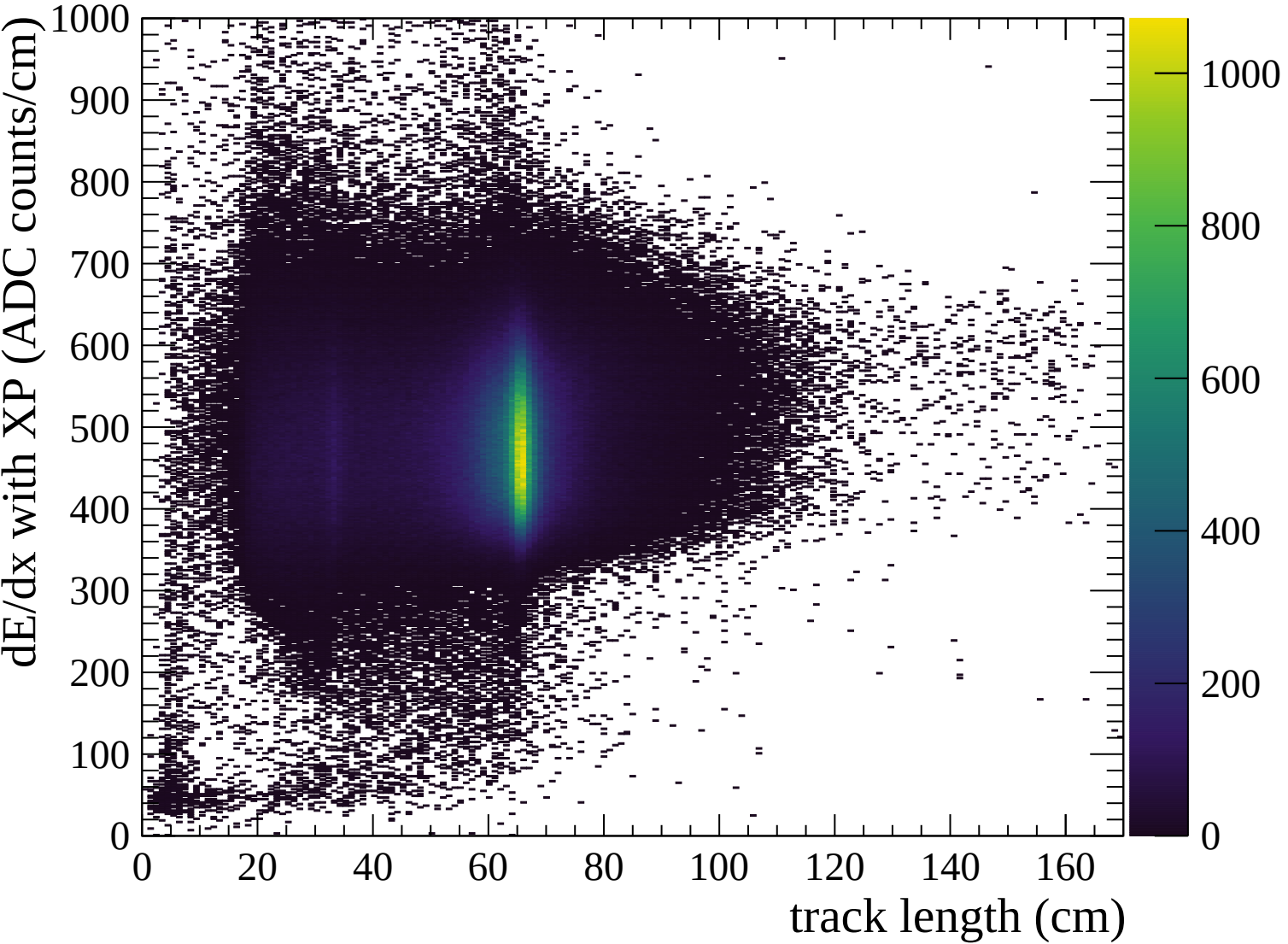


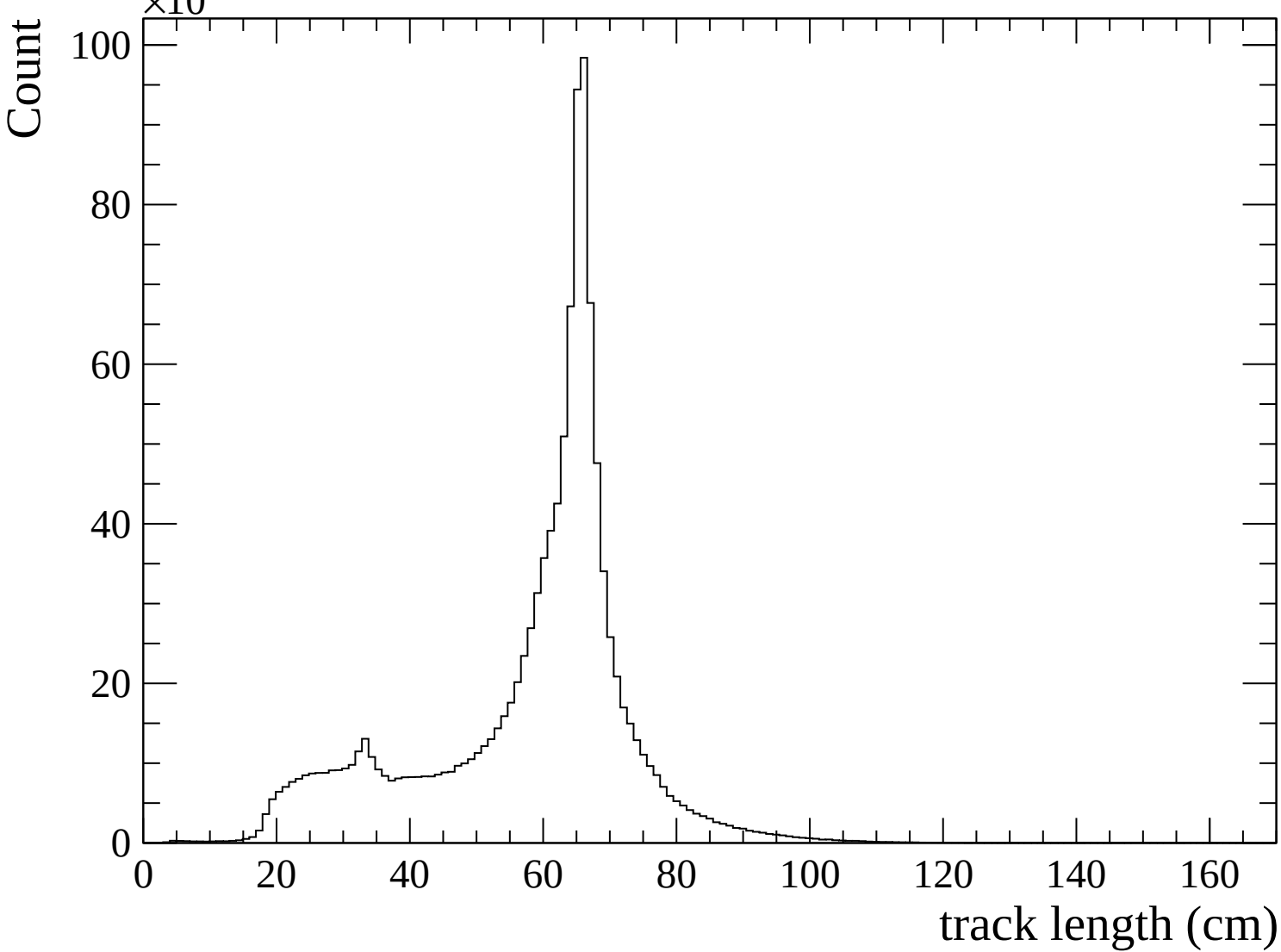


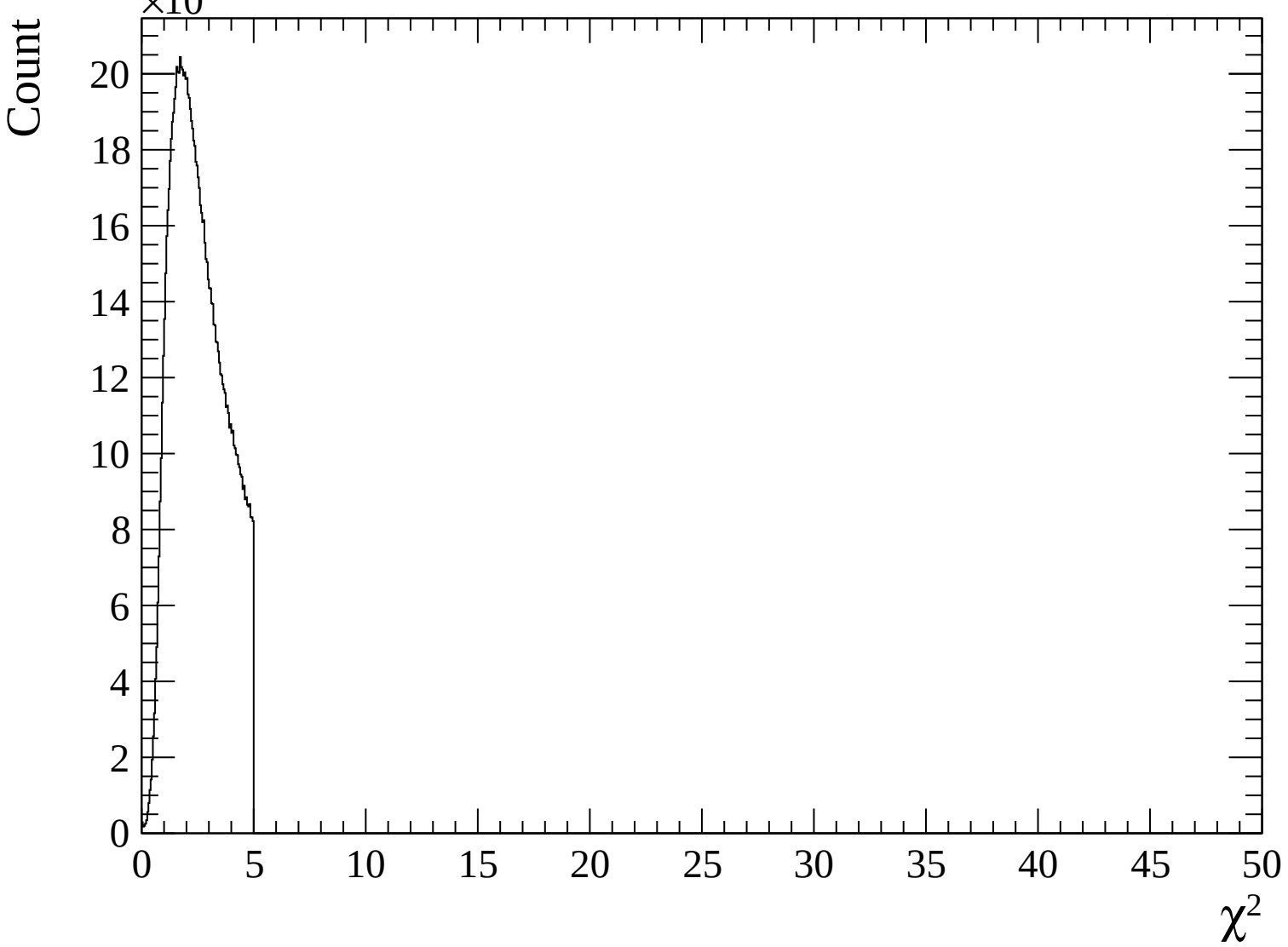


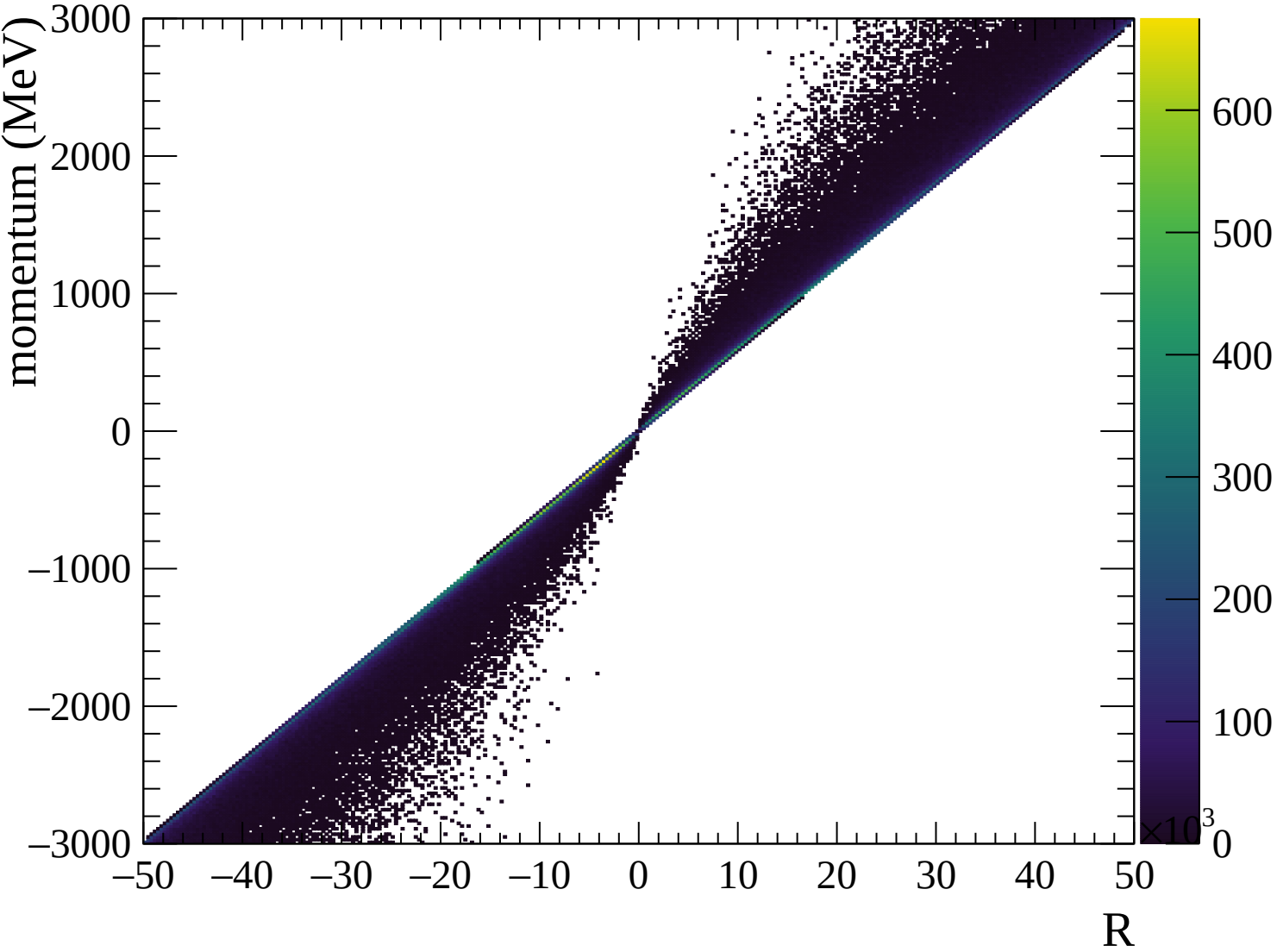


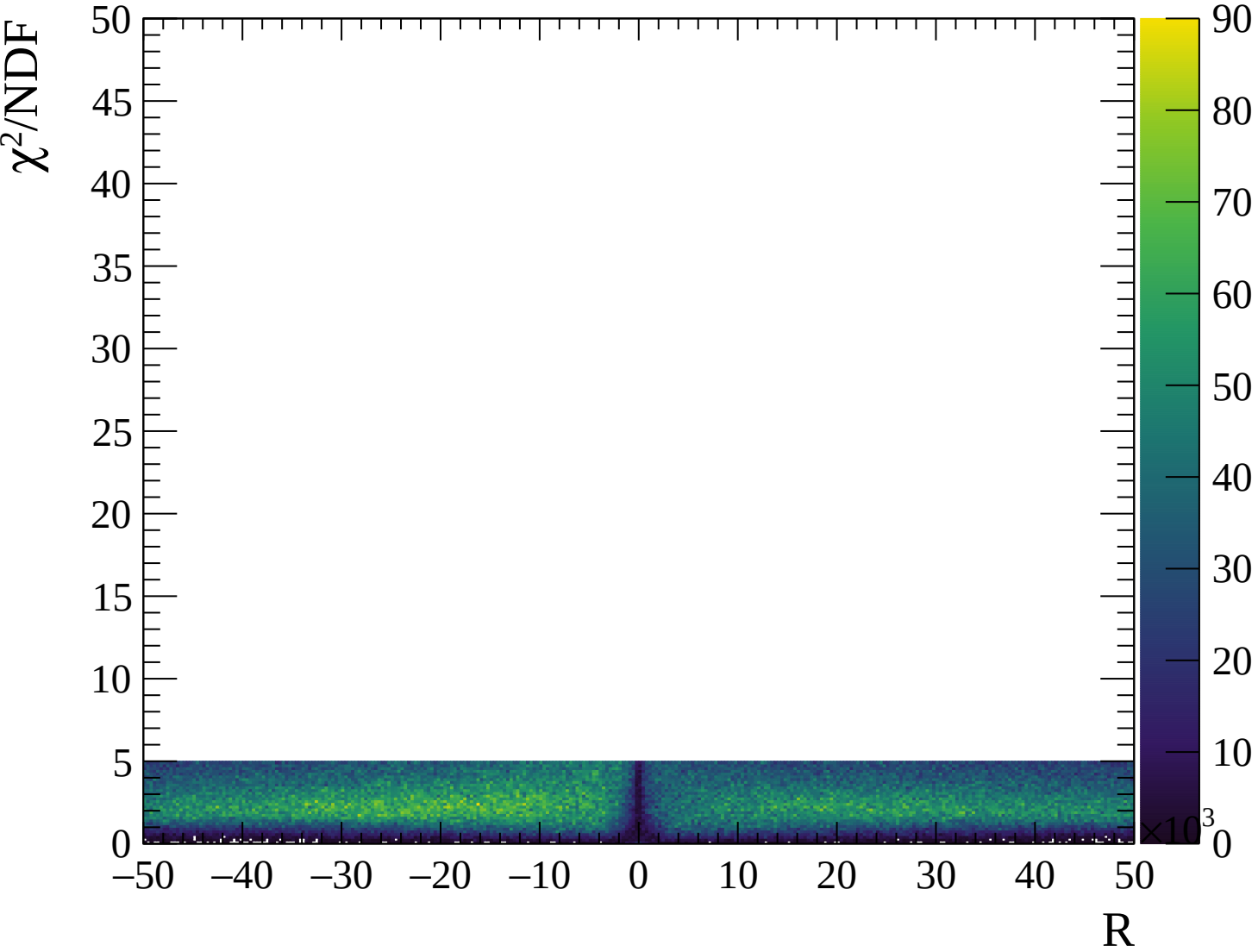






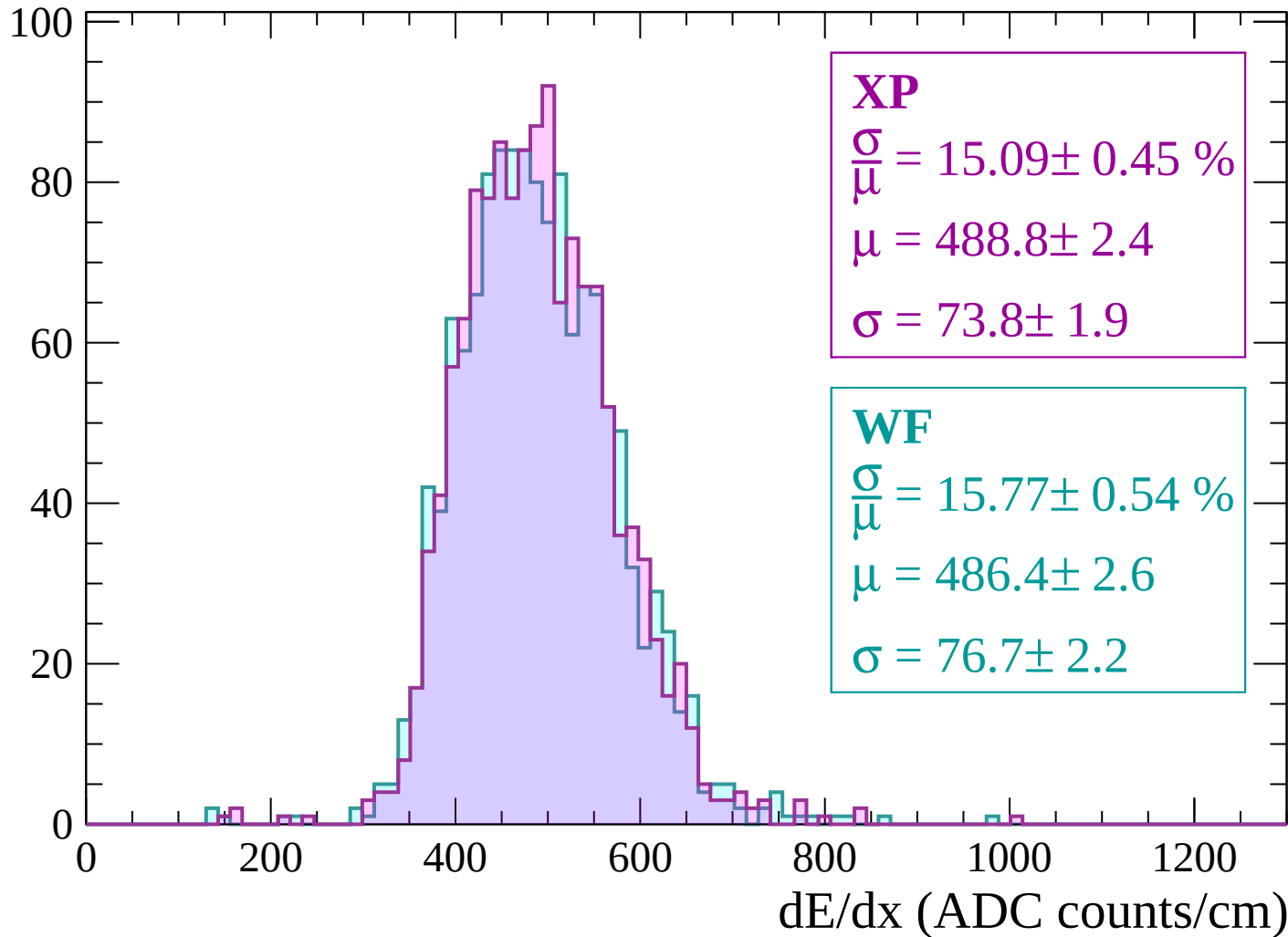






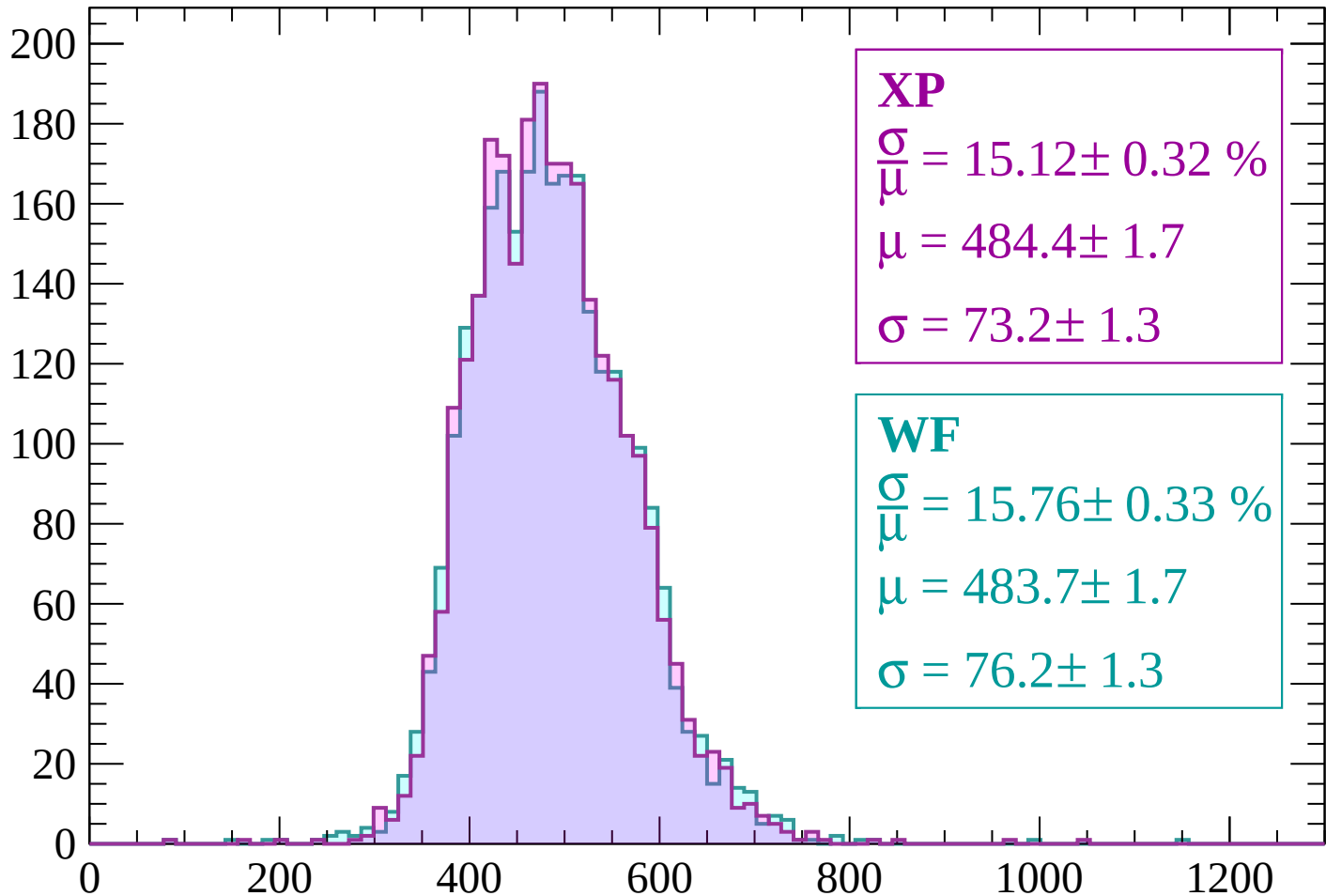
Energy loss | $-3000 < p < -2940$

Count



Energy loss | $-2940 < p < -2880$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 15.12 \pm 0.32 \%$$

$$\mu = 484.4 \pm 1.7$$

$$\sigma = 73.2 \pm 1.3$$

WF

$$\frac{\sigma}{\bar{\mu}} = 15.76 \pm 0.33 \%$$

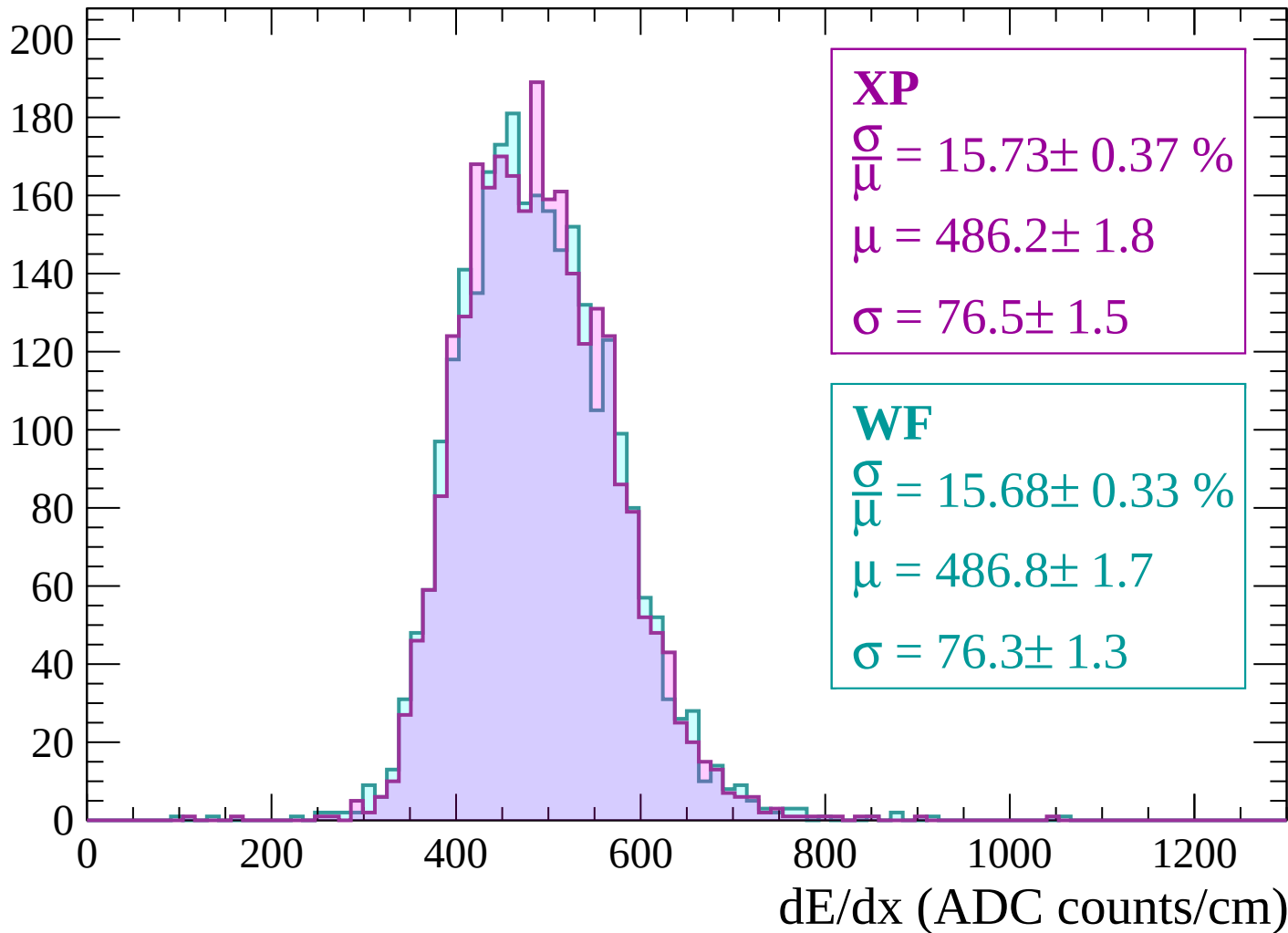
$$\mu = 483.7 \pm 1.7$$

$$\sigma = 76.2 \pm 1.3$$

dE/dx (ADC counts/cm)

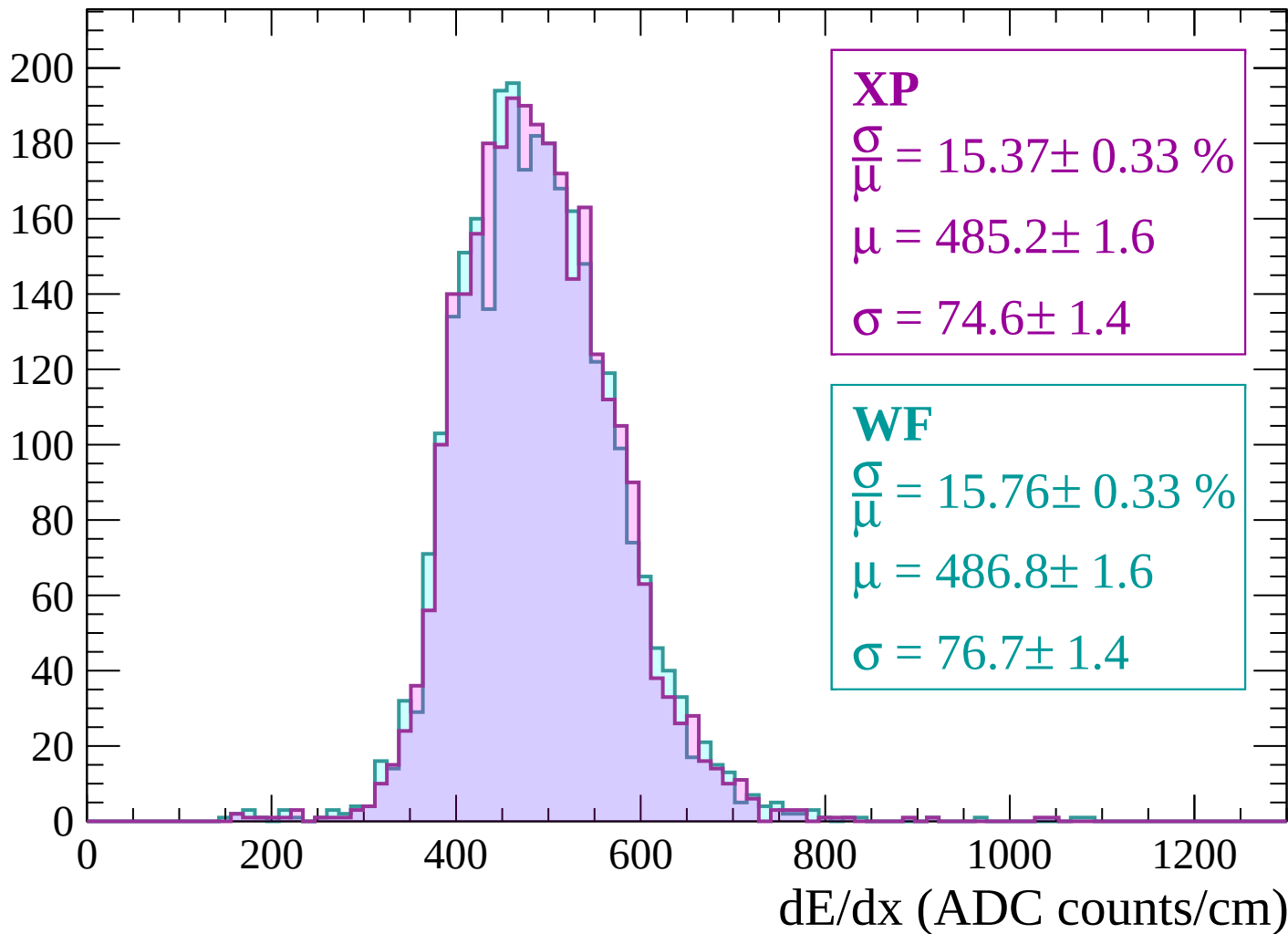
Energy loss | $-2880 < p < -2820$

Count



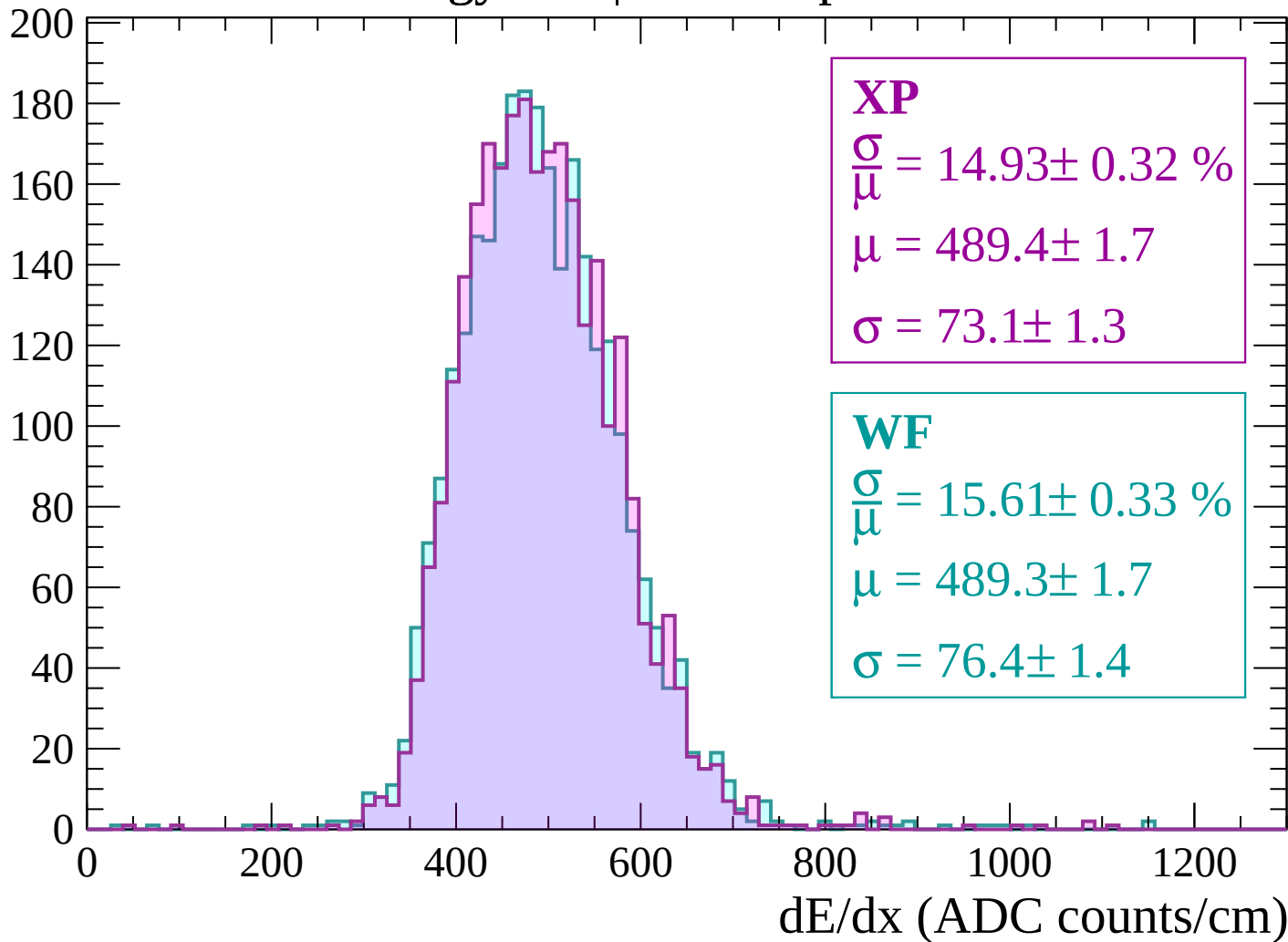
Energy loss | $-2820 < p < -2760$

Count



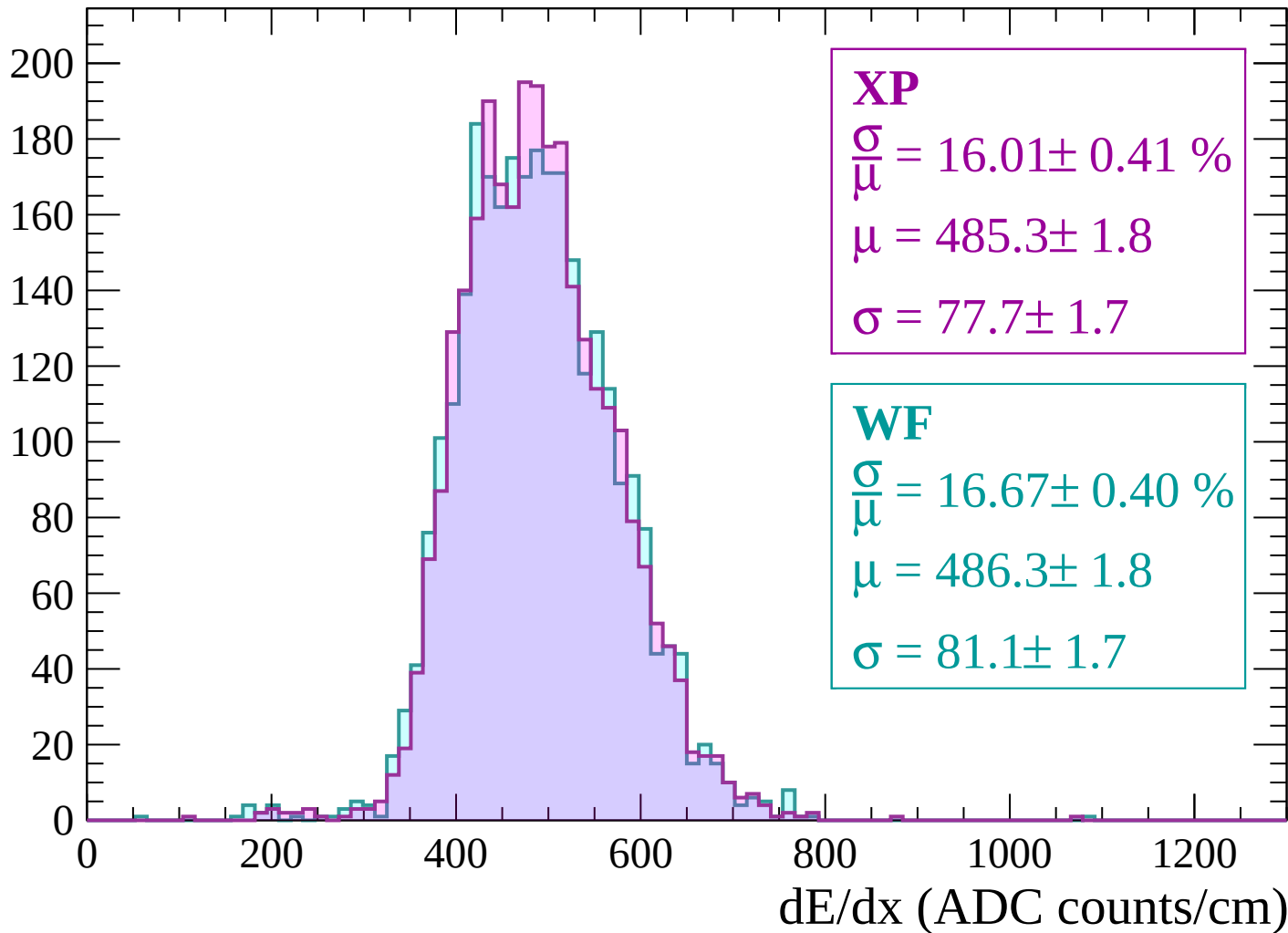
Energy loss | $-2760 < p < -2700$

Count



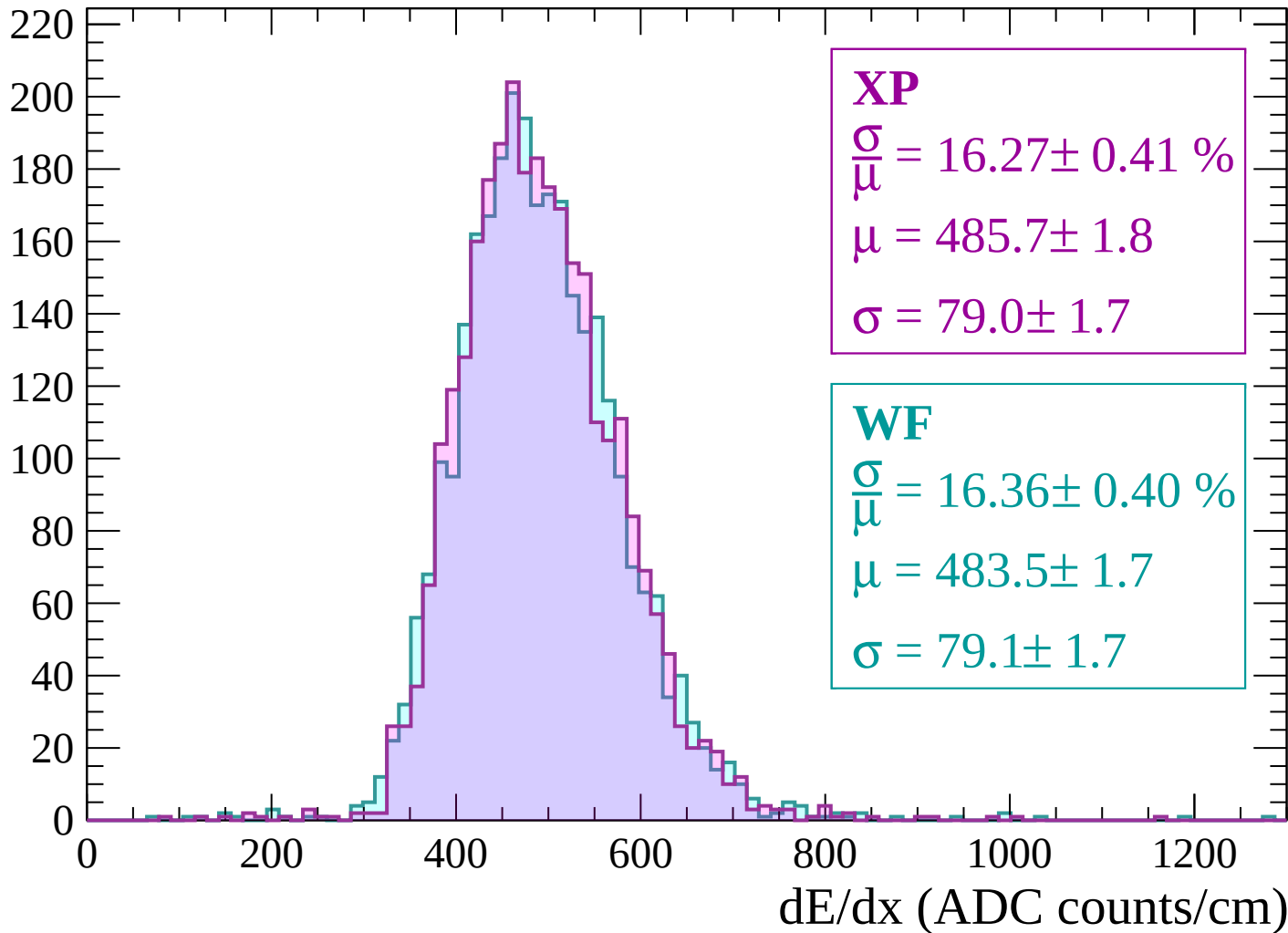
Energy loss | -2700 < p < -2640

Count



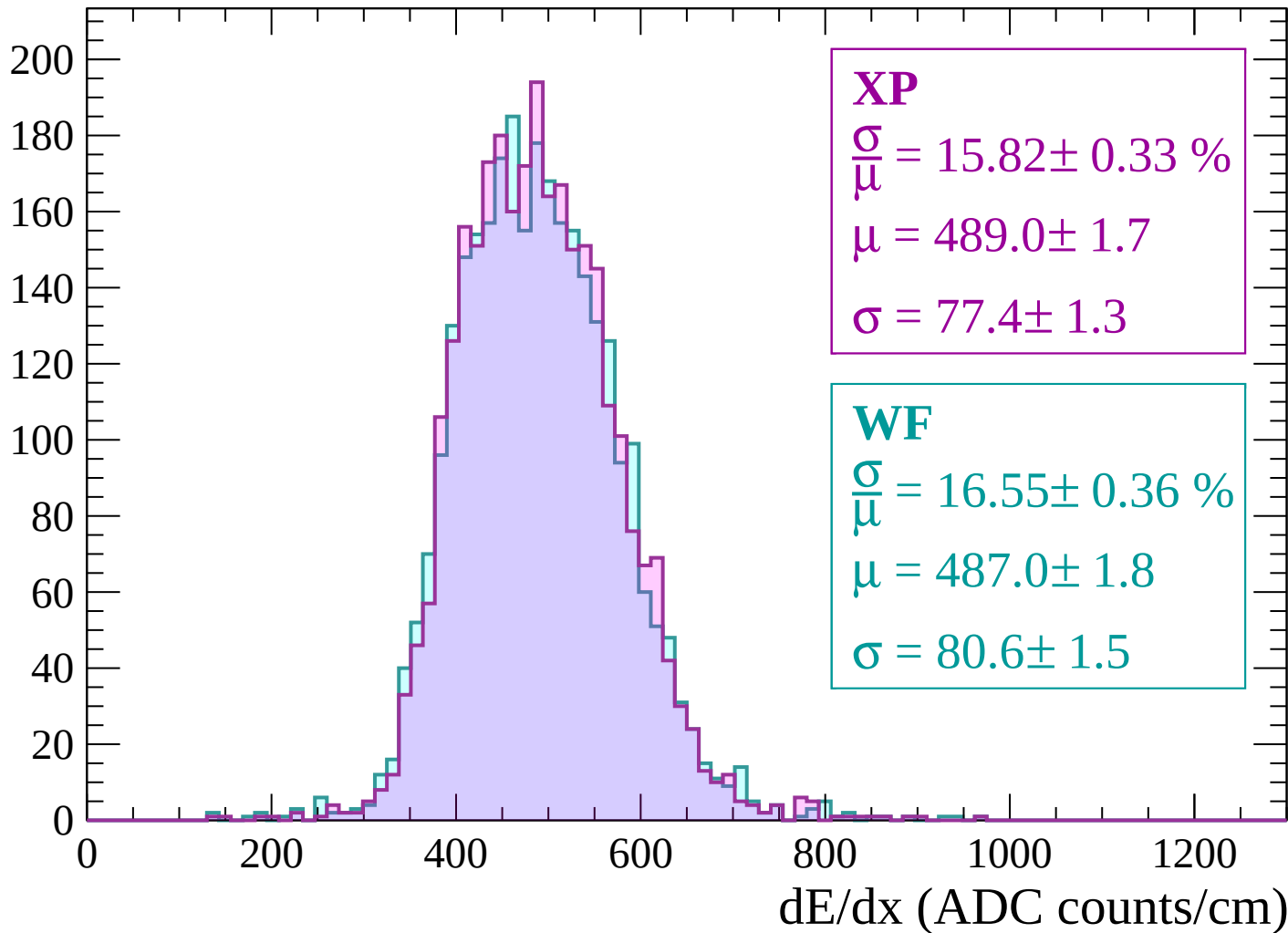
Energy loss | $-2640 < p < -2580$

Count



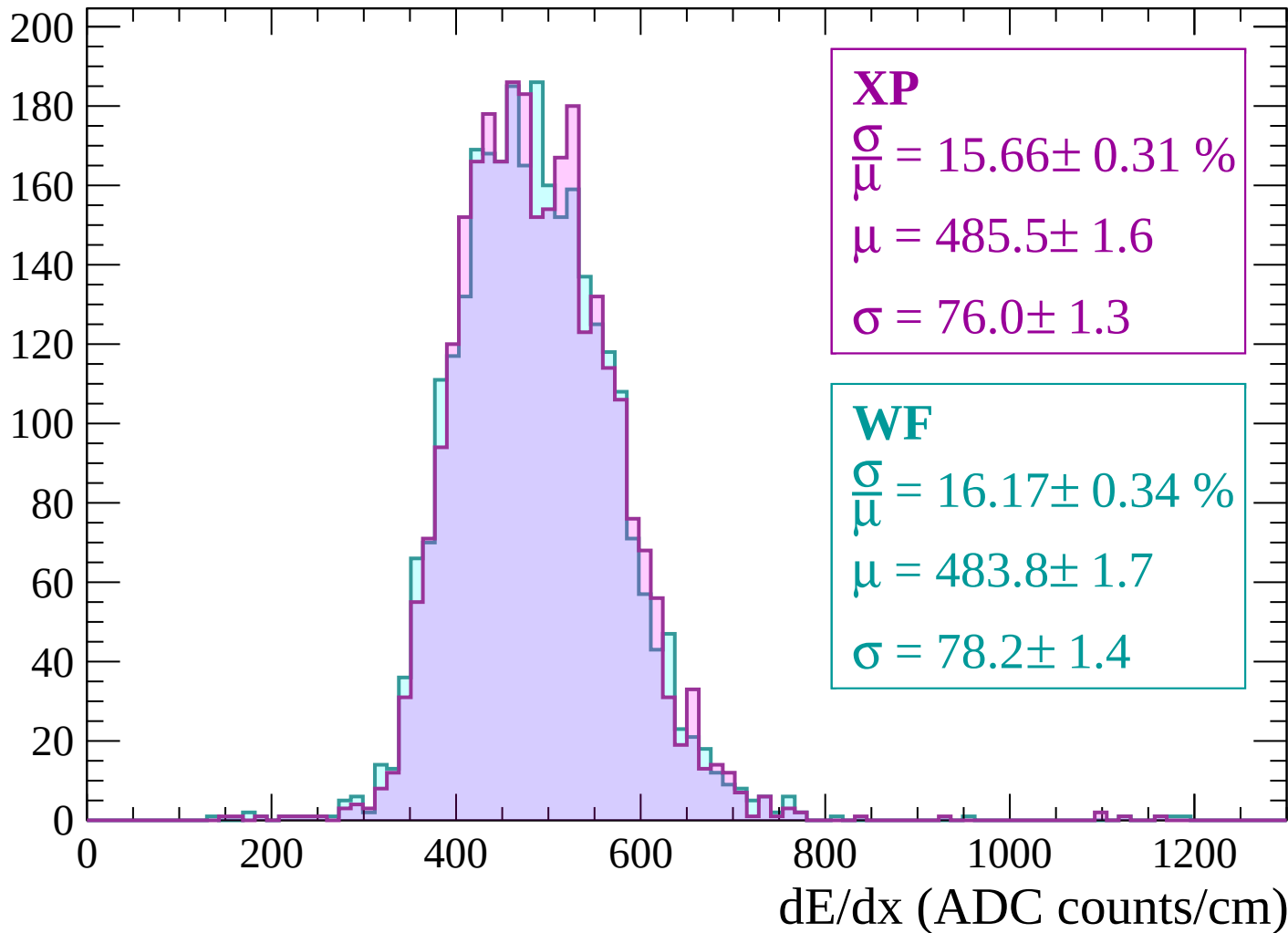
Energy loss | -2580 < p < -2520

Count



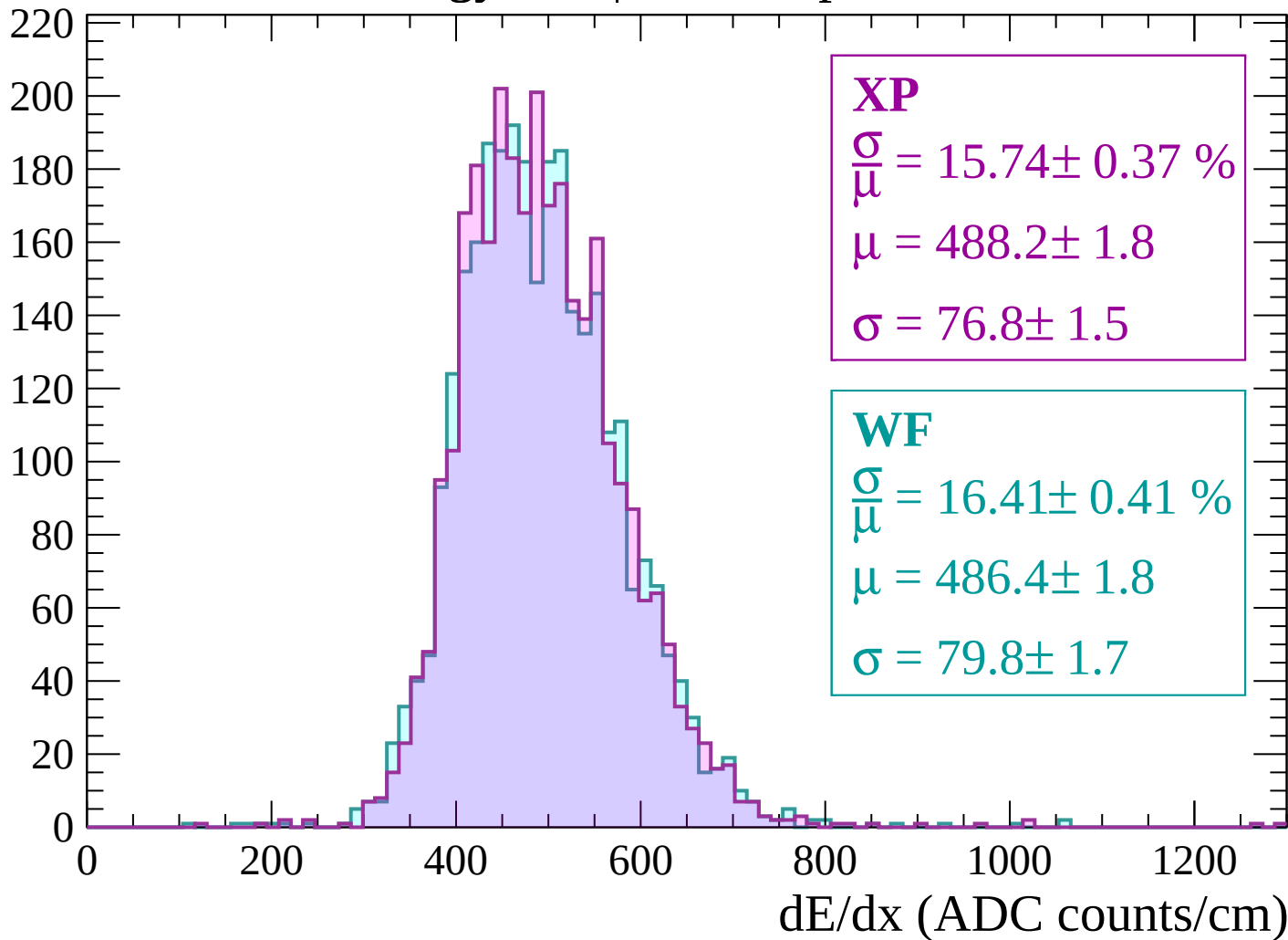
Energy loss | $-2520 < p < -2460$

Count



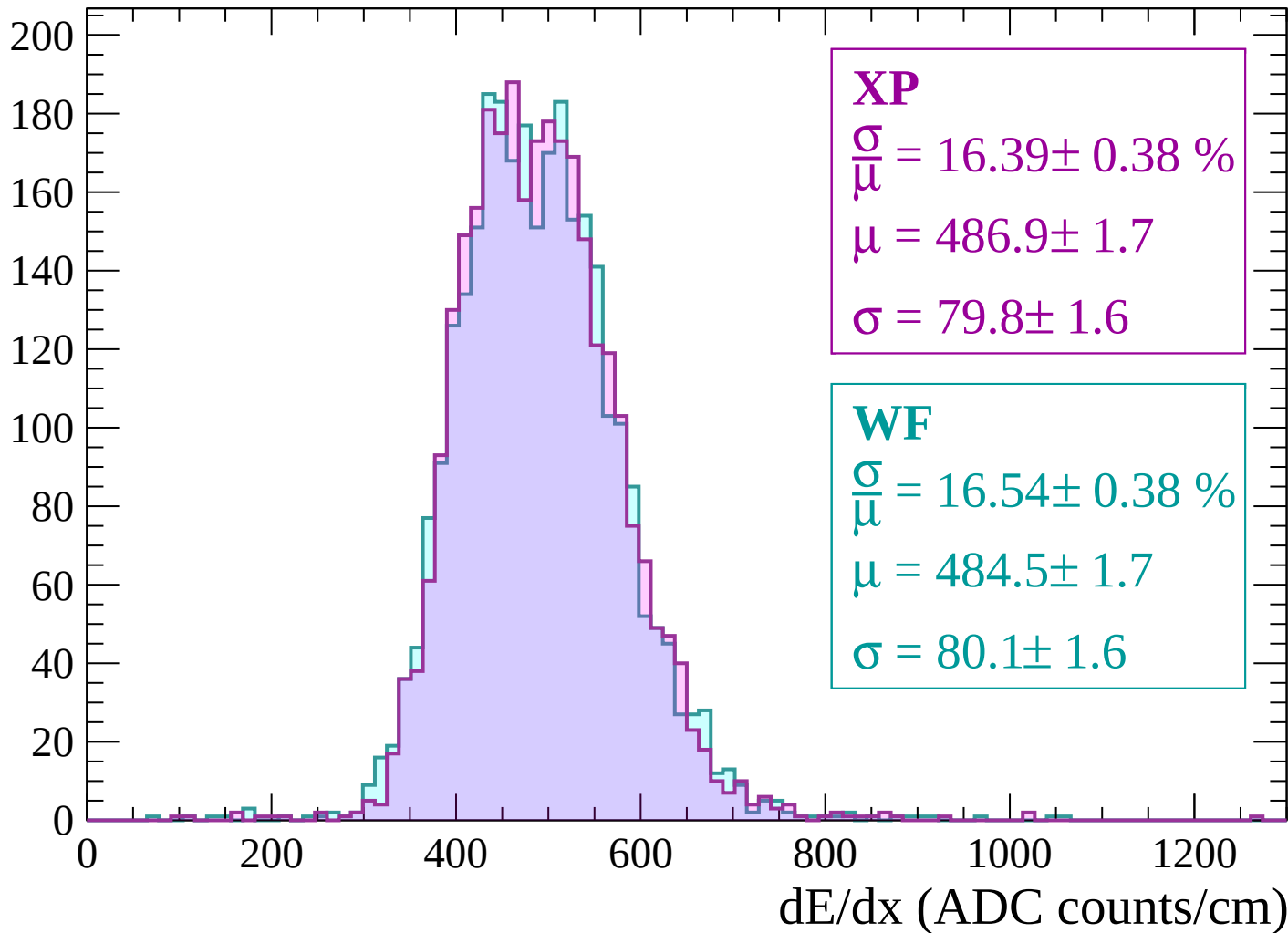
Energy loss | -2460 < p < -2400

Count



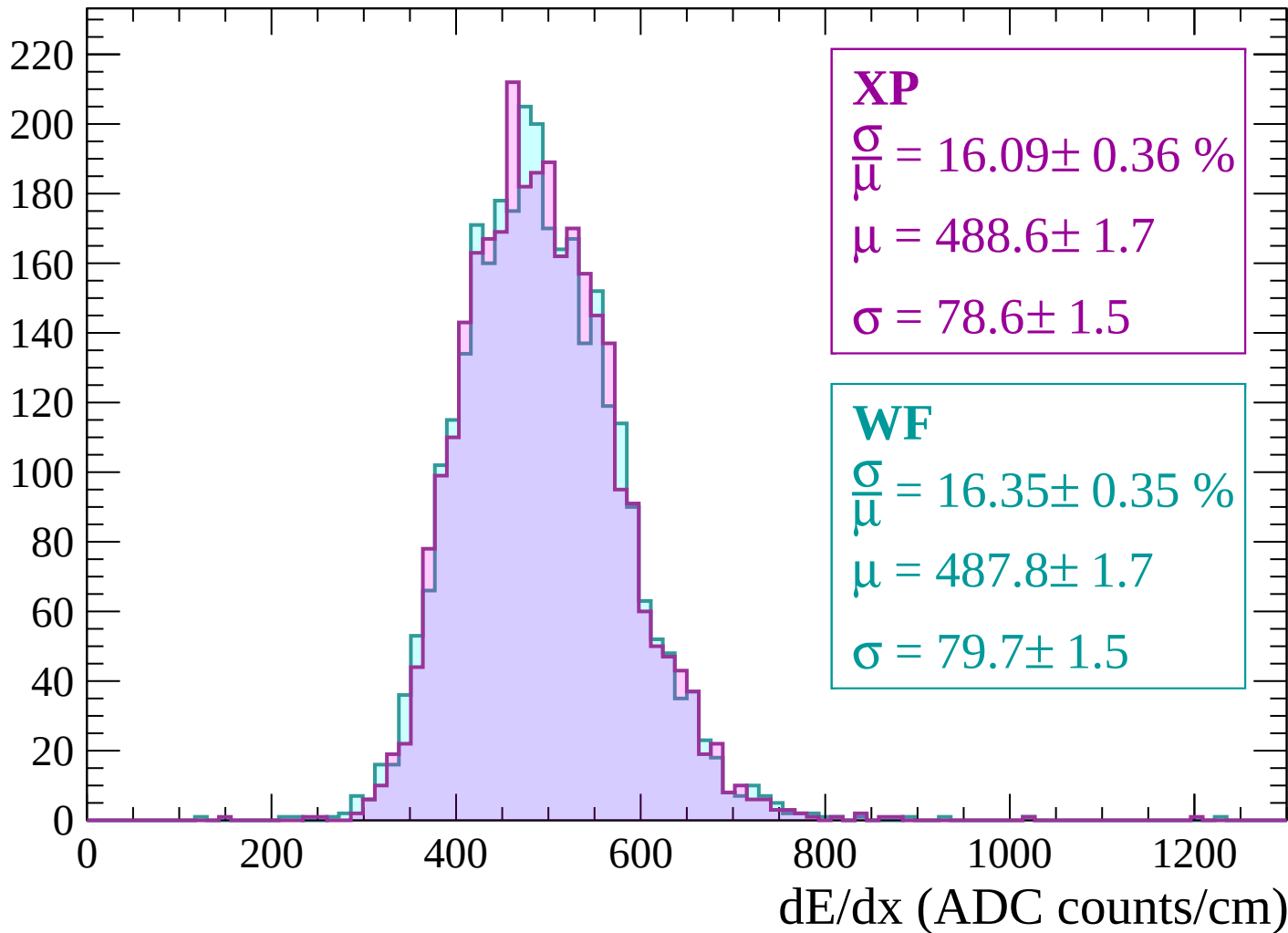
Energy loss | -2400 < p < -2340

Count



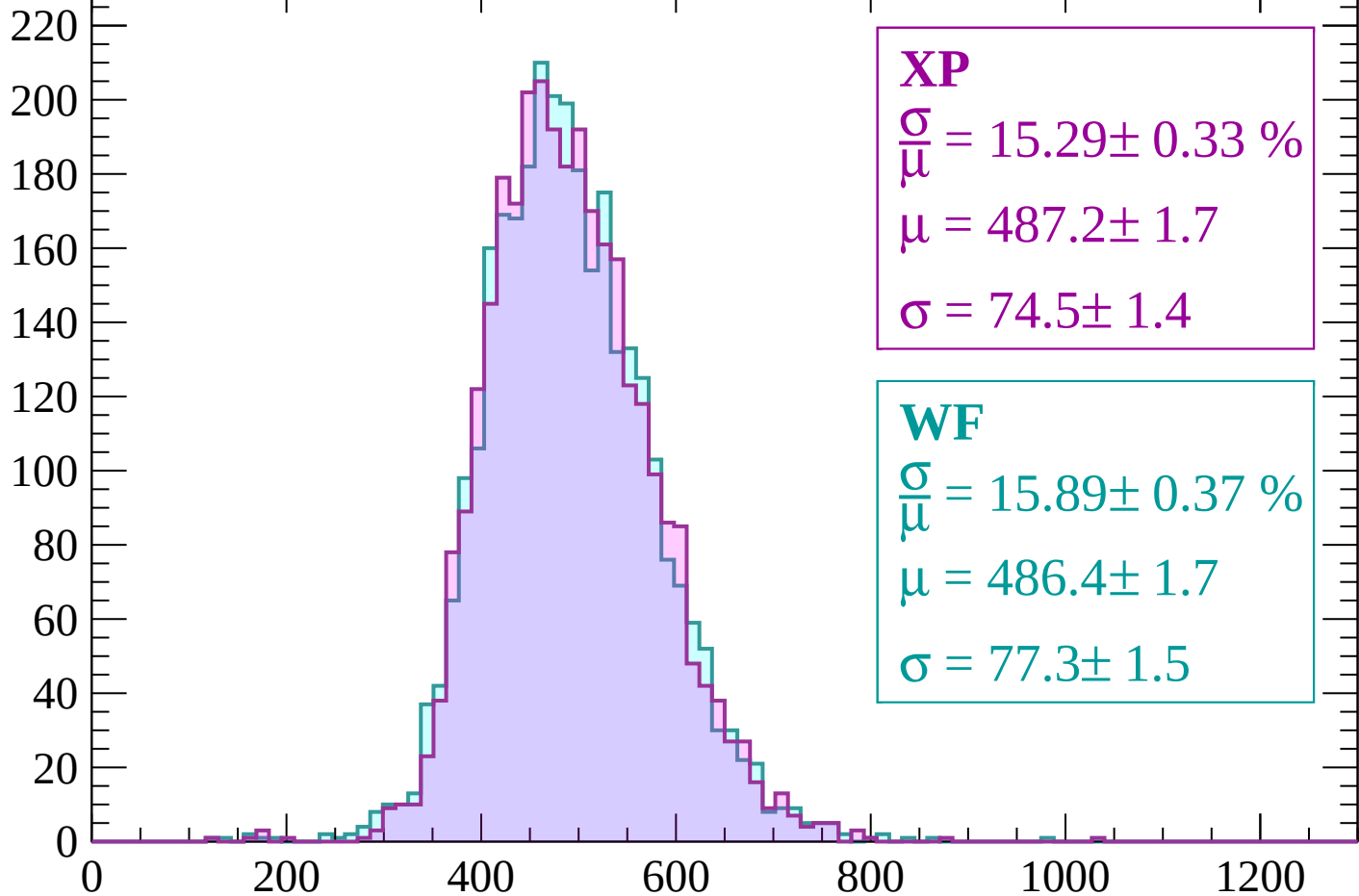
Energy loss | $-2340 < p < -2280$

Count



Energy loss | -2280 < p < -2220

Count



XP

$$\frac{\sigma}{\mu} = 15.29 \pm 0.33 \%$$

$$\mu = 487.2 \pm 1.7$$

$$\sigma = 74.5 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 15.89 \pm 0.37 \%$$

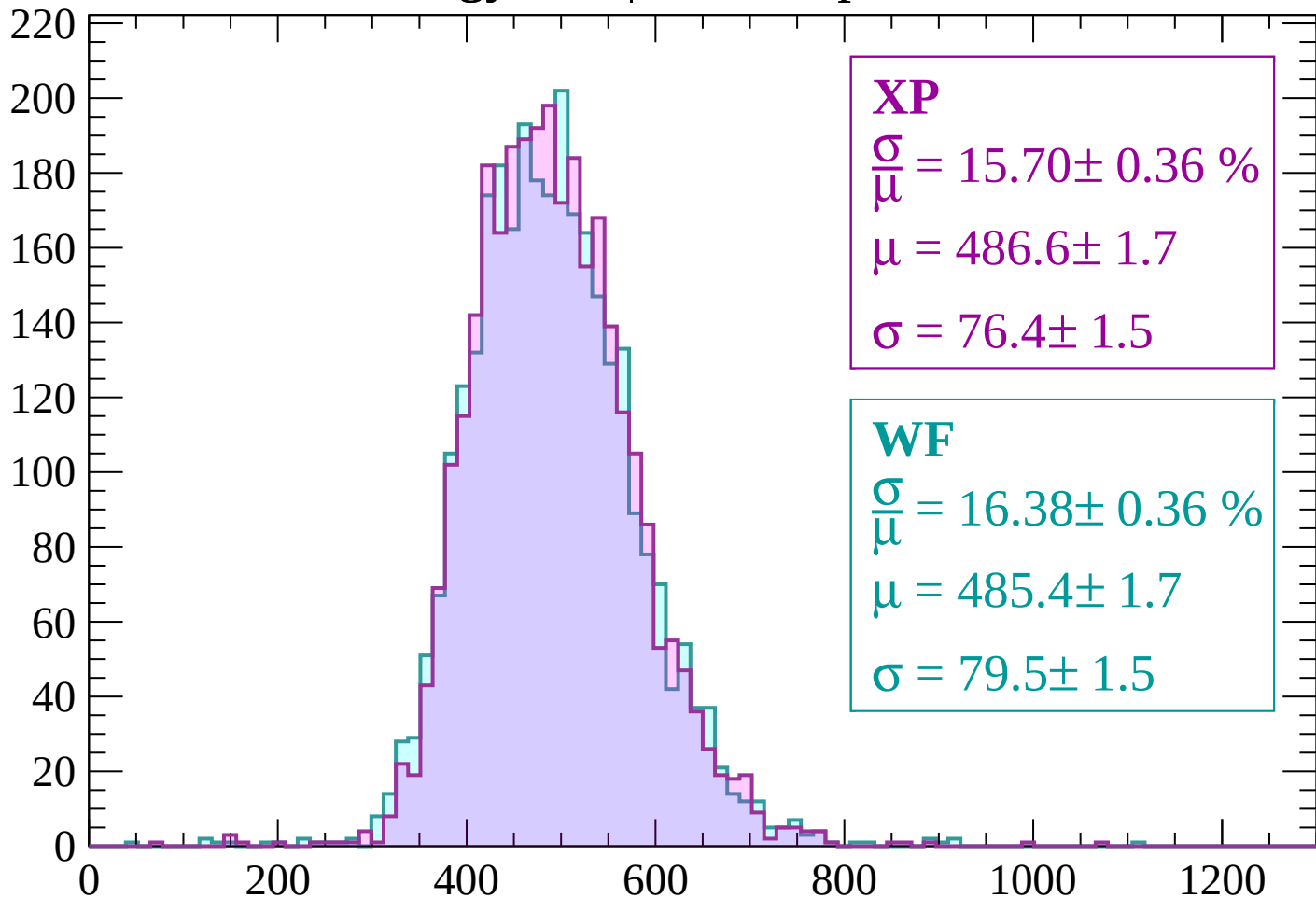
$$\mu = 486.4 \pm 1.7$$

$$\sigma = 77.3 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -2220 < p < -2160

Count



XP

$$\frac{\sigma}{\mu} = 15.70 \pm 0.36 \%$$

$$\mu = 486.6 \pm 1.7$$

$$\sigma = 76.4 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 16.38 \pm 0.36 \%$$

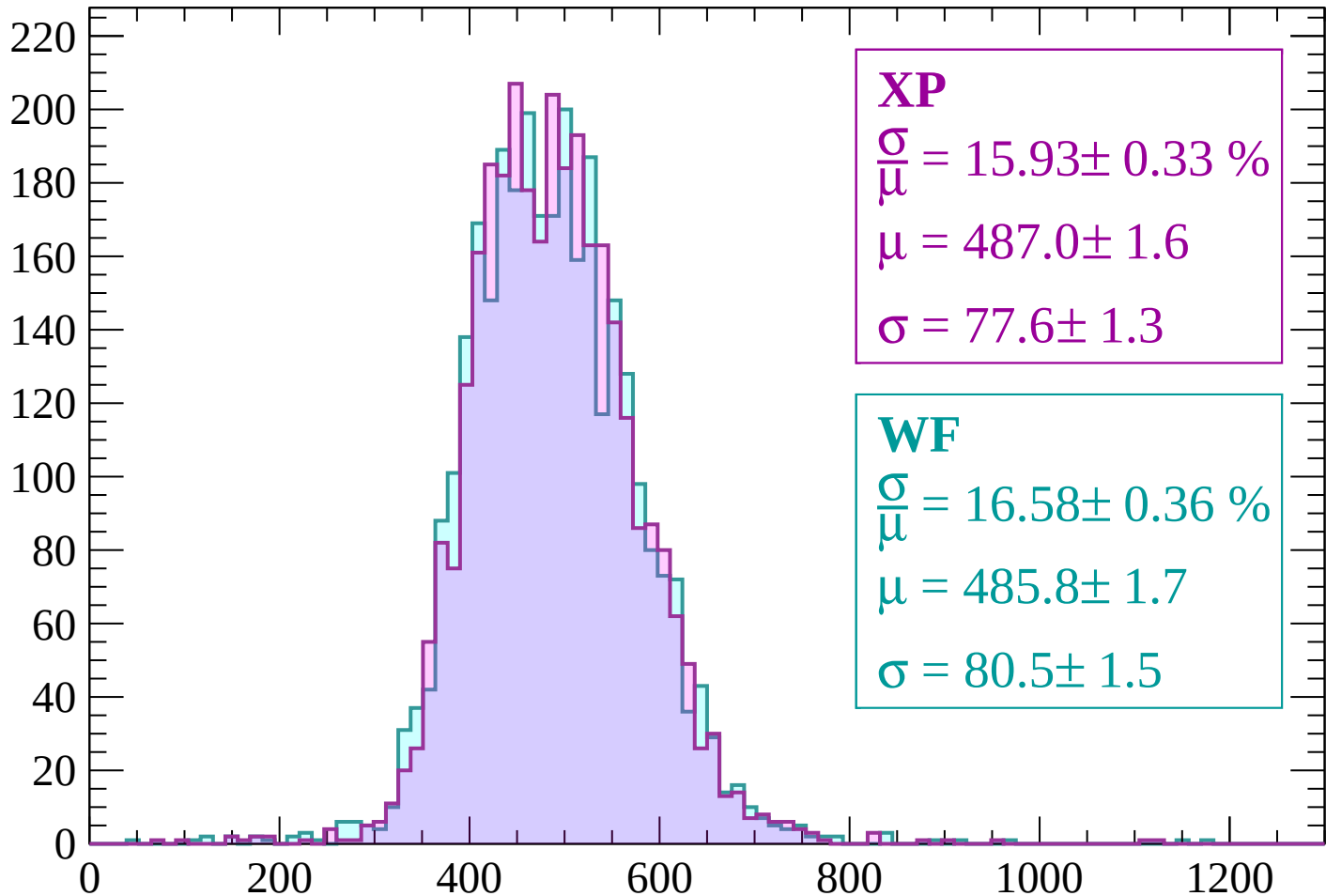
$$\mu = 485.4 \pm 1.7$$

$$\sigma = 79.5 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -2160 < p < -2100

Count



XP

$$\frac{\sigma}{\mu} = 15.93 \pm 0.33 \%$$

$$\mu = 487.0 \pm 1.6$$

$$\sigma = 77.6 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 16.58 \pm 0.36 \%$$

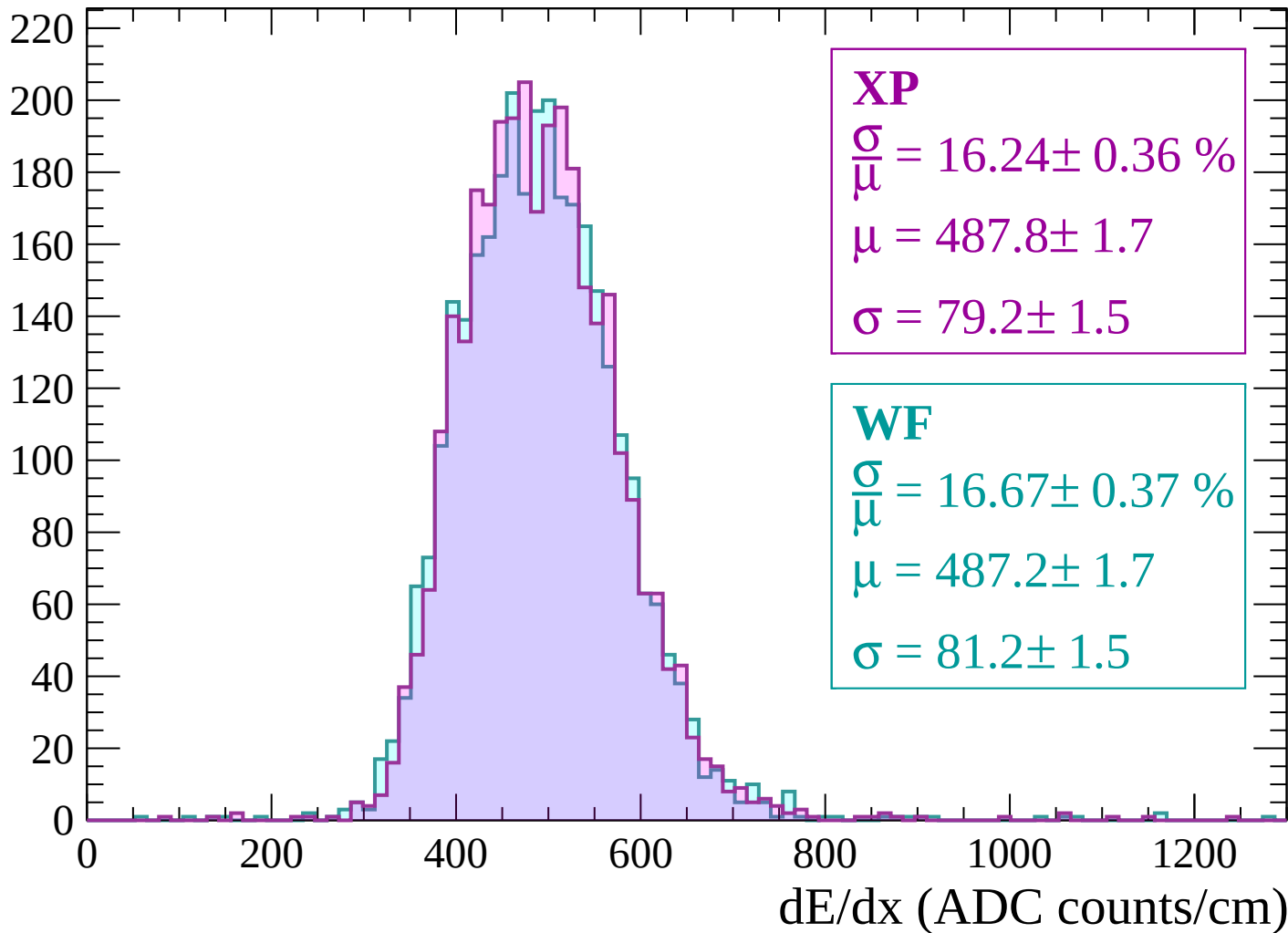
$$\mu = 485.8 \pm 1.7$$

$$\sigma = 80.5 \pm 1.5$$

dE/dx (ADC counts/cm)

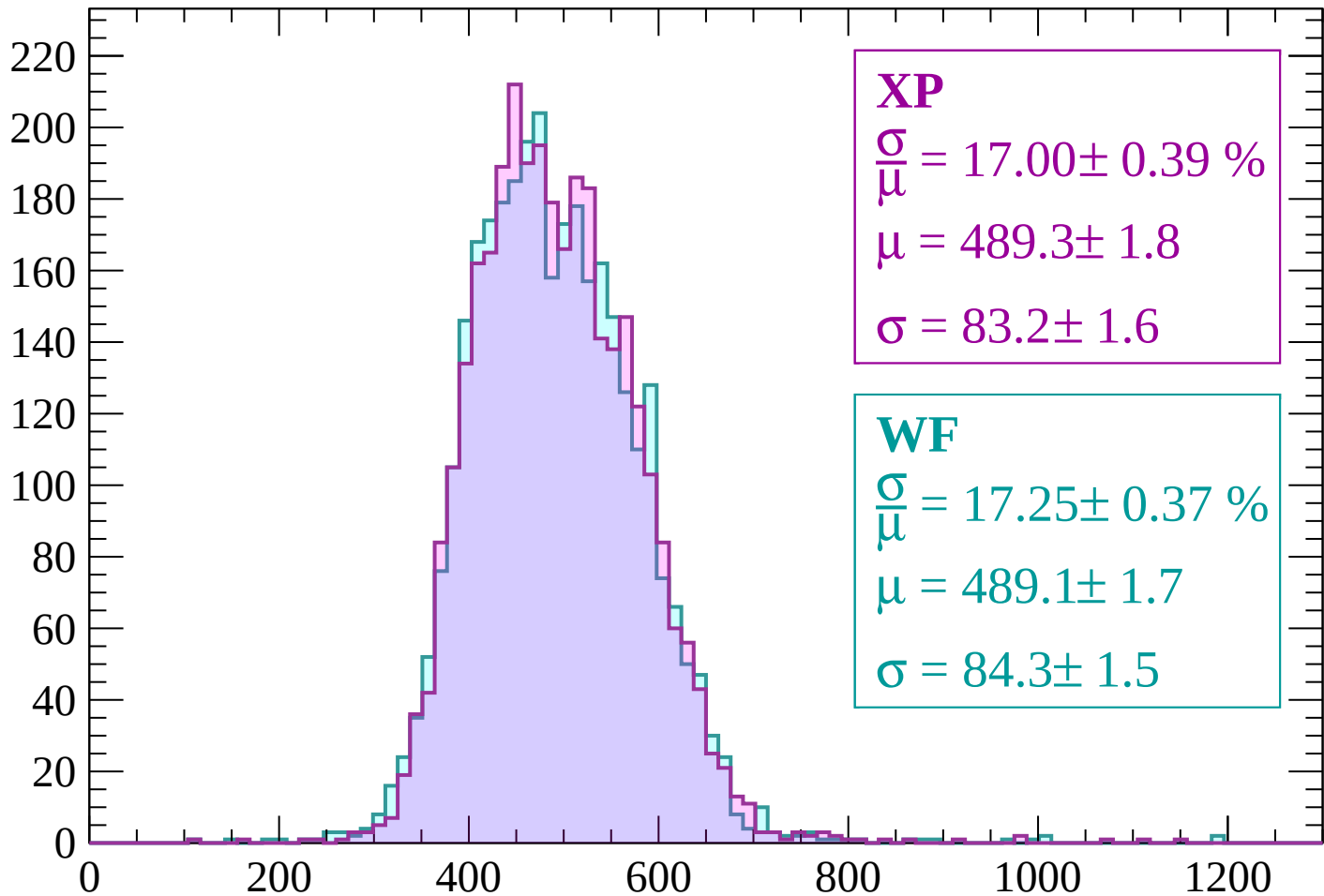
Energy loss | -2100 < p < -2040

Count



Energy loss | -2040 < p < -1980

Count



XP

$$\frac{\sigma}{\mu} = 17.00 \pm 0.39 \%$$

$$\mu = 489.3 \pm 1.8$$

$$\sigma = 83.2 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 17.25 \pm 0.37 \%$$

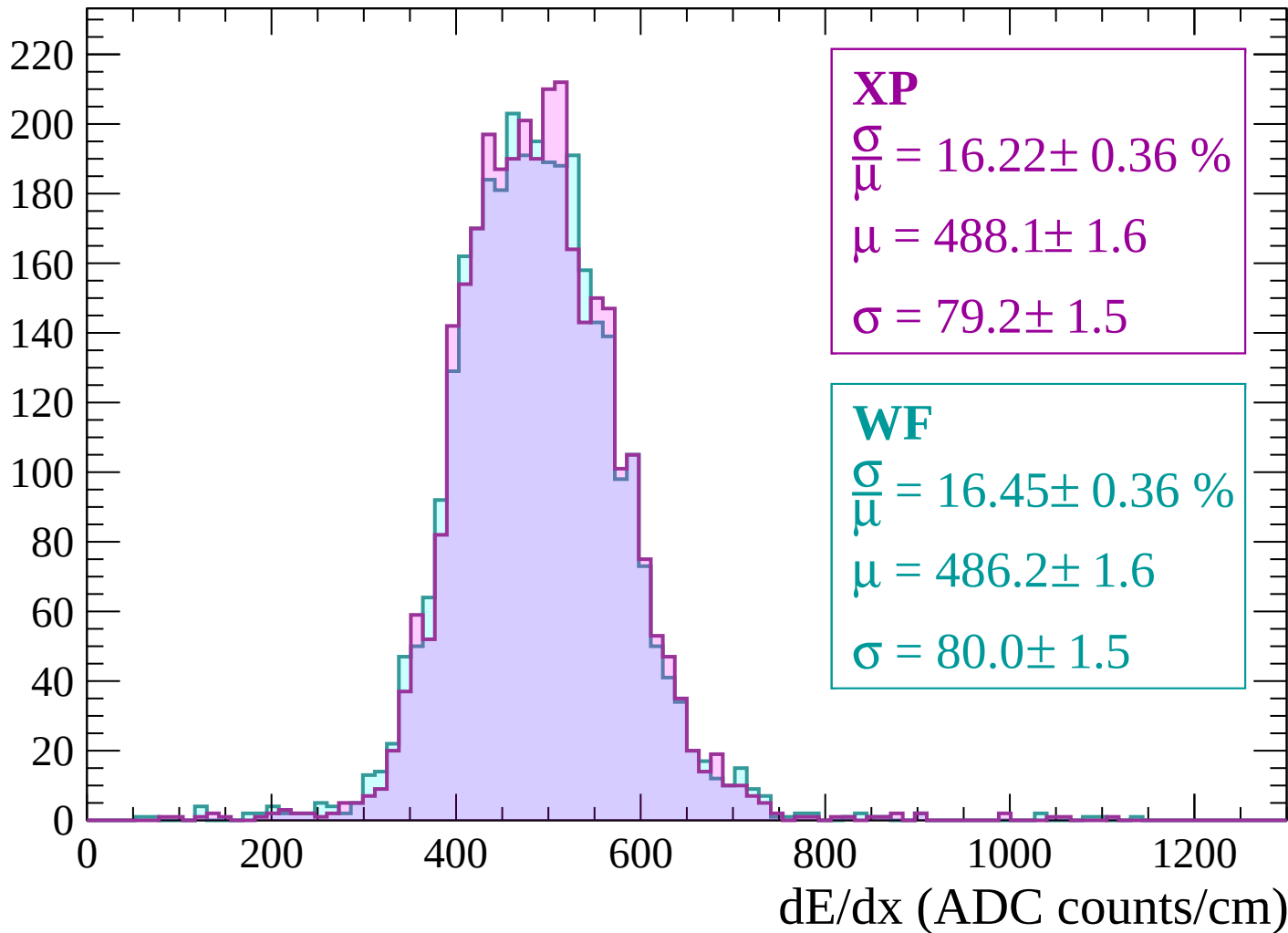
$$\mu = 489.1 \pm 1.7$$

$$\sigma = 84.3 \pm 1.5$$

dE/dx (ADC counts/cm)

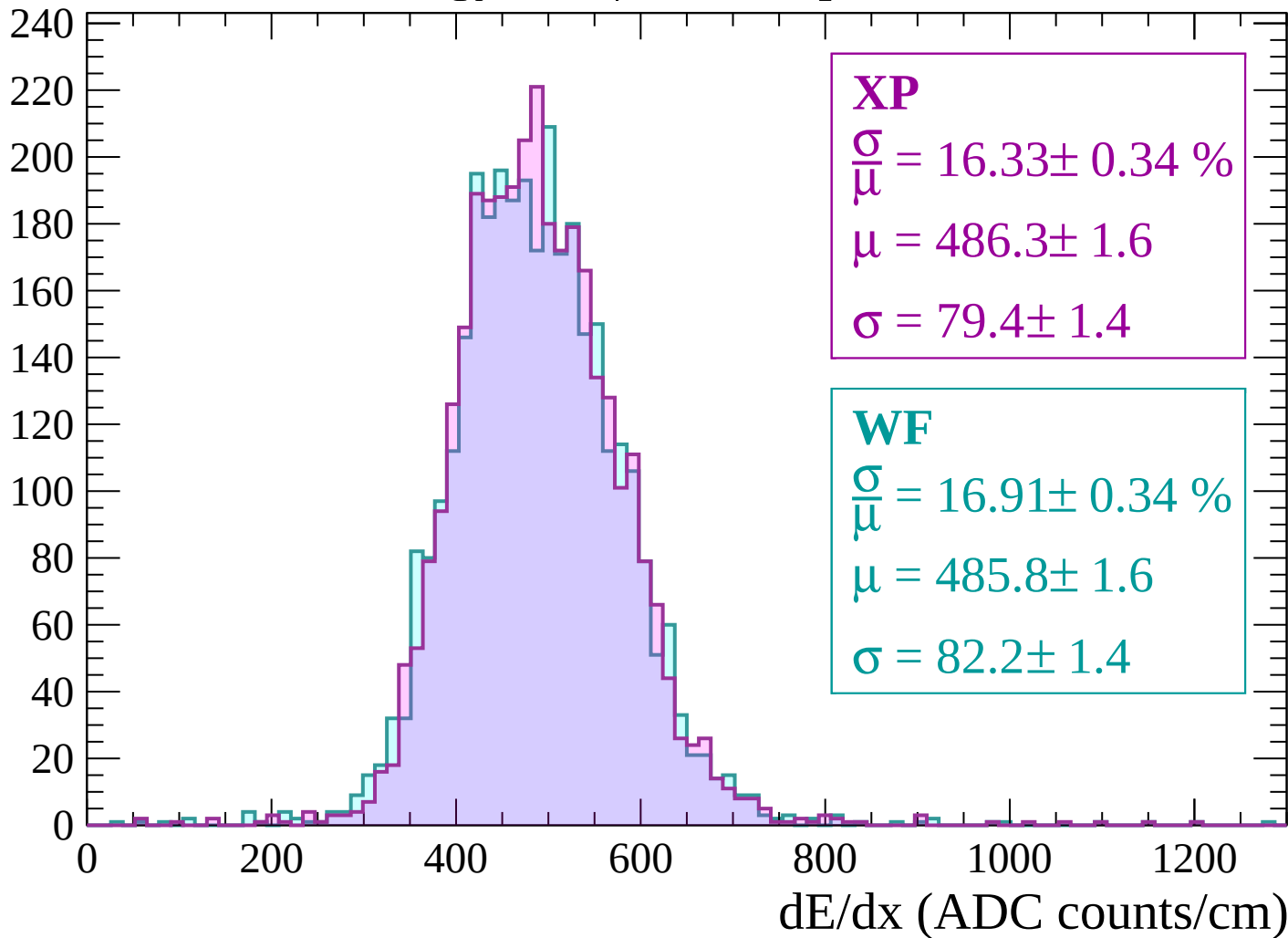
Energy loss | -1980 < p < -1920

Count



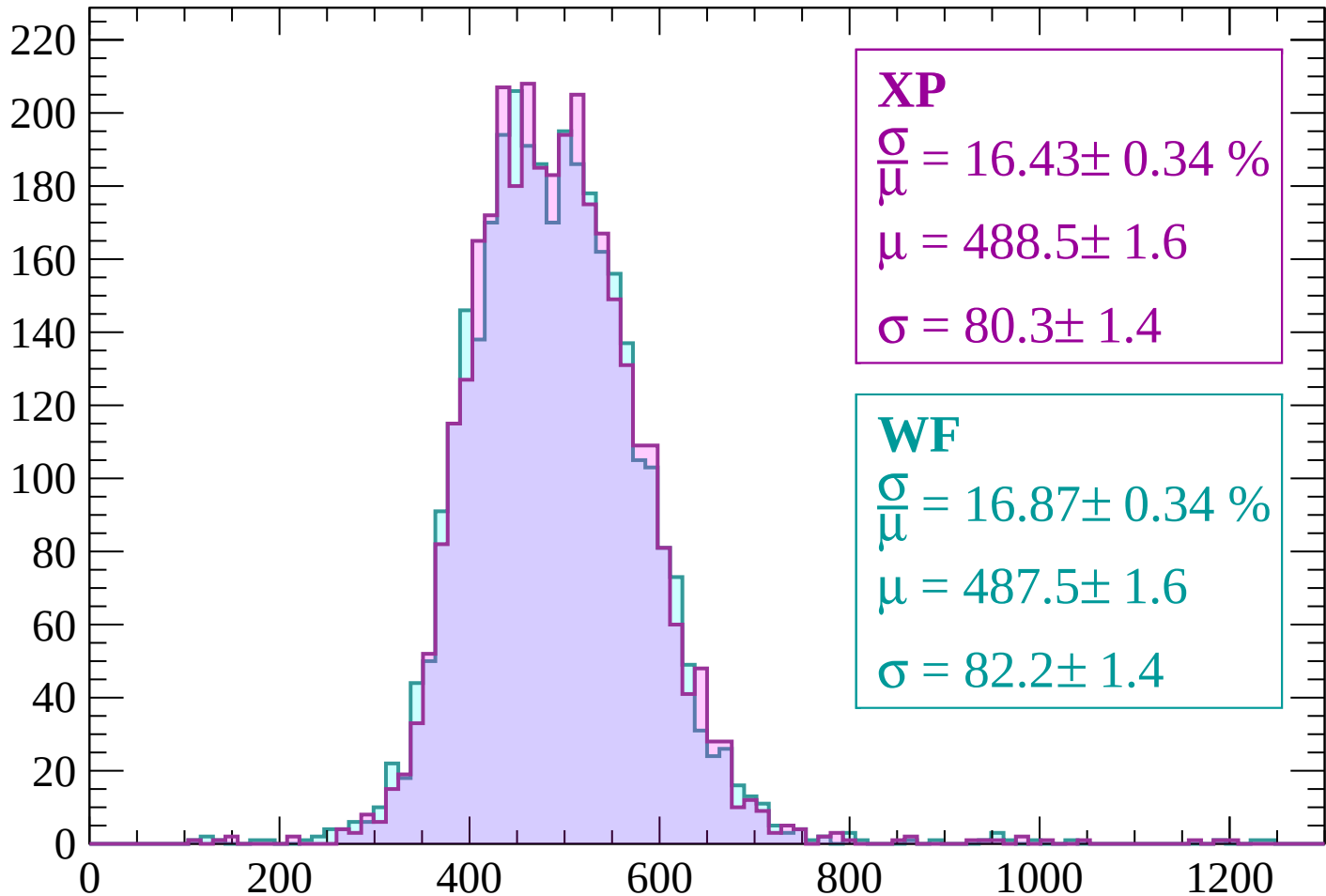
Energy loss | -1920 < p < -1860

Count



Energy loss | -1860 < p < -1800

Count



XP

$$\frac{\sigma}{\mu} = 16.43 \pm 0.34 \%$$

$$\mu = 488.5 \pm 1.6$$

$$\sigma = 80.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 16.87 \pm 0.34 \%$$

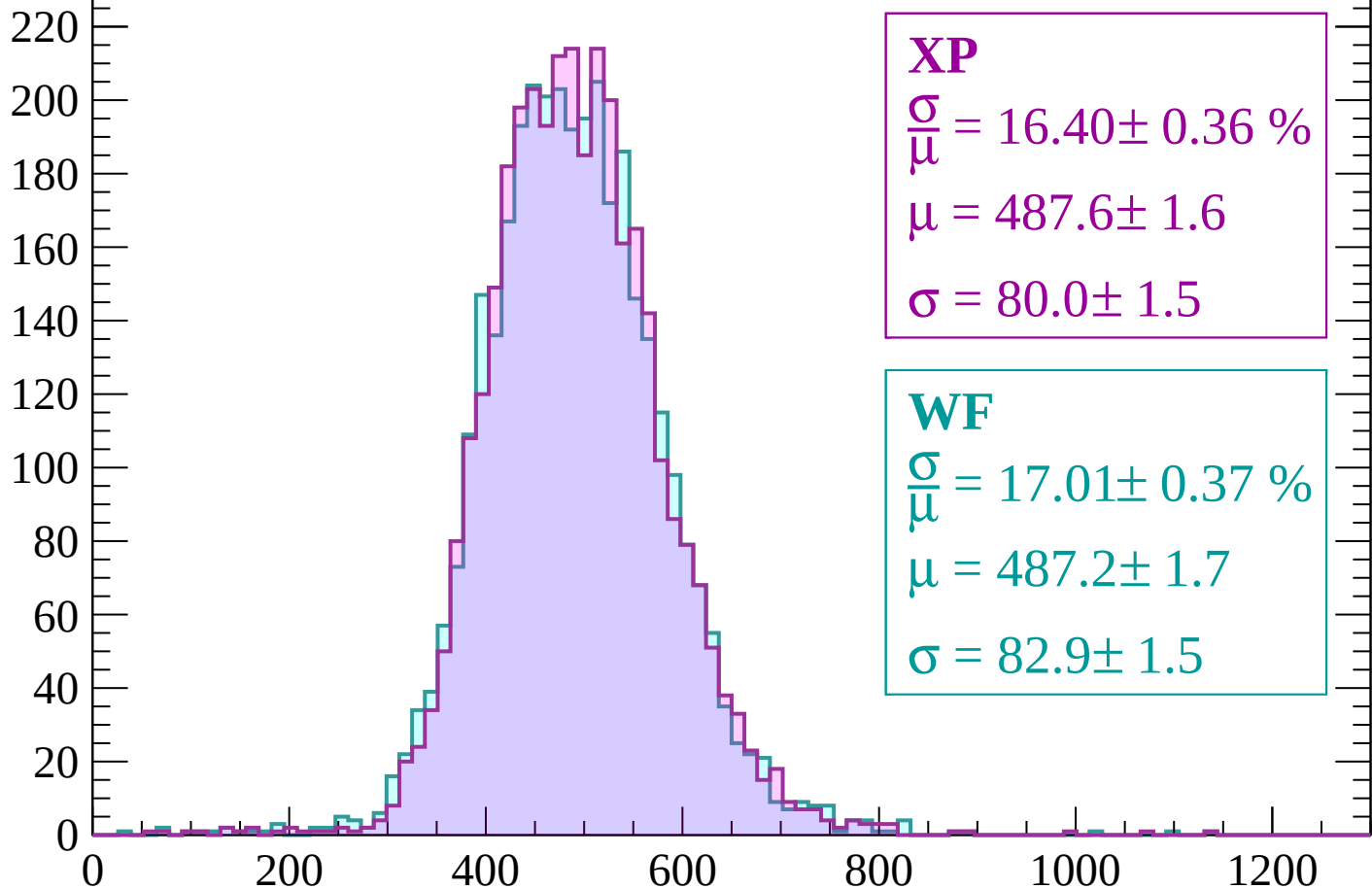
$$\mu = 487.5 \pm 1.6$$

$$\sigma = 82.2 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | -1800 < p < -1740

Count



XP

$$\frac{\sigma}{\mu} = 16.40 \pm 0.36 \%$$

$$\mu = 487.6 \pm 1.6$$

$$\sigma = 80.0 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 17.01 \pm 0.37 \%$$

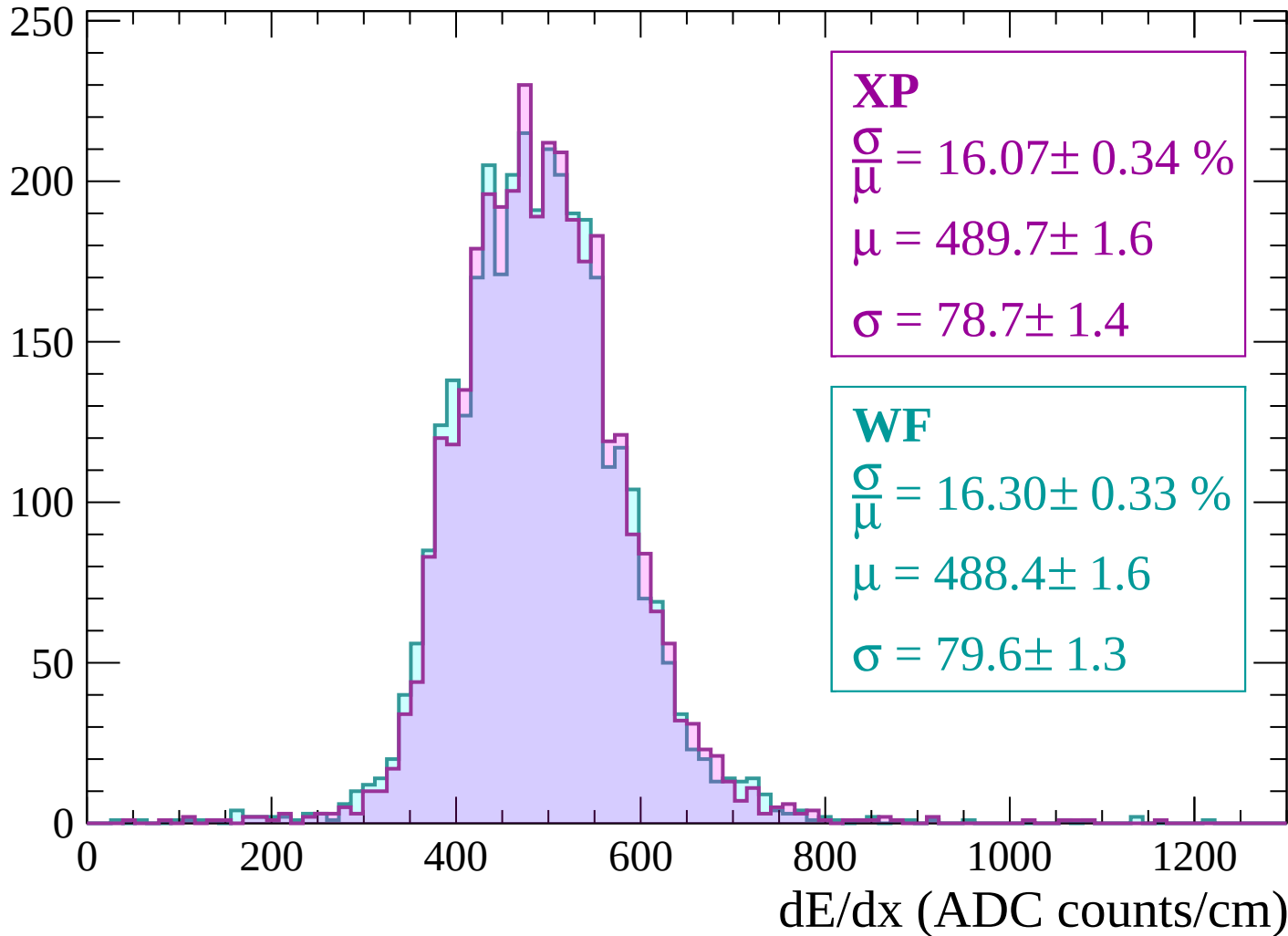
$$\mu = 487.2 \pm 1.7$$

$$\sigma = 82.9 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1740 < p < -1680$

Count



Energy loss | -1680 < p < -1620

Count

220
200
180
160
140
120
100
80
60
40
20
0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 16.20 \pm 0.32 \%$$

$$\mu = 489.2 \pm 1.6$$

$$\sigma = 79.2 \pm 1.3$$

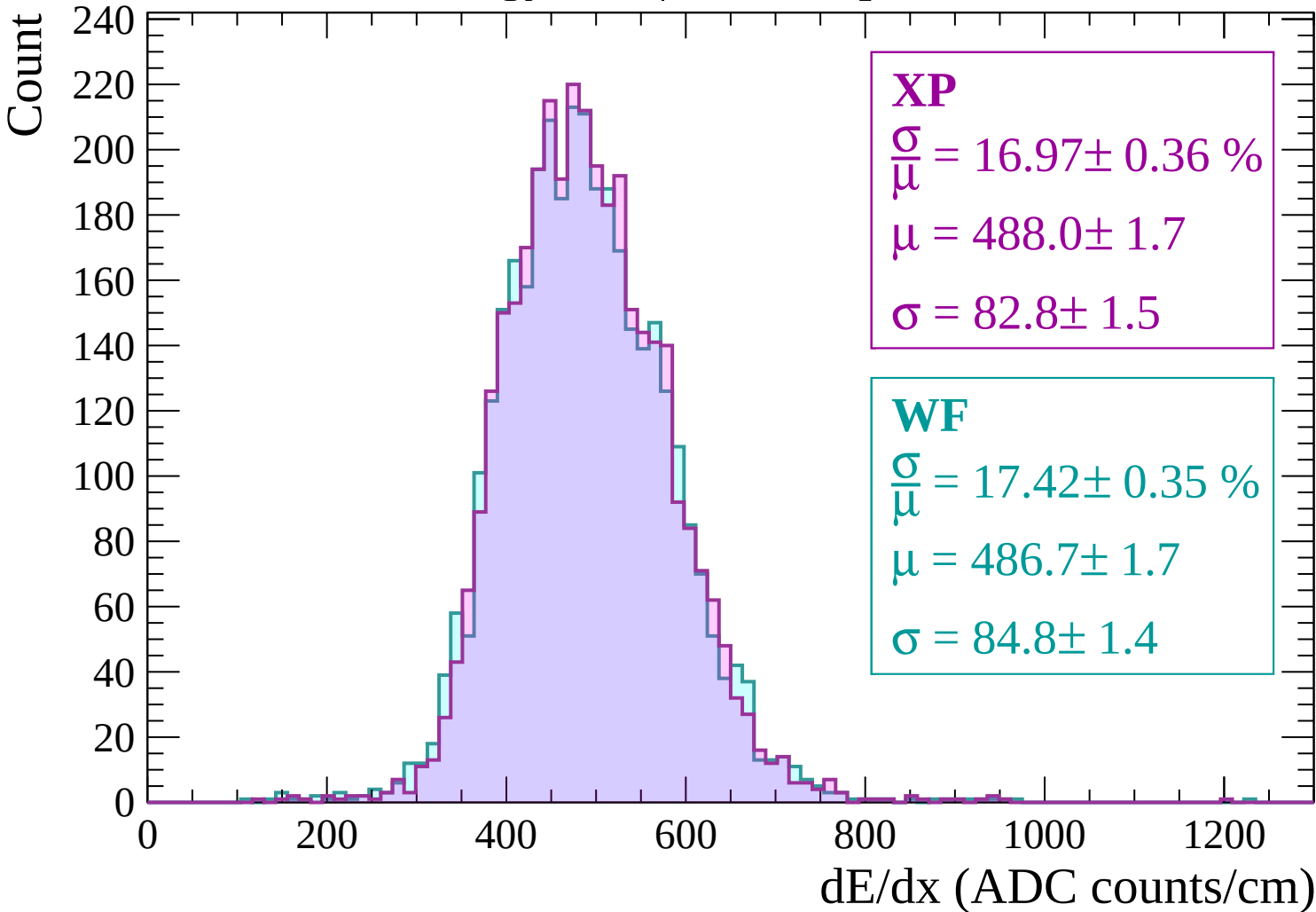
WF

$$\frac{\sigma}{\mu} = 17.20 \pm 0.35 \%$$

$$\mu = 486.9 \pm 1.7$$

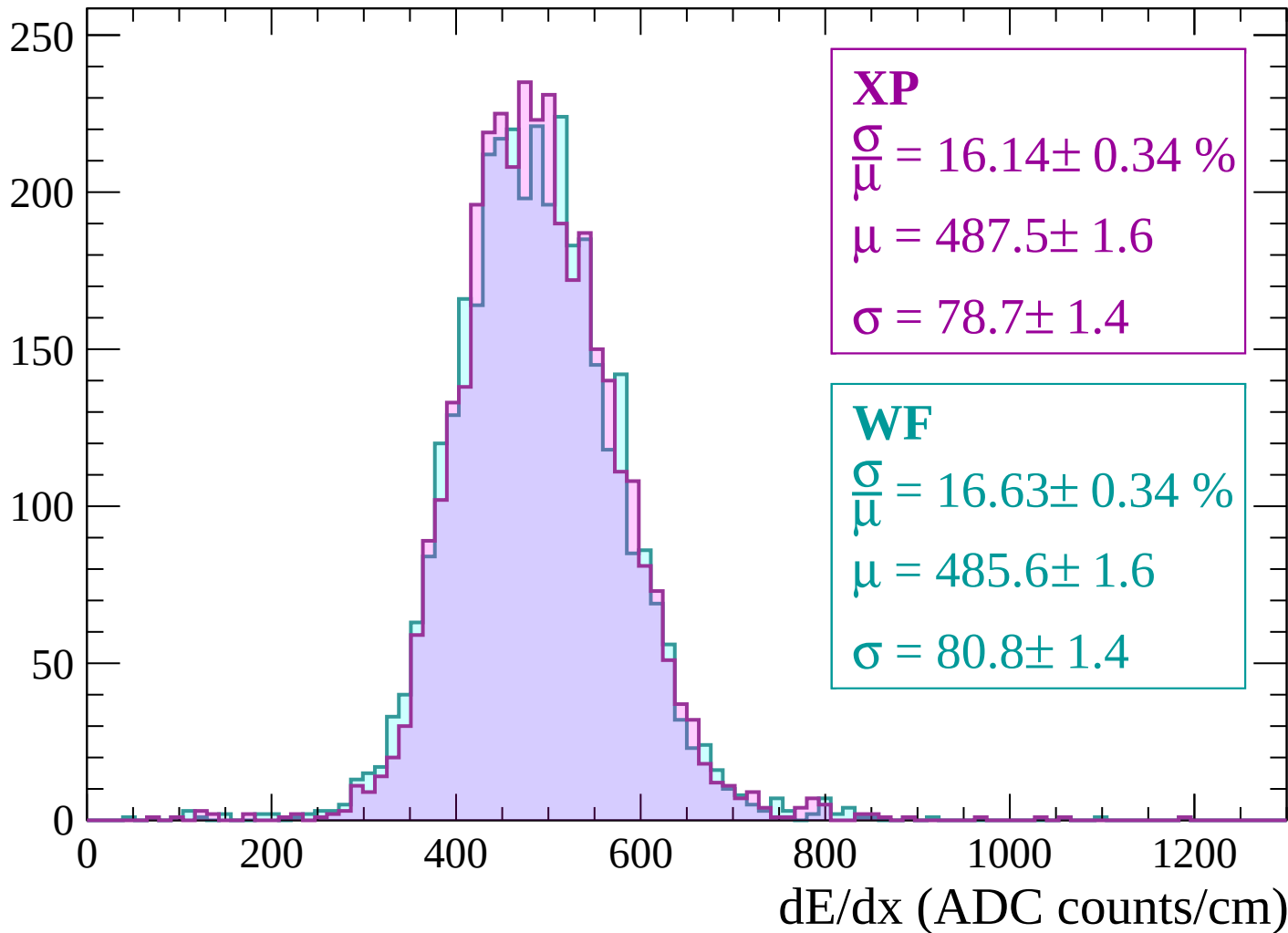
$$\sigma = 83.7 \pm 1.4$$

Energy loss | -1620 < p < -1560



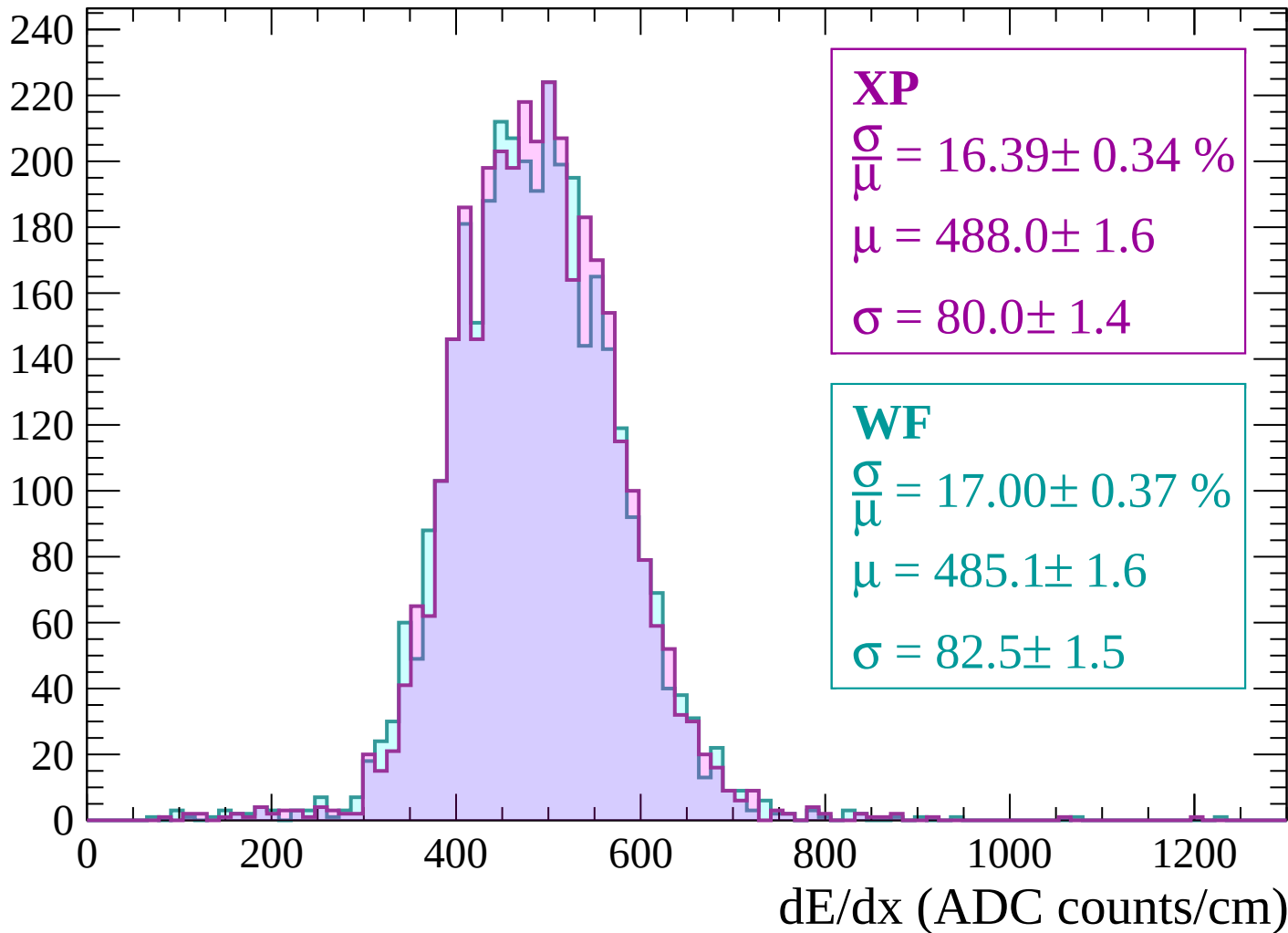
Energy loss | -1560 < p < -1500

Count



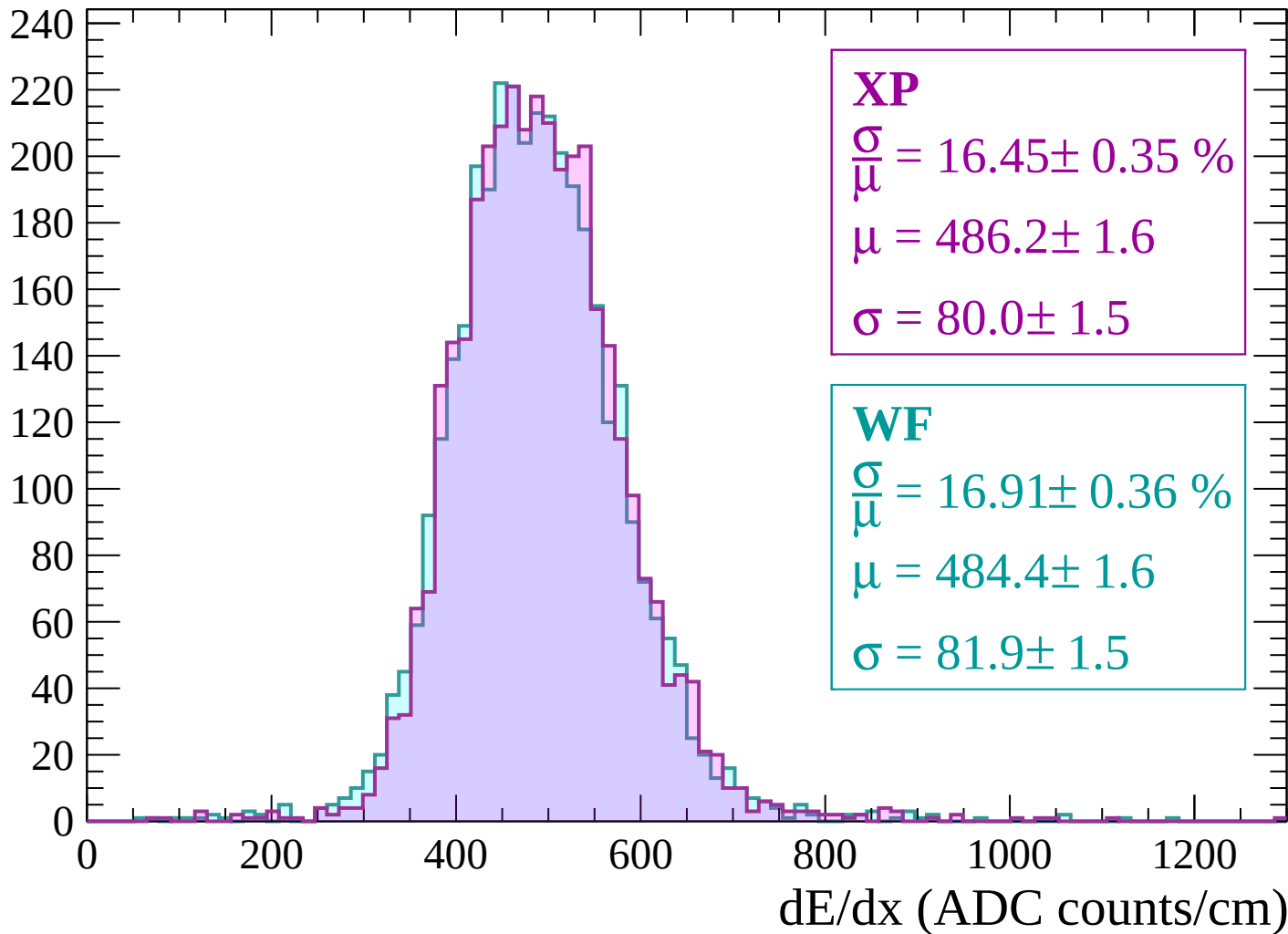
Energy loss | -1500 < p < -1440

Count

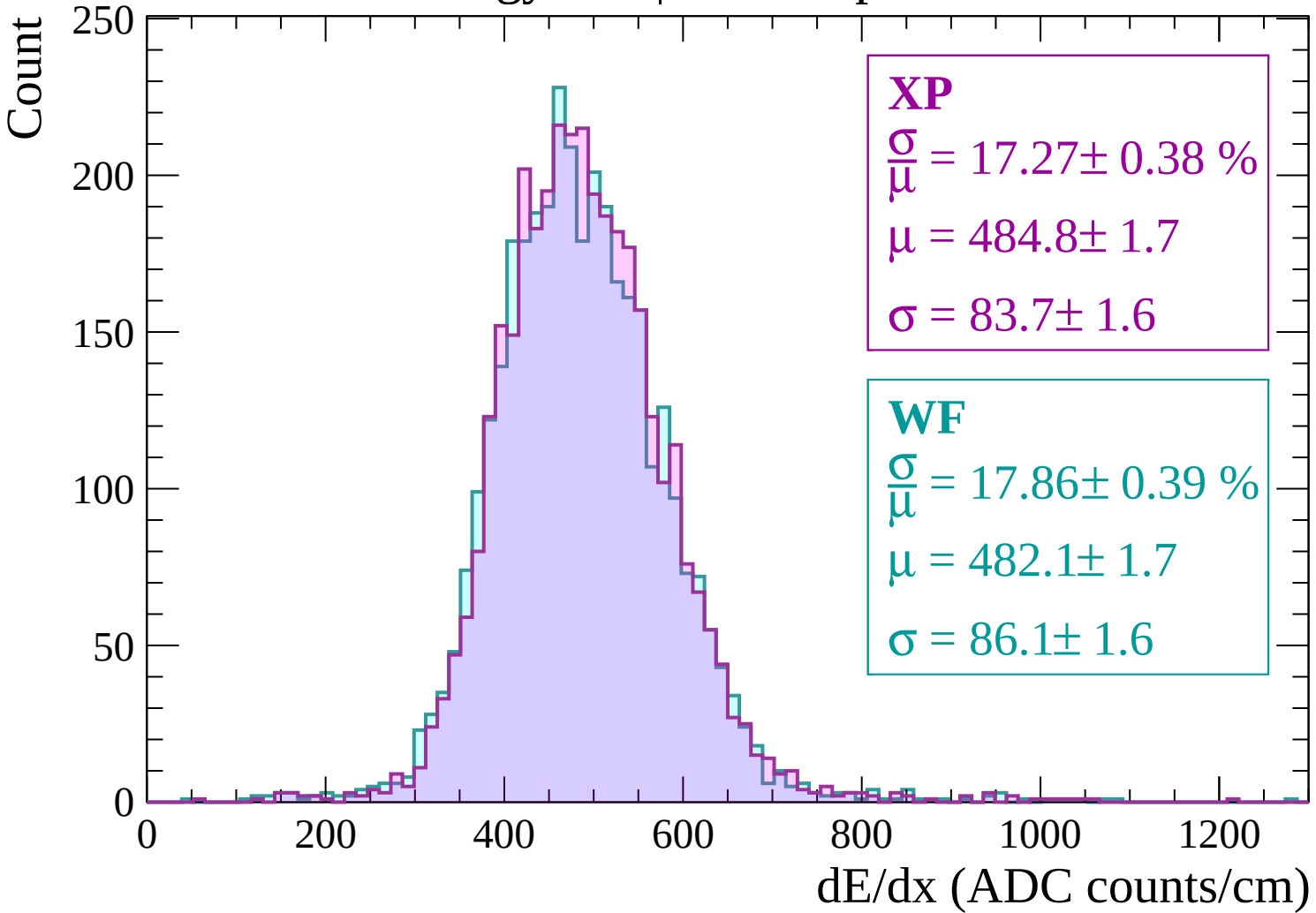


Energy loss | $-1440 < p < -1380$

Count

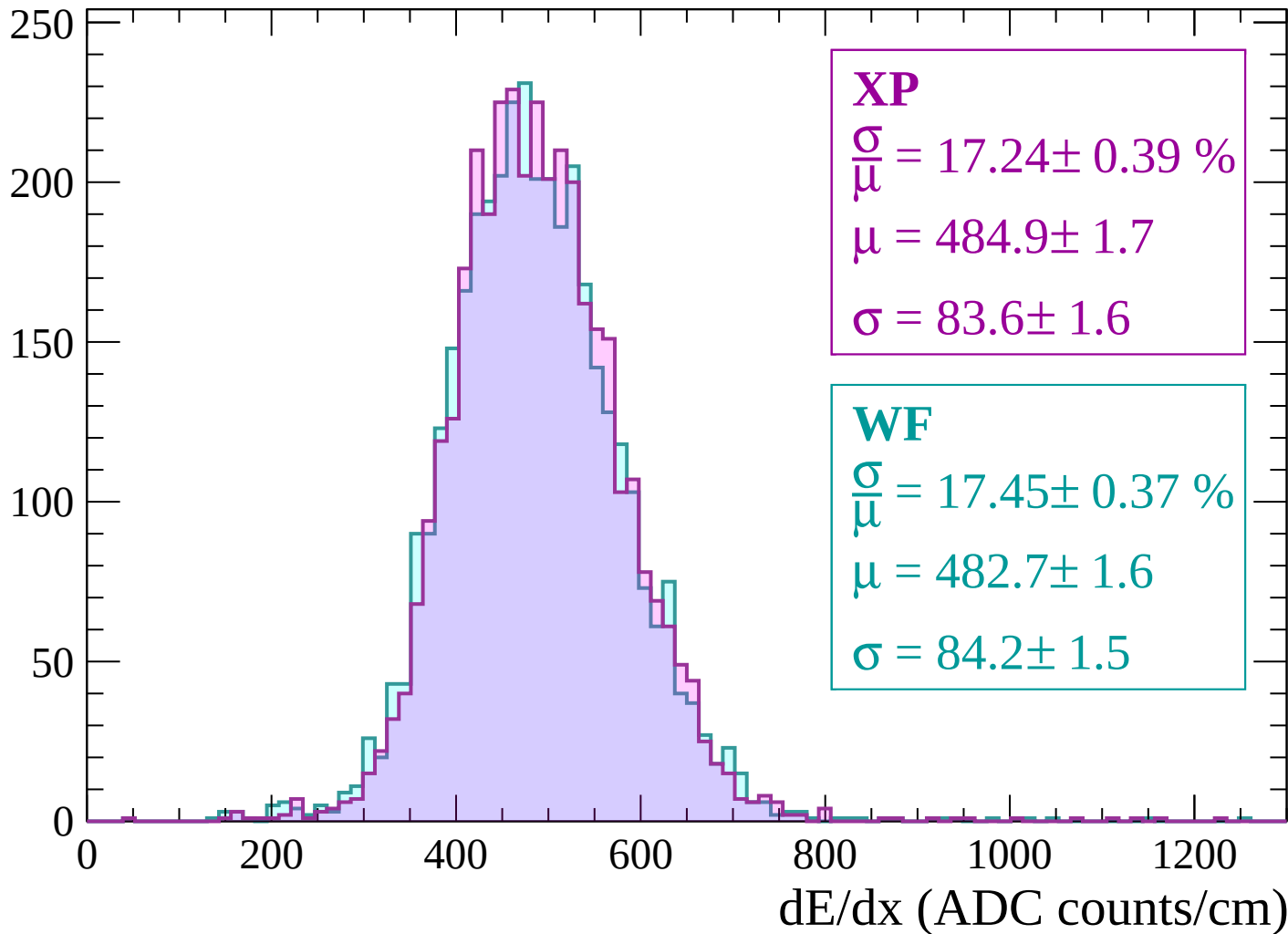


Energy loss | -1380 < p < -1320



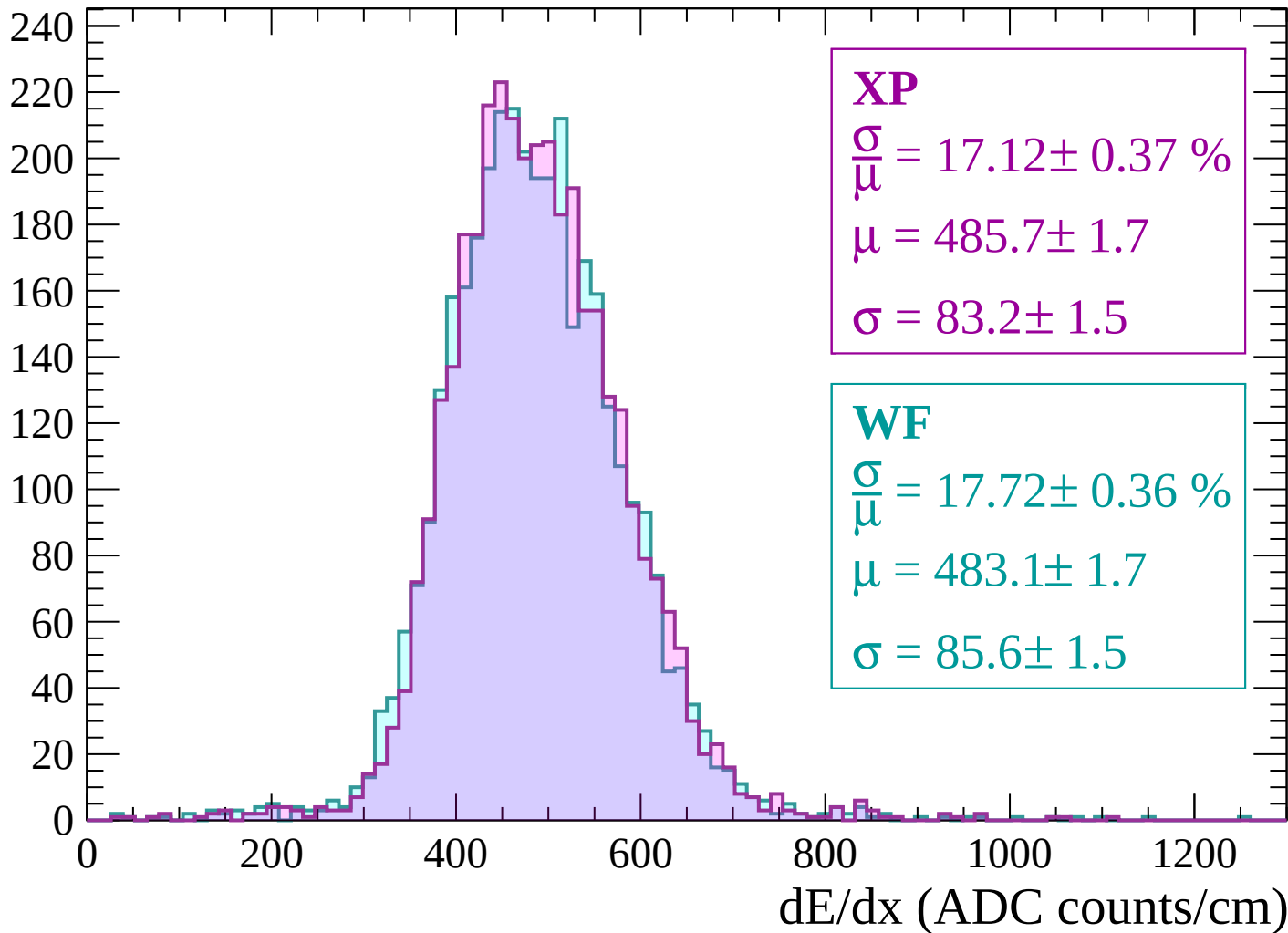
Energy loss | -1320 < p < -1260

Count



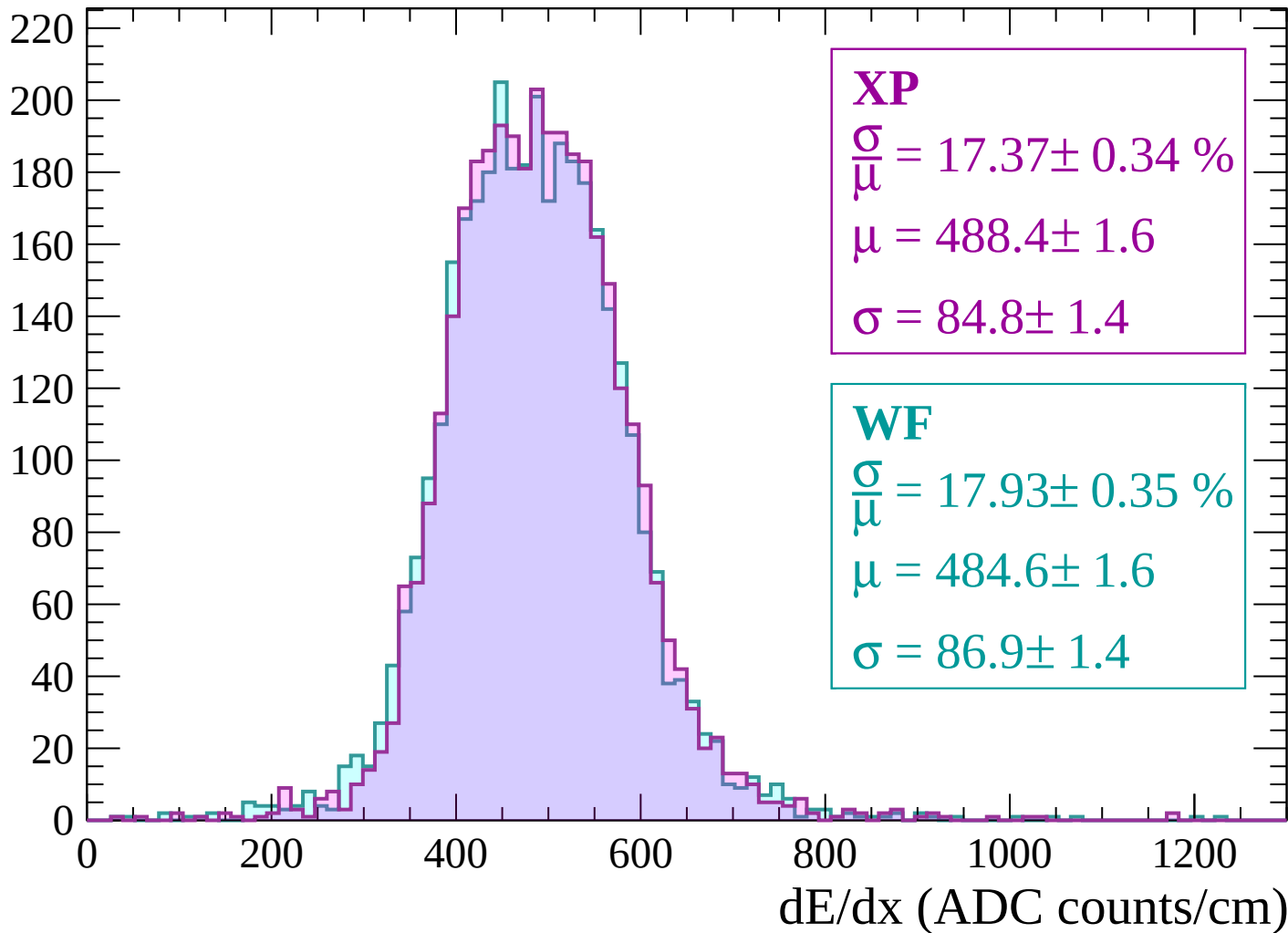
Energy loss | -1260 < p < -1200

Count



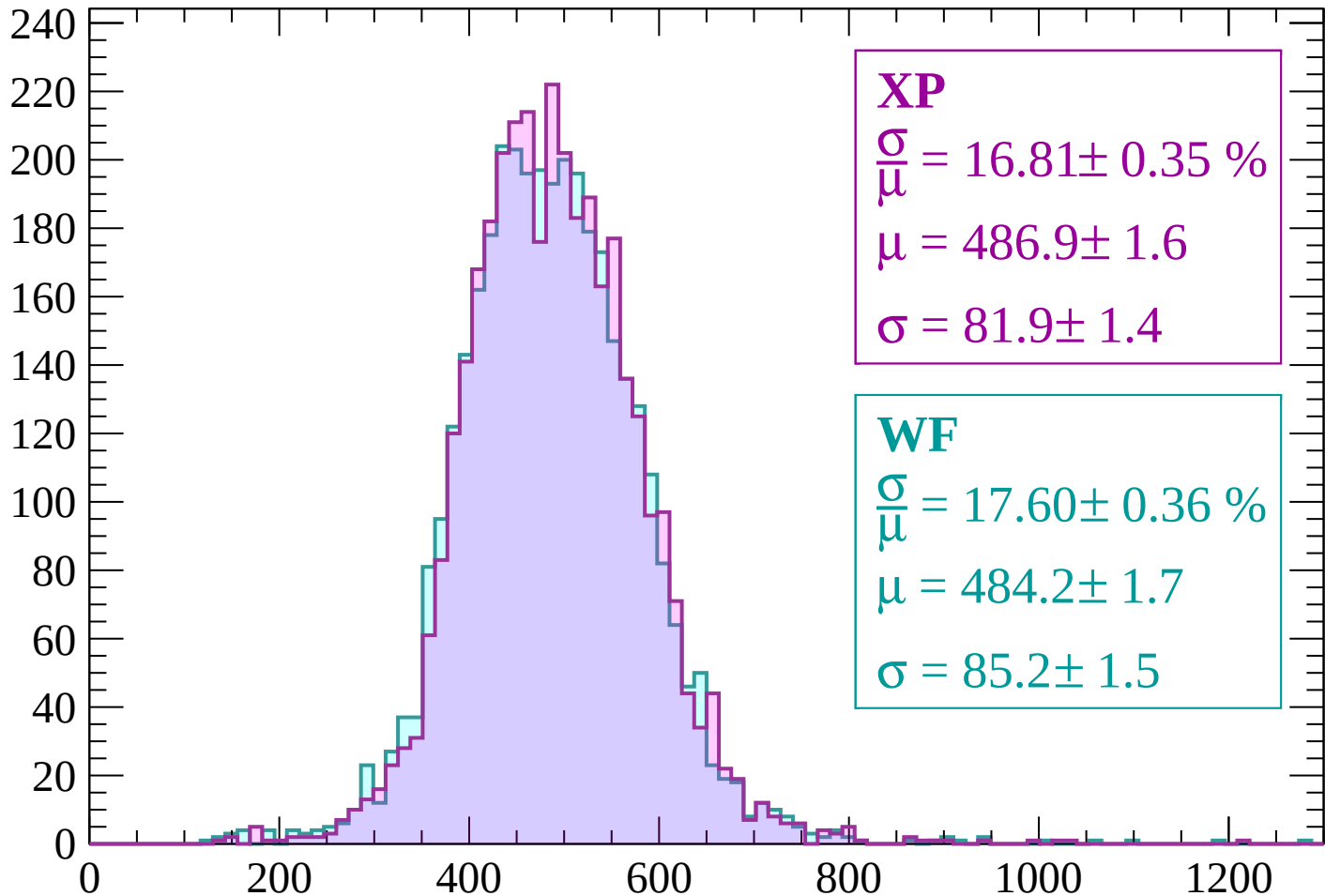
Energy loss | -1200 < p < -1140

Count



Energy loss | -1140 < p < -1080

Count



XP

$$\frac{\sigma}{\mu} = 16.81 \pm 0.35 \%$$

$$\mu = 486.9 \pm 1.6$$

$$\sigma = 81.9 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 17.60 \pm 0.36 \%$$

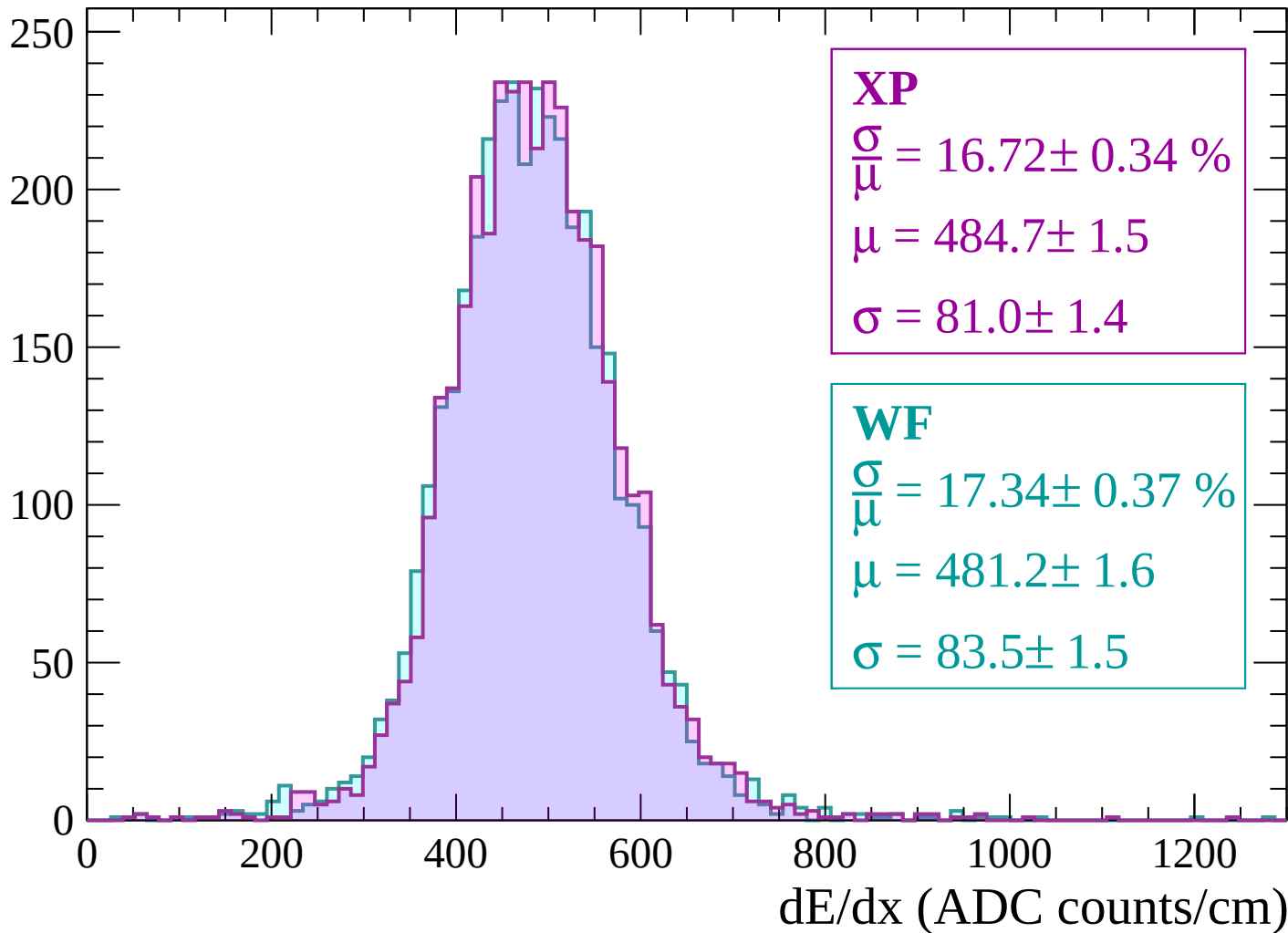
$$\mu = 484.2 \pm 1.7$$

$$\sigma = 85.2 \pm 1.5$$

dE/dx (ADC counts/cm)

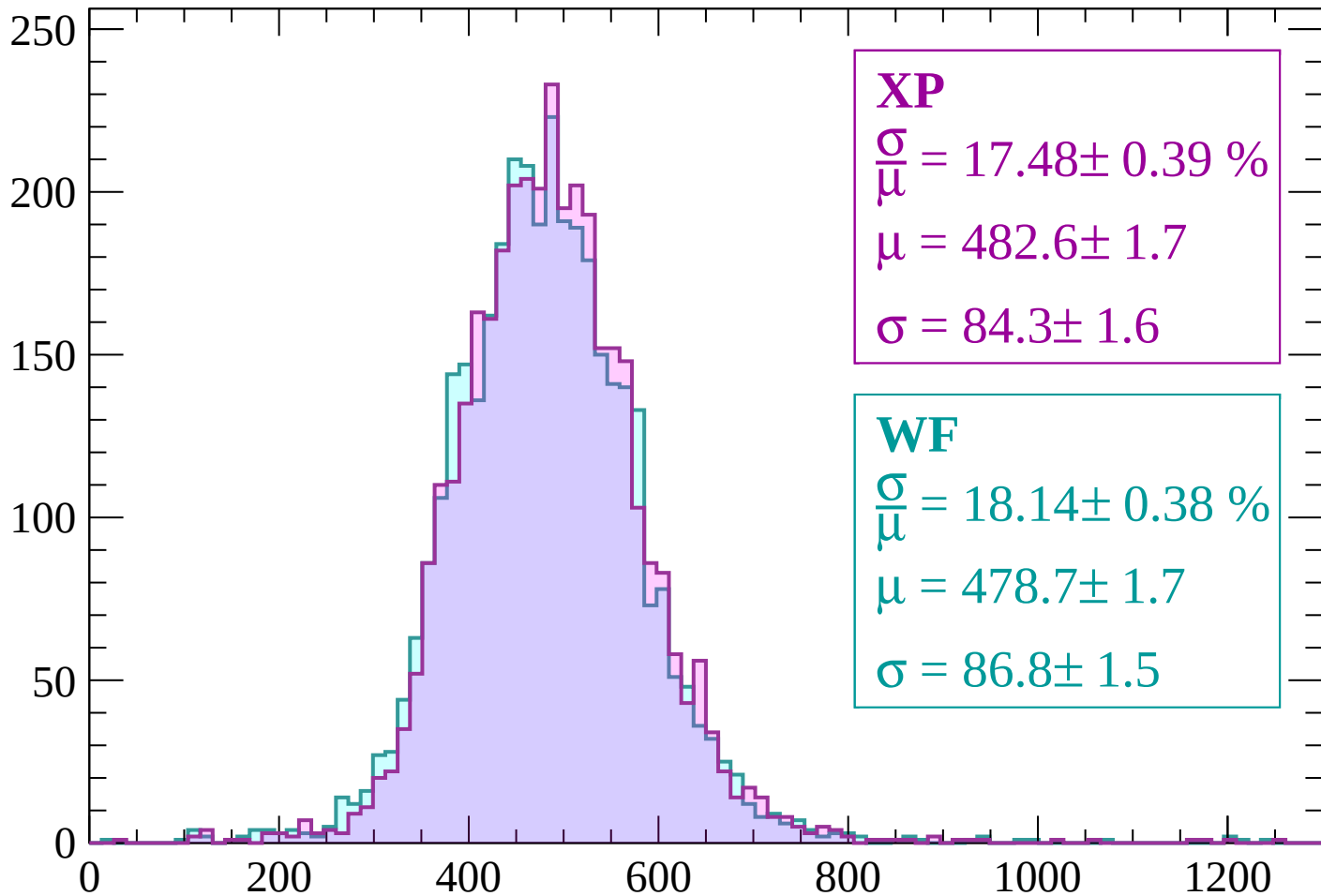
Energy loss | -1080 < p < -1020

Count



Energy loss | $-1020 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 17.48 \pm 0.39 \%$$

$$\mu = 482.6 \pm 1.7$$

$$\sigma = 84.3 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 18.14 \pm 0.38 \%$$

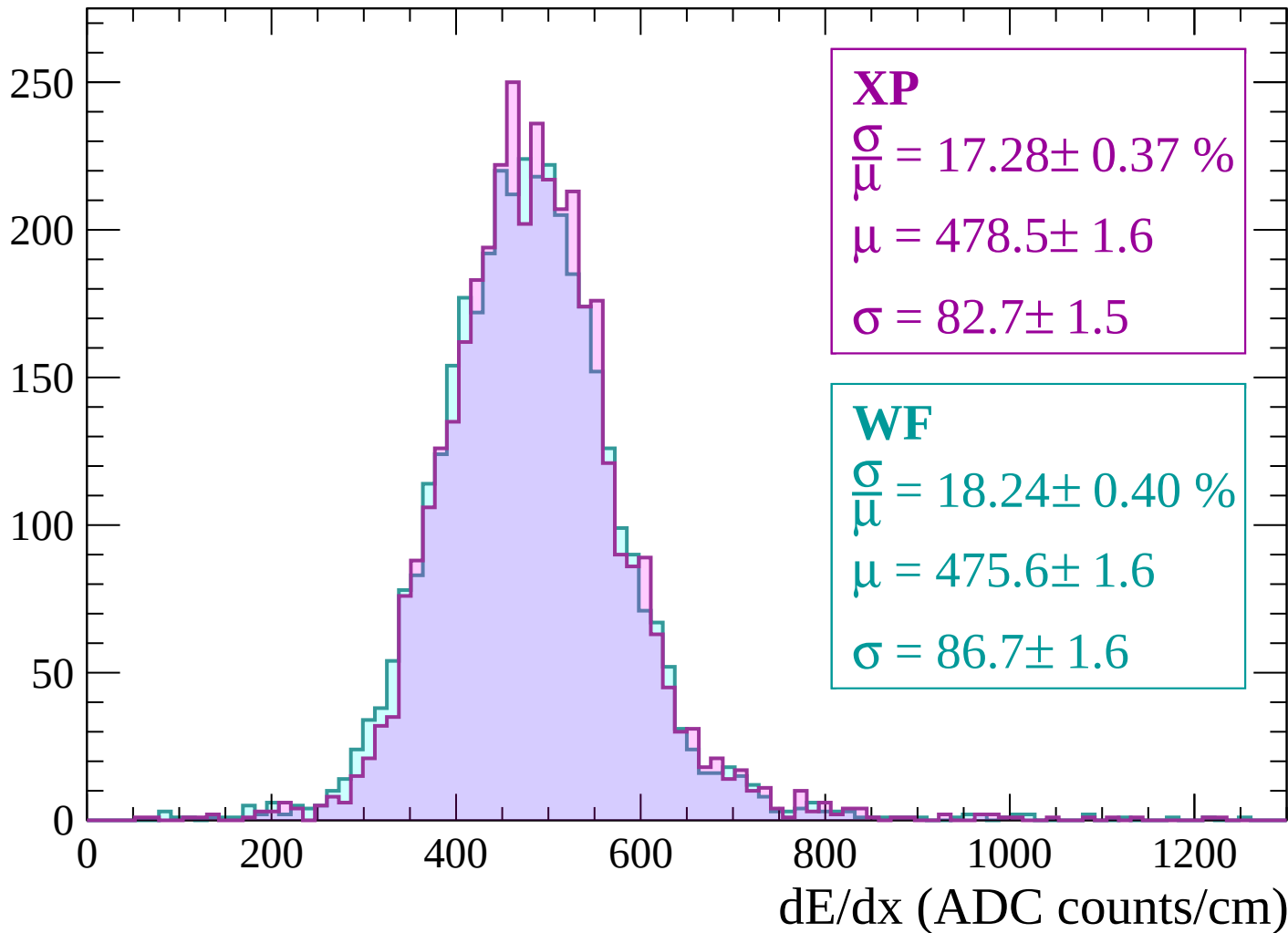
$$\mu = 478.7 \pm 1.7$$

$$\sigma = 86.8 \pm 1.5$$

dE/dx (ADC counts/cm)

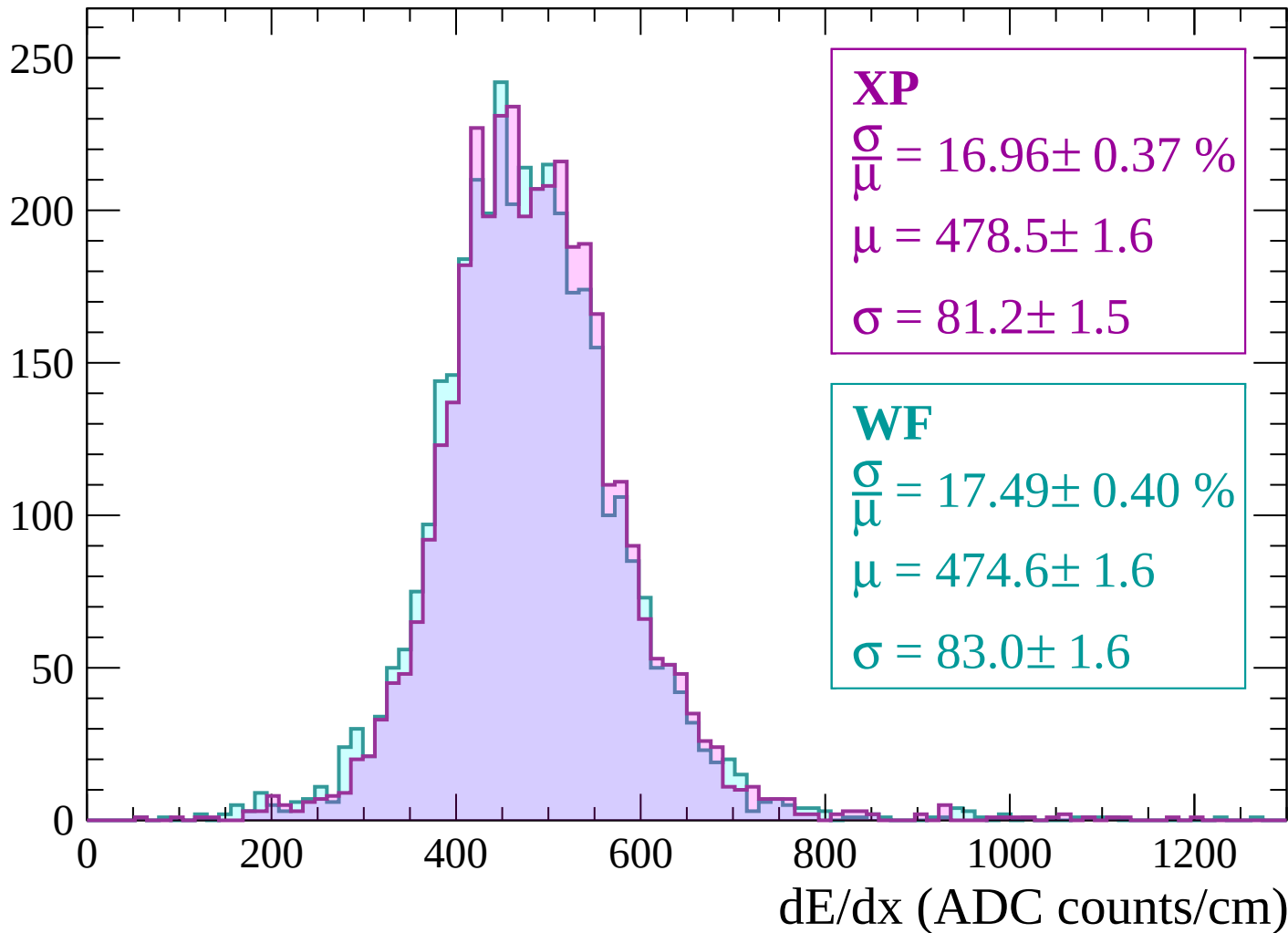
Energy loss | $-960 < p < -900$

Count



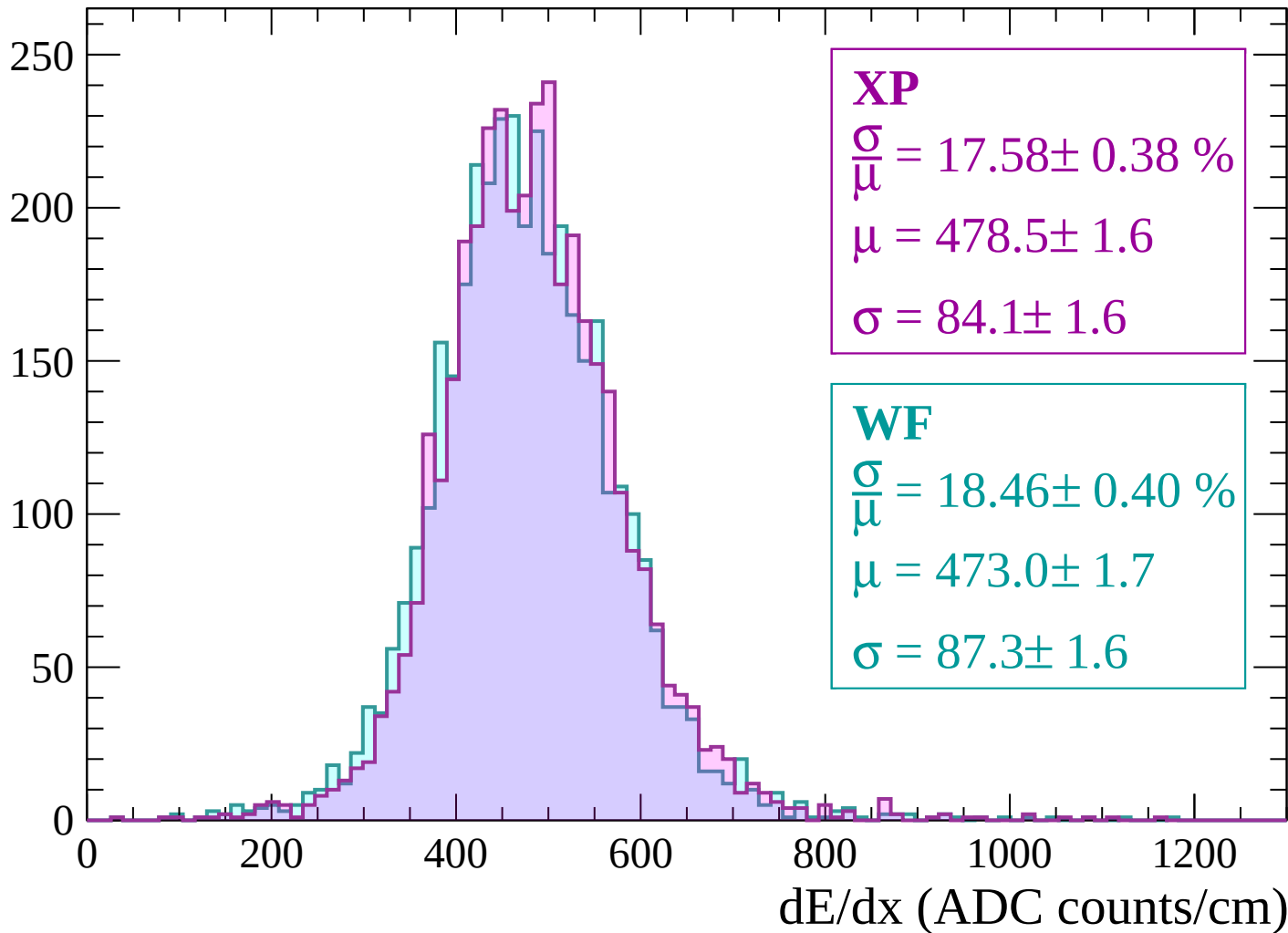
Energy loss | $-900 < p < -840$

Count



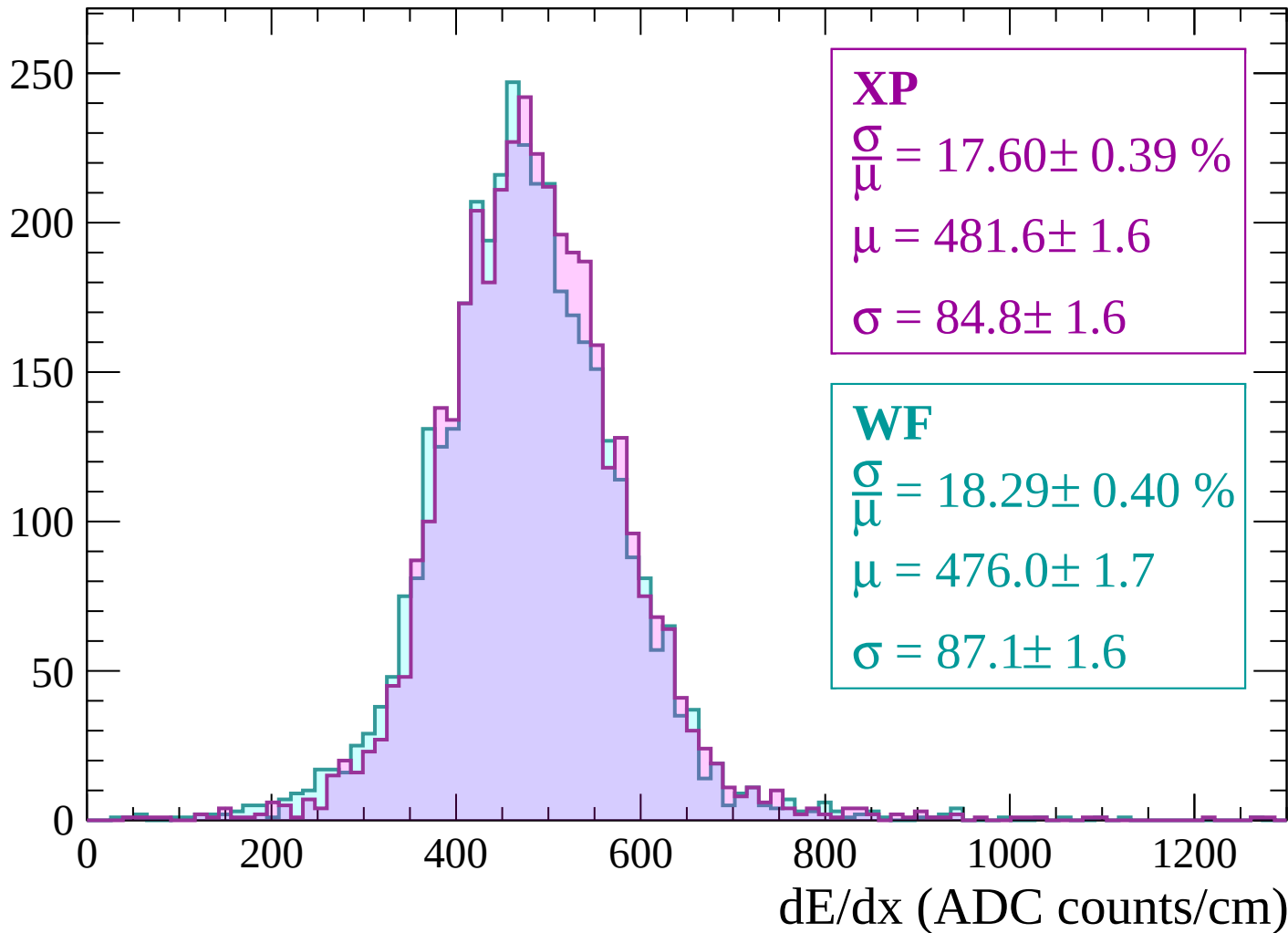
Energy loss | $-840 < p < -780$

Count



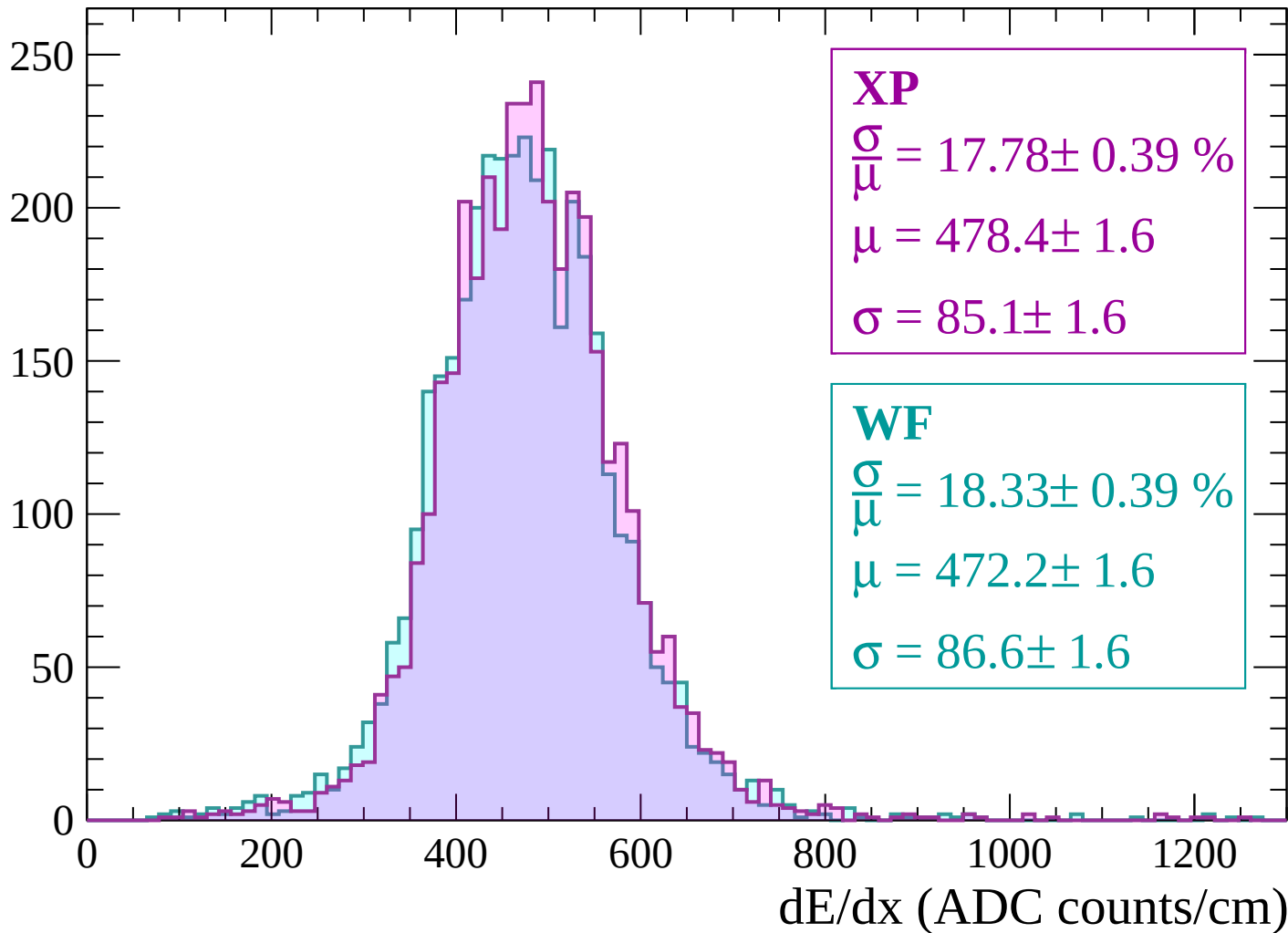
Energy loss | $-780 < p < -720$

Count



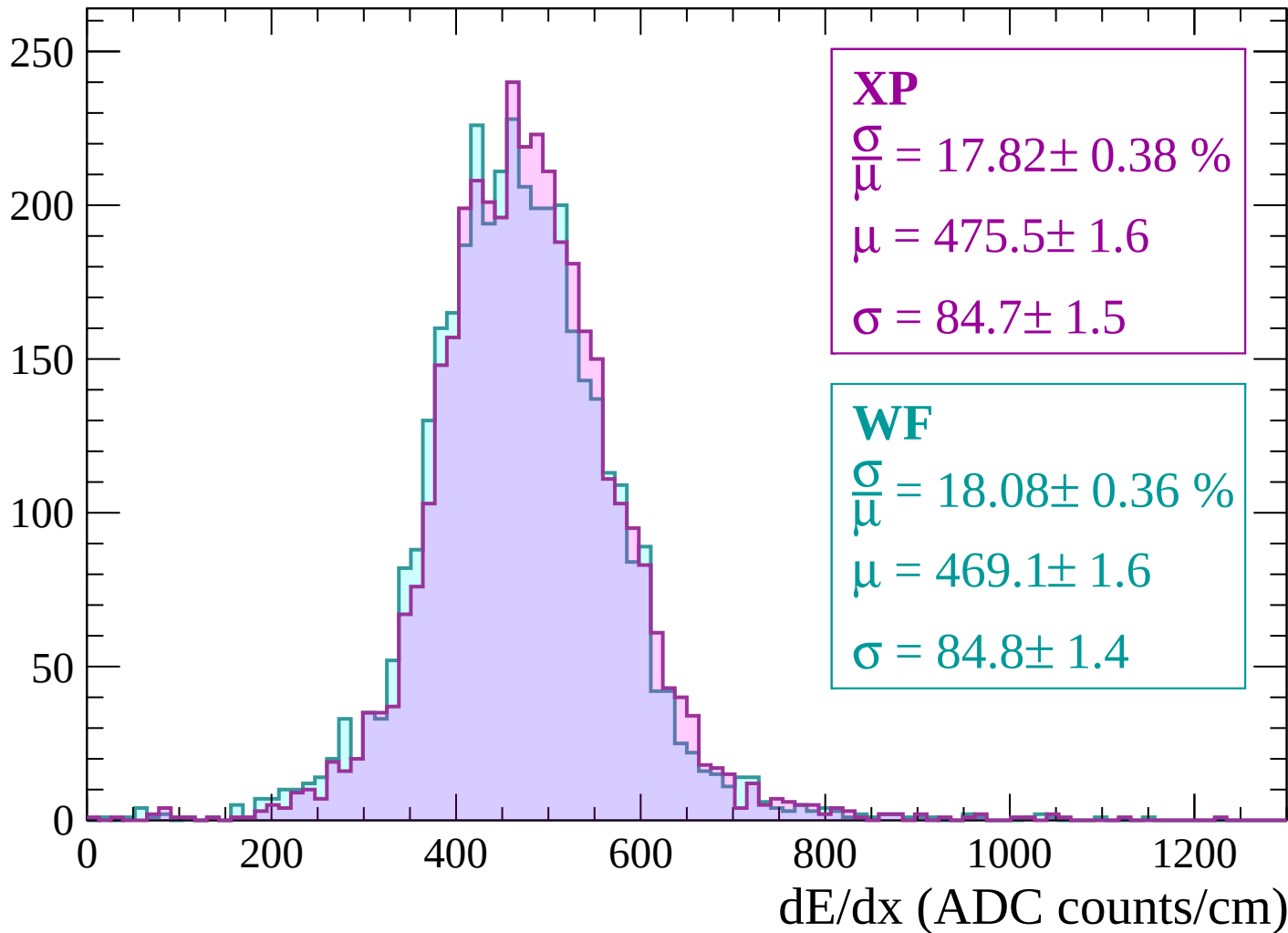
Energy loss | $-720 < p < -660$

Count



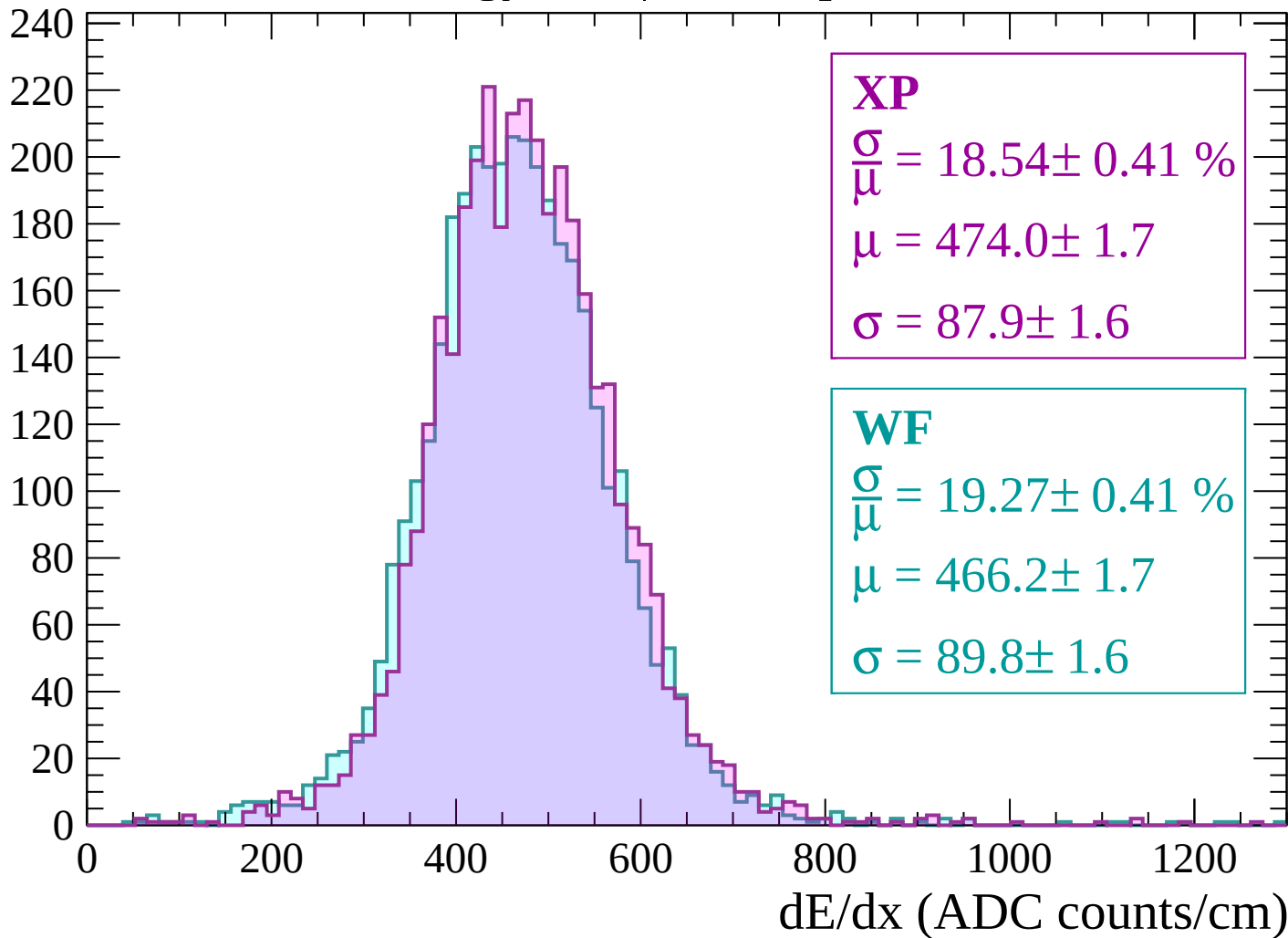
Energy loss | $-660 < p < -600$

Count



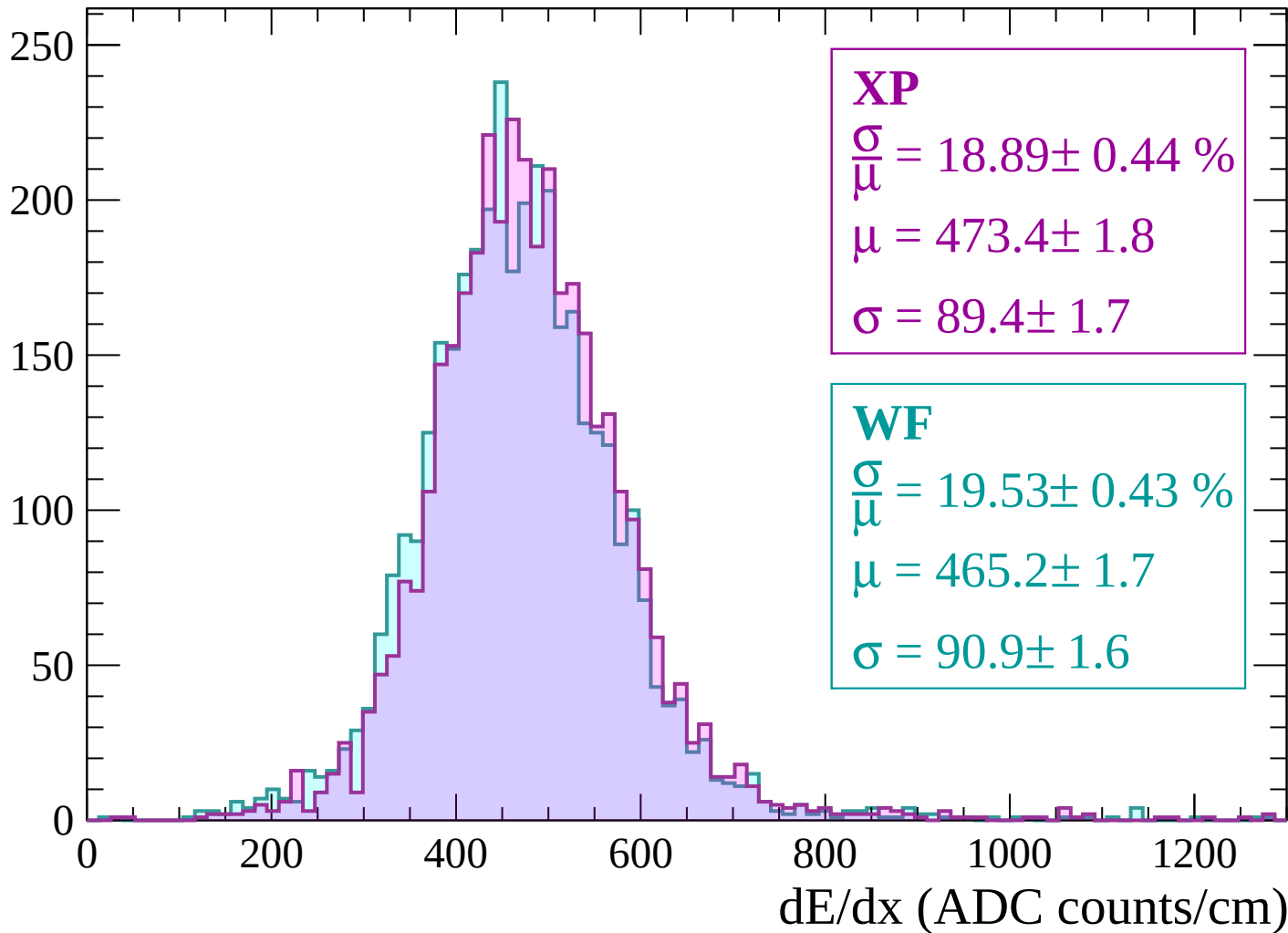
Energy loss | -600 < p < -540

Count



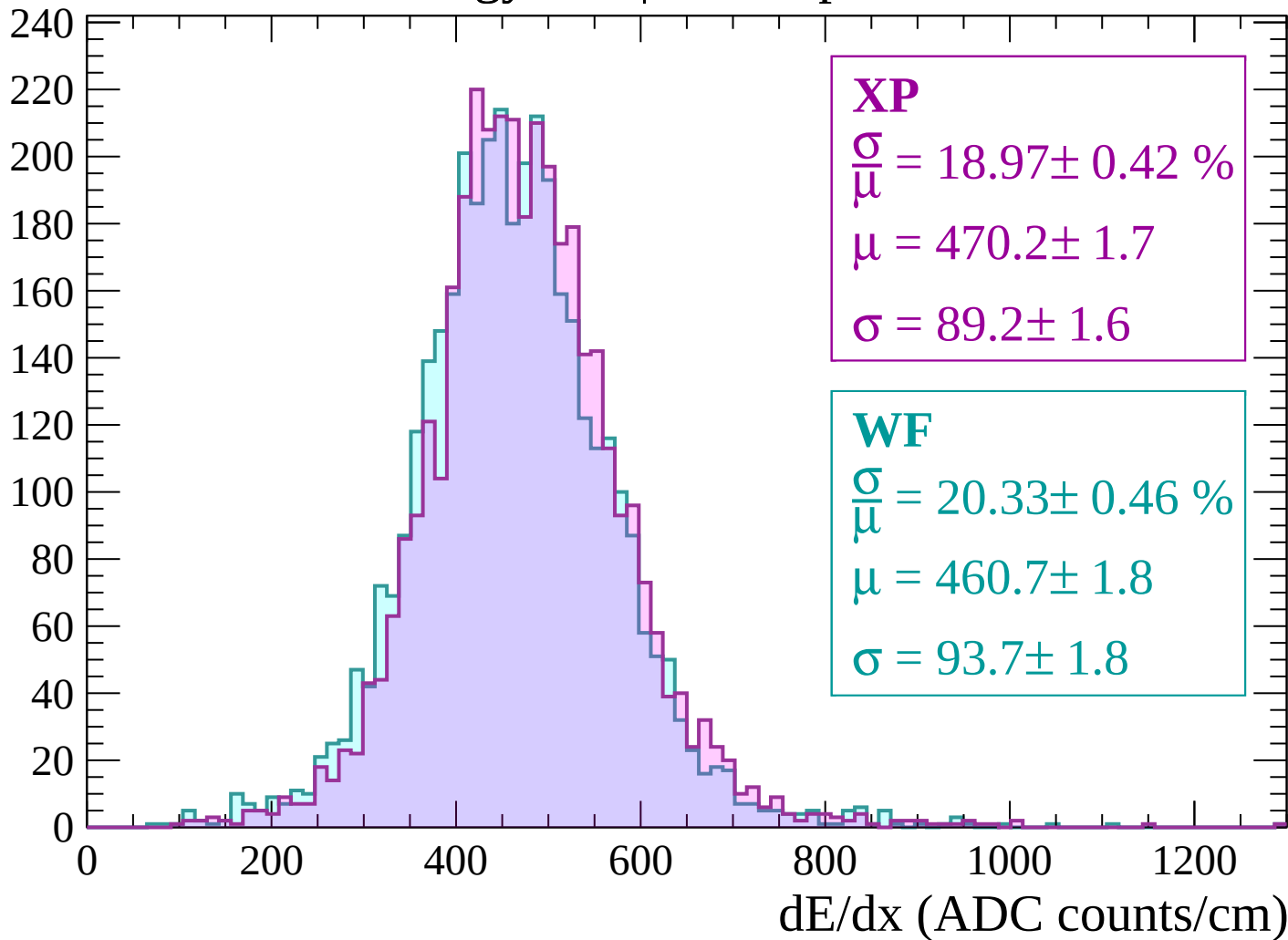
Energy loss | $-540 < p < -480$

Count



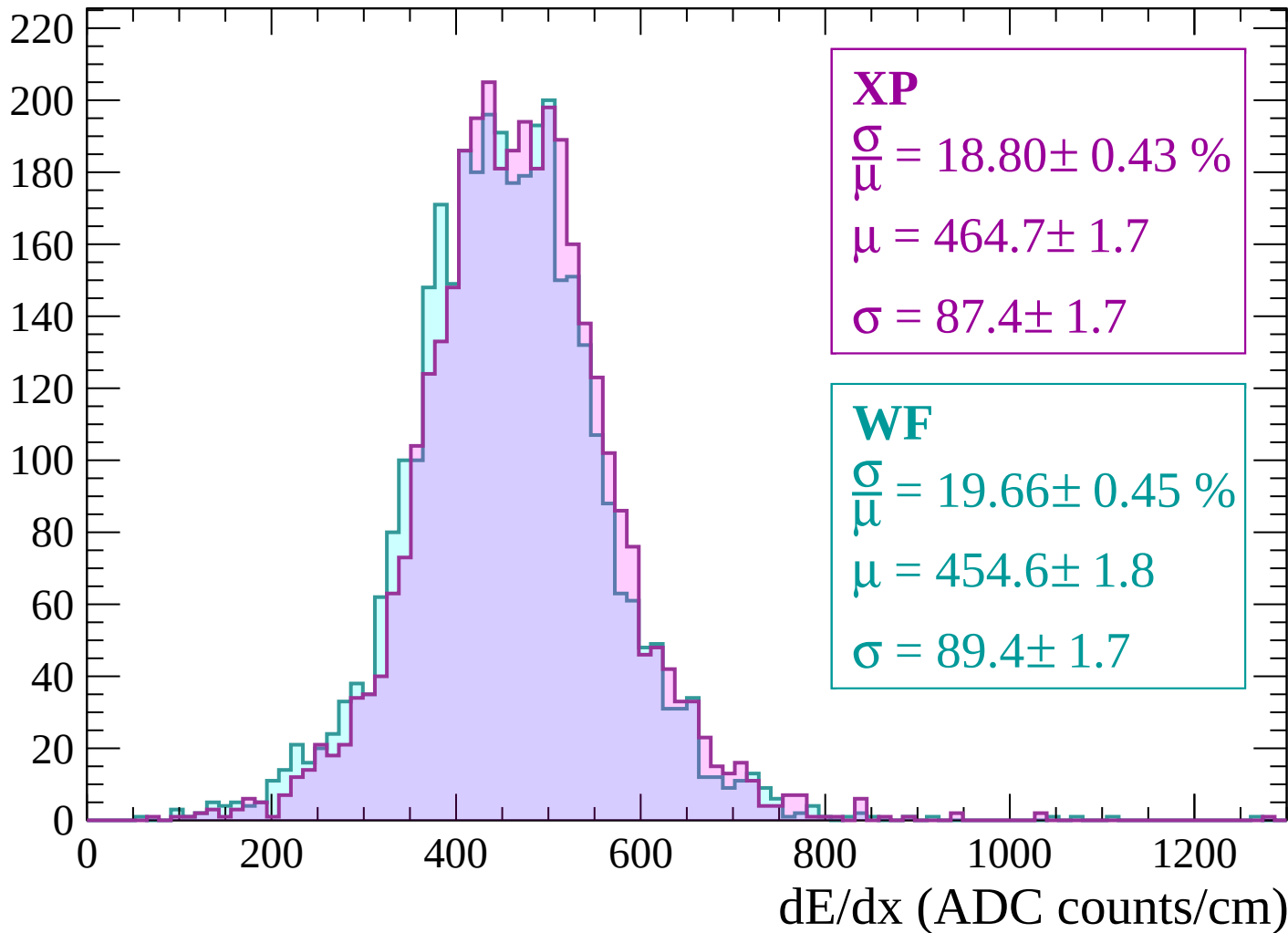
Energy loss | $-480 < p < -420$

Count



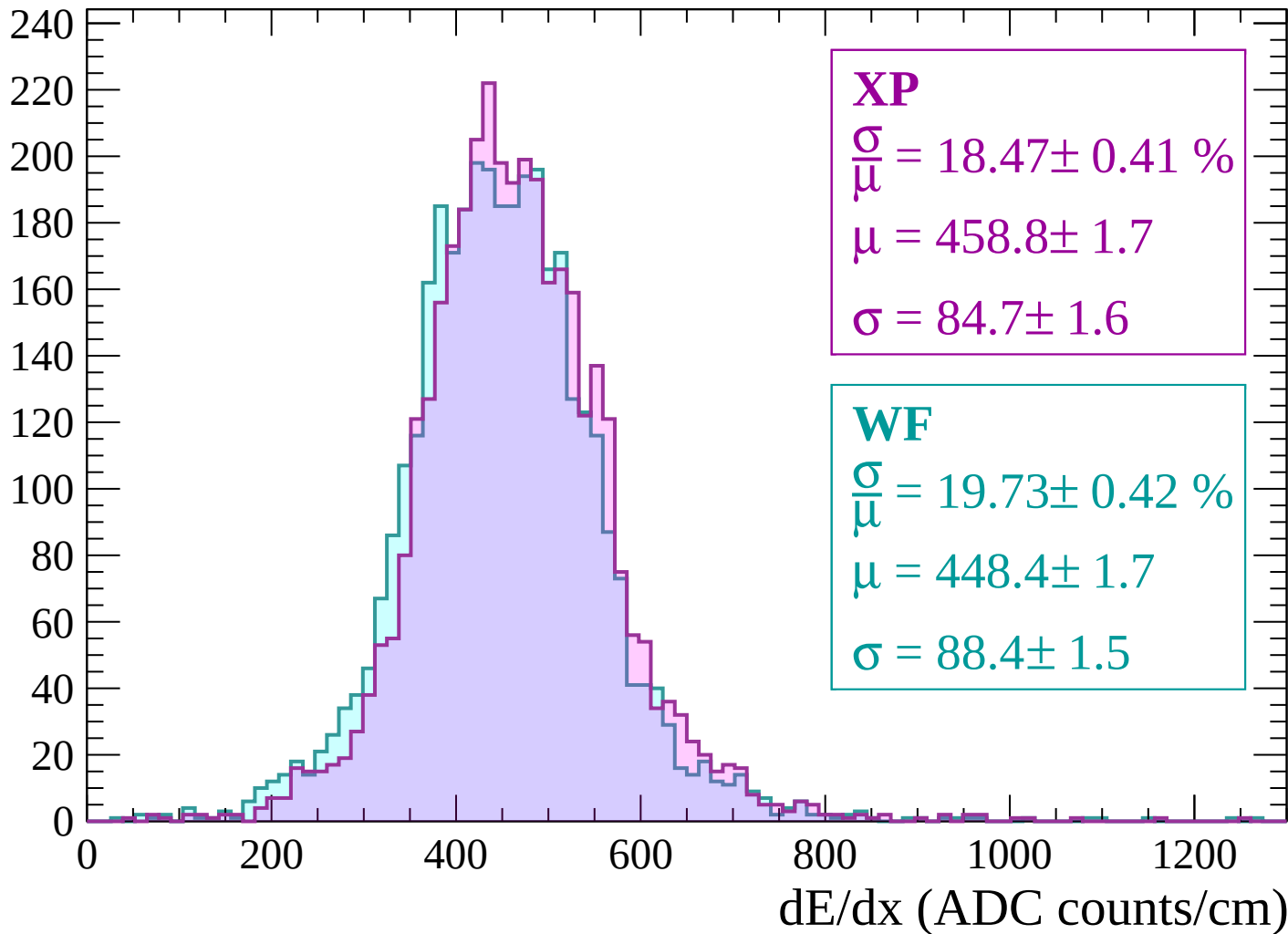
Energy loss | $-420 < p < -360$

Count



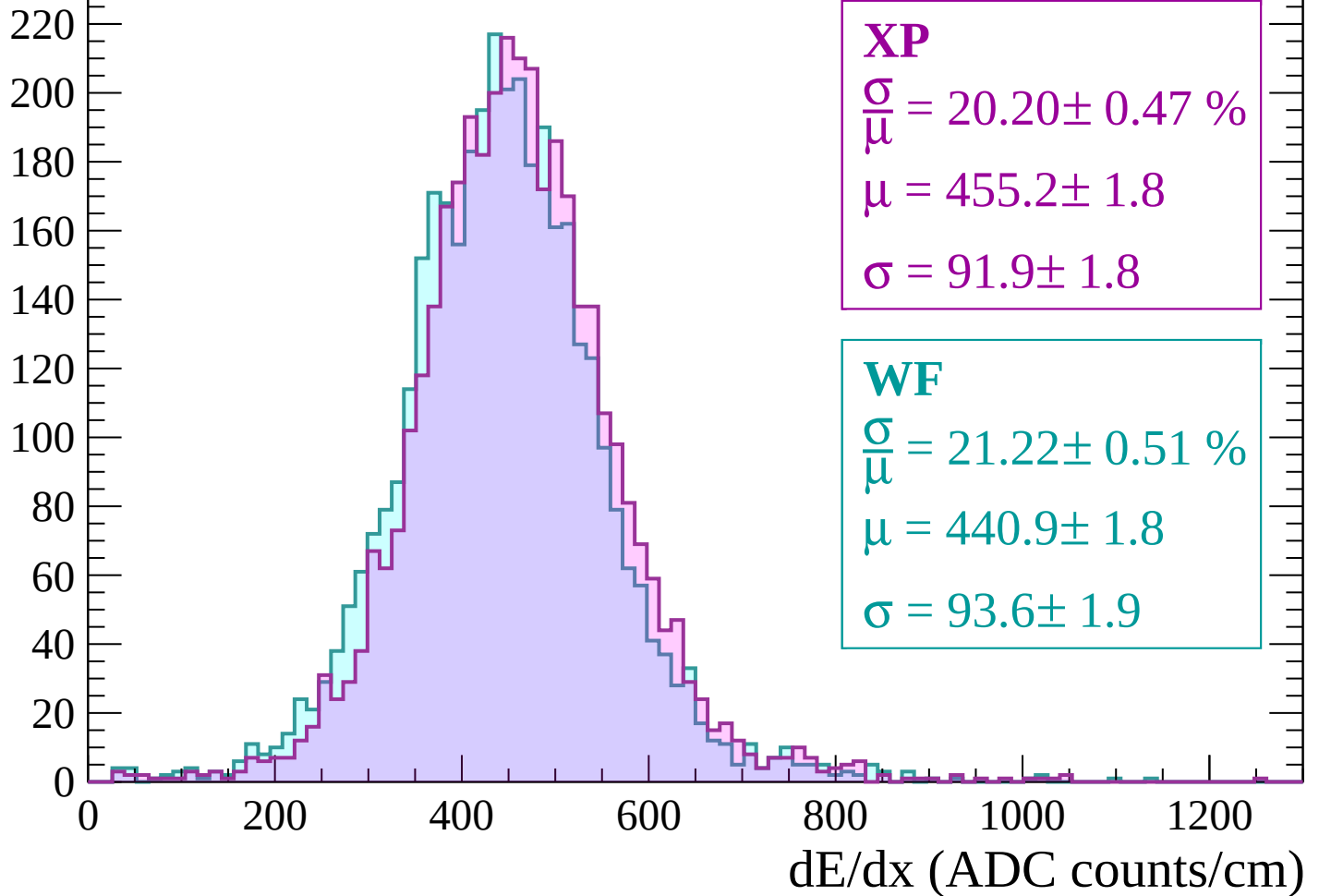
Energy loss | $-360 < p < -300$

Count



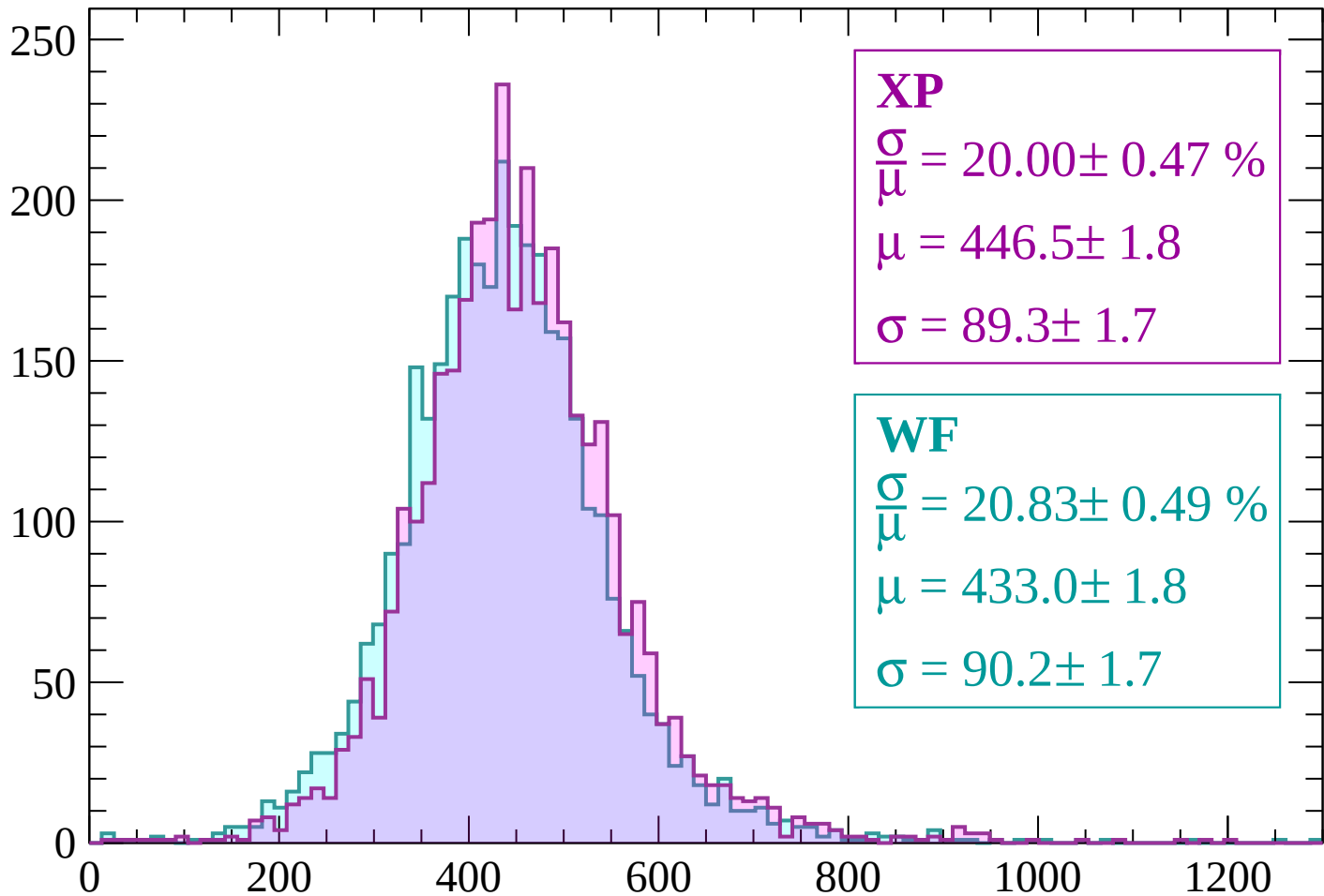
Energy loss | $-300 < p < -240$

Count



Energy loss | $-240 < p < -180$

Count



XP

$$\frac{\sigma}{\mu} = 20.00 \pm 0.47 \%$$

$$\mu = 446.5 \pm 1.8$$

$$\sigma = 89.3 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 20.83 \pm 0.49 \%$$

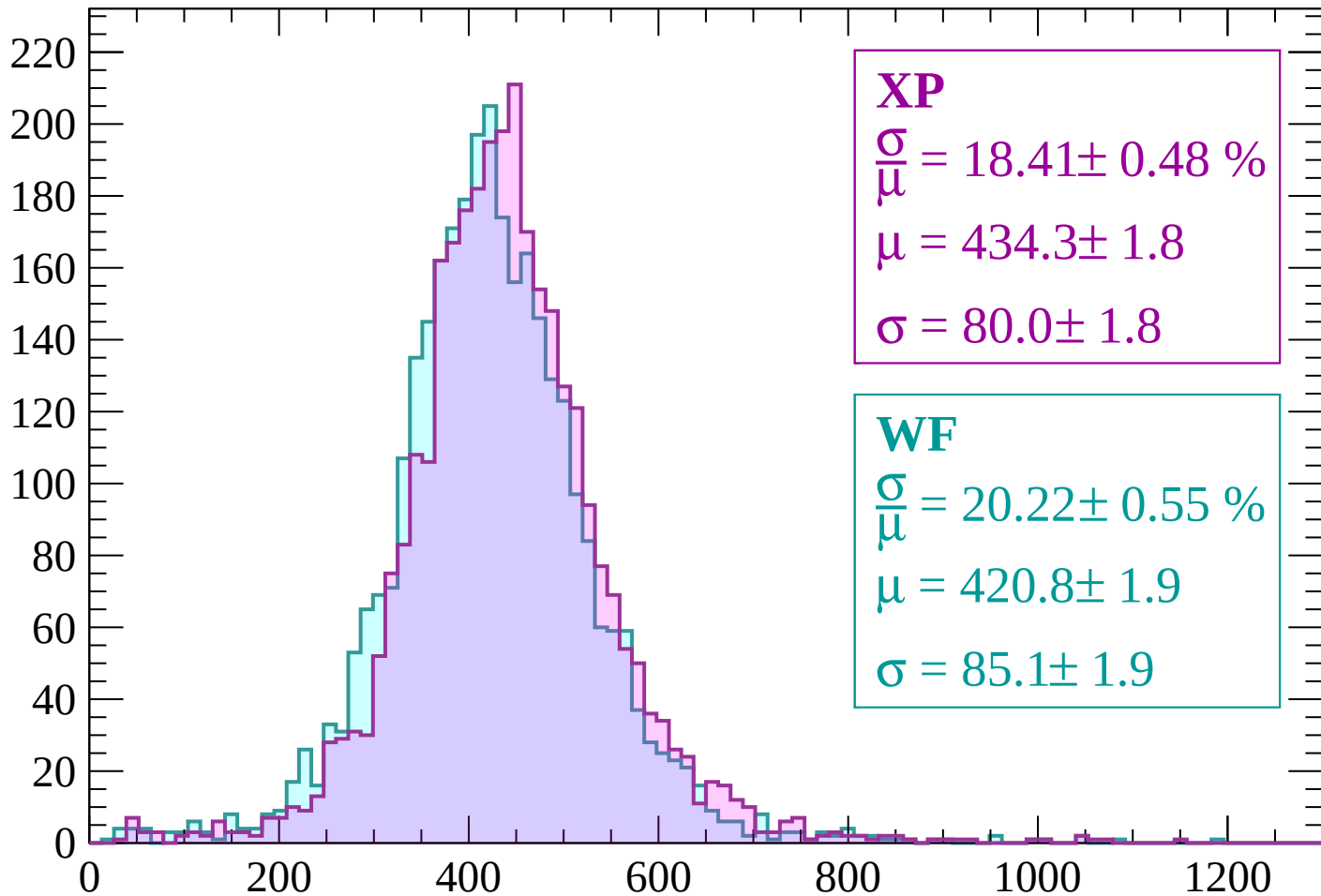
$$\mu = 433.0 \pm 1.8$$

$$\sigma = 90.2 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $-180 < p < -120$

Count



XP

$$\frac{\sigma}{\mu} = 18.41 \pm 0.48 \%$$

$$\mu = 434.3 \pm 1.8$$

$$\sigma = 80.0 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 20.22 \pm 0.55 \%$$

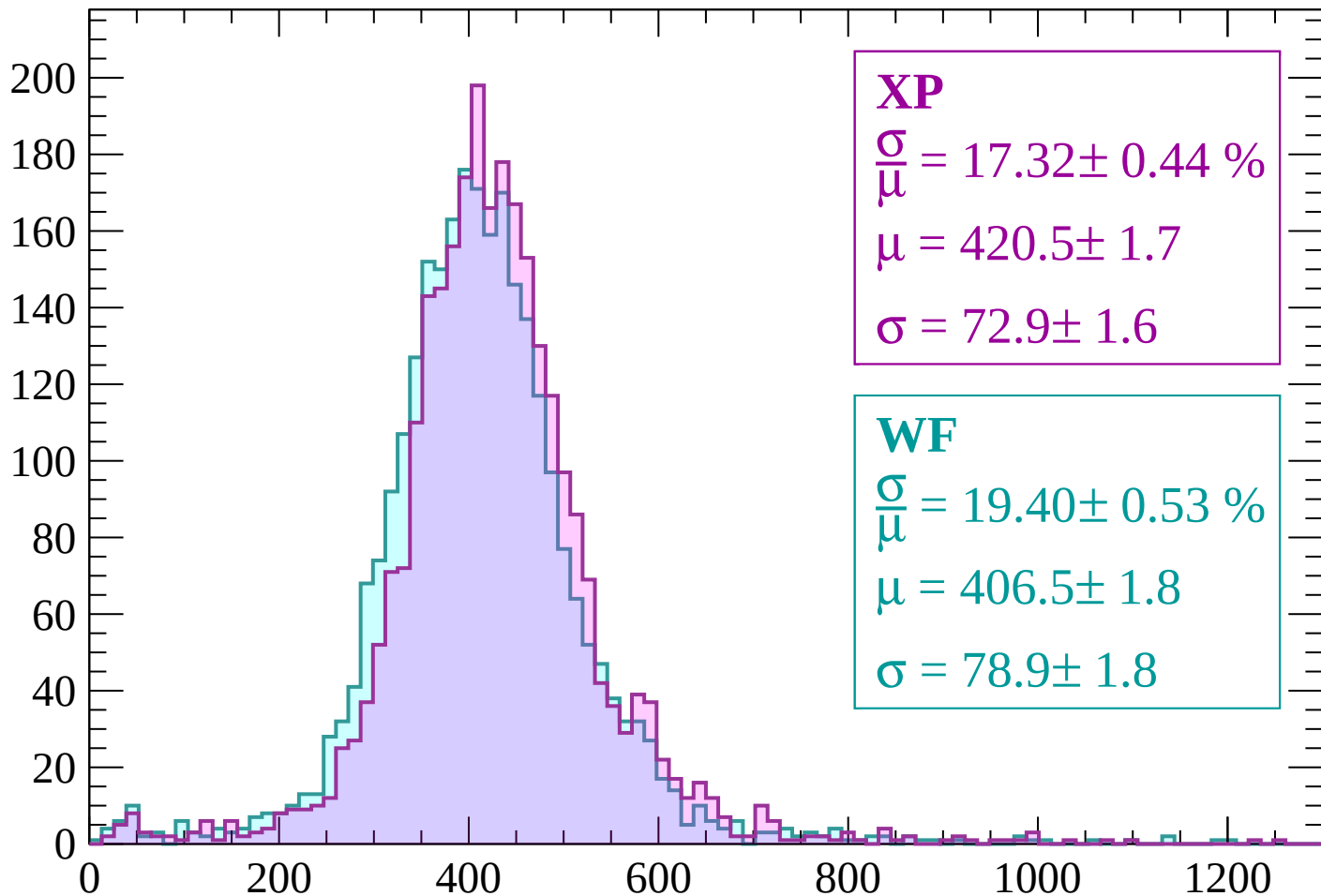
$$\mu = 420.8 \pm 1.9$$

$$\sigma = 85.1 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -60$

Count



XP

$$\frac{\sigma}{\mu} = 17.32 \pm 0.44 \%$$

$$\mu = 420.5 \pm 1.7$$

$$\sigma = 72.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 19.40 \pm 0.53 \%$$

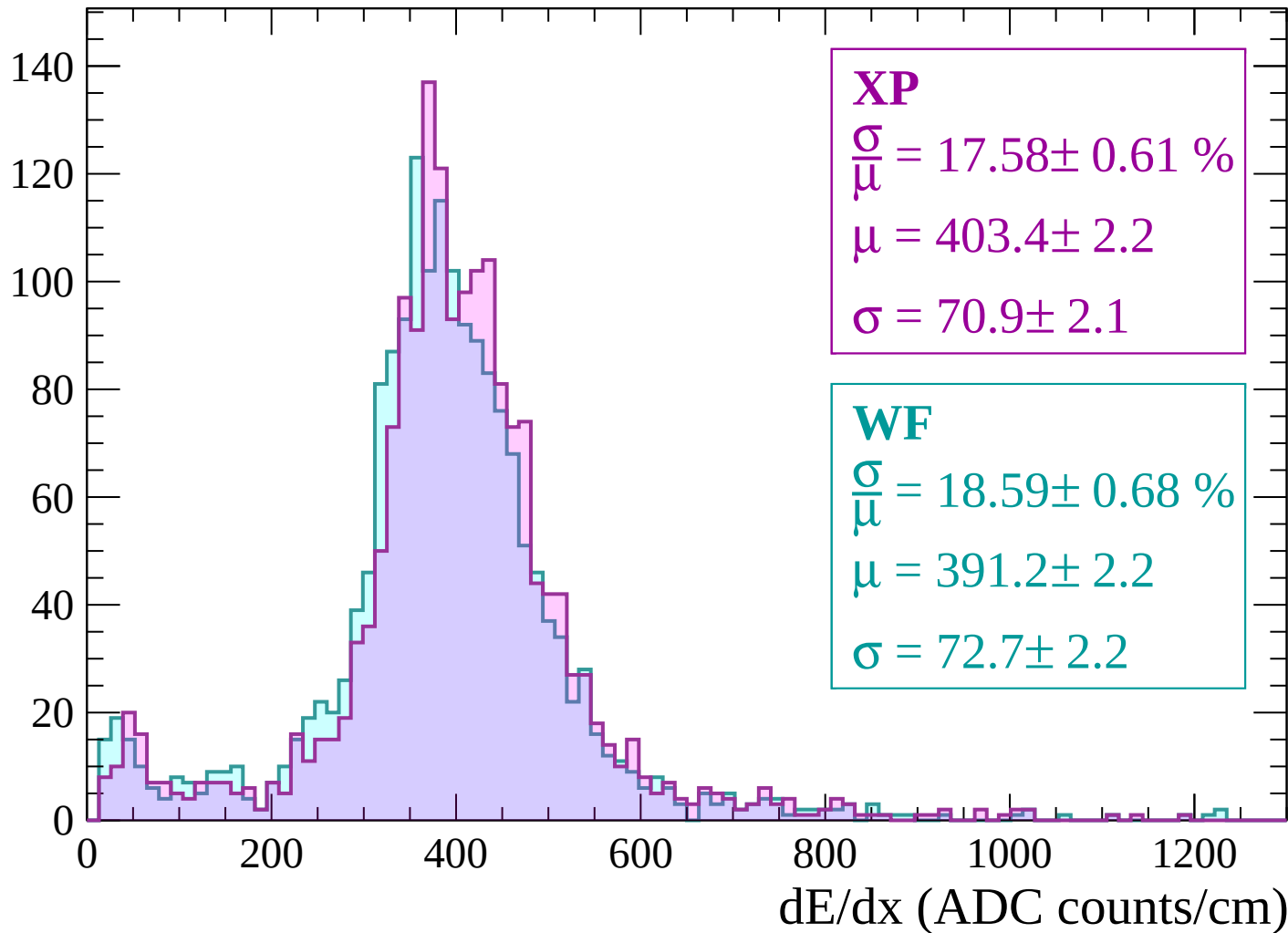
$$\mu = 406.5 \pm 1.8$$

$$\sigma = 78.9 \pm 1.8$$

dE/dx (ADC counts/cm)

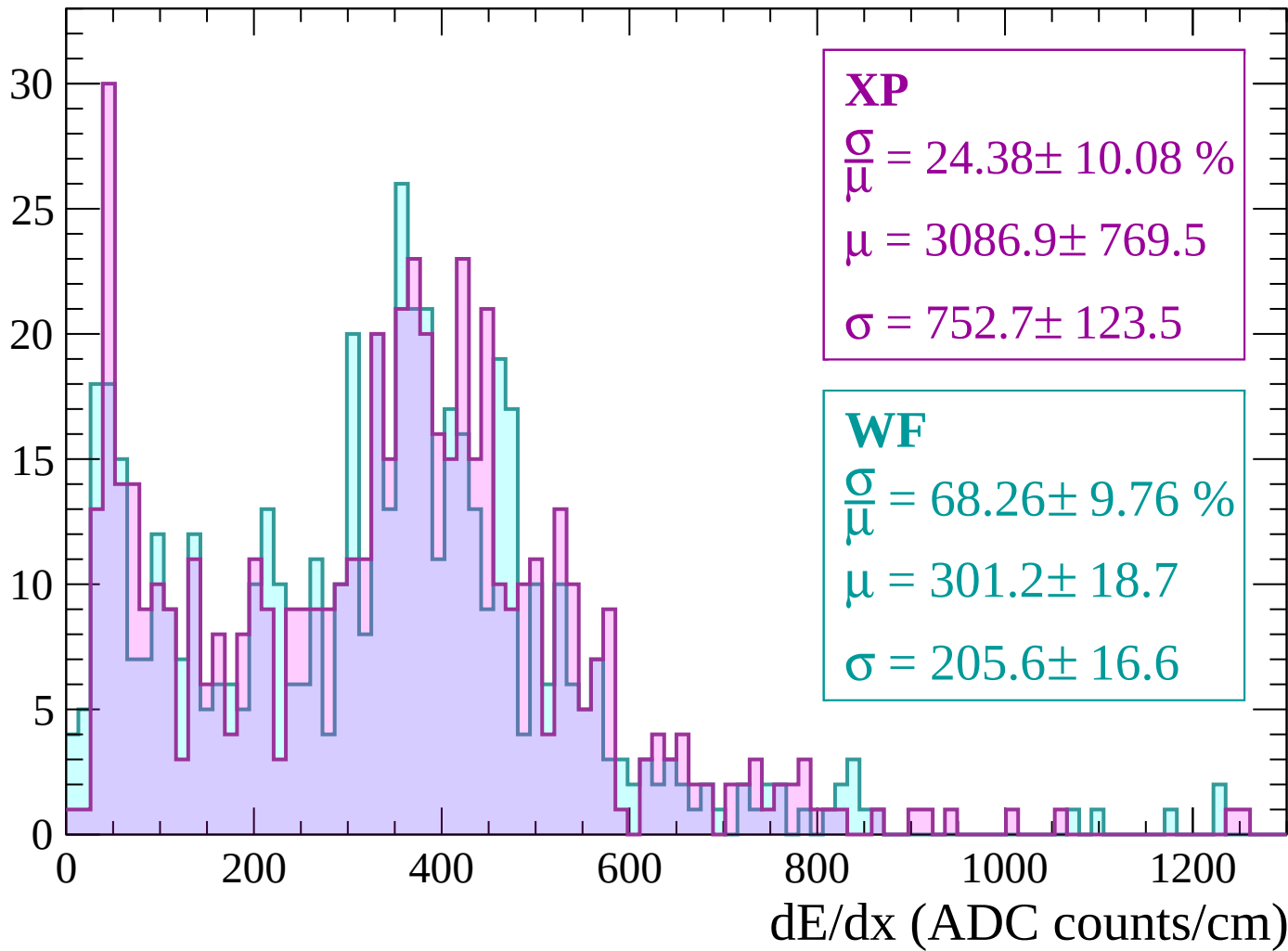
Energy loss | $-60 < p < 0$

Count



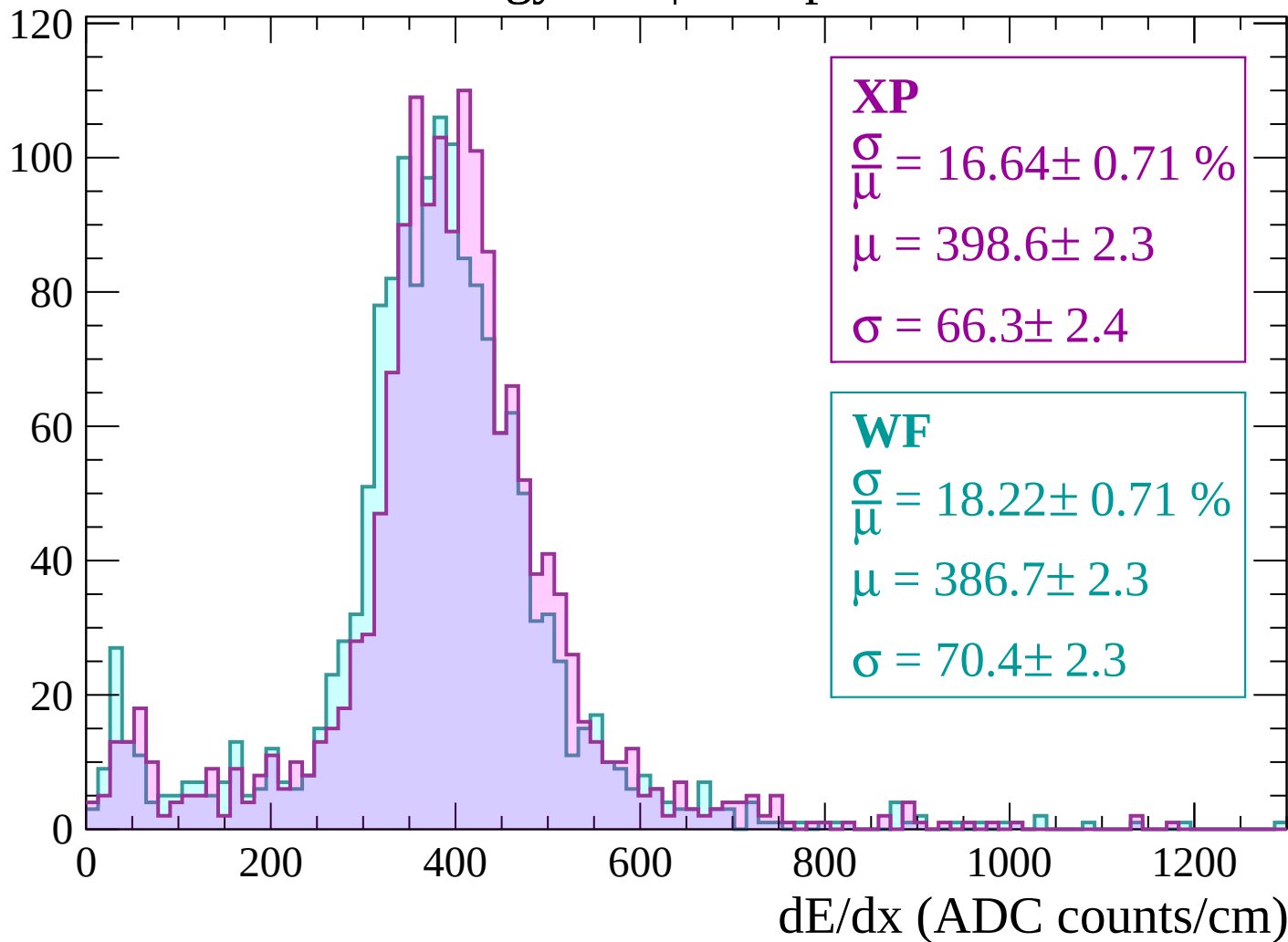
Energy loss | $0 < p < 60$

Count



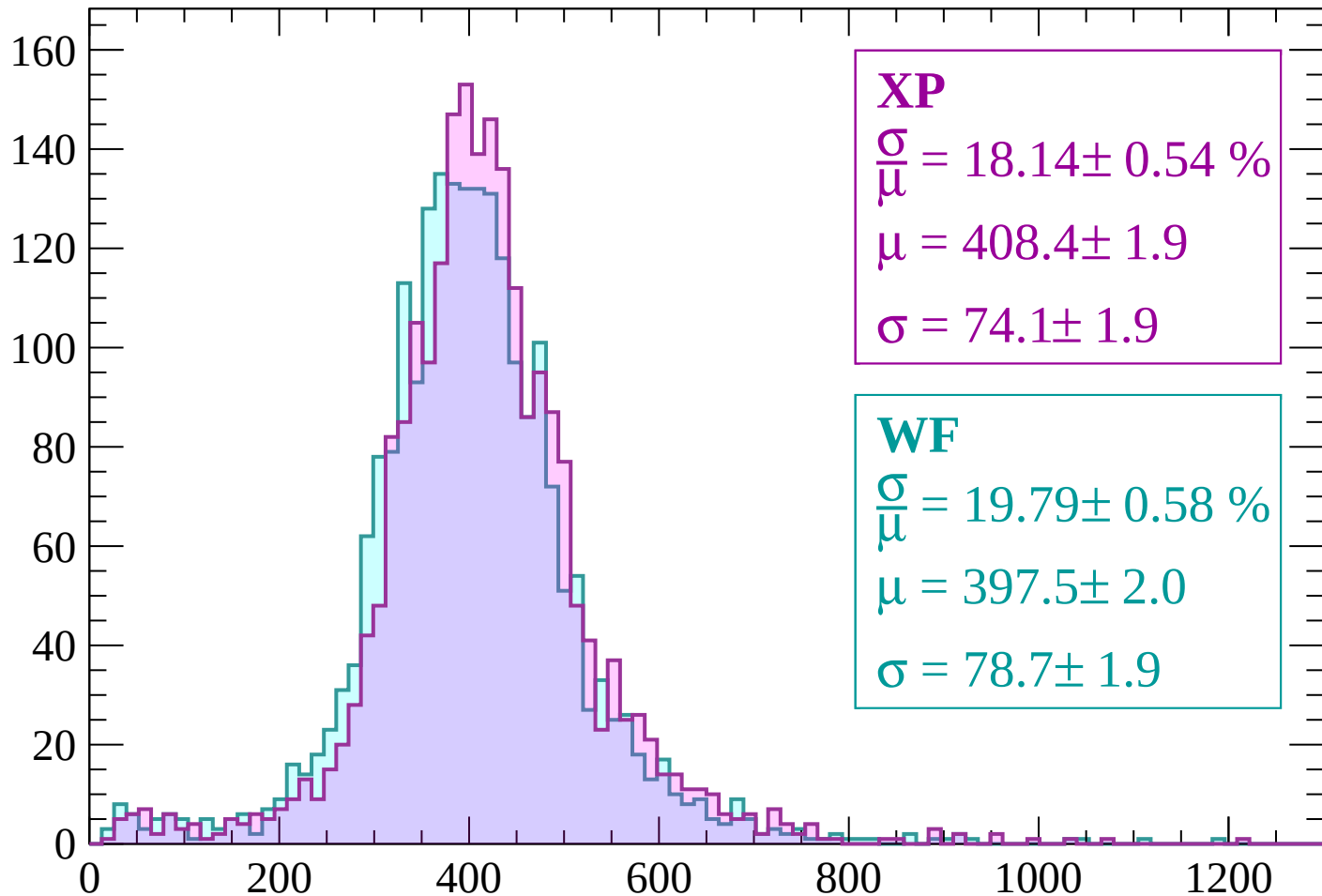
Energy loss | $60 < p < 120$

Count



Energy loss | $120 < p < 180$

Count



XP

$$\frac{\sigma}{\mu} = 18.14 \pm 0.54 \%$$

$$\mu = 408.4 \pm 1.9$$

$$\sigma = 74.1 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 19.79 \pm 0.58 \%$$

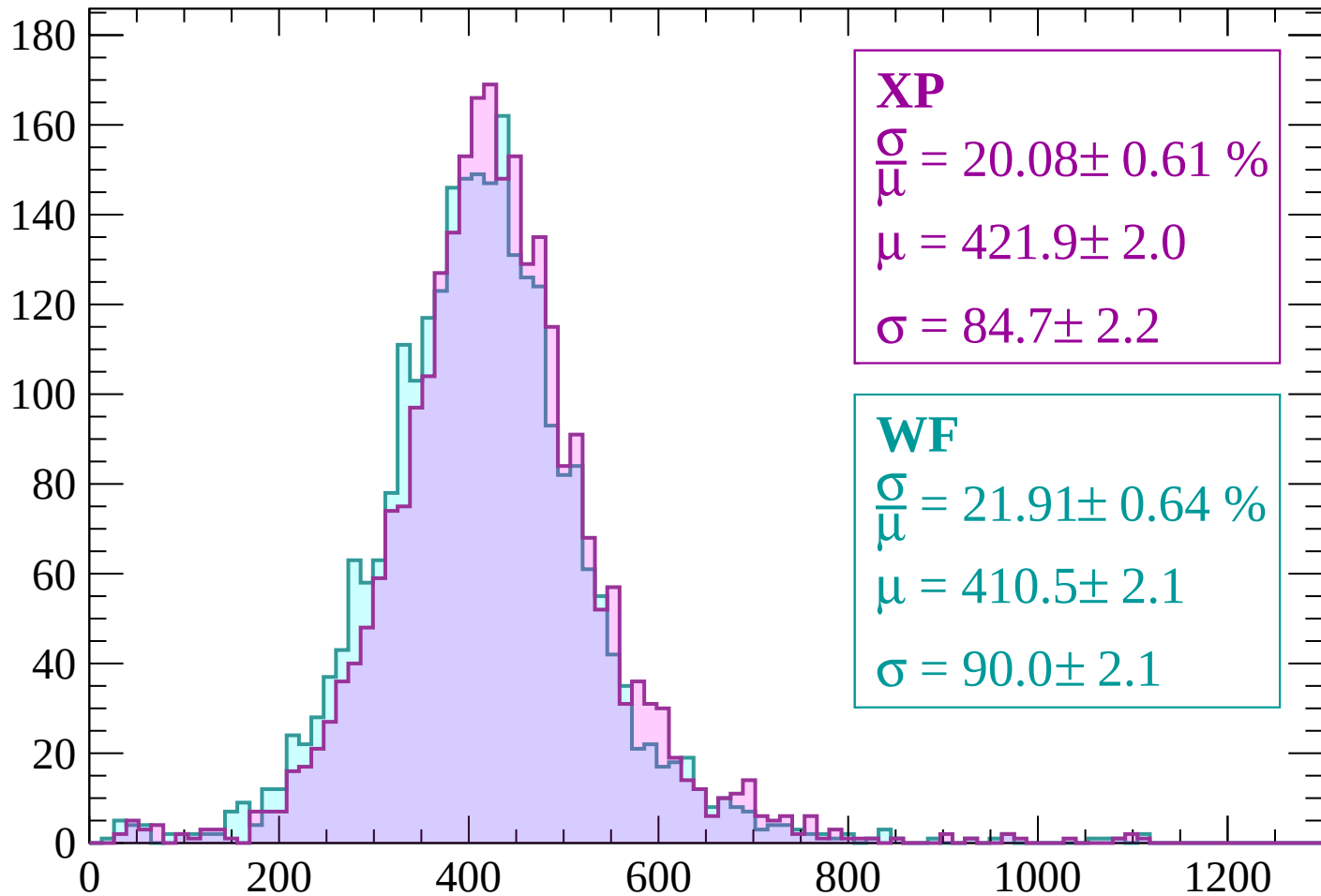
$$\mu = 397.5 \pm 2.0$$

$$\sigma = 78.7 \pm 1.9$$

dE/dx (ADC counts/cm)

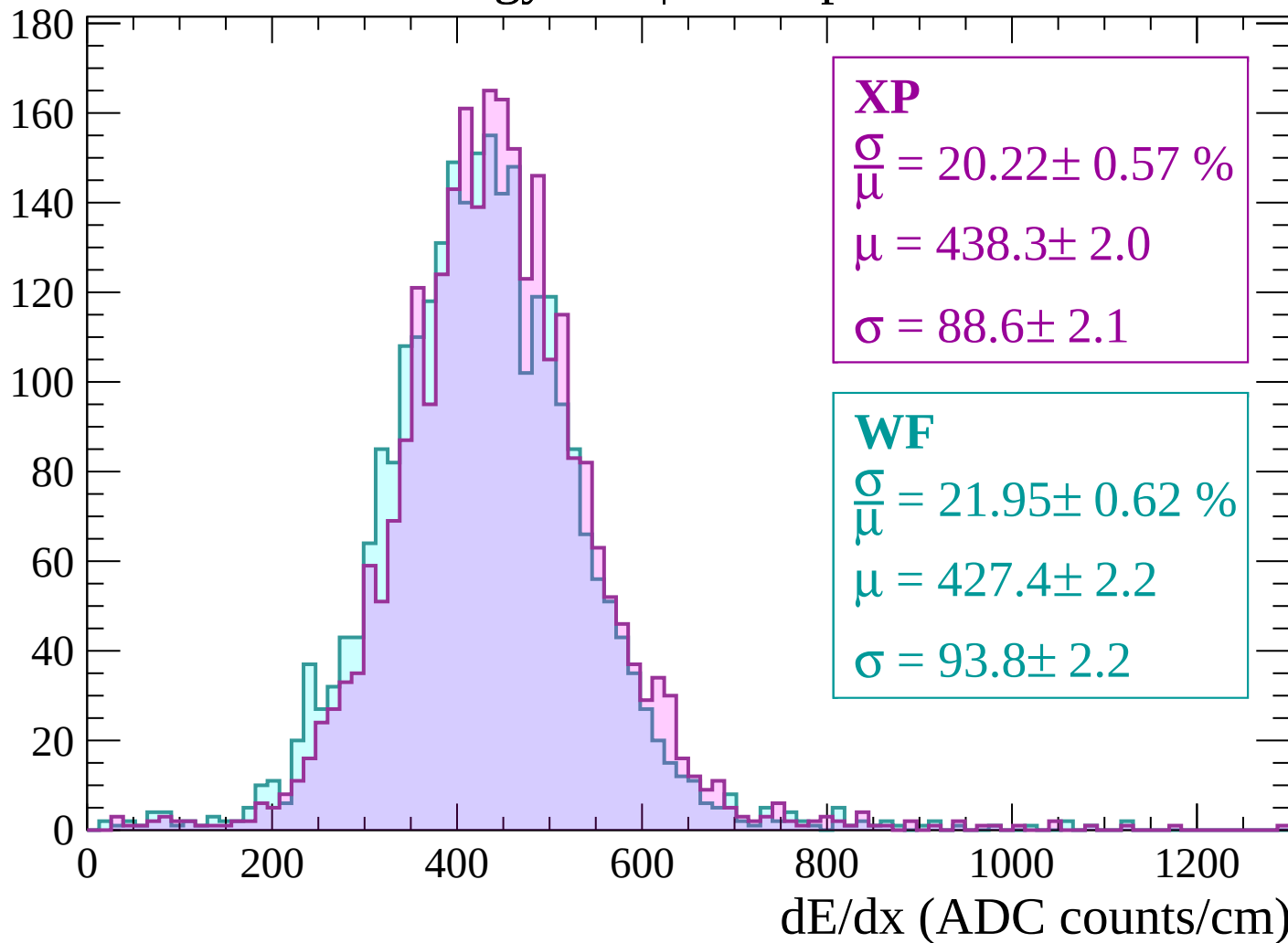
Energy loss | $180 < p < 240$

Count



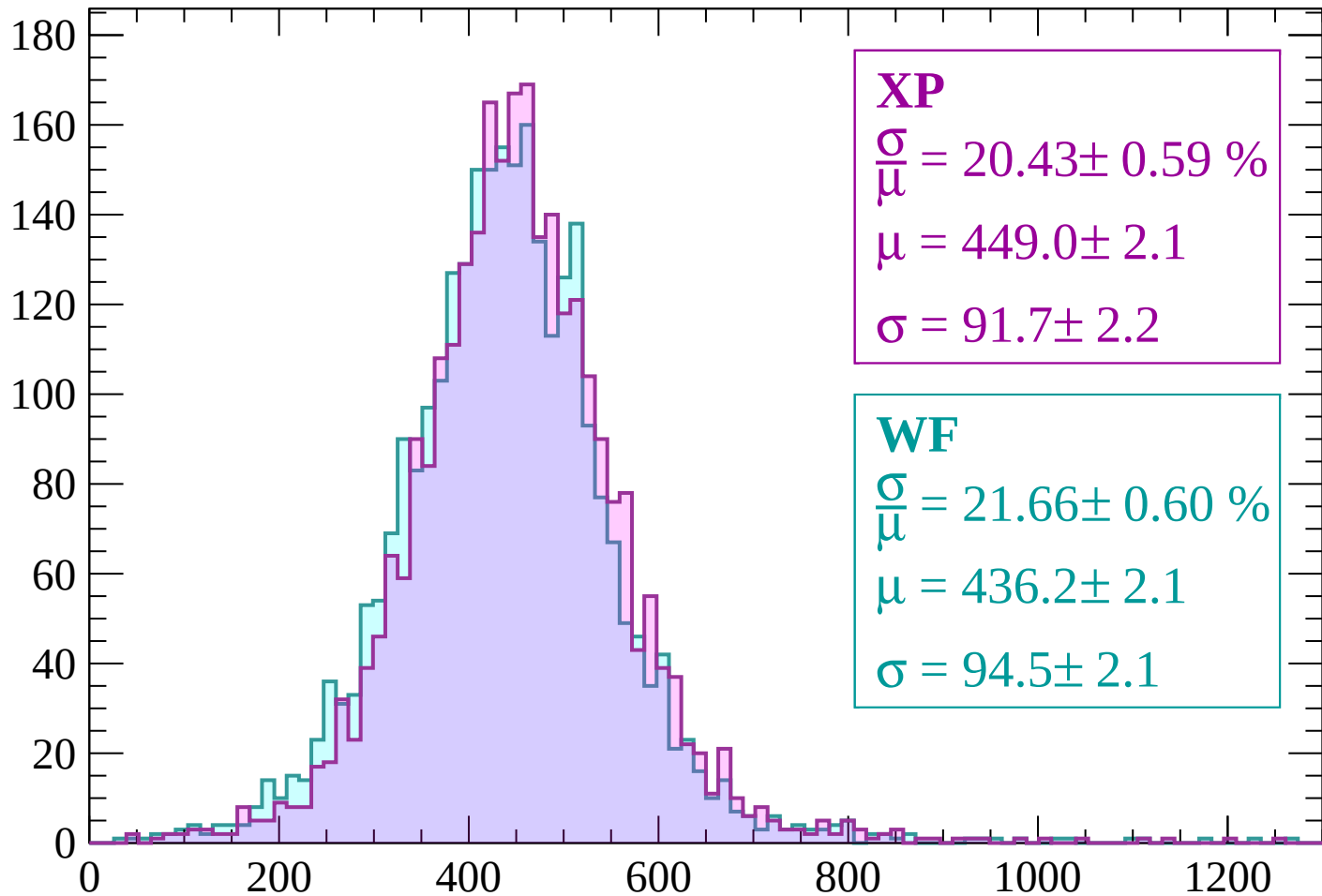
Energy loss | $240 < p < 300$

Count



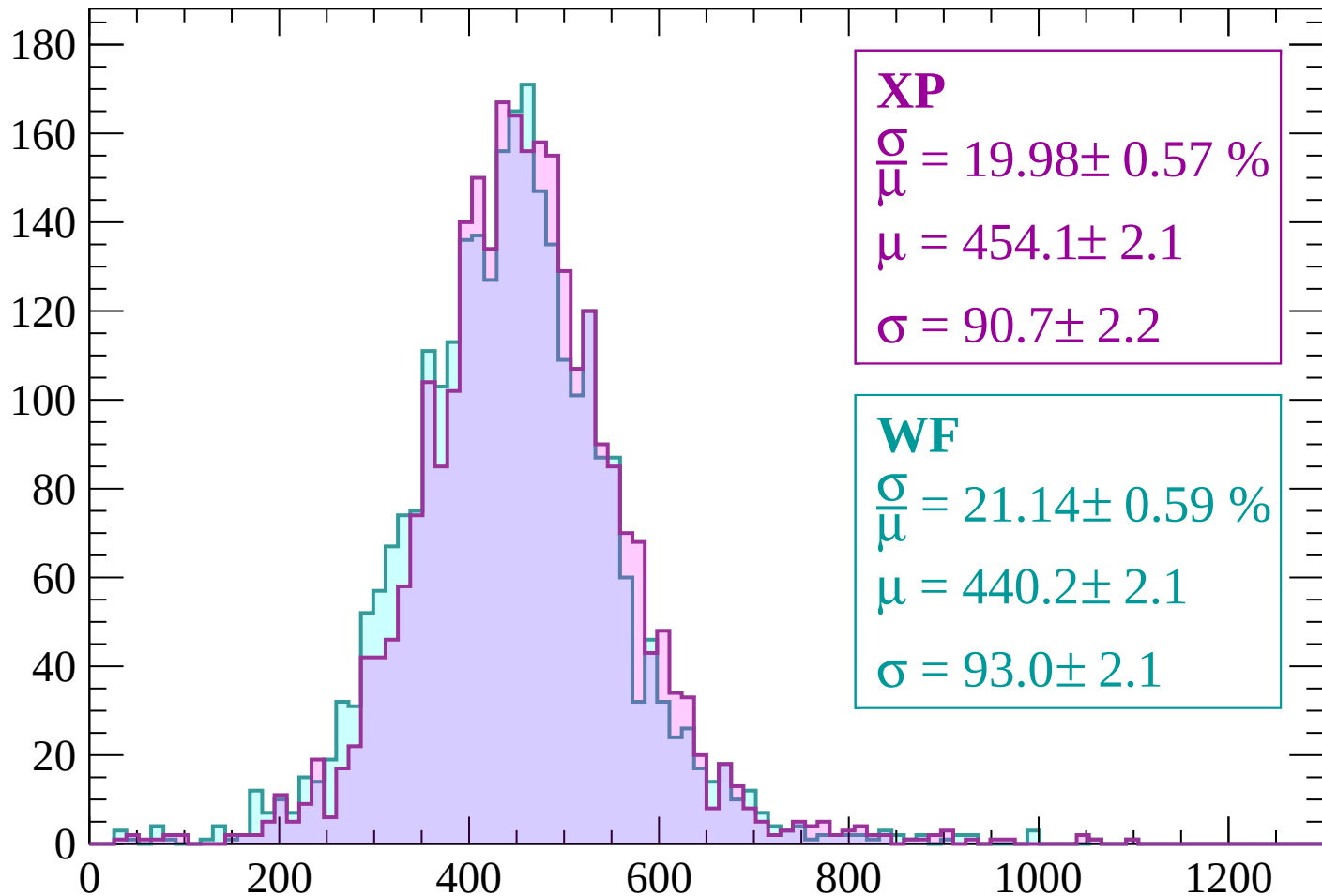
Energy loss | $300 < p < 360$

Count



Energy loss | $360 < p < 420$

Count



XP

$$\frac{\sigma}{\mu} = 19.98 \pm 0.57 \%$$

$$\mu = 454.1 \pm 2.1$$

$$\sigma = 90.7 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 21.14 \pm 0.59 \%$$

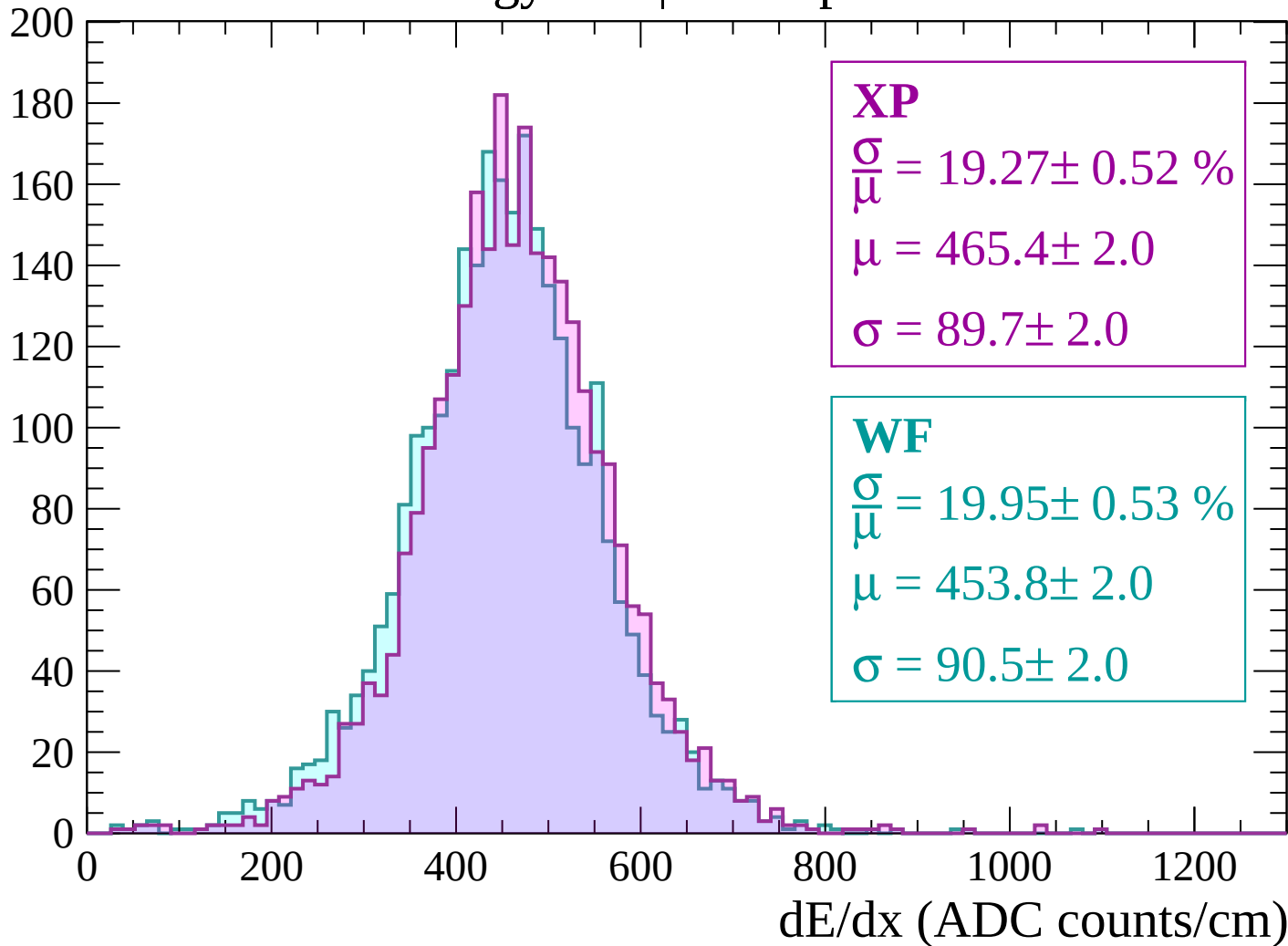
$$\mu = 440.2 \pm 2.1$$

$$\sigma = 93.0 \pm 2.1$$

dE/dx (ADC counts/cm)

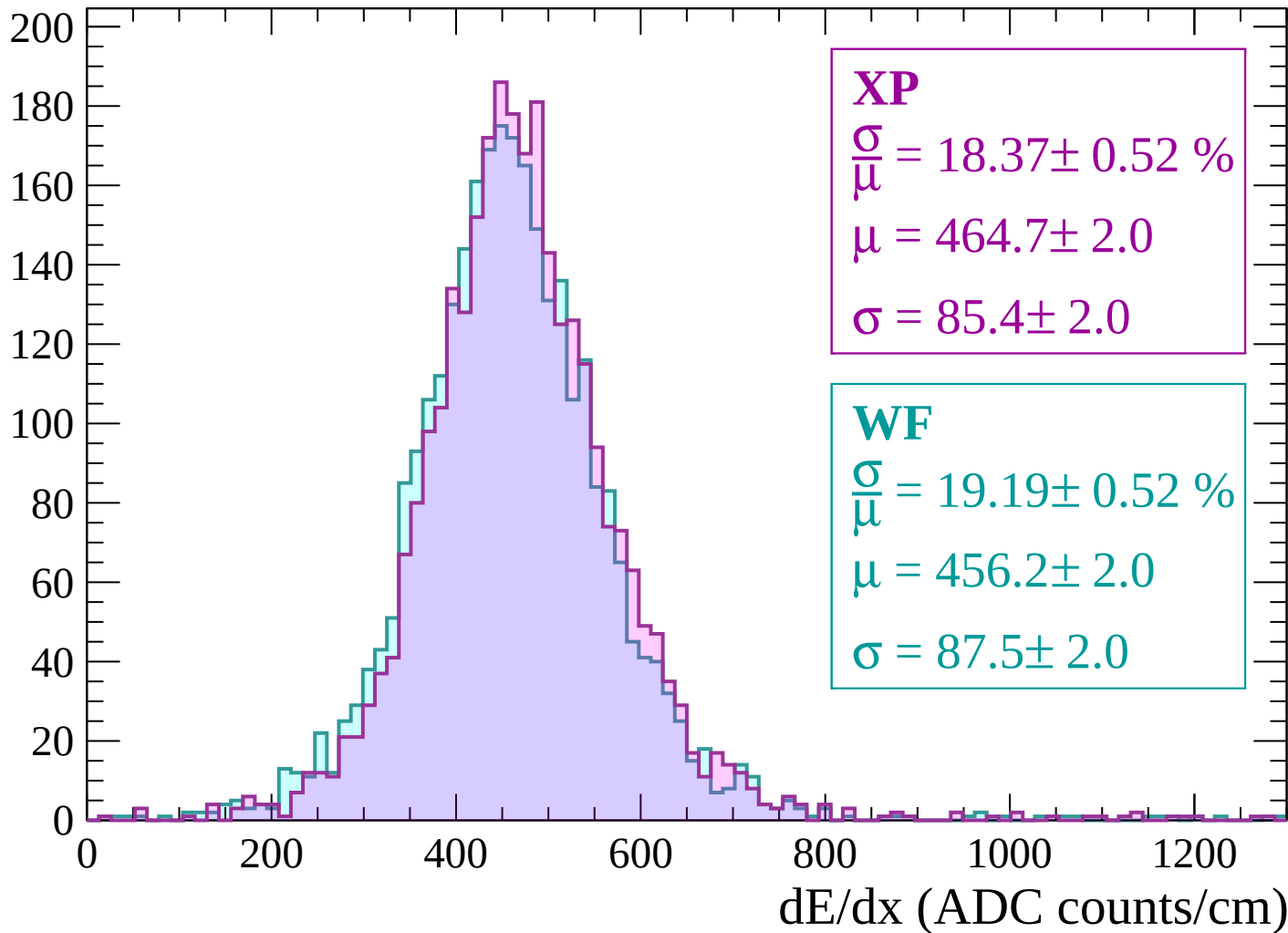
Energy loss | $420 < p < 480$

Count



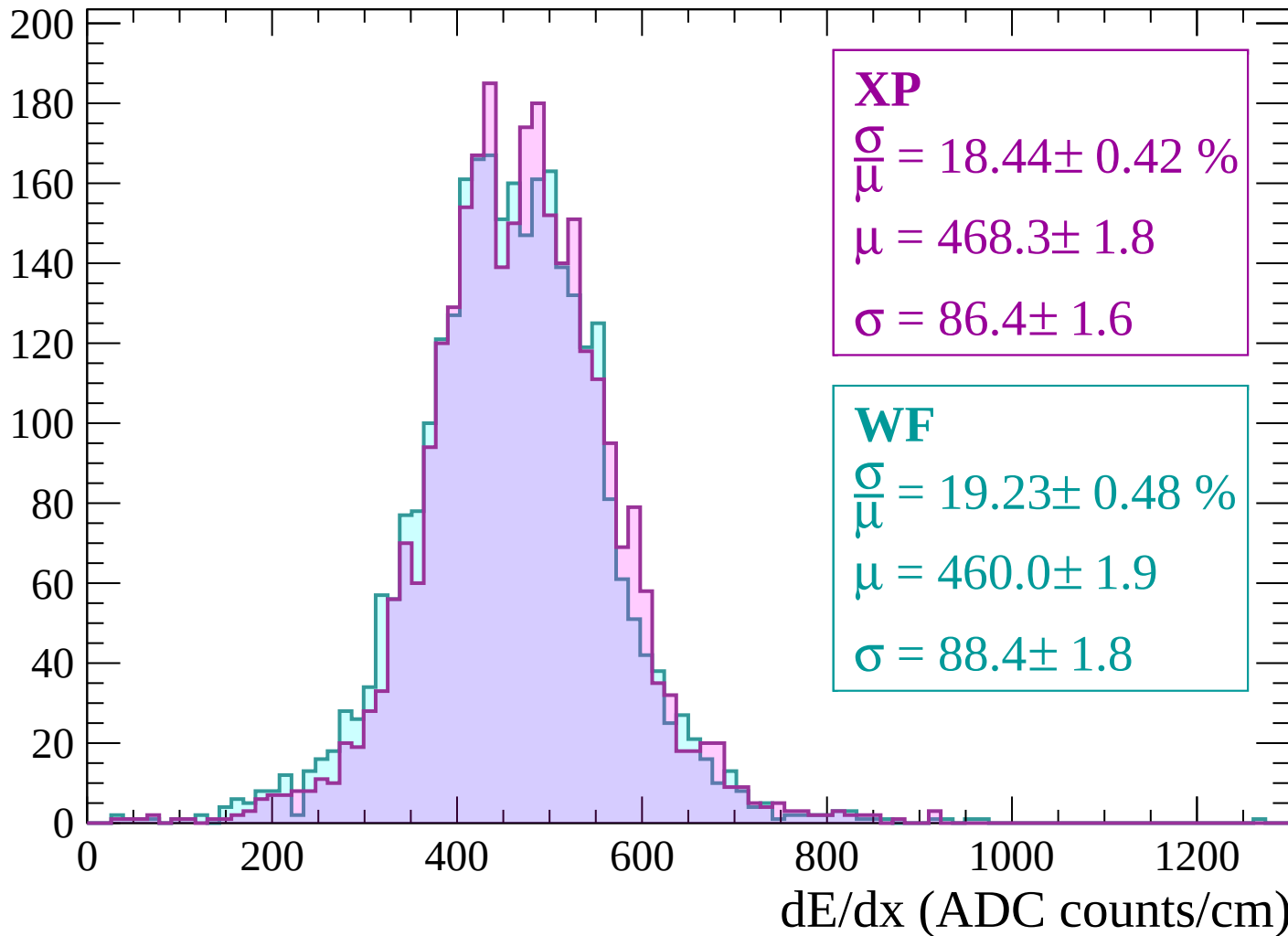
Energy loss | $480 < p < 540$

Count



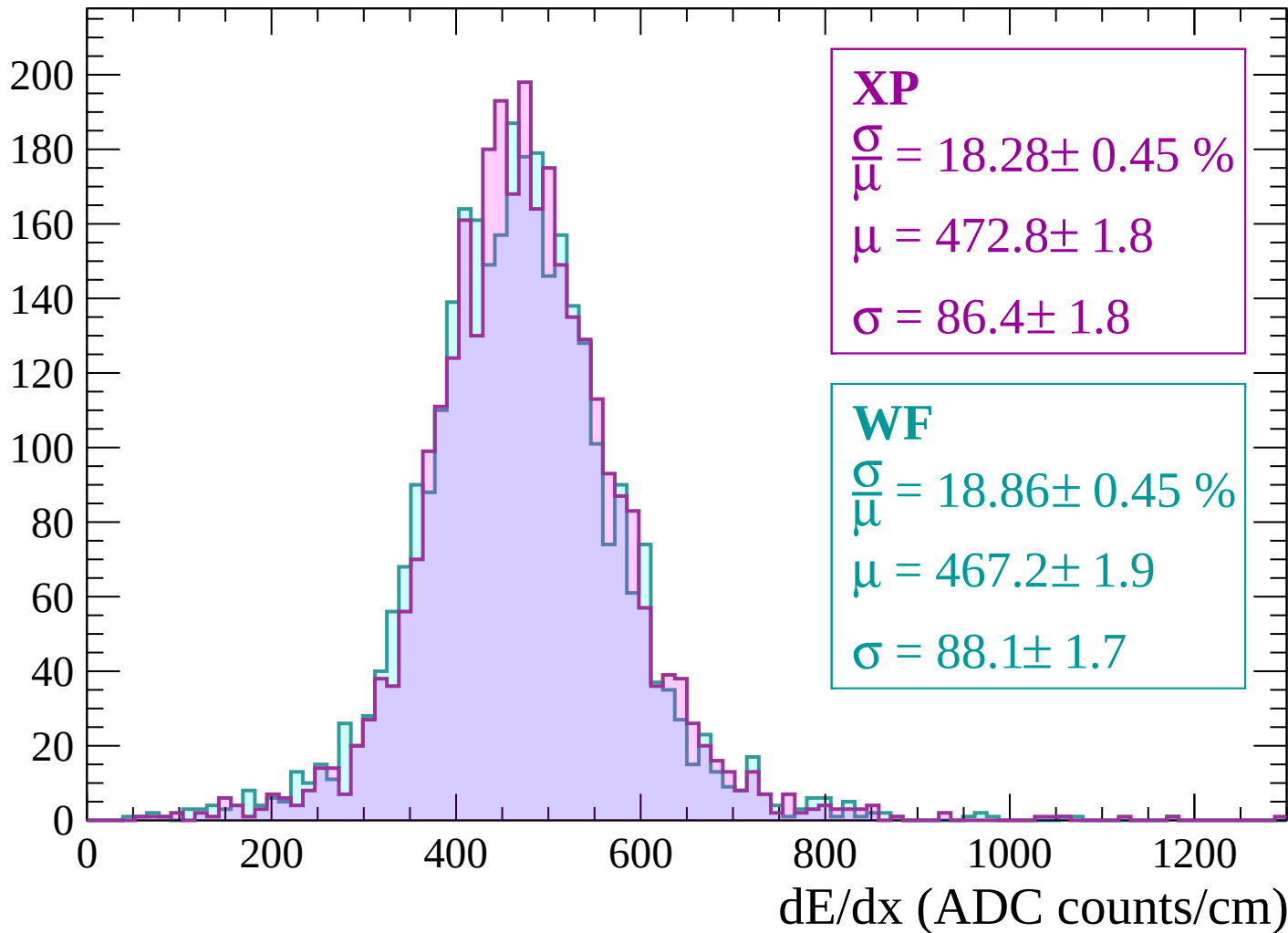
Energy loss | $540 < p < 600$

Count



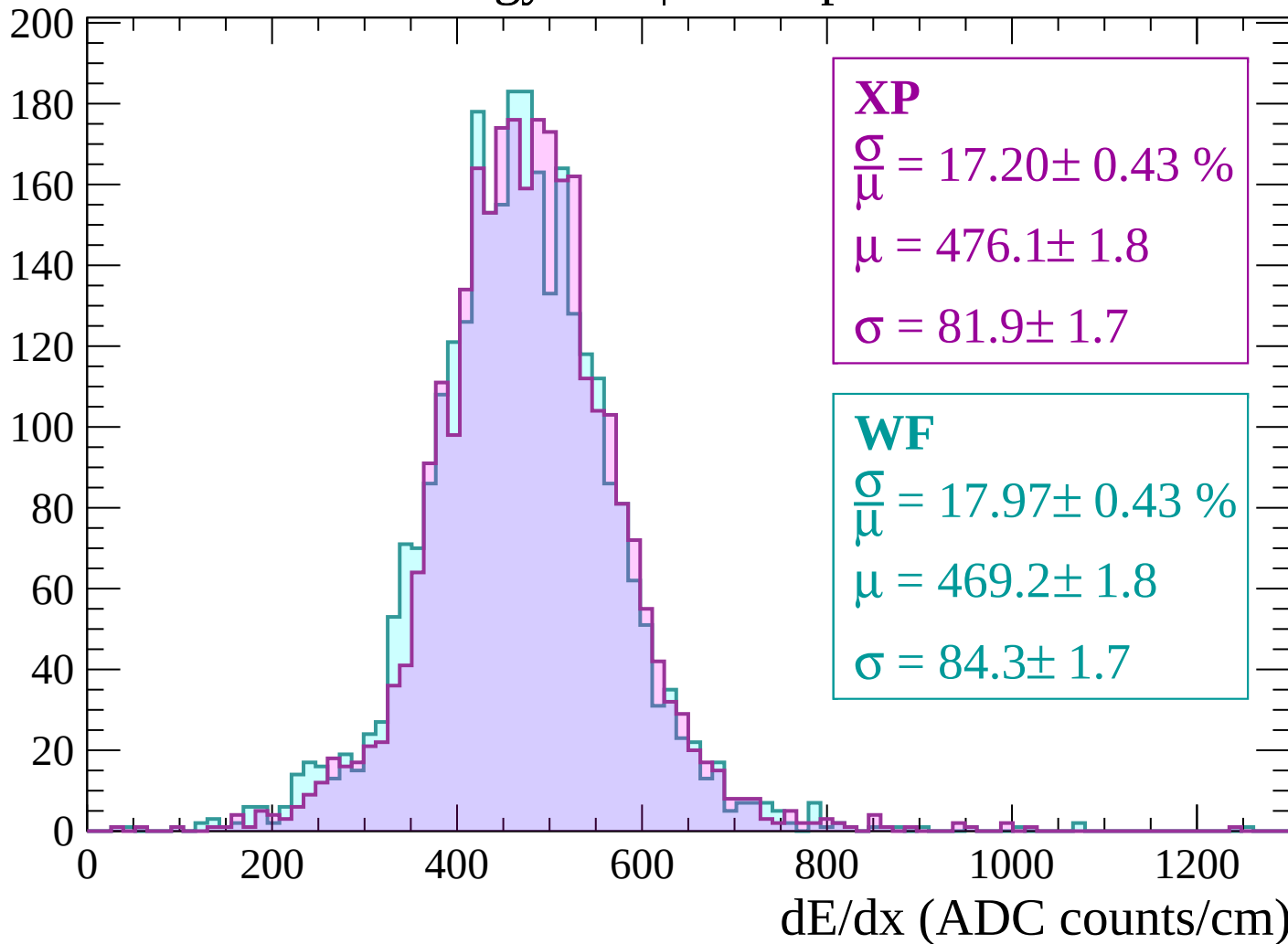
Energy loss | $600 < p < 660$

Count



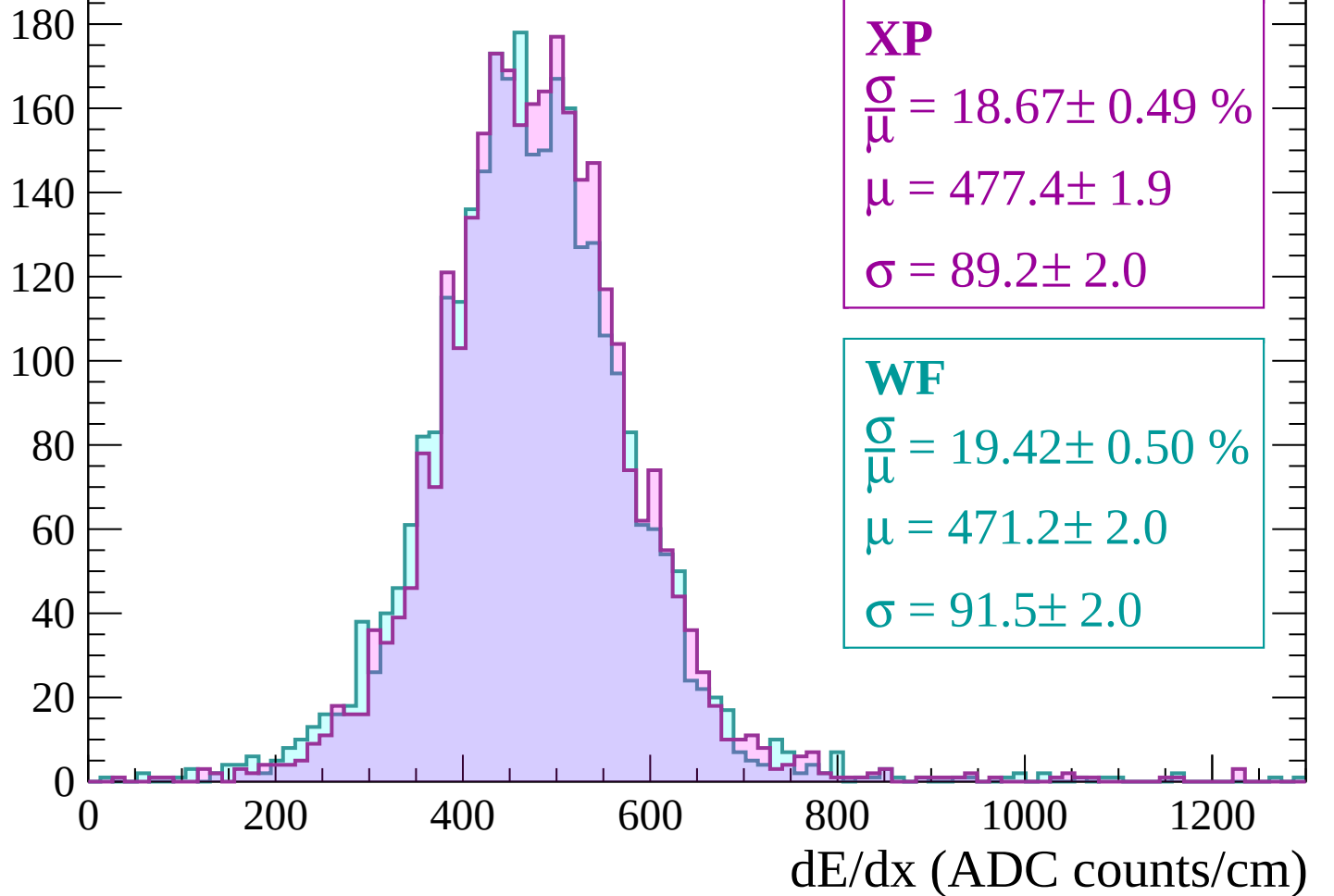
Energy loss | $660 < p < 720$

Count



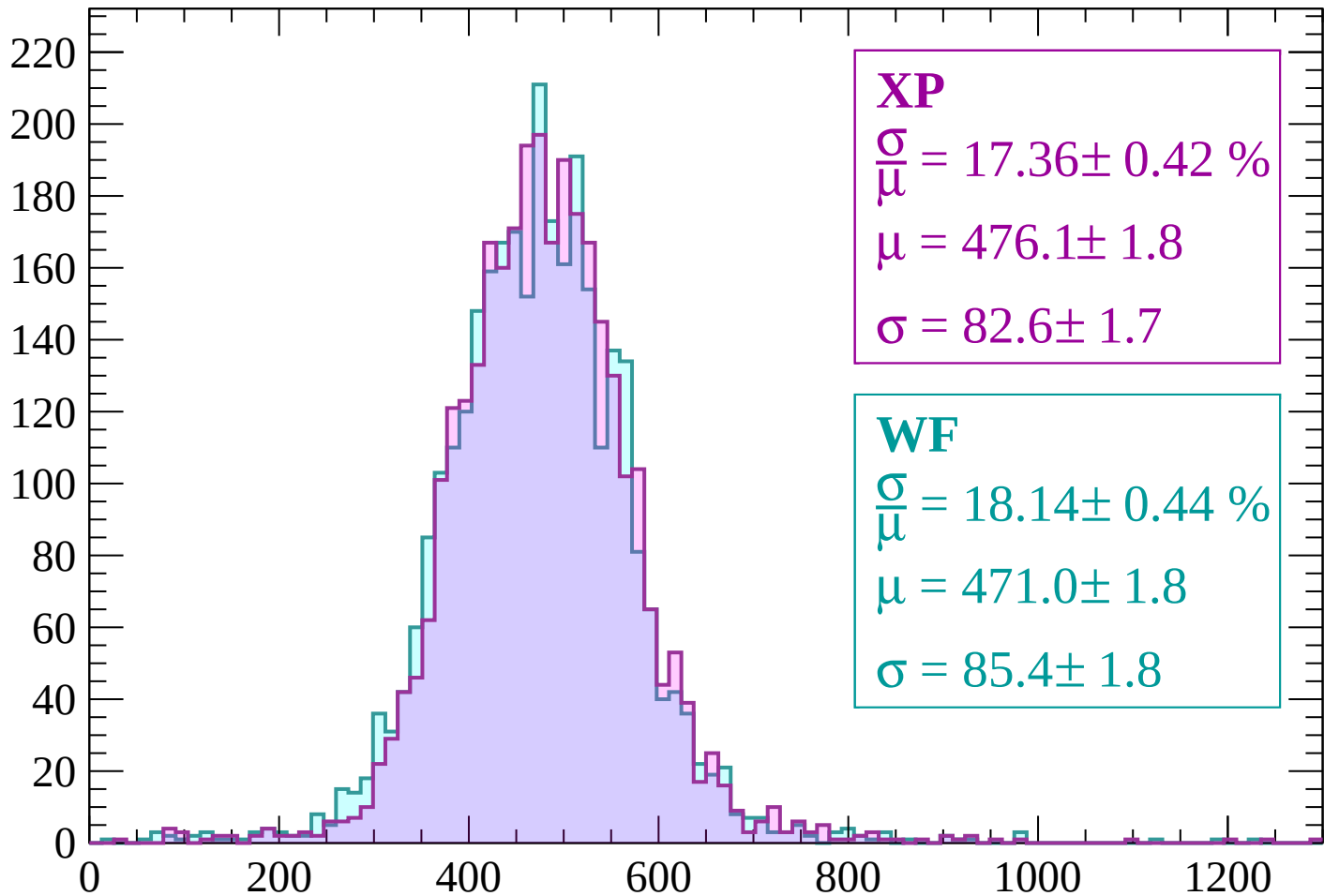
Energy loss | $720 < p < 780$

Count



Energy loss | $780 < p < 840$

Count



XP

$$\frac{\sigma}{\mu} = 17.36 \pm 0.42 \%$$

$$\mu = 476.1 \pm 1.8$$

$$\sigma = 82.6 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 18.14 \pm 0.44 \%$$

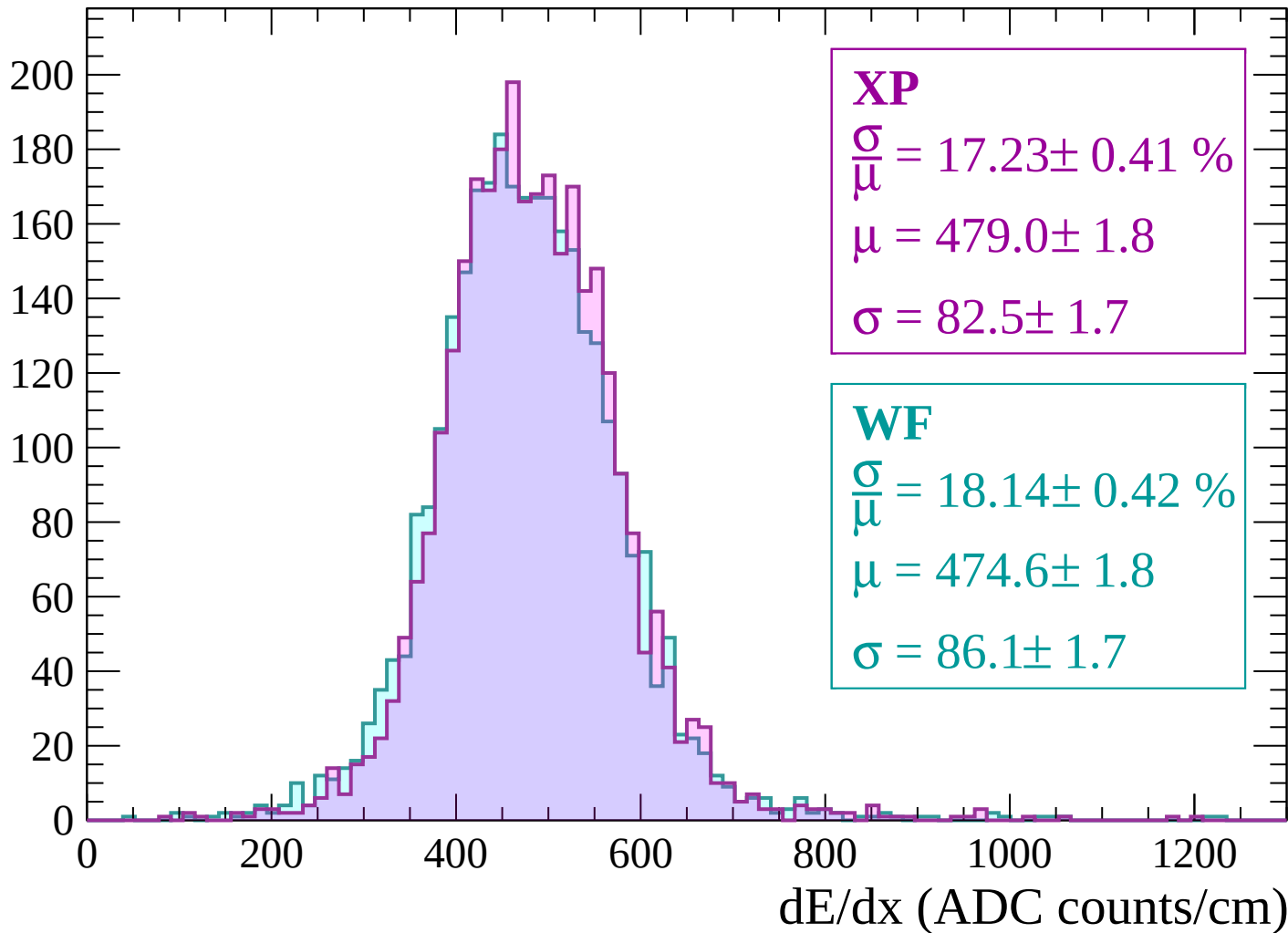
$$\mu = 471.0 \pm 1.8$$

$$\sigma = 85.4 \pm 1.8$$

dE/dx (ADC counts/cm)

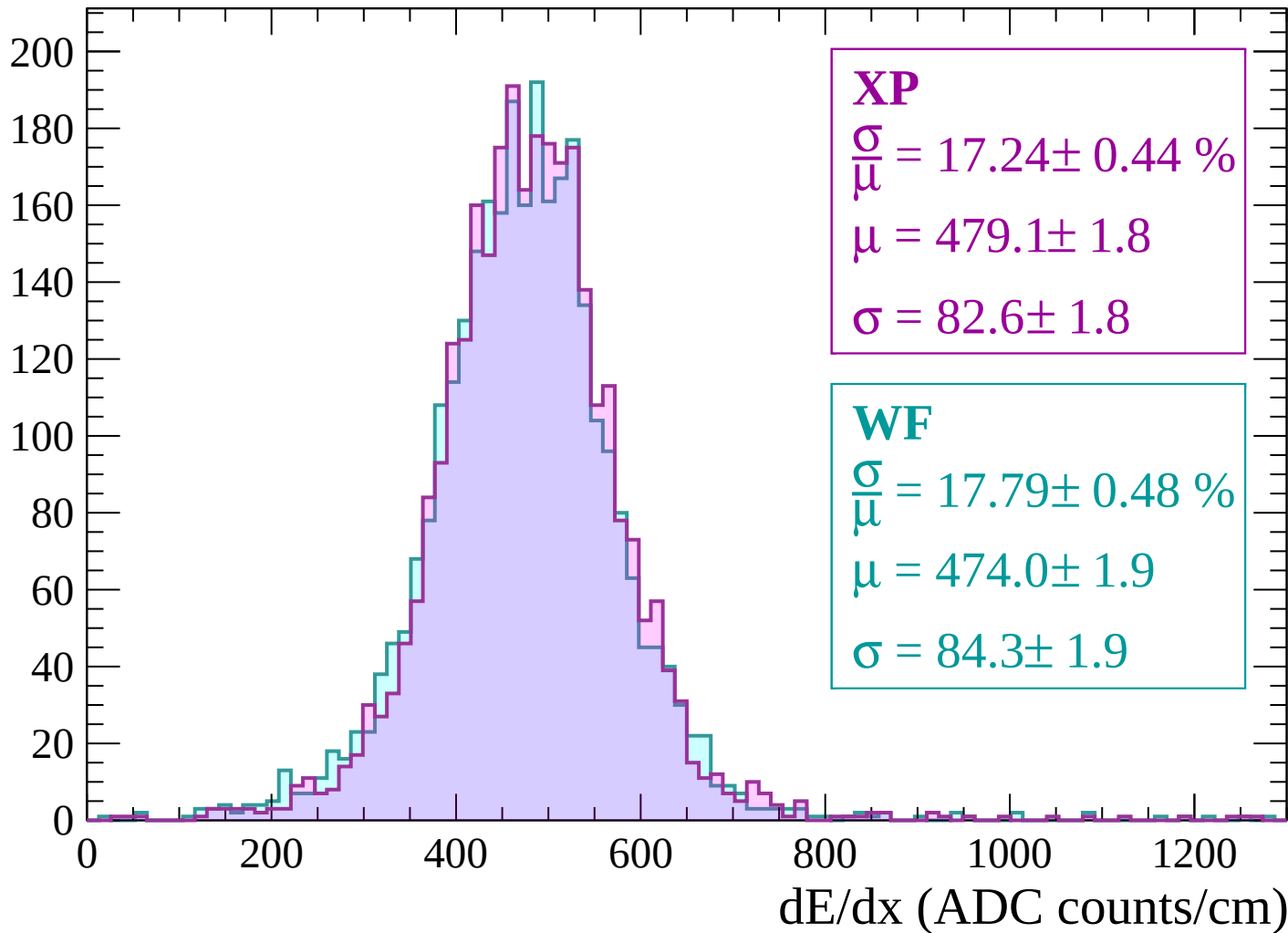
Energy loss | $840 < p < 900$

Count



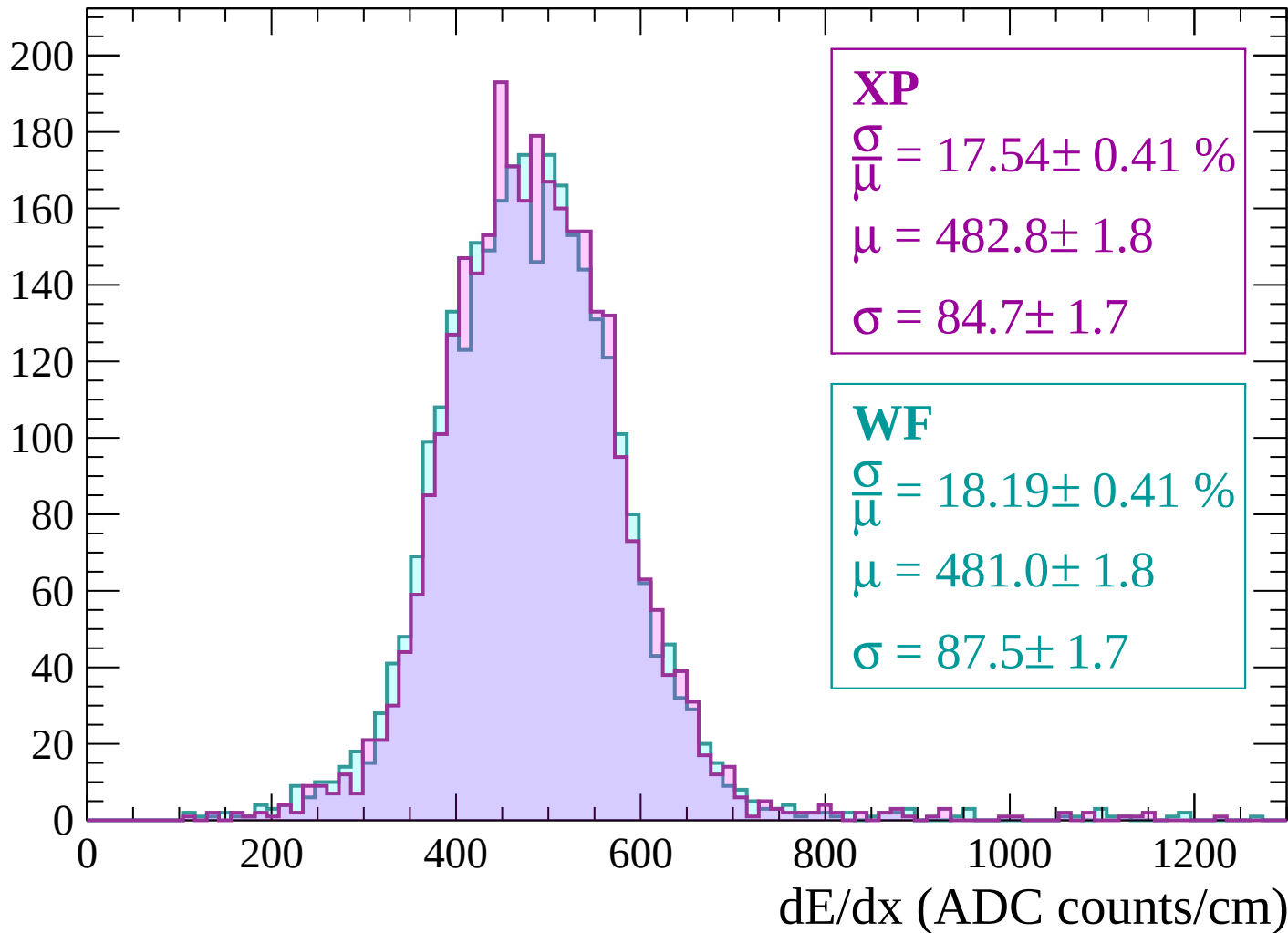
Energy loss | $900 < p < 960$

Count



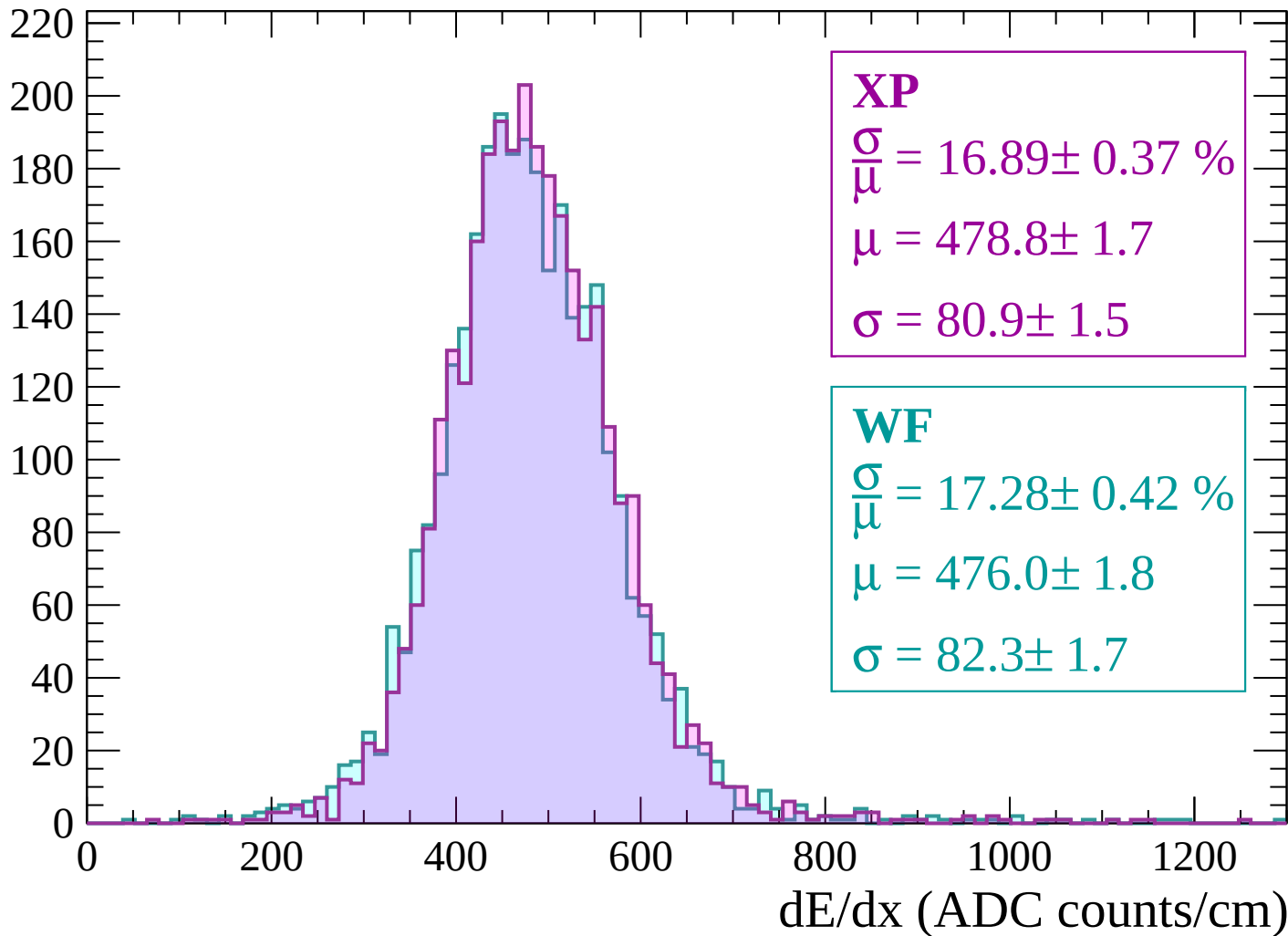
Energy loss | $960 < p < 1020$

Count



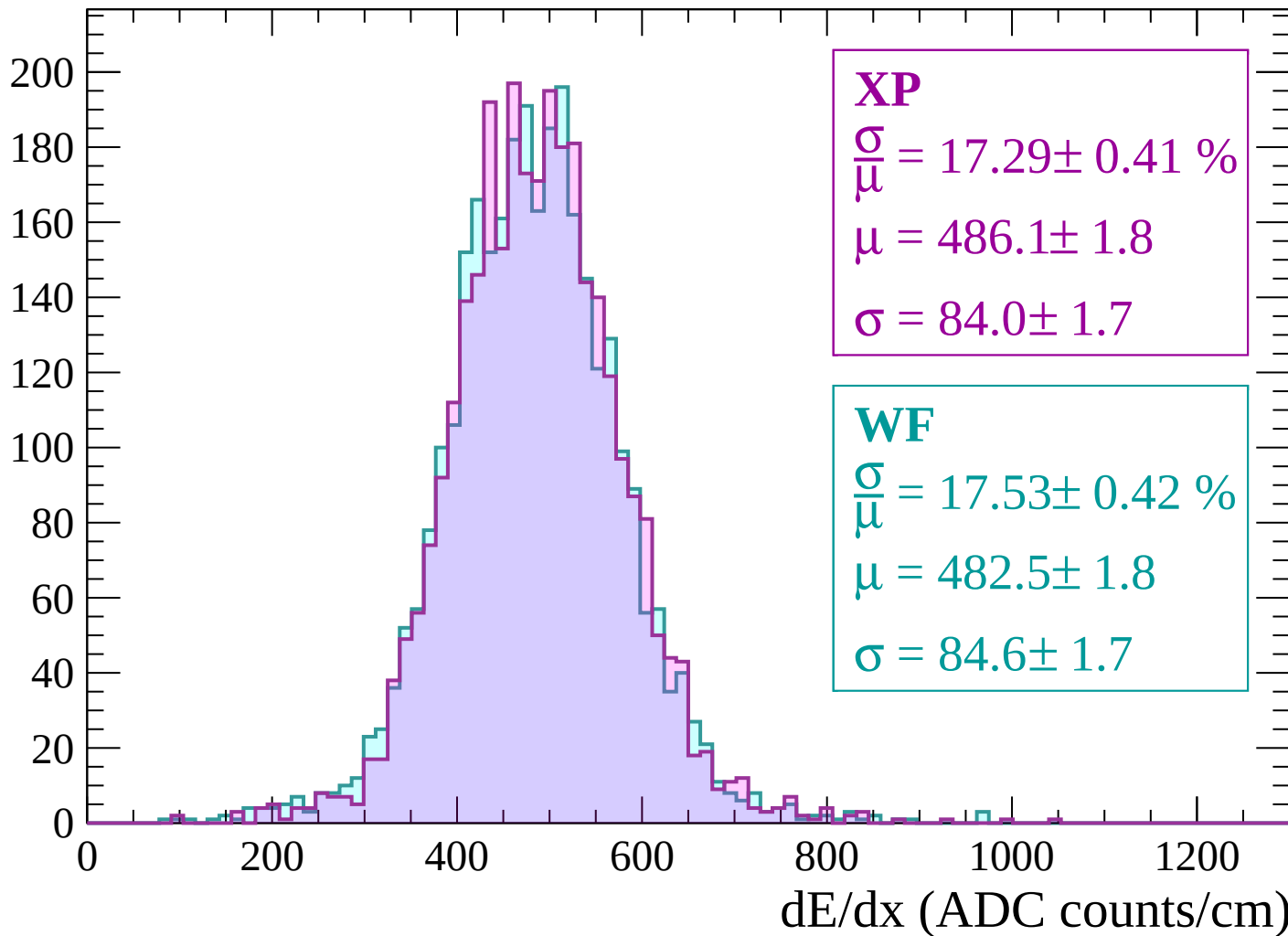
Energy loss | $1020 < p < 1080$

Count



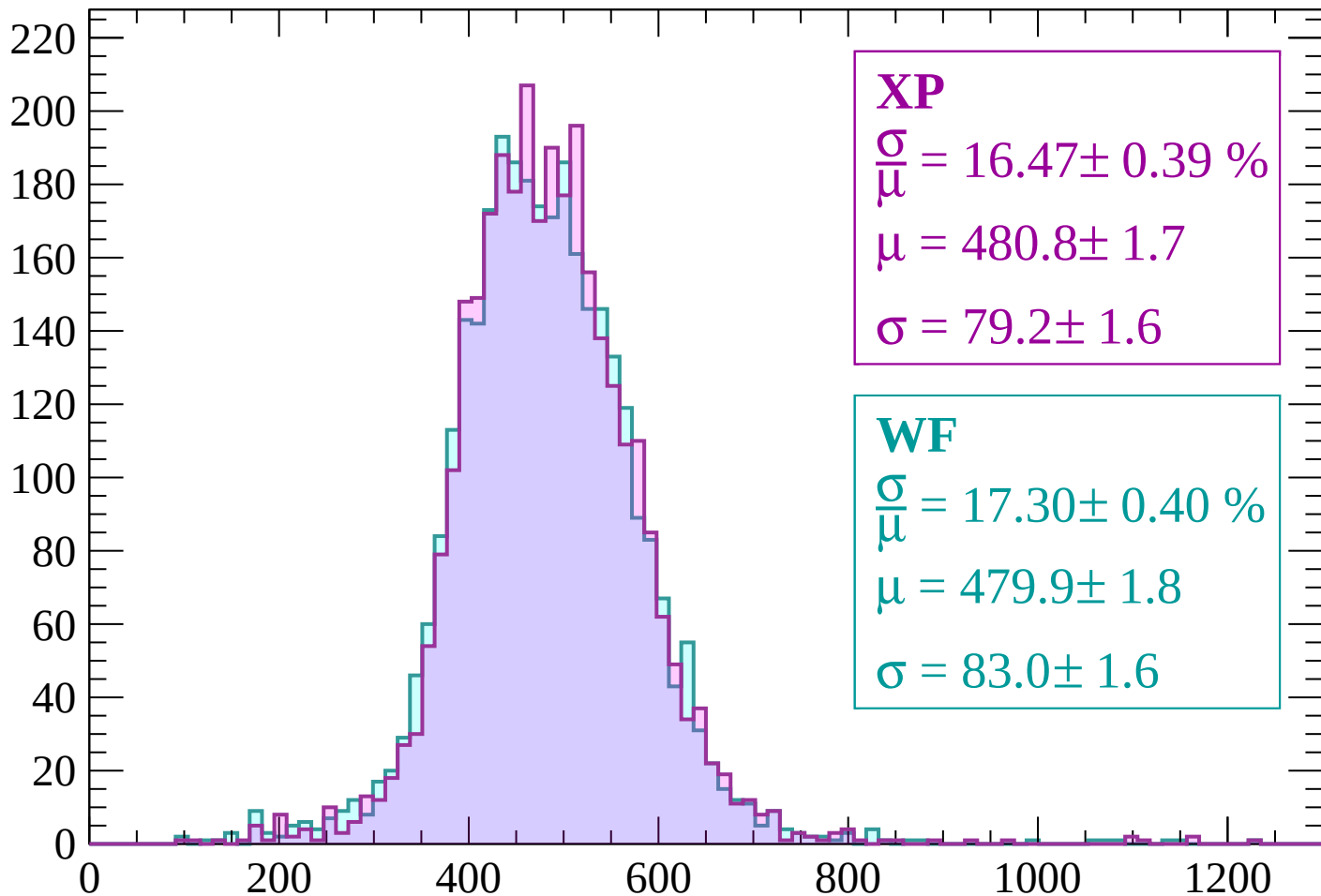
Energy loss | $1080 < p < 1140$

Count



Energy loss | $1140 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 16.47 \pm 0.39 \%$$

$$\mu = 480.8 \pm 1.7$$

$$\sigma = 79.2 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 17.30 \pm 0.40 \%$$

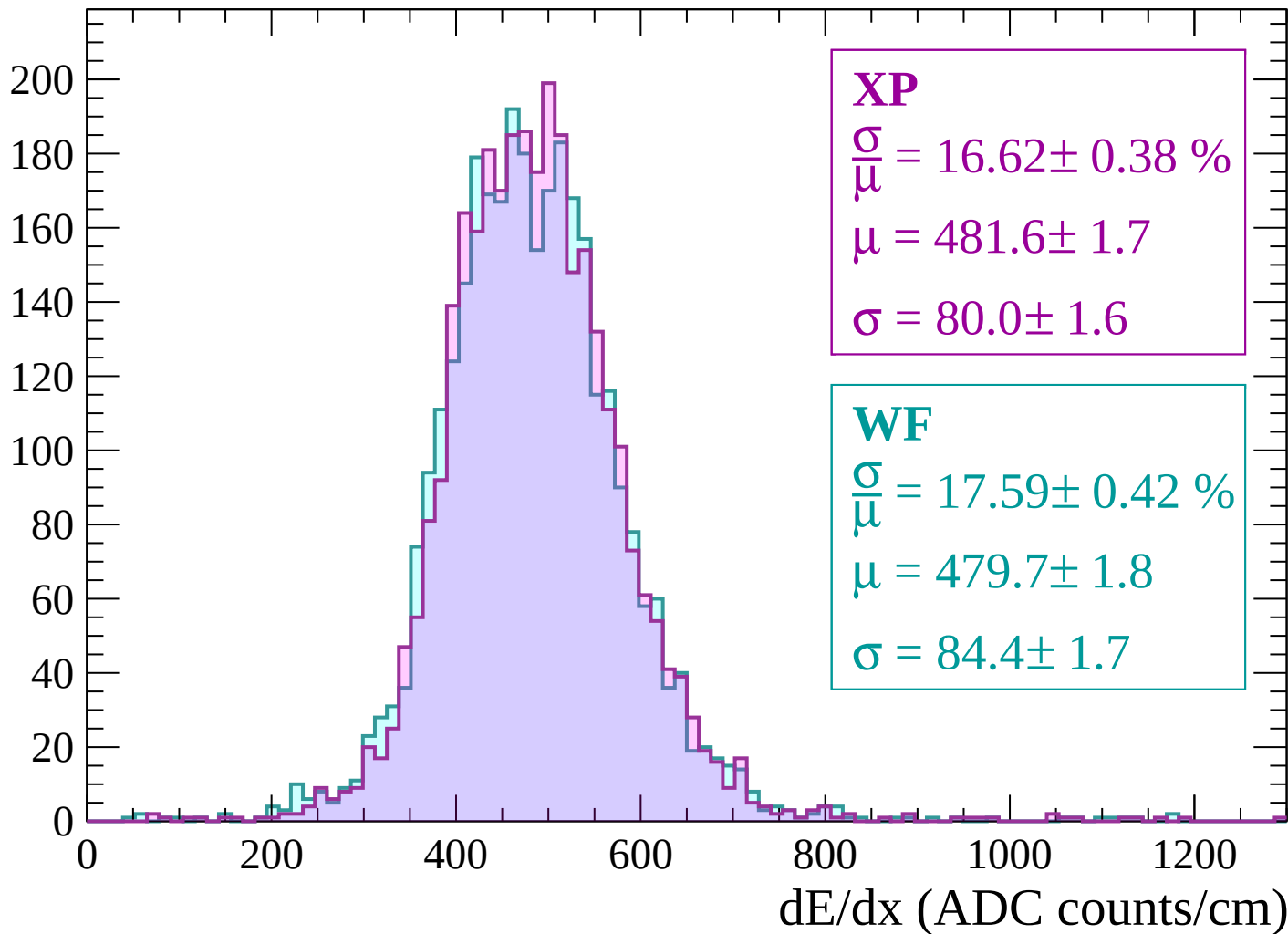
$$\mu = 479.9 \pm 1.8$$

$$\sigma = 83.0 \pm 1.6$$

dE/dx (ADC counts/cm)

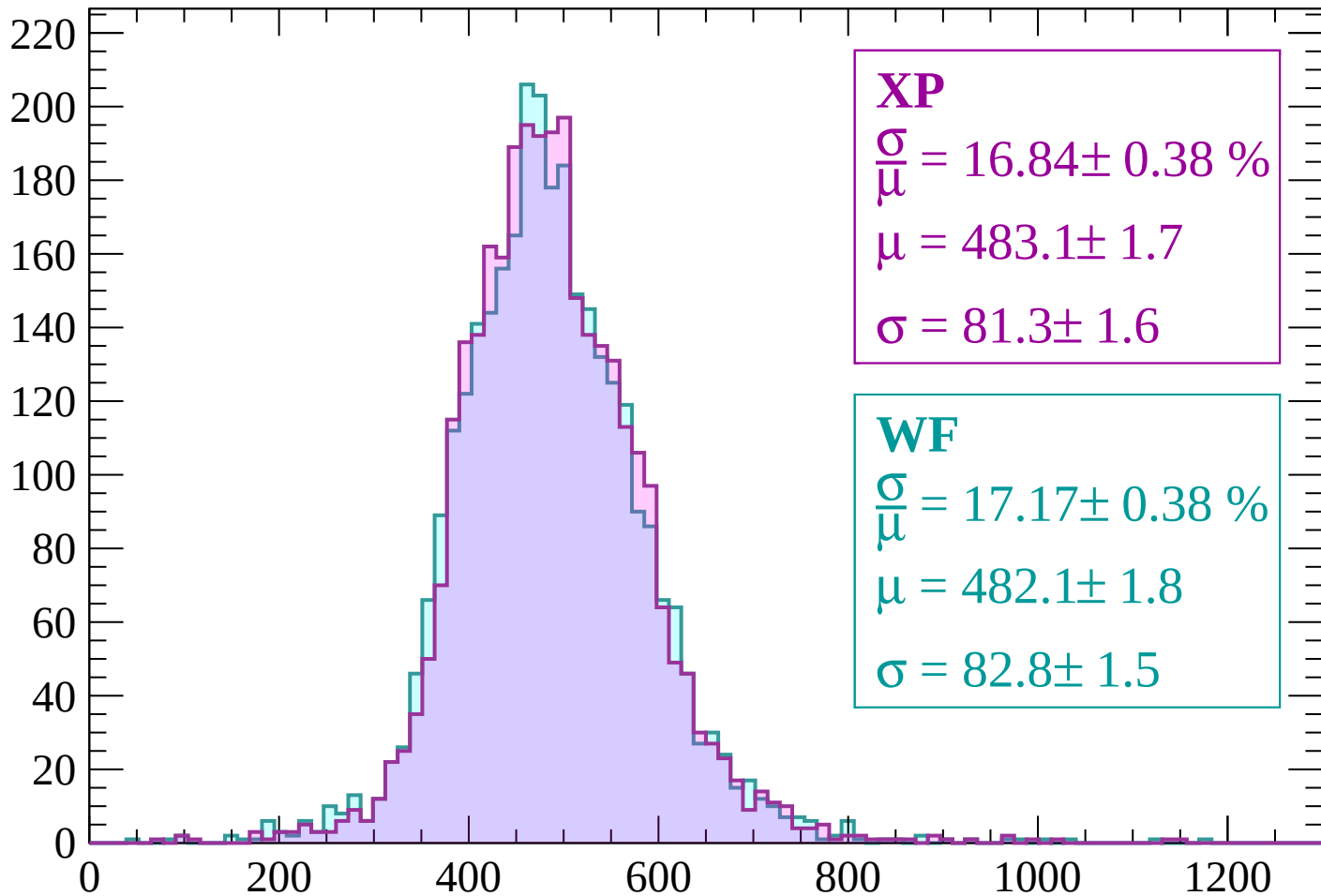
Energy loss | $1200 < p < 1260$

Count



Energy loss | $1260 < p < 1320$

Count



XP

$$\frac{\sigma}{\mu} = 16.84 \pm 0.38 \%$$

$$\mu = 483.1 \pm 1.7$$

$$\sigma = 81.3 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 17.17 \pm 0.38 \%$$

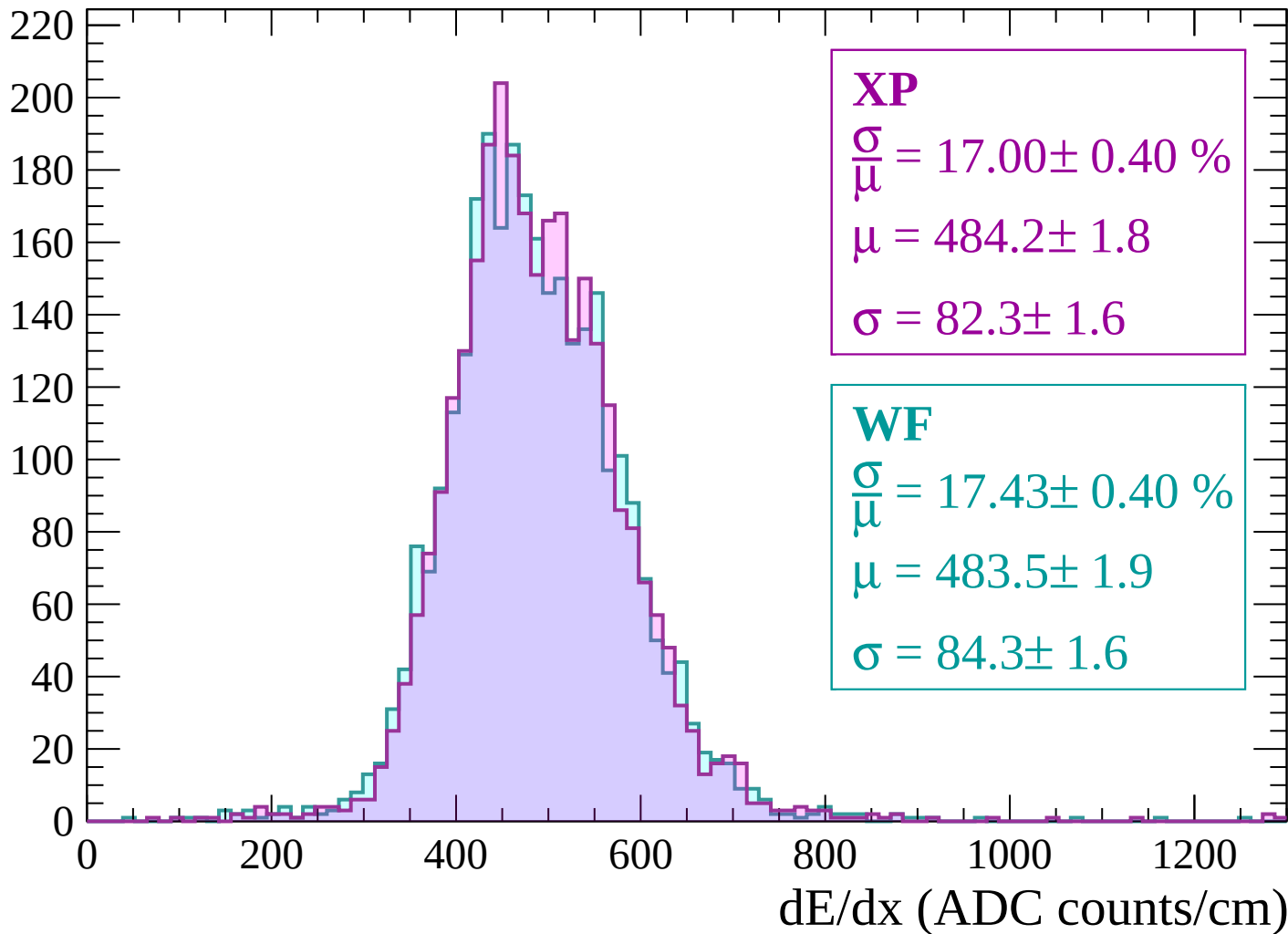
$$\mu = 482.1 \pm 1.8$$

$$\sigma = 82.8 \pm 1.5$$

dE/dx (ADC counts/cm)

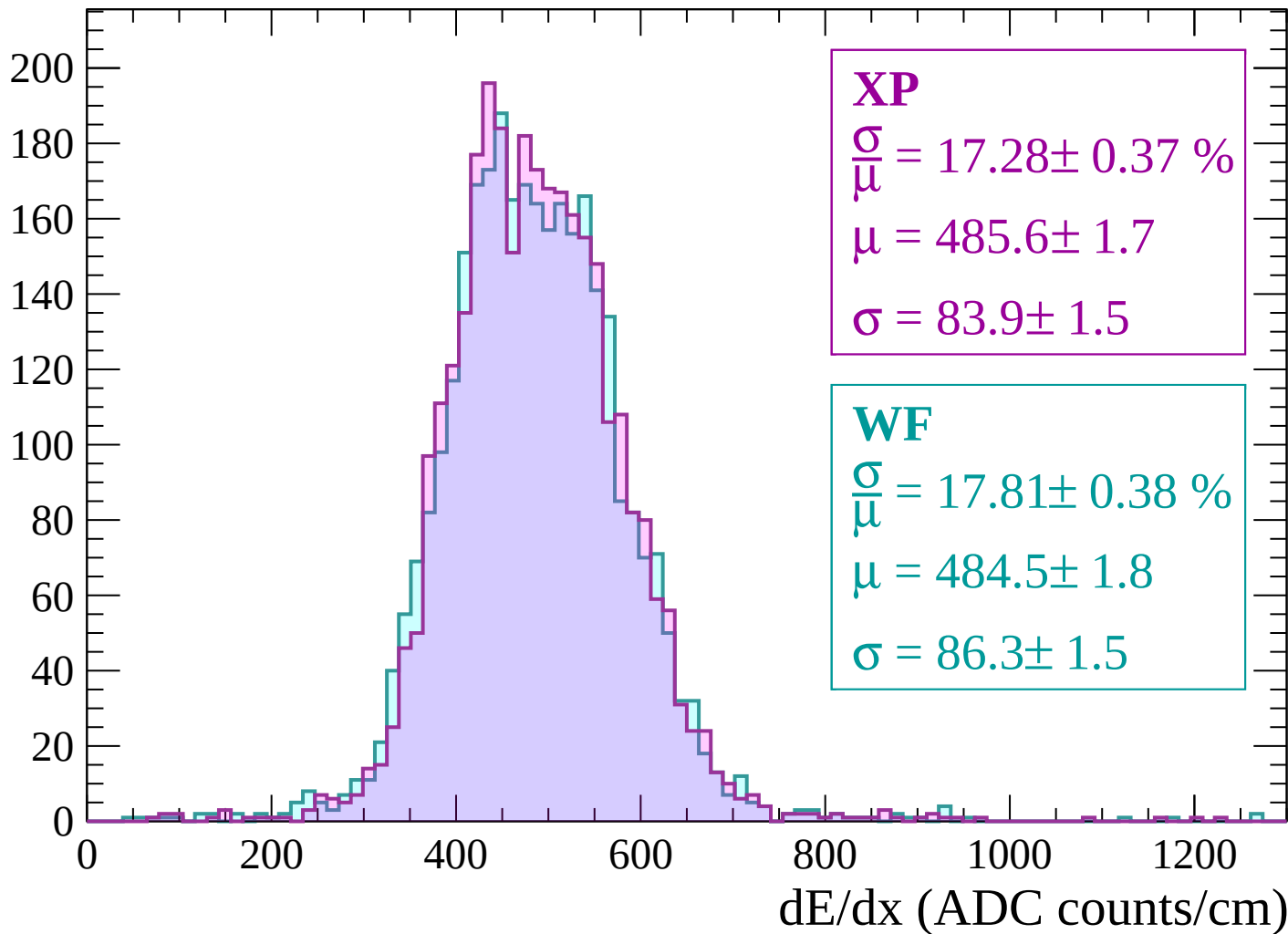
Energy loss | $1320 < p < 1380$

Count



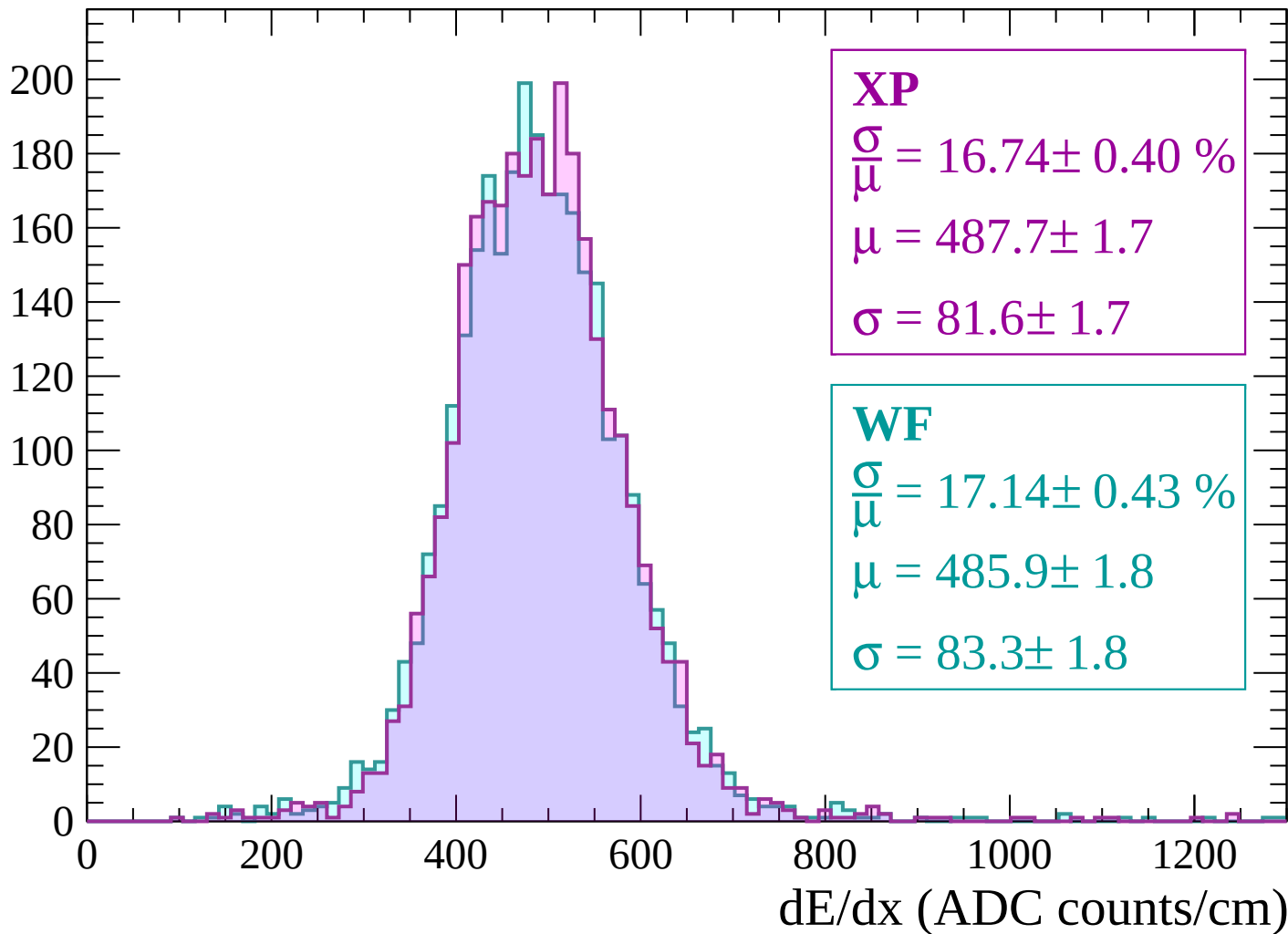
Energy loss | $1380 < p < 1440$

Count



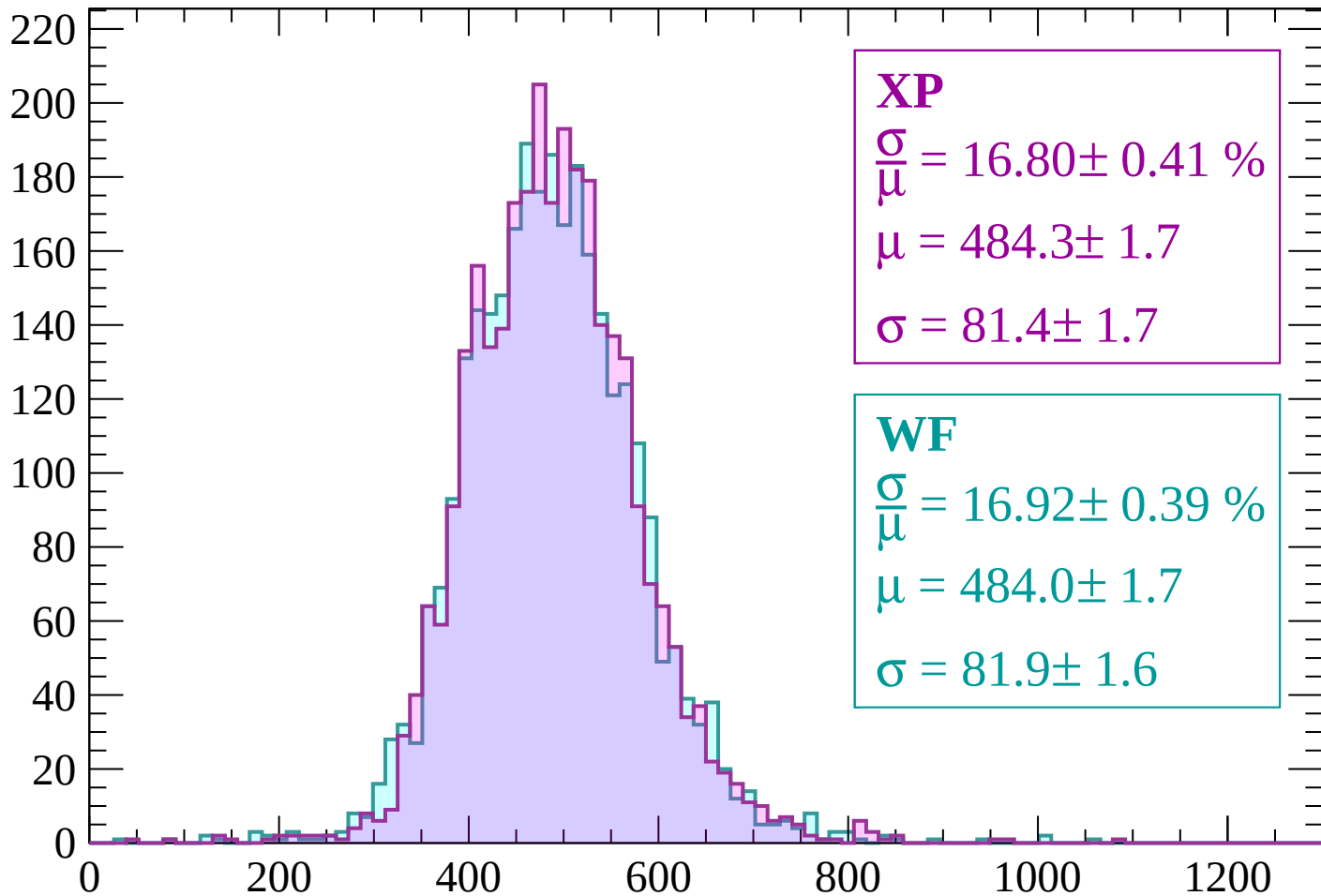
Energy loss | $1440 < p < 1500$

Count



Energy loss | $1500 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 16.80 \pm 0.41 \%$$

$$\mu = 484.3 \pm 1.7$$

$$\sigma = 81.4 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 16.92 \pm 0.39 \%$$

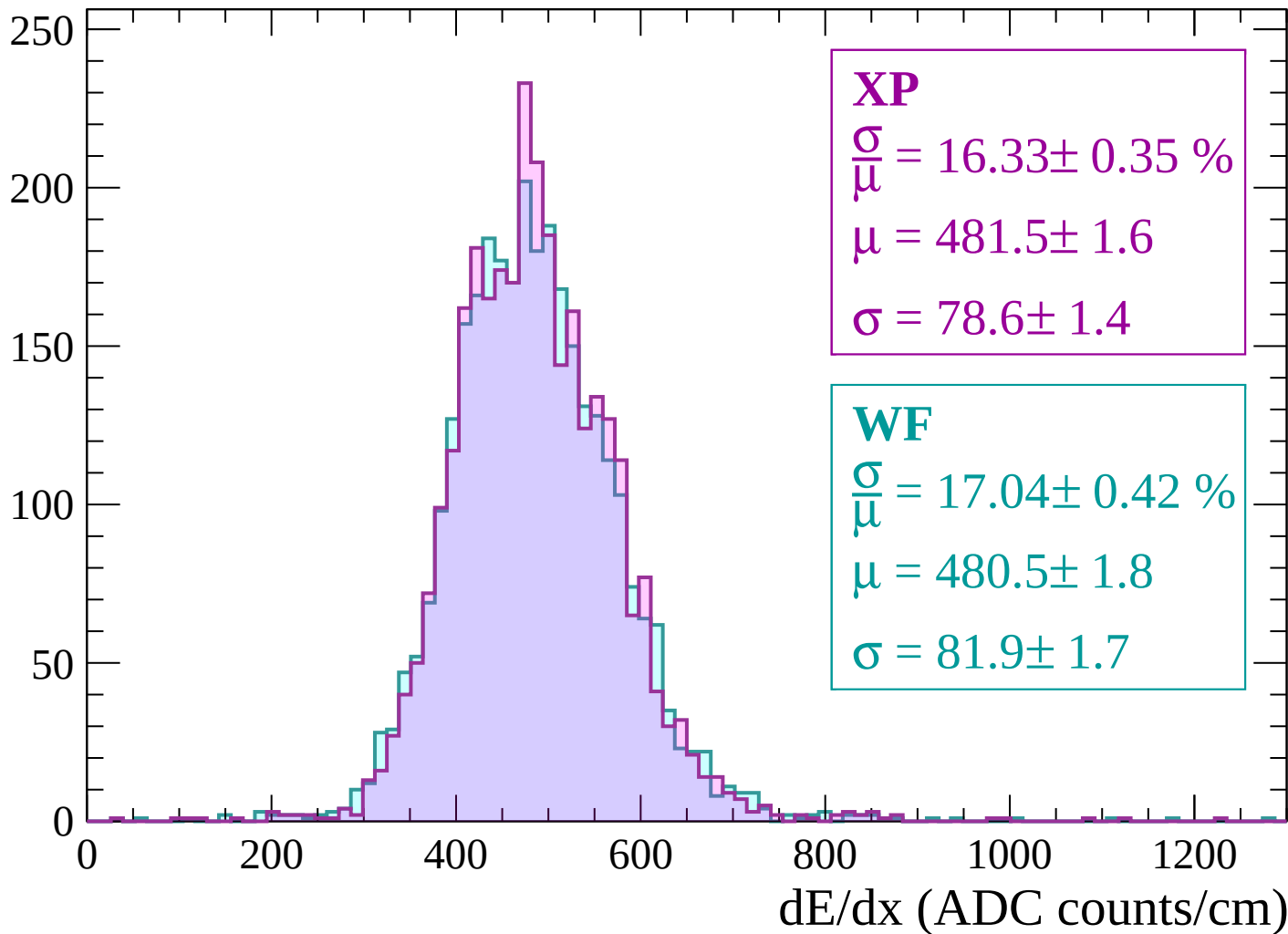
$$\mu = 484.0 \pm 1.7$$

$$\sigma = 81.9 \pm 1.6$$

dE/dx (ADC counts/cm)

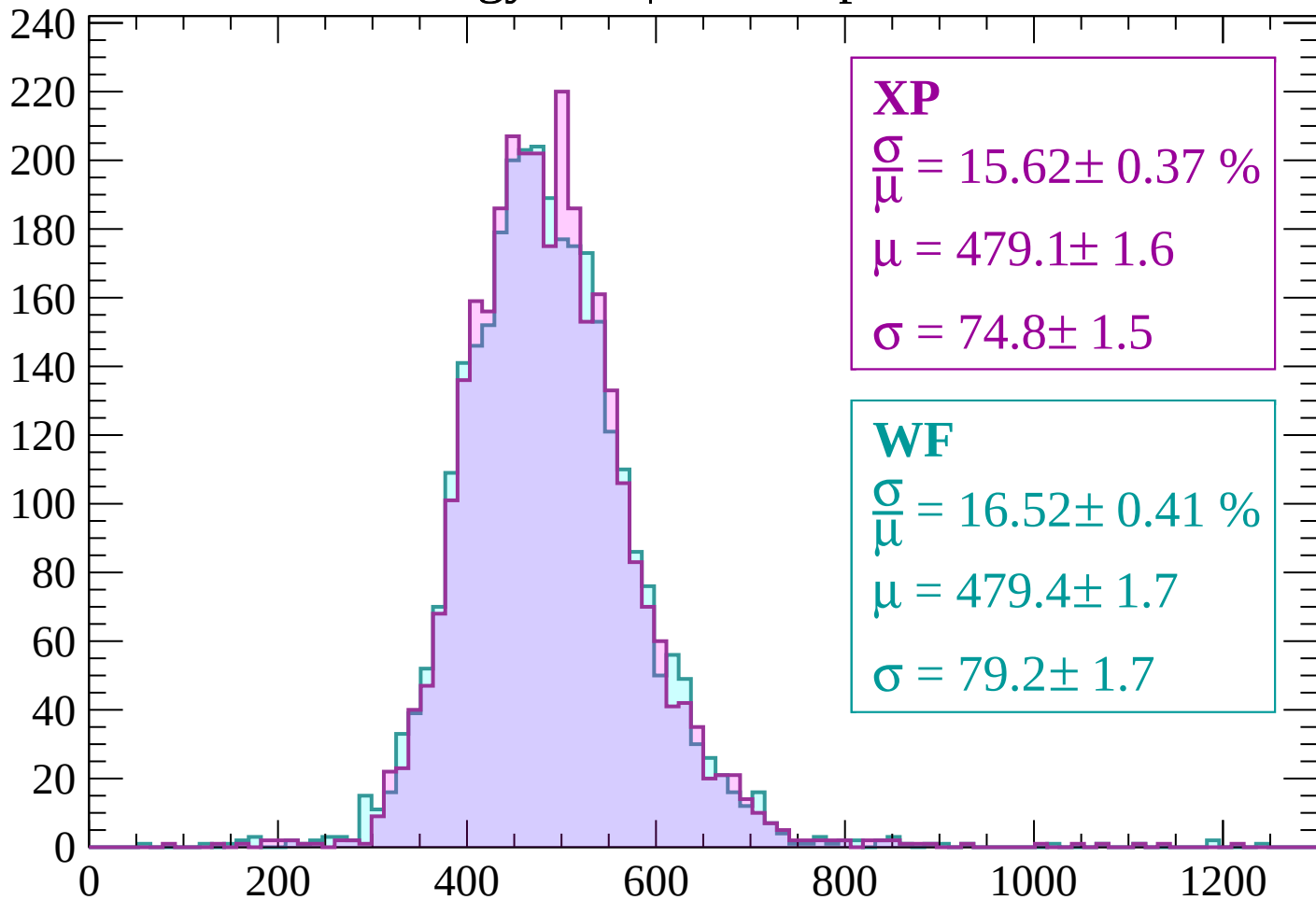
Energy loss | $1560 < p < 1620$

Count



Energy loss | $1620 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 15.62 \pm 0.37 \%$$

$$\mu = 479.1 \pm 1.6$$

$$\sigma = 74.8 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 16.52 \pm 0.41 \%$$

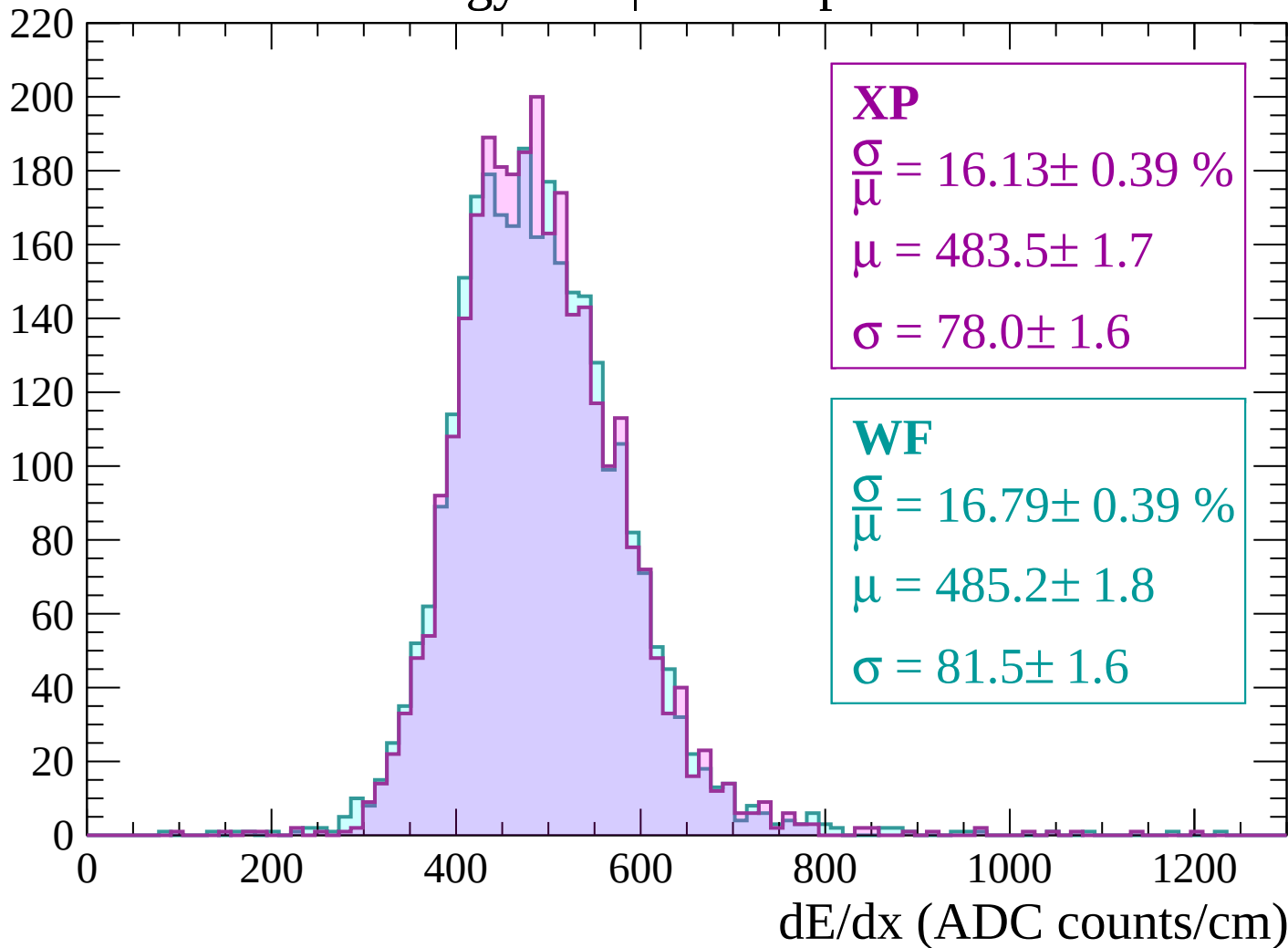
$$\mu = 479.4 \pm 1.7$$

$$\sigma = 79.2 \pm 1.7$$

dE/dx (ADC counts/cm)

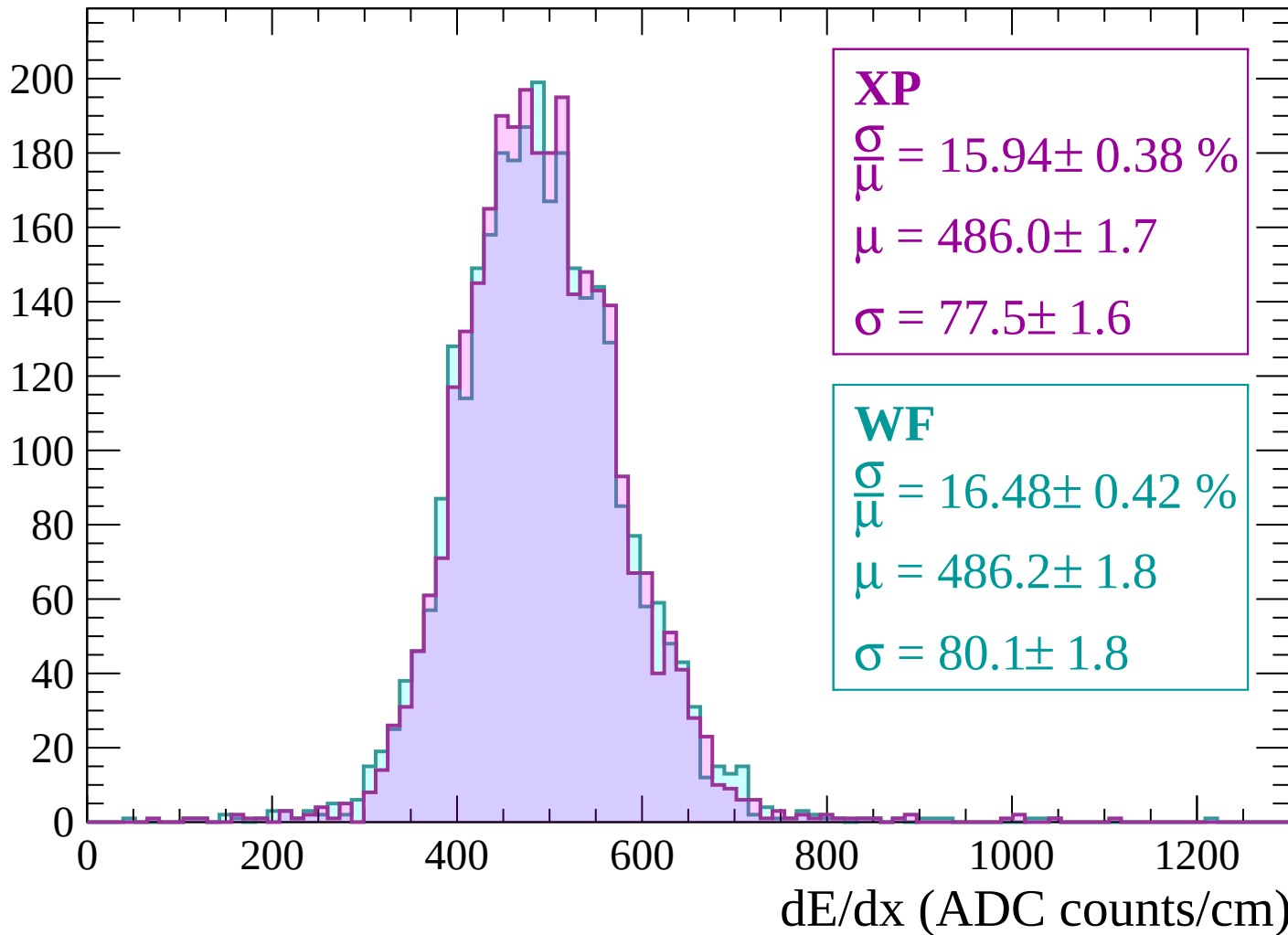
Energy loss | $1680 < p < 1740$

Count



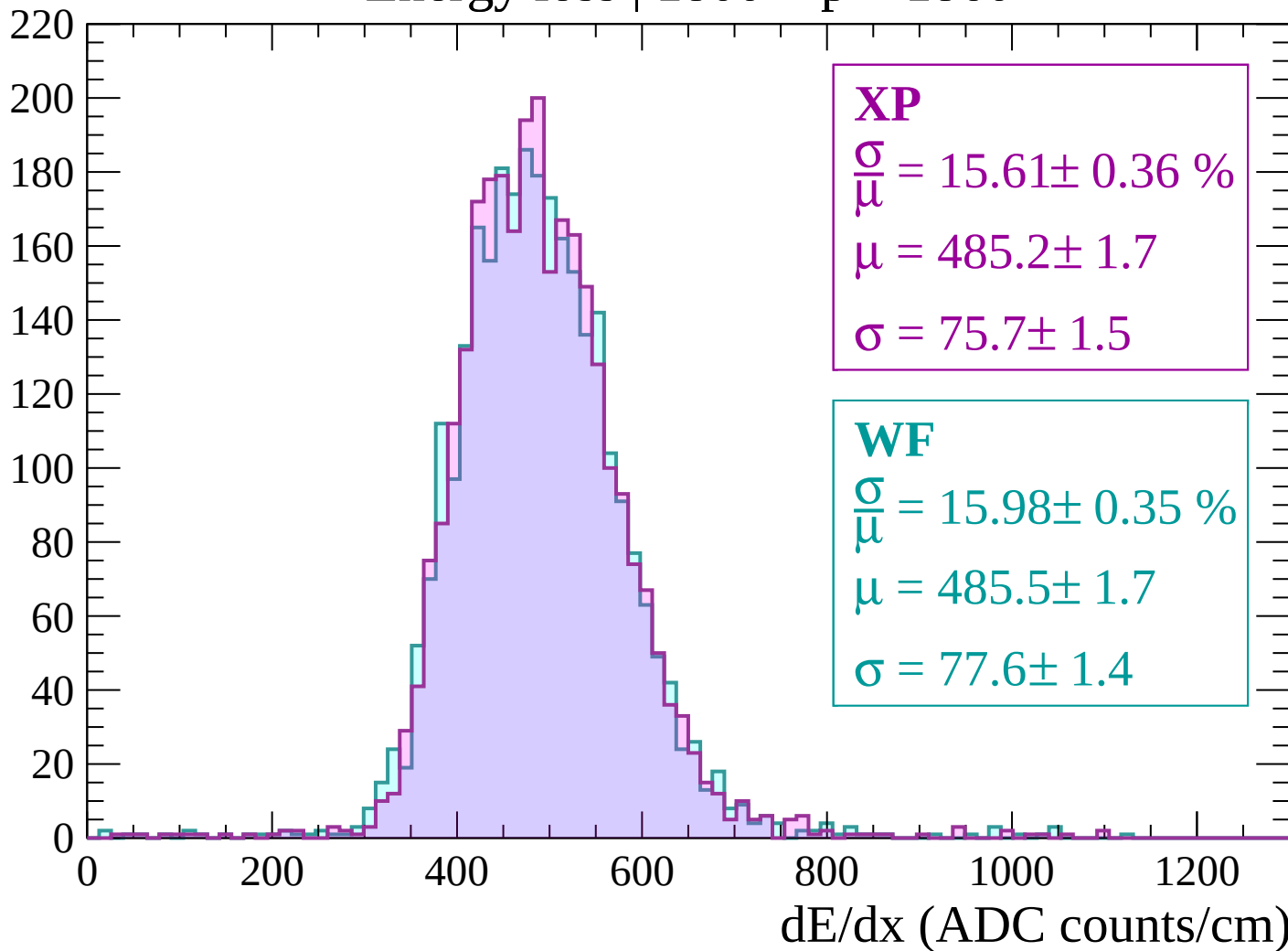
Energy loss | $1740 < p < 1800$

Count



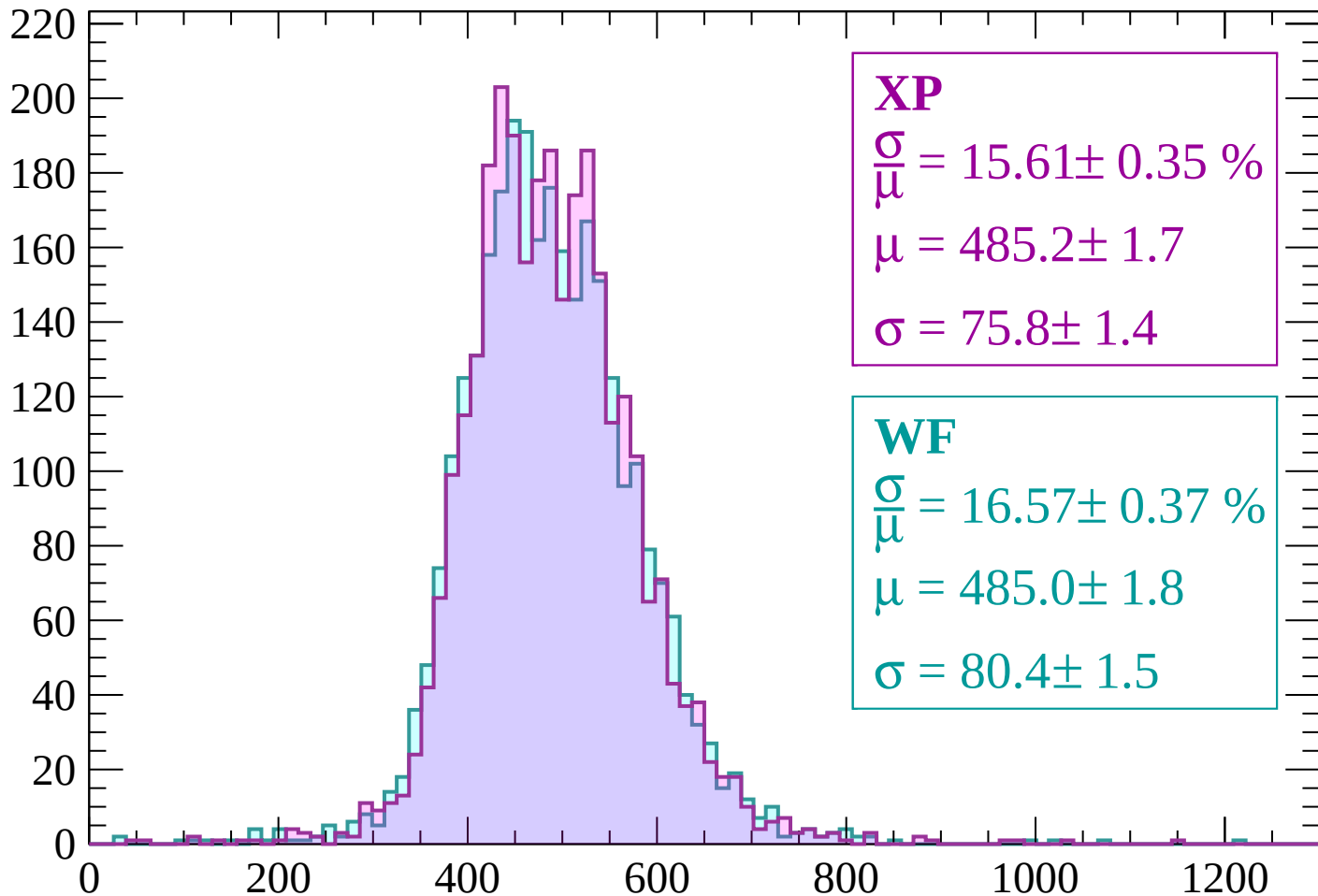
Energy loss | $1800 < p < 1860$

Count



Energy loss | $1860 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 15.61 \pm 0.35 \%$$

$$\mu = 485.2 \pm 1.7$$

$$\sigma = 75.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 16.57 \pm 0.37 \%$$

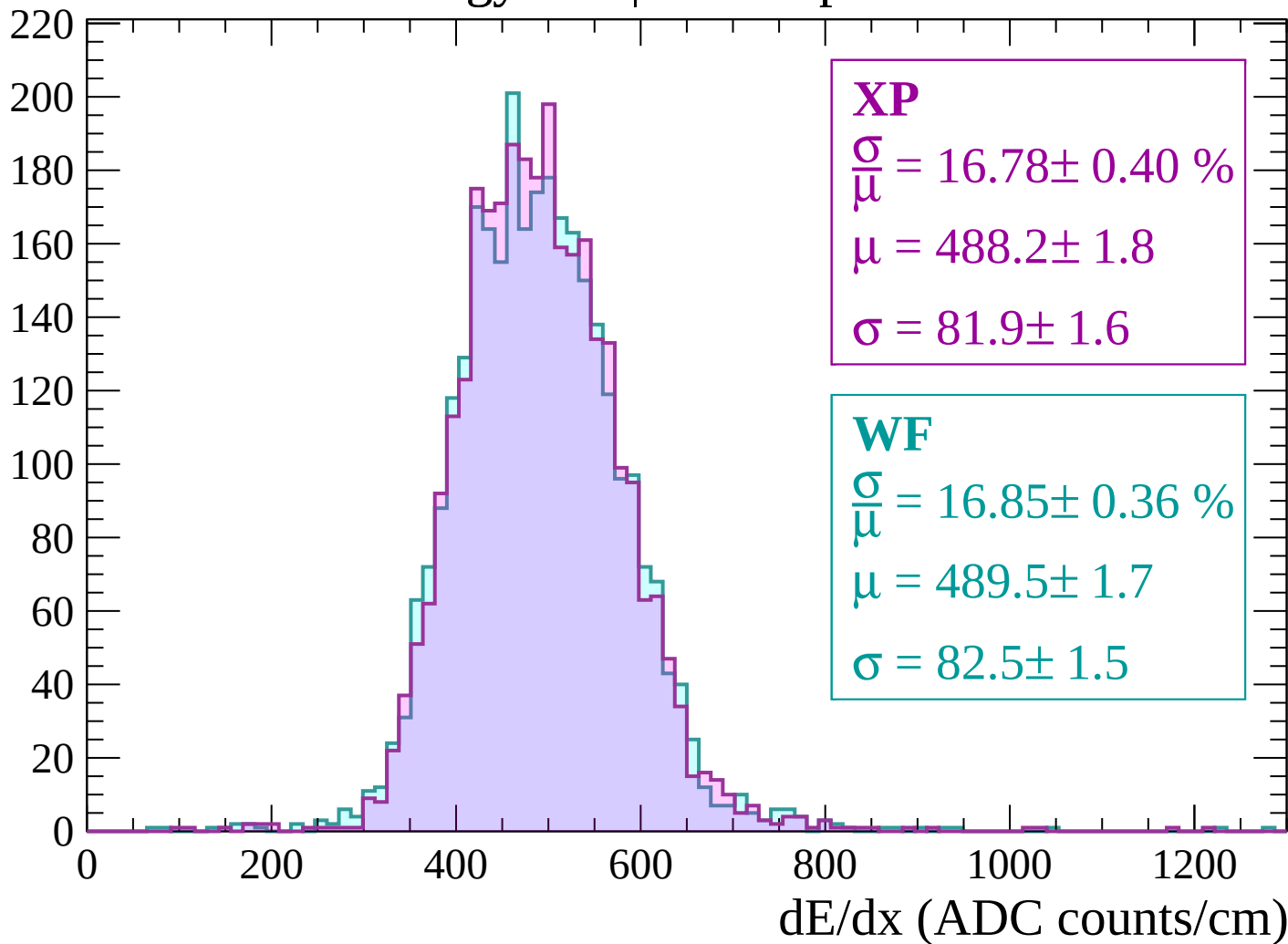
$$\mu = 485.0 \pm 1.8$$

$$\sigma = 80.4 \pm 1.5$$

dE/dx (ADC counts/cm)

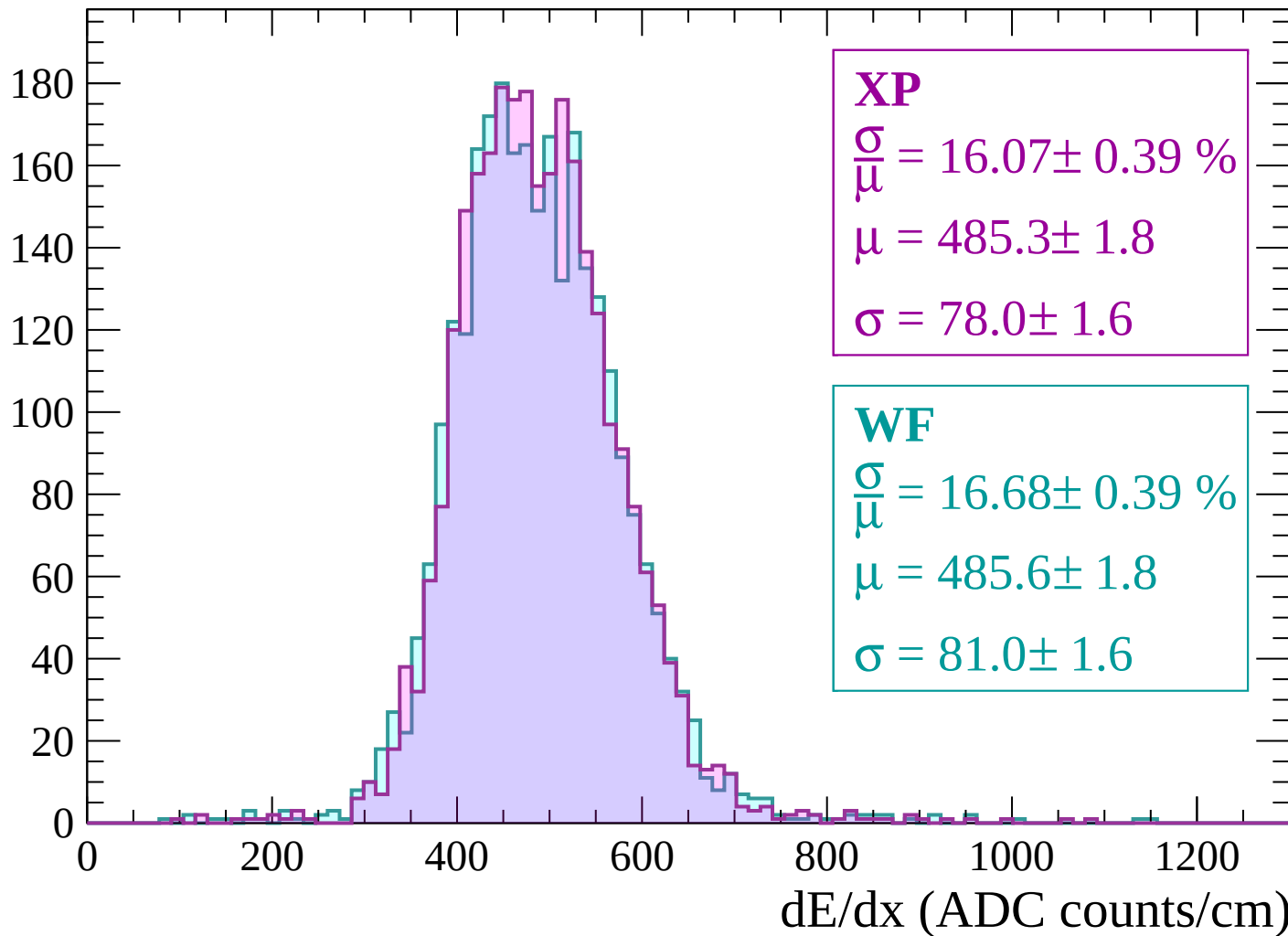
Energy loss | $1920 < p < 1980$

Count



Energy loss | $1980 < p < 2040$

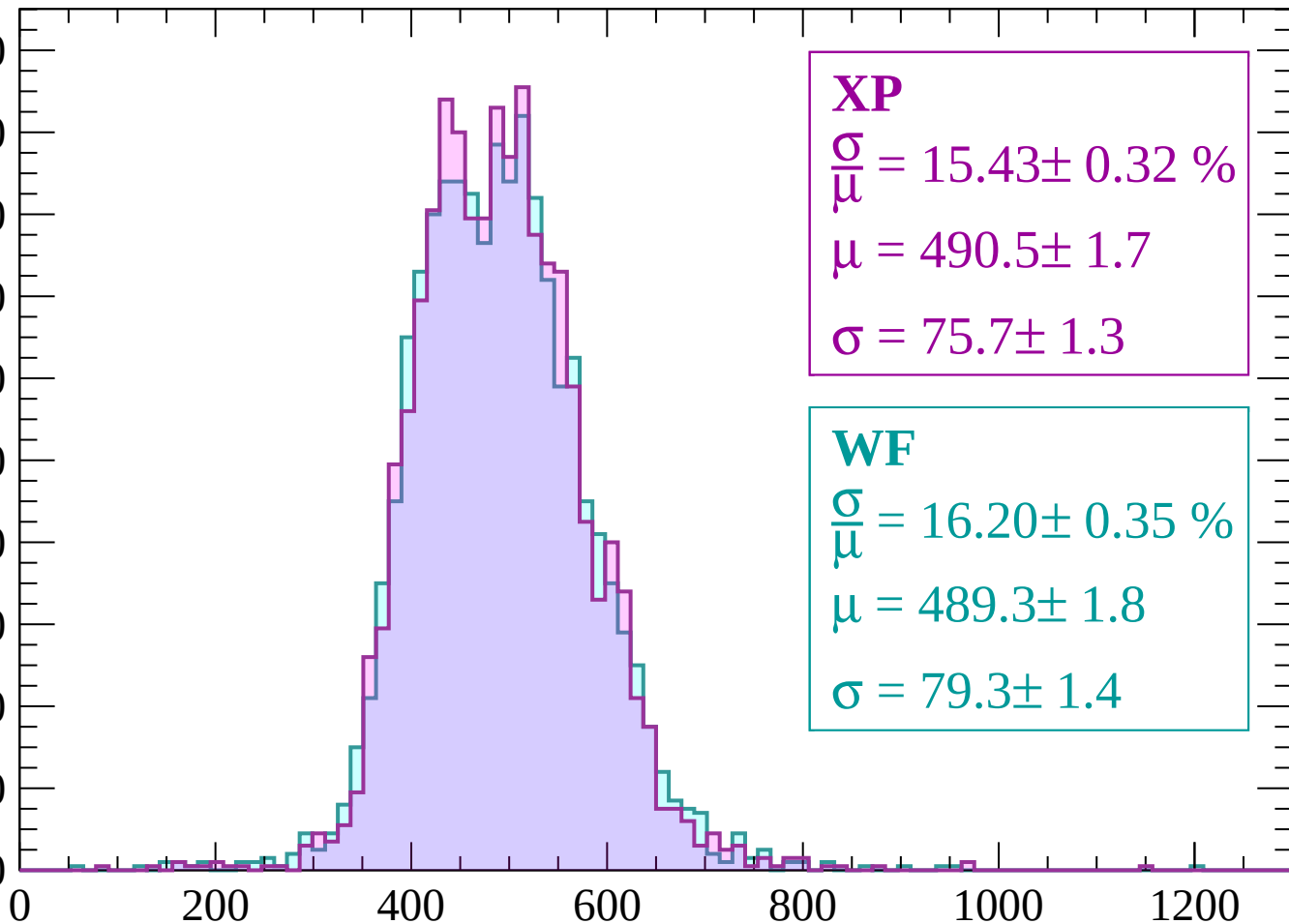
Count



Energy loss | $2040 < p < 2100$

Count

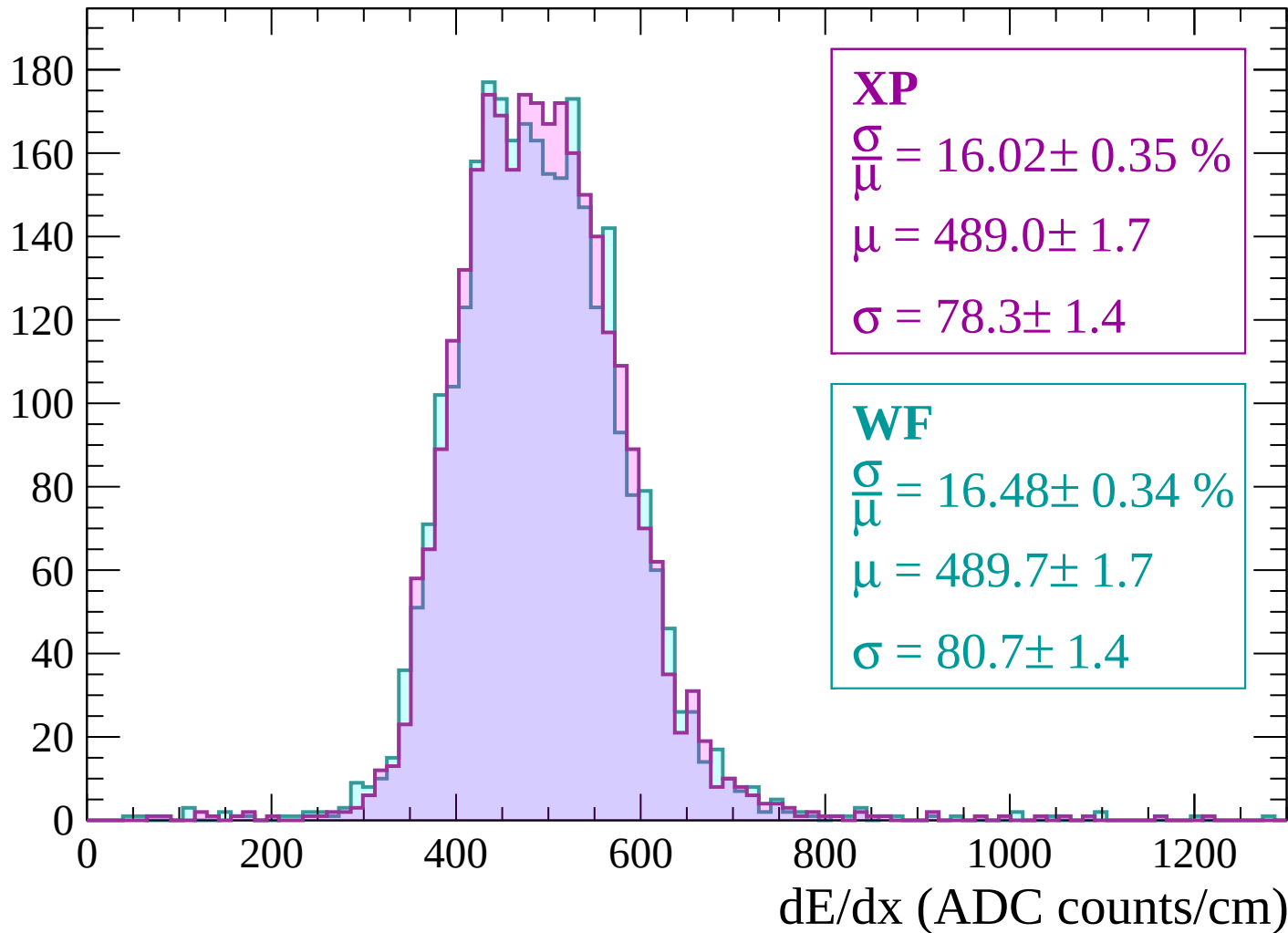
200
180
160
140
120
100
80
60
40
20
0



dE/dx (ADC counts/cm)

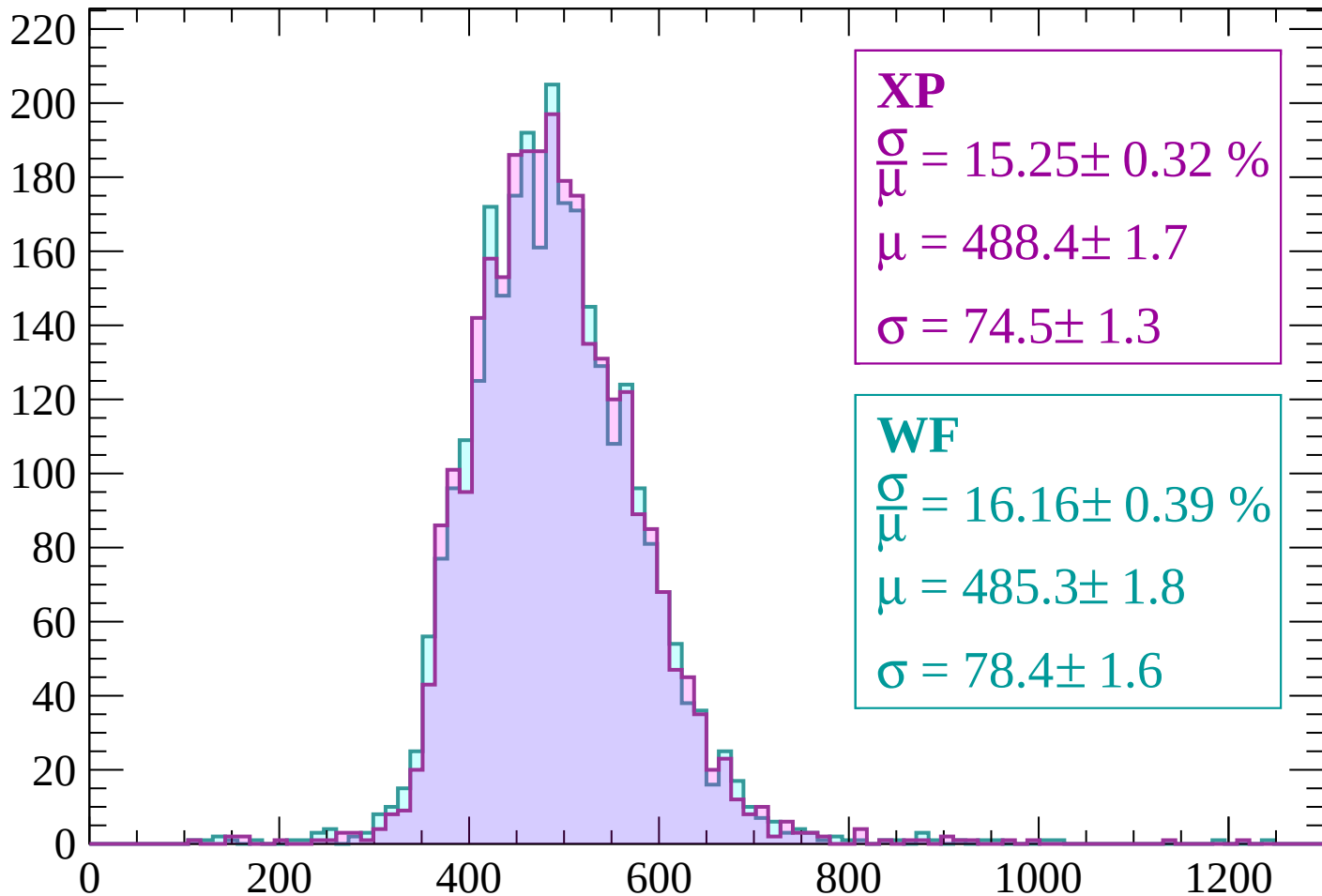
Energy loss | $2100 < p < 2160$

Count



Energy loss | $2160 < p < 2220$

Count



XP

$$\frac{\sigma}{\mu} = 15.25 \pm 0.32 \%$$

$$\mu = 488.4 \pm 1.7$$

$$\sigma = 74.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 16.16 \pm 0.39 \%$$

$$\mu = 485.3 \pm 1.8$$

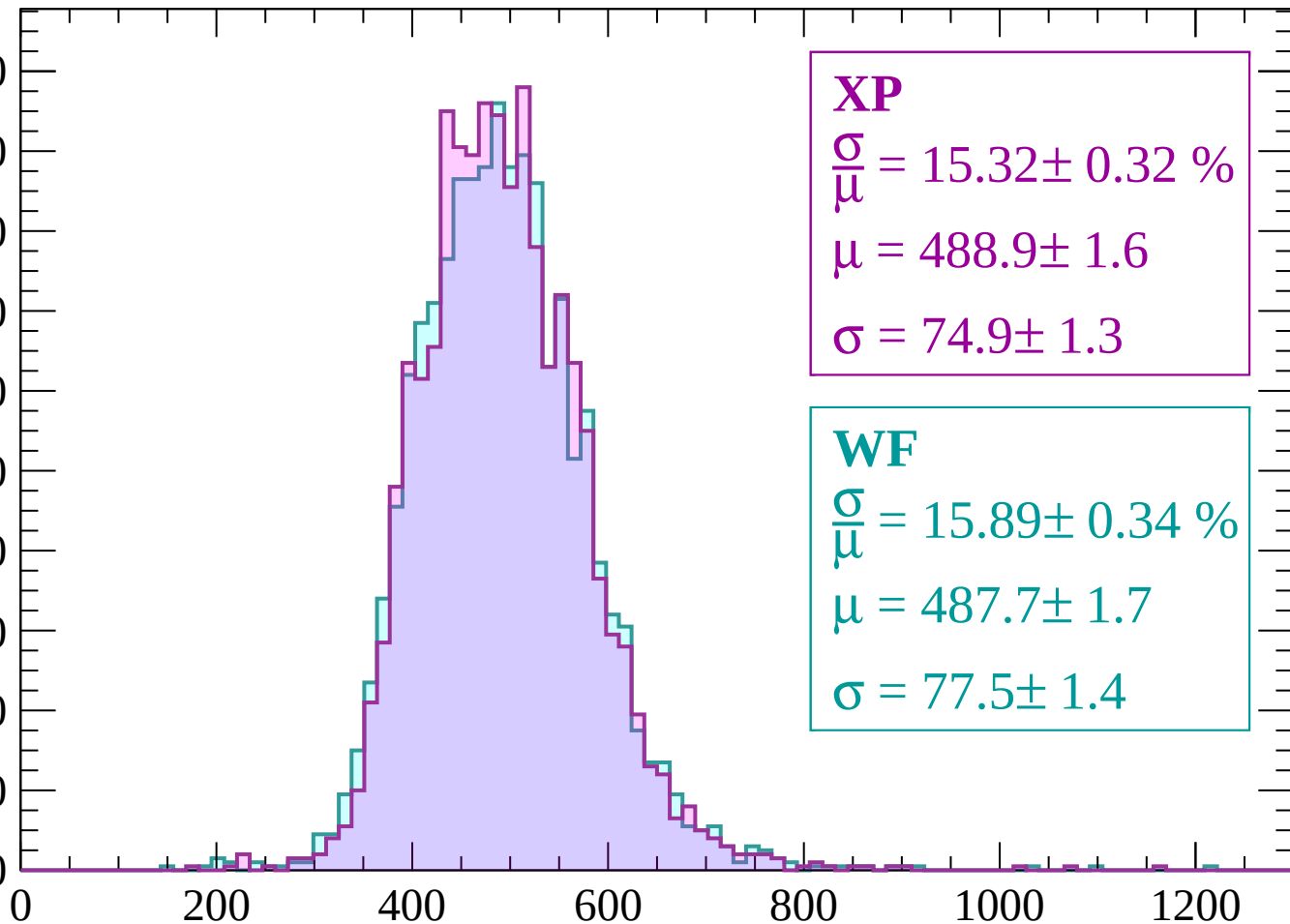
$$\sigma = 78.4 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count

200
180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 15.32 \pm 0.32 \%$$

$$\mu = 488.9 \pm 1.6$$

$$\sigma = 74.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 15.89 \pm 0.34 \%$$

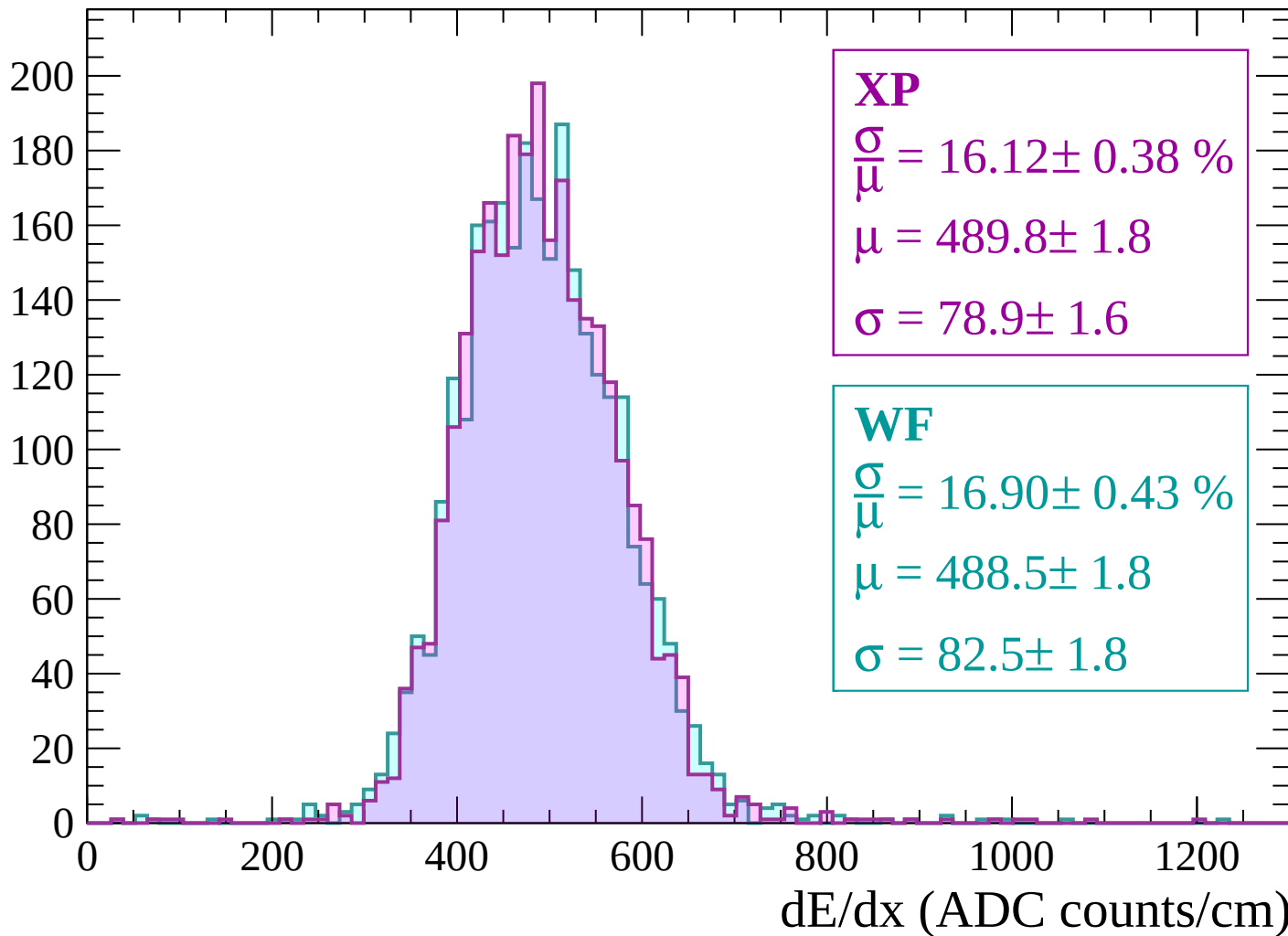
$$\mu = 487.7 \pm 1.7$$

$$\sigma = 77.5 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

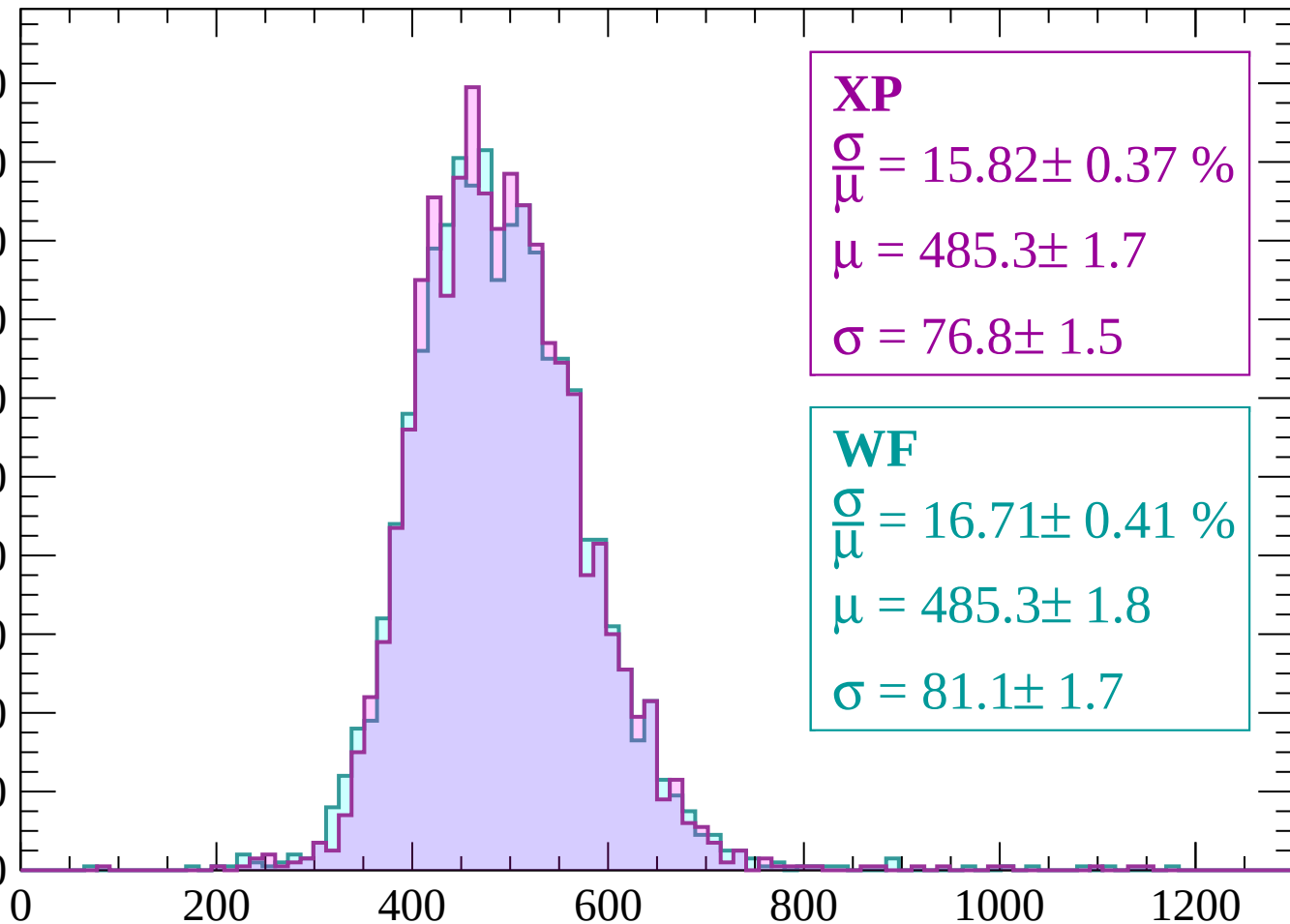
Count



Energy loss | $2340 < p < 2400$

Count

200
180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 15.82 \pm 0.37 \%$$

$$\mu = 485.3 \pm 1.7$$

$$\sigma = 76.8 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 16.71 \pm 0.41 \%$$

$$\mu = 485.3 \pm 1.8$$

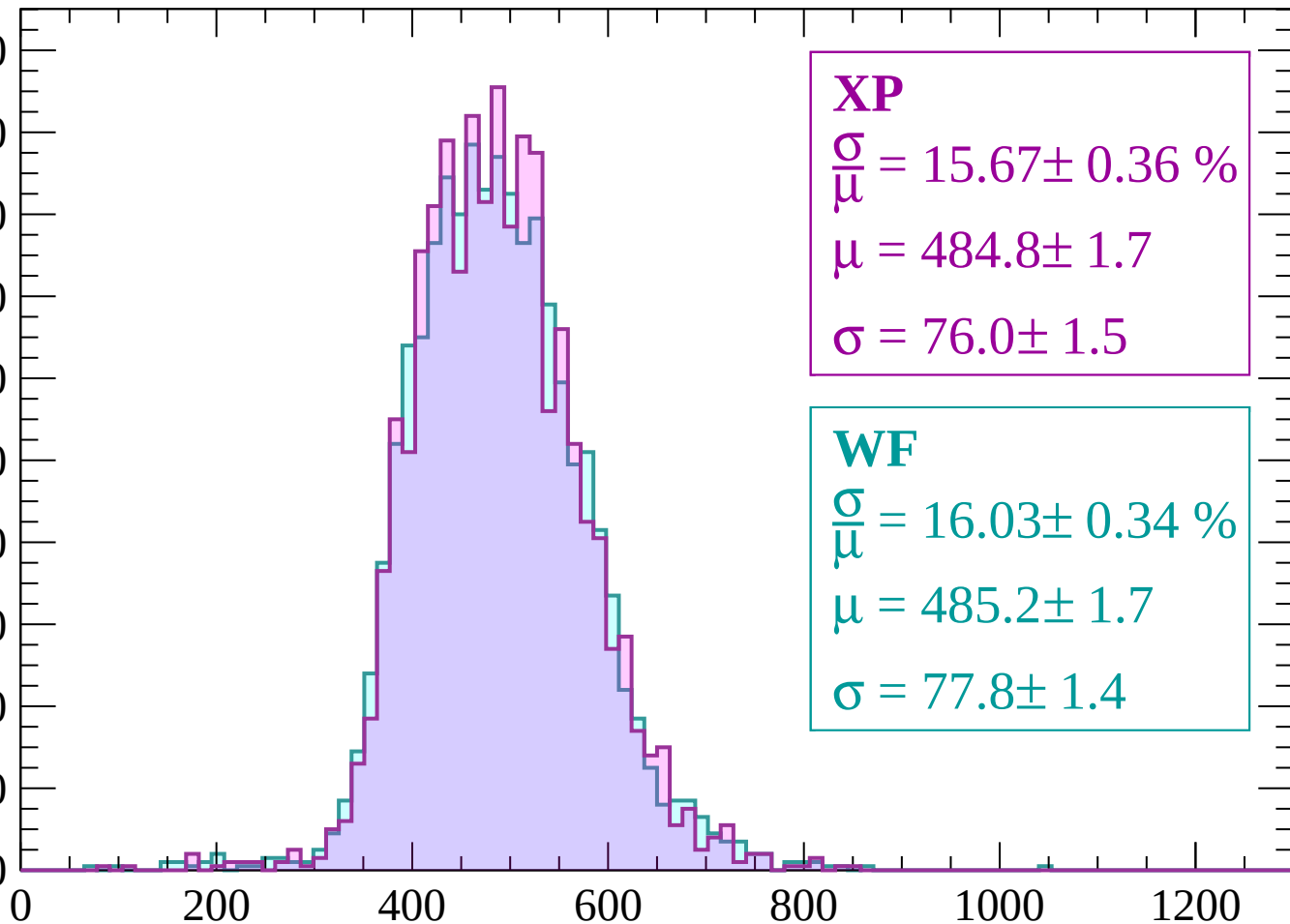
$$\sigma = 81.1 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $2400 < p < 2460$

Count

200
180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 15.67 \pm 0.36 \%$$

$$\mu = 484.8 \pm 1.7$$

$$\sigma = 76.0 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 16.03 \pm 0.34 \%$$

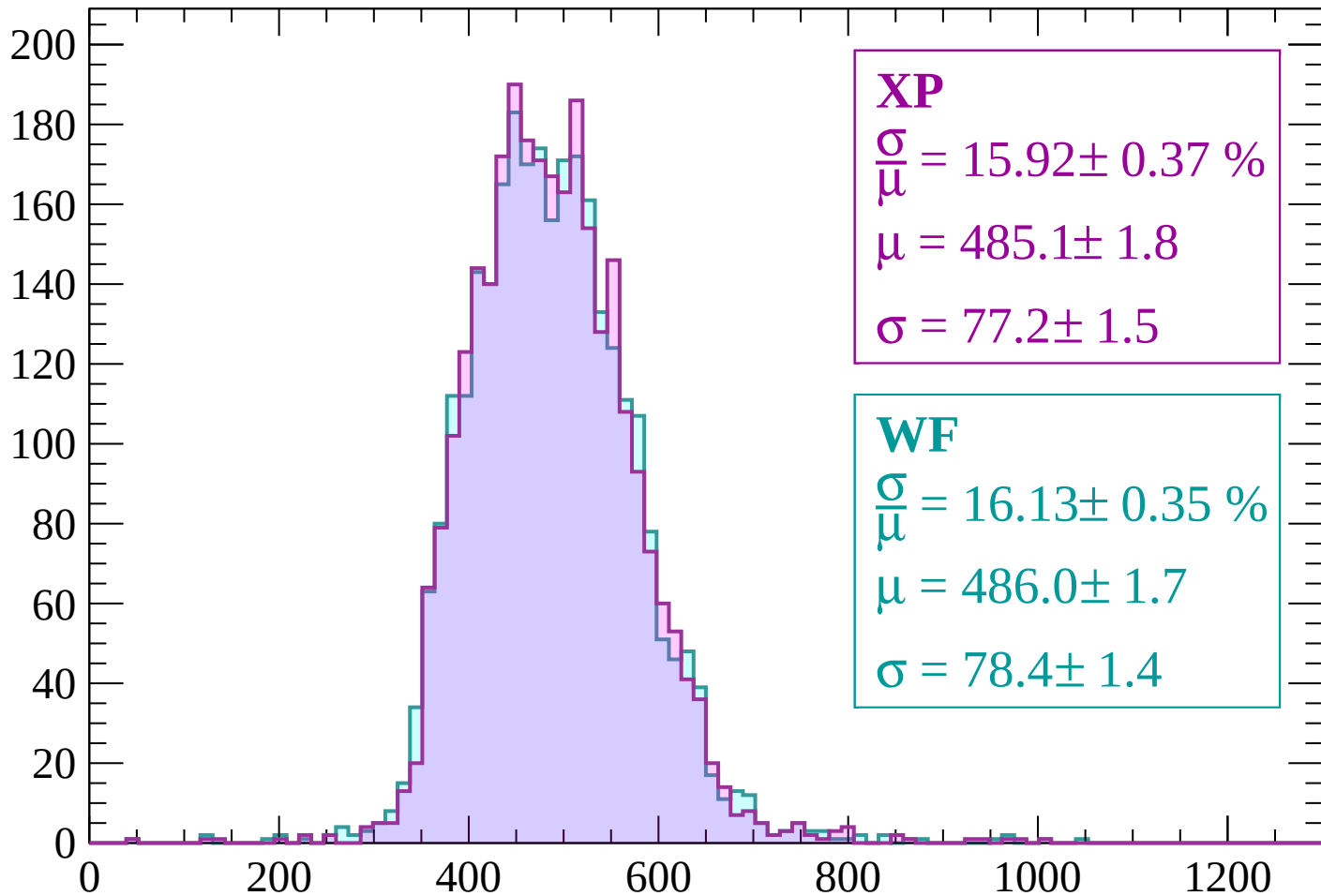
$$\mu = 485.2 \pm 1.7$$

$$\sigma = 77.8 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $2460 < p < 2520$

Count



XP

$$\frac{\sigma}{\mu} = 15.92 \pm 0.37 \%$$

$$\mu = 485.1 \pm 1.8$$

$$\sigma = 77.2 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 16.13 \pm 0.35 \%$$

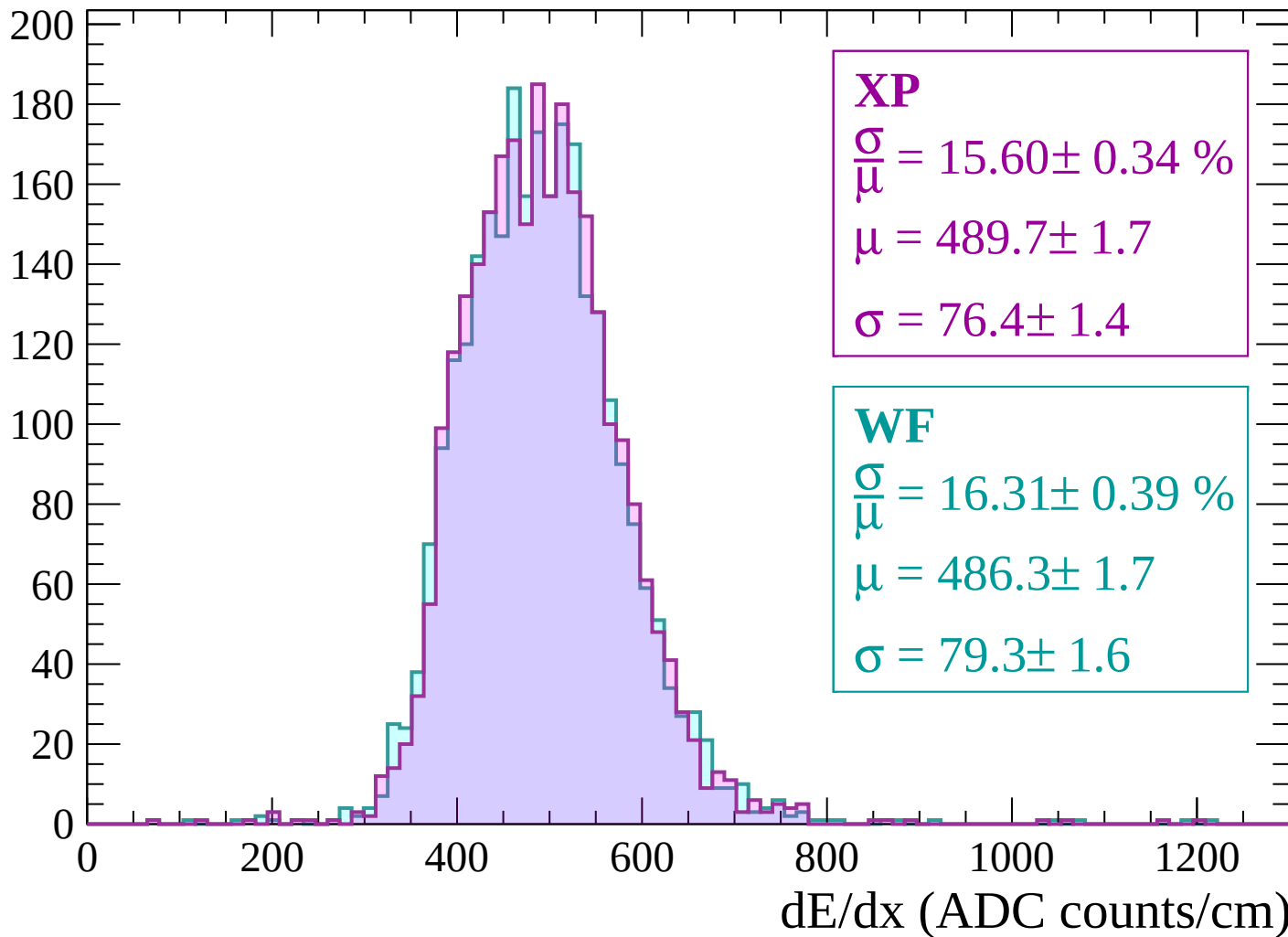
$$\mu = 486.0 \pm 1.7$$

$$\sigma = 78.4 \pm 1.4$$

dE/dx (ADC counts/cm)

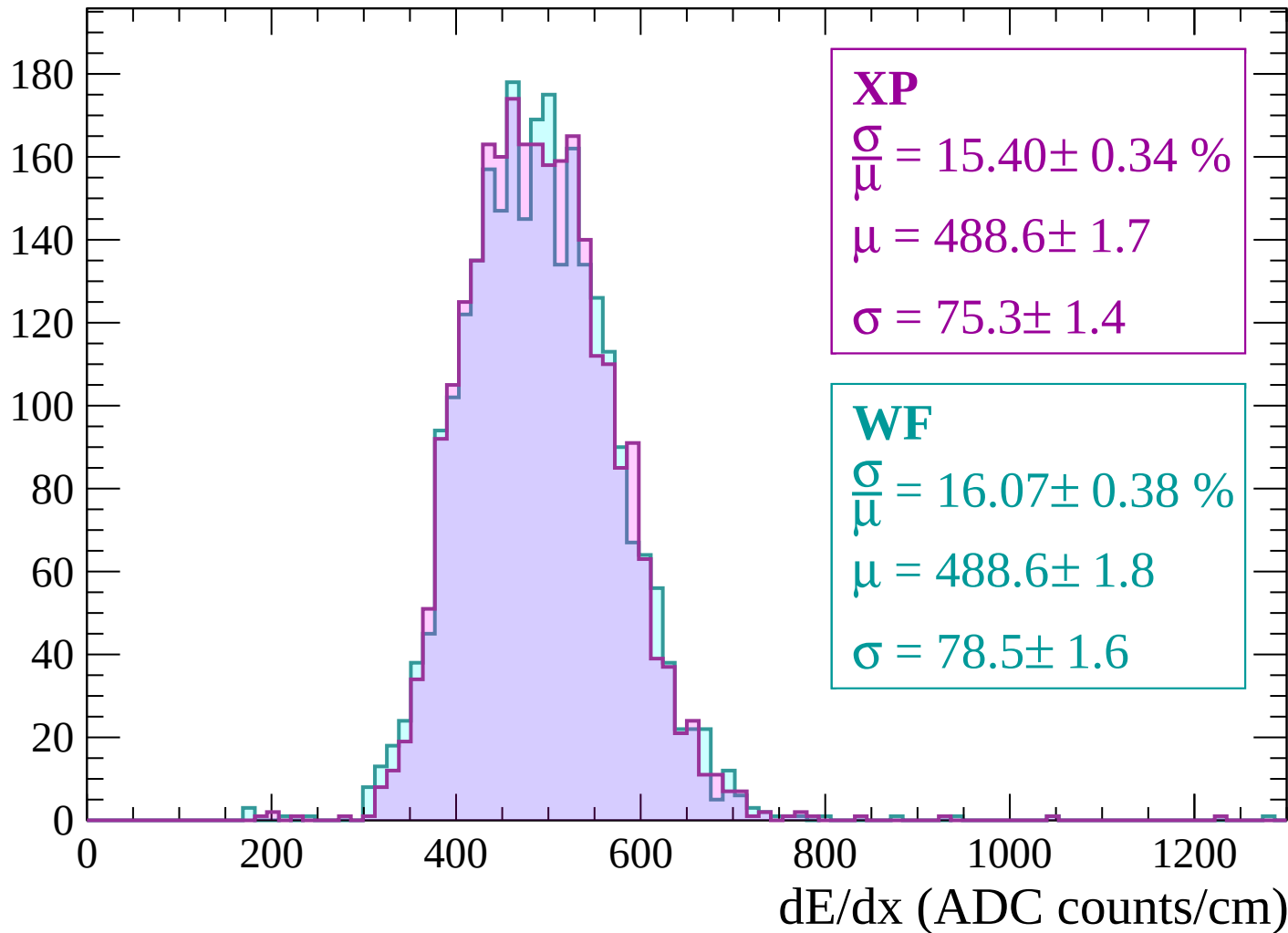
Energy loss | $2520 < p < 2580$

Count



Energy loss | $2580 < p < 2640$

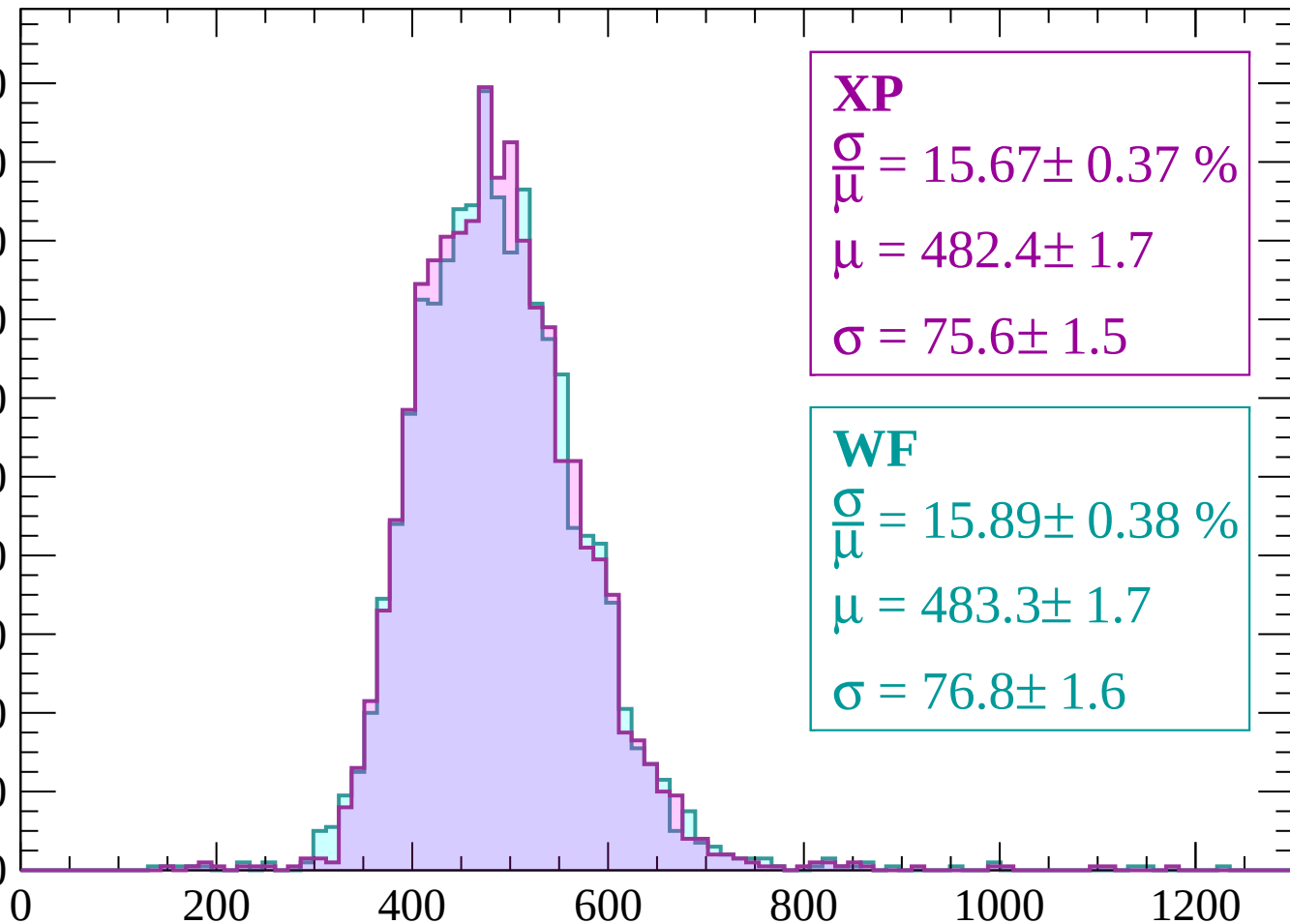
Count



Energy loss | $2640 < p < 2700$

Count

200
180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 15.67 \pm 0.37 \%$$

$$\mu = 482.4 \pm 1.7$$

$$\sigma = 75.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 15.89 \pm 0.38 \%$$

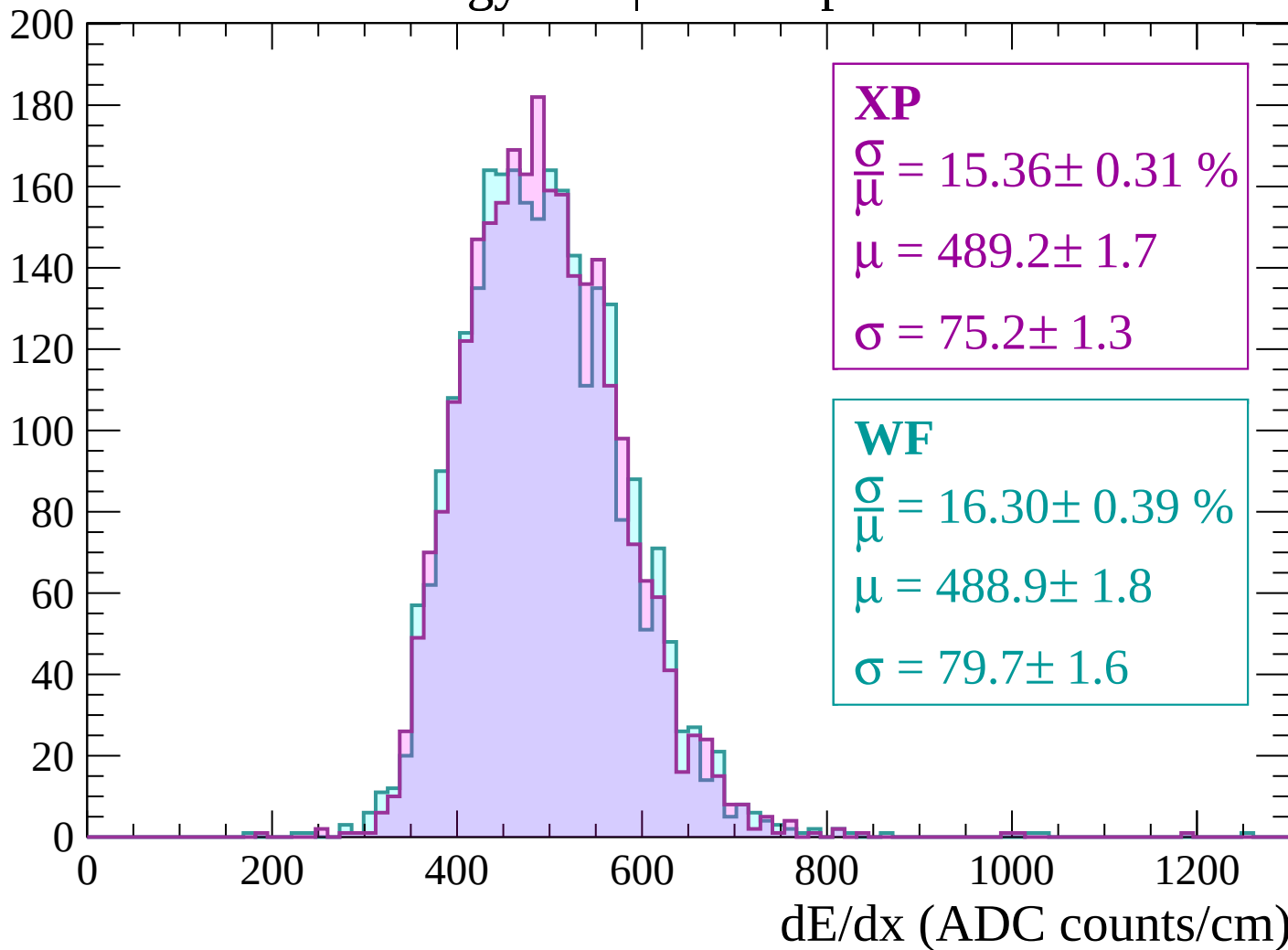
$$\mu = 483.3 \pm 1.7$$

$$\sigma = 76.8 \pm 1.6$$

dE/dx (ADC counts/cm)

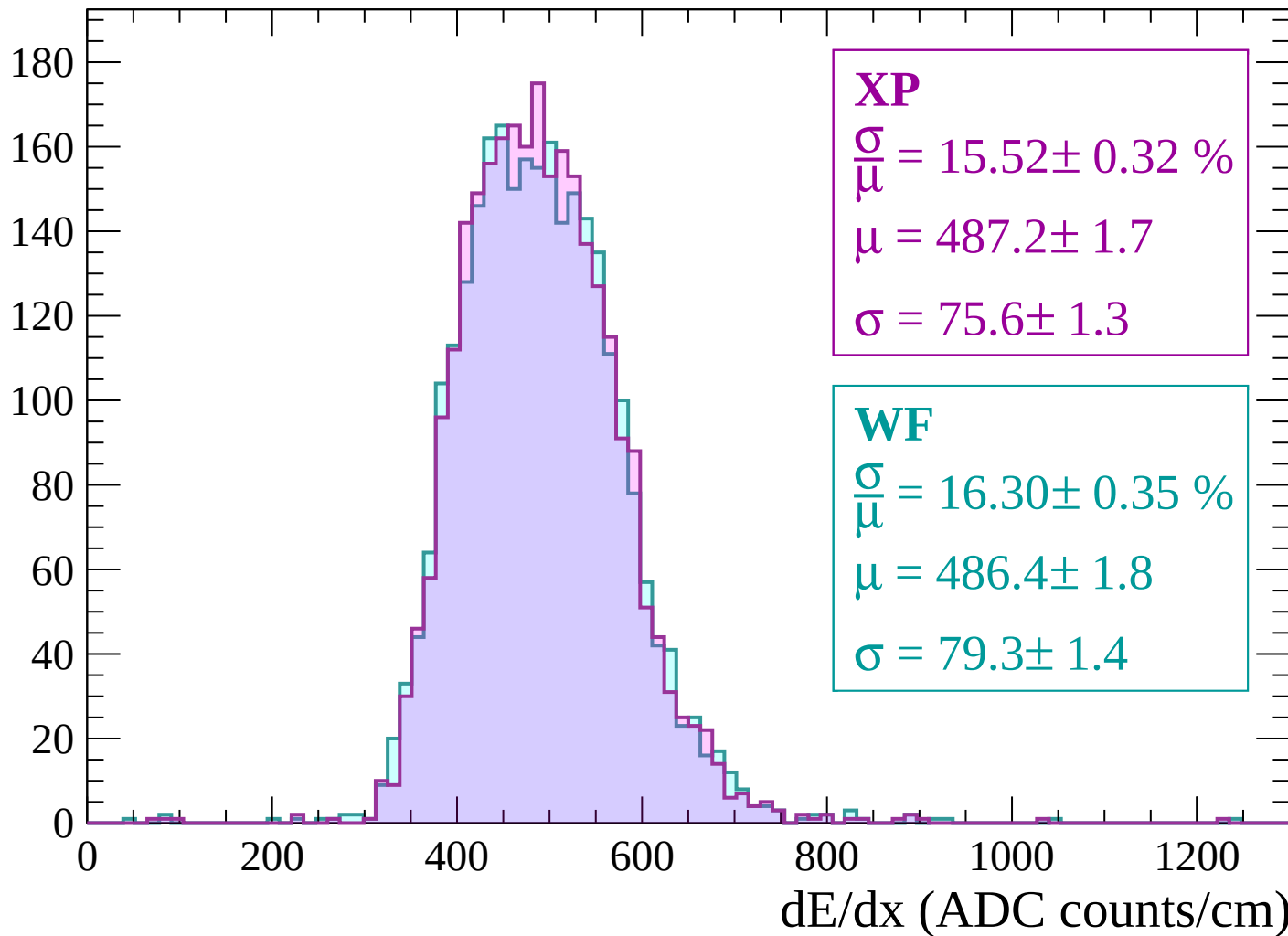
Energy loss | $2700 < p < 2760$

Count



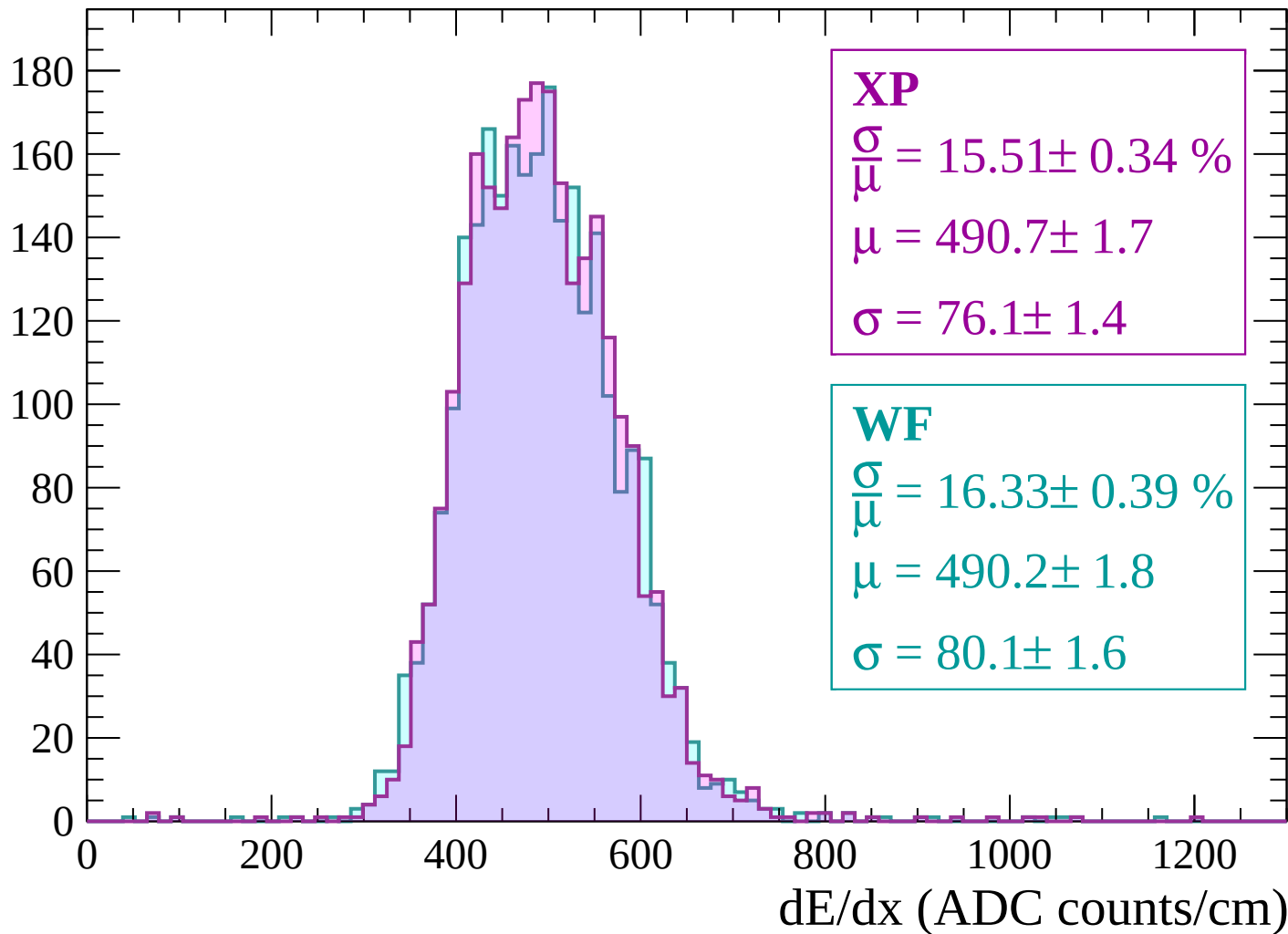
Energy loss | $2760 < p < 2820$

Count



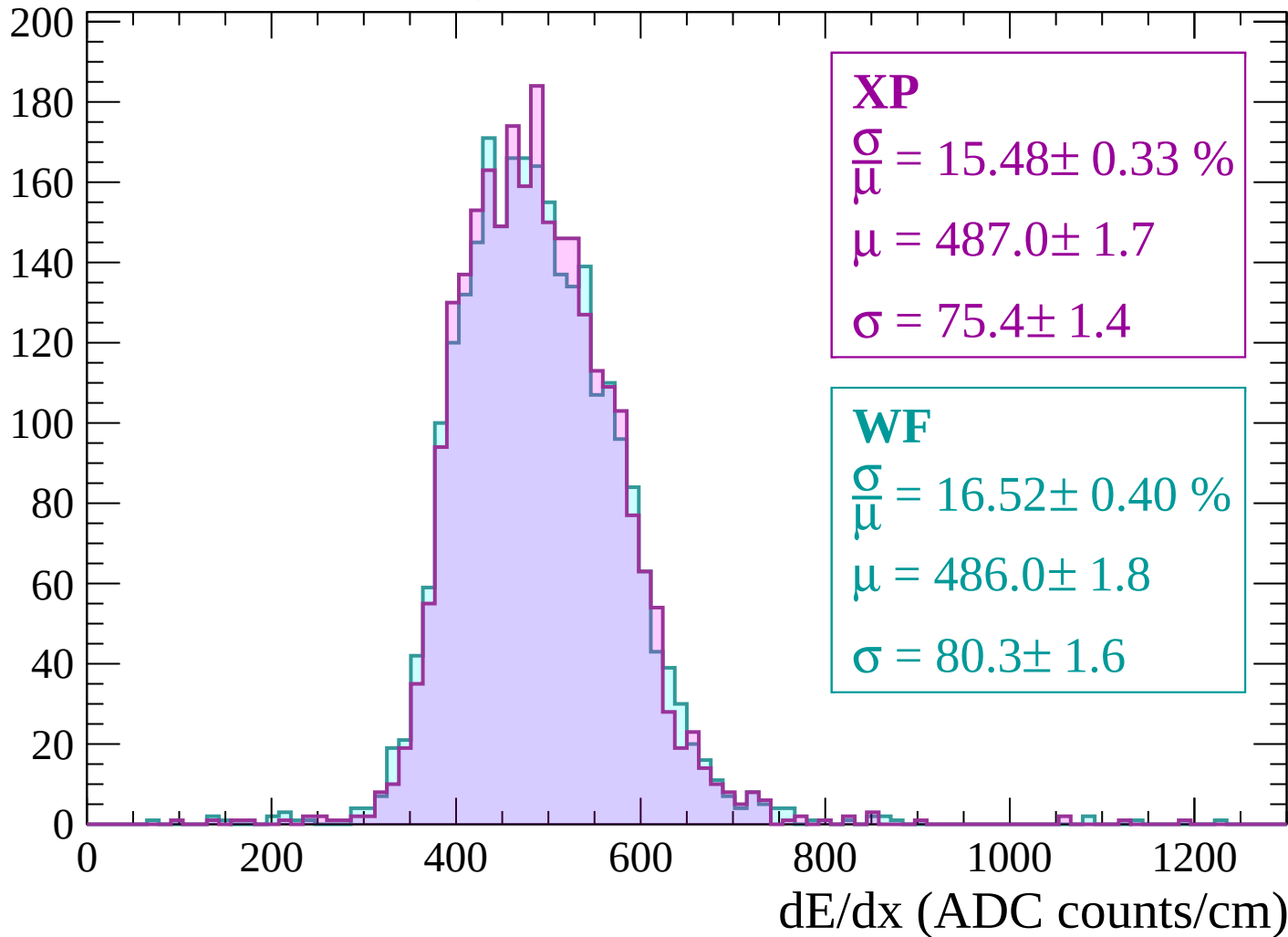
Energy loss | $2820 < p < 2880$

Count



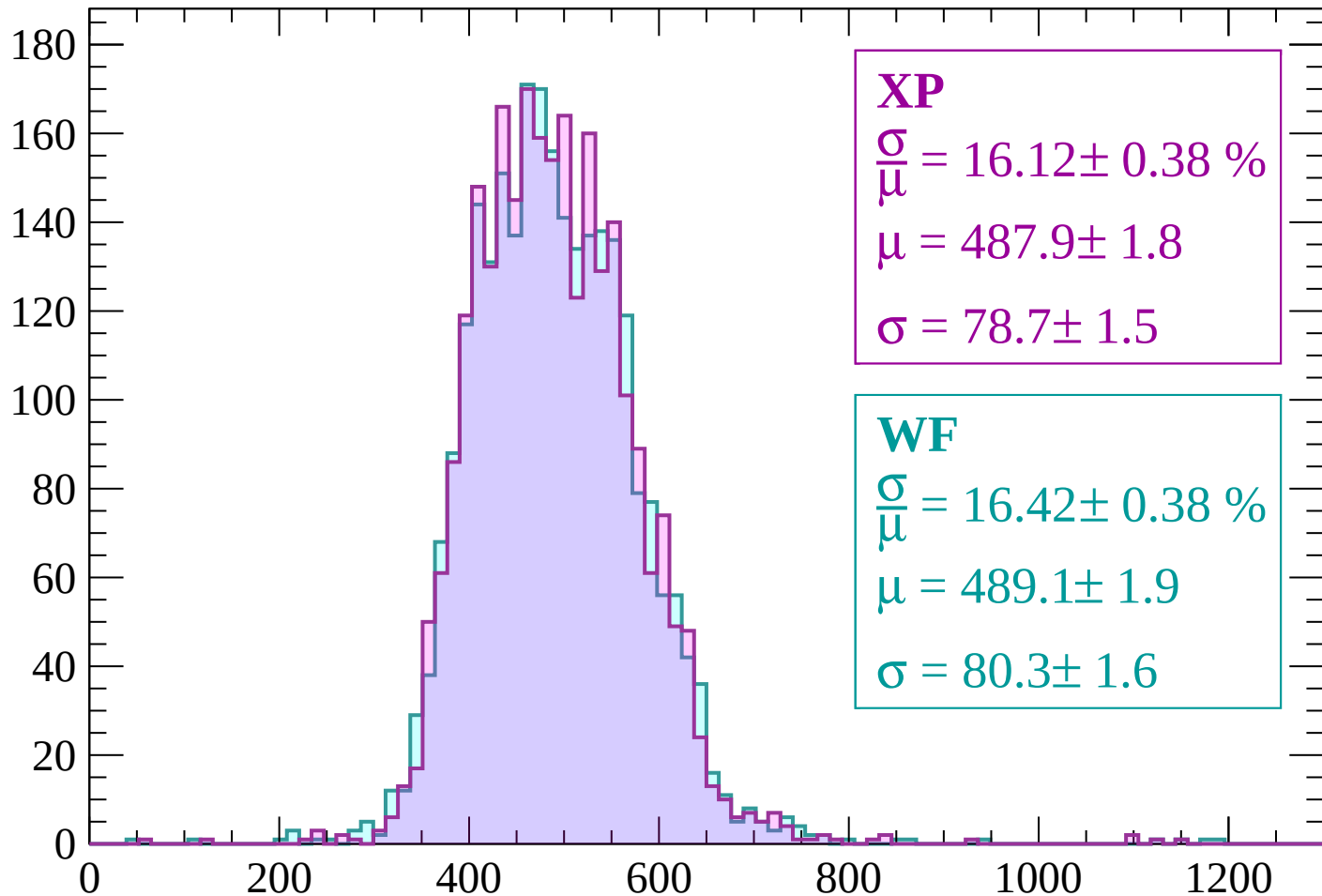
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



XP

$$\frac{\sigma}{\mu} = 16.12 \pm 0.38 \%$$

$$\mu = 487.9 \pm 1.8$$

$$\sigma = 78.7 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 16.42 \pm 0.38 \%$$

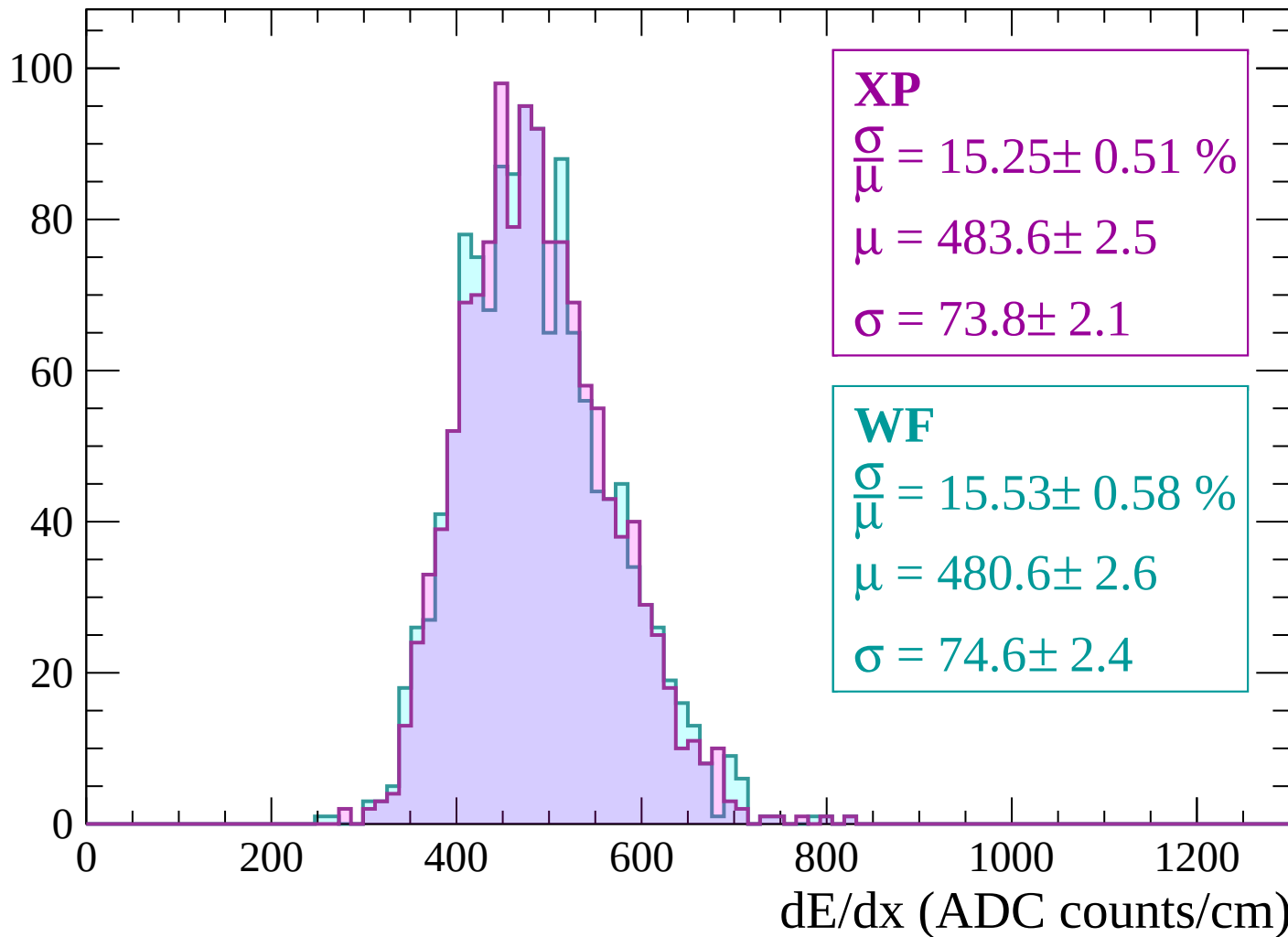
$$\mu = 489.1 \pm 1.9$$

$$\sigma = 80.3 \pm 1.6$$

dE/dx (ADC counts/cm)

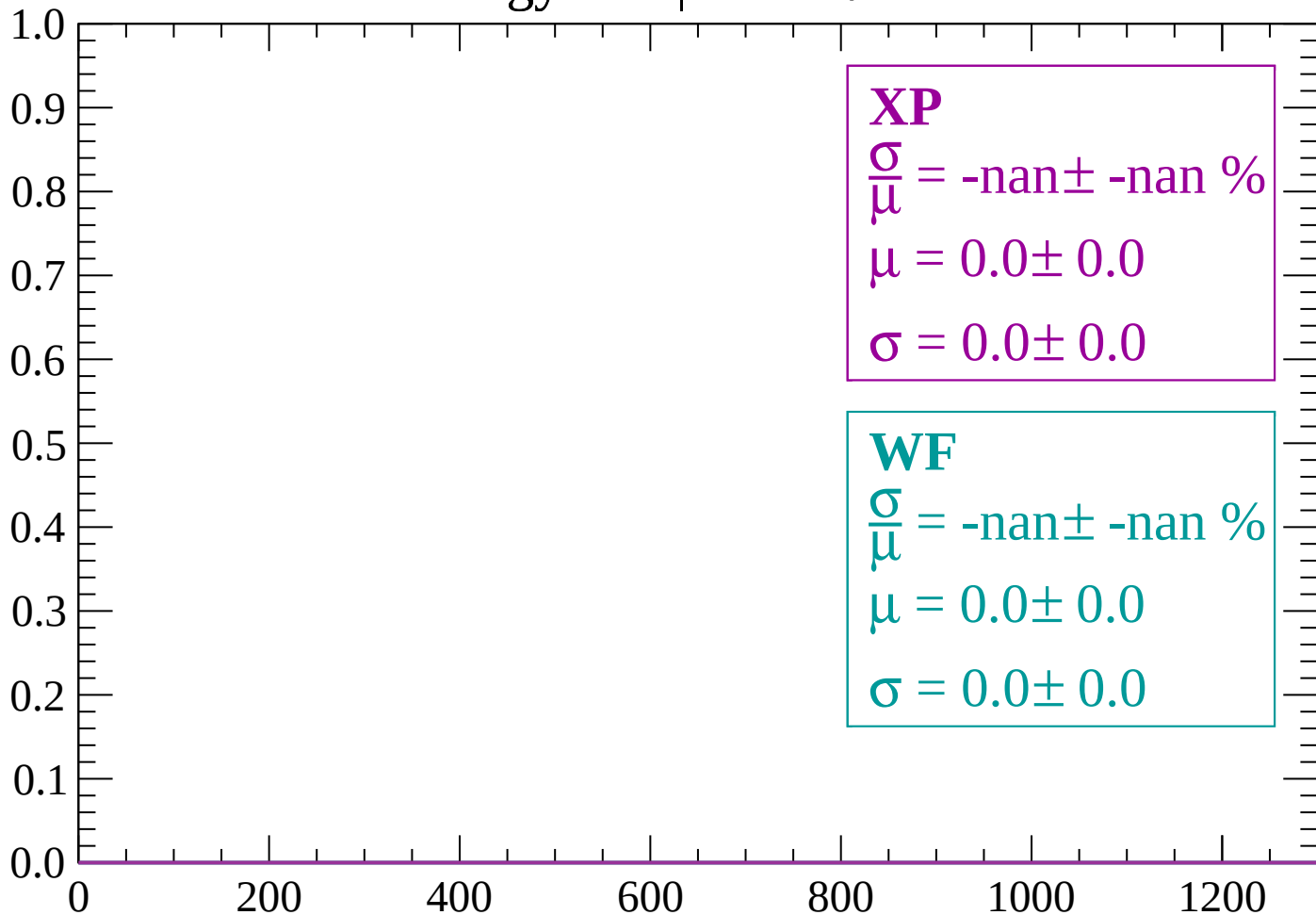
Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

WF

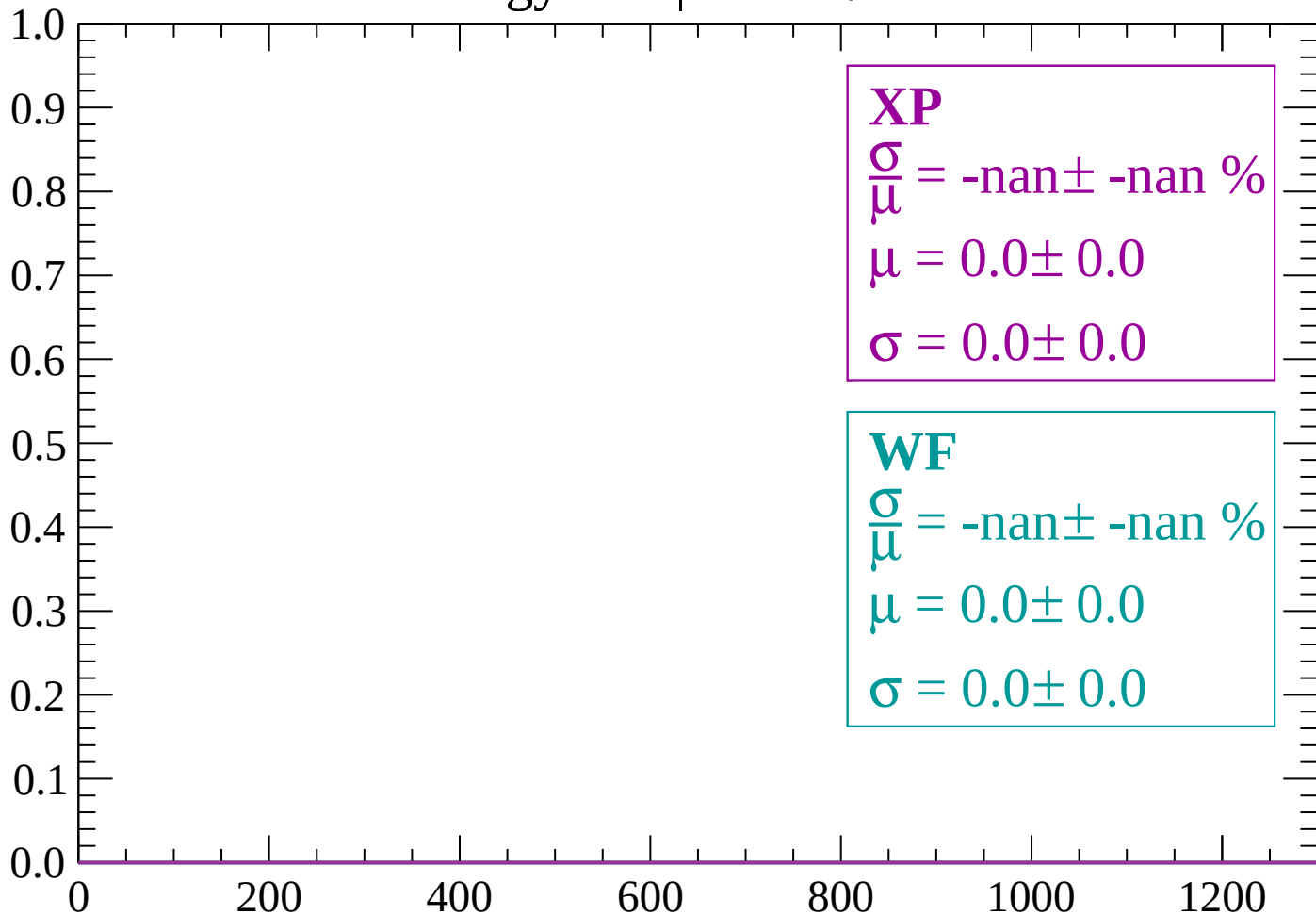
$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

Energy loss | $-86 < \theta < -84$

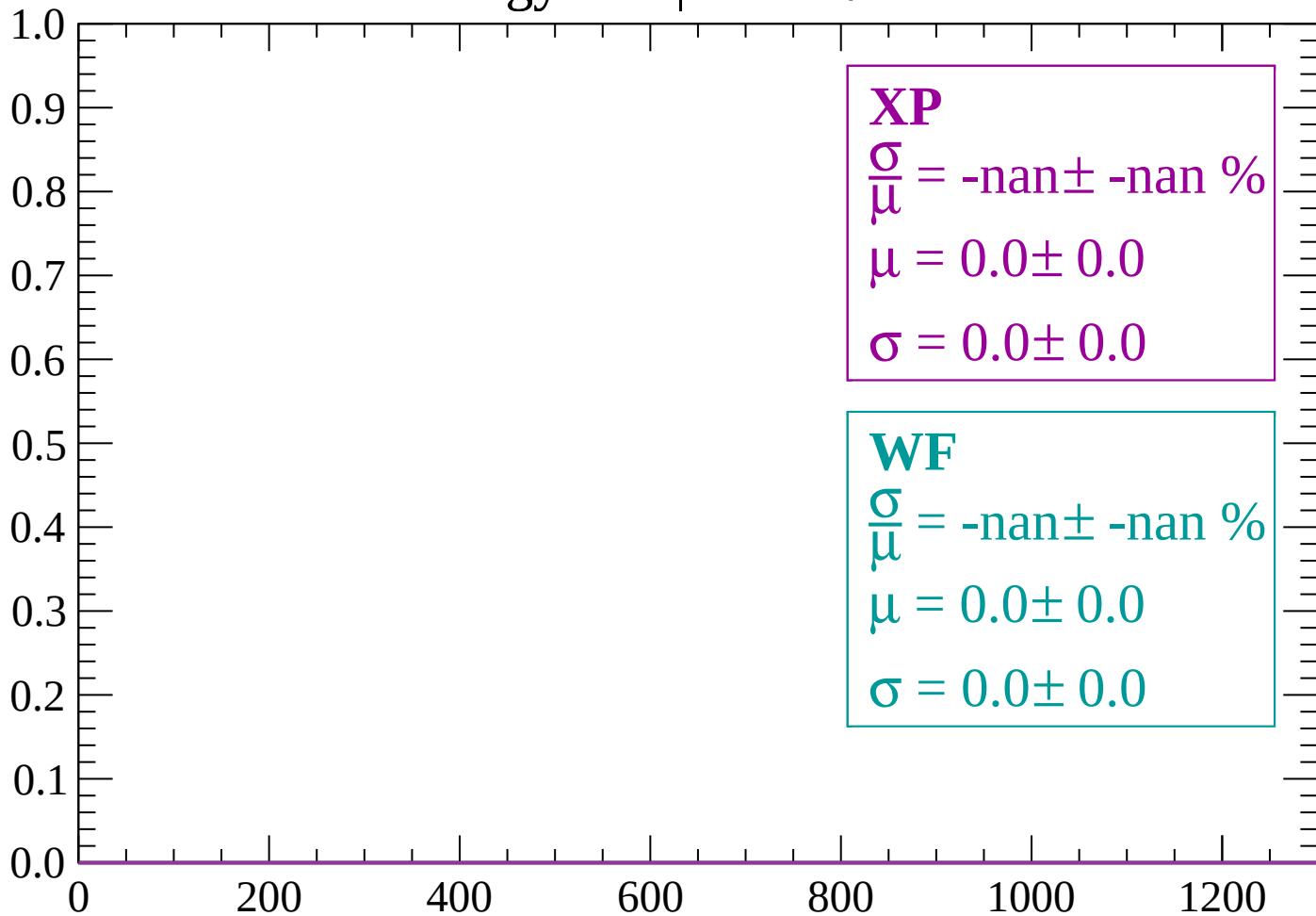
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

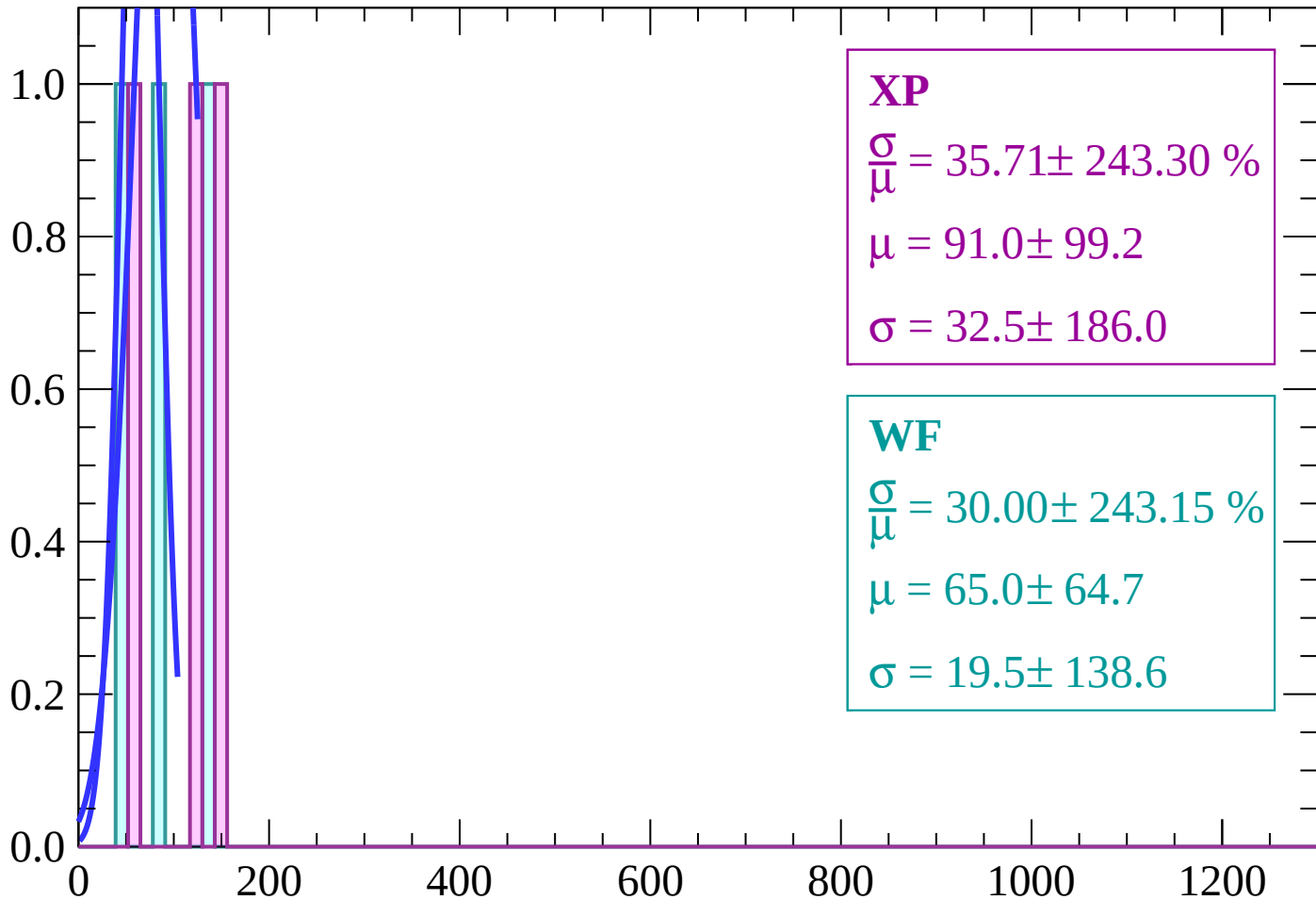
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$

Count



XP

$$\frac{\sigma}{\mu} = 35.71 \pm 243.30 \%$$

$$\mu = 91.0 \pm 99.2$$

$$\sigma = 32.5 \pm 186.0$$

WF

$$\frac{\sigma}{\mu} = 30.00 \pm 243.15 \%$$

$$\mu = 65.0 \pm 64.7$$

$$\sigma = 19.5 \pm 138.6$$

dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

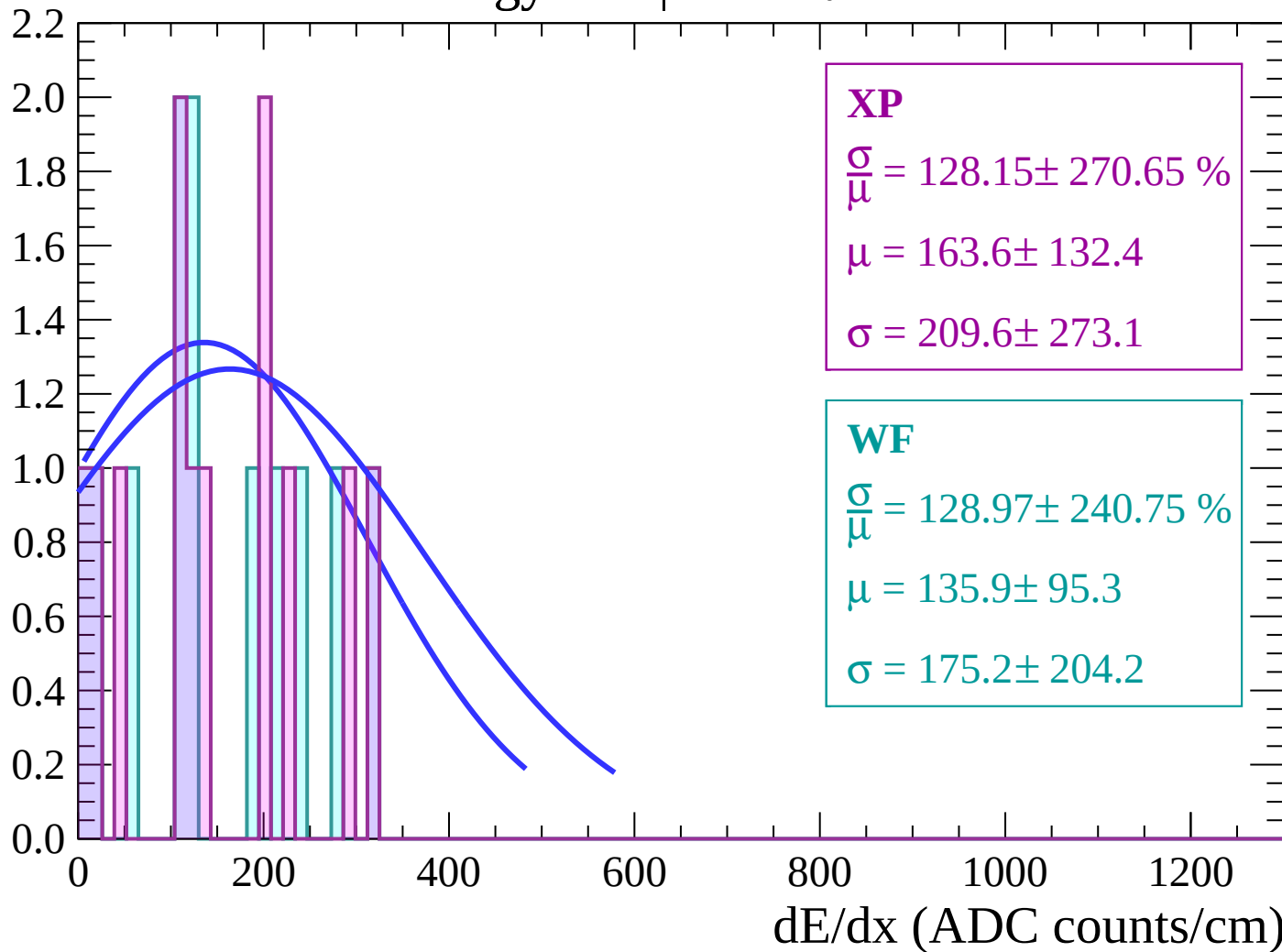
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

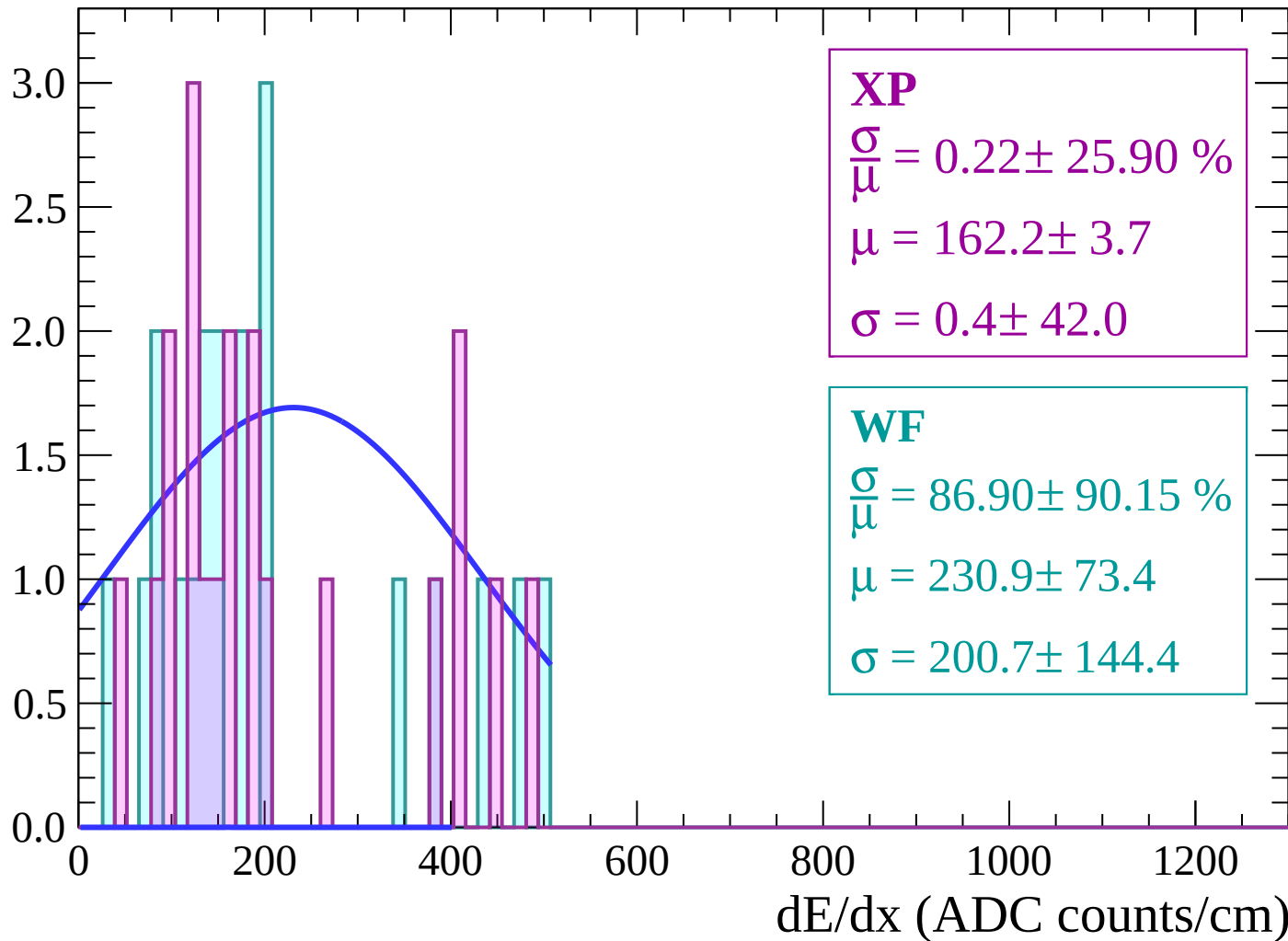
Energy loss | $-76 < \theta < -74$

Count



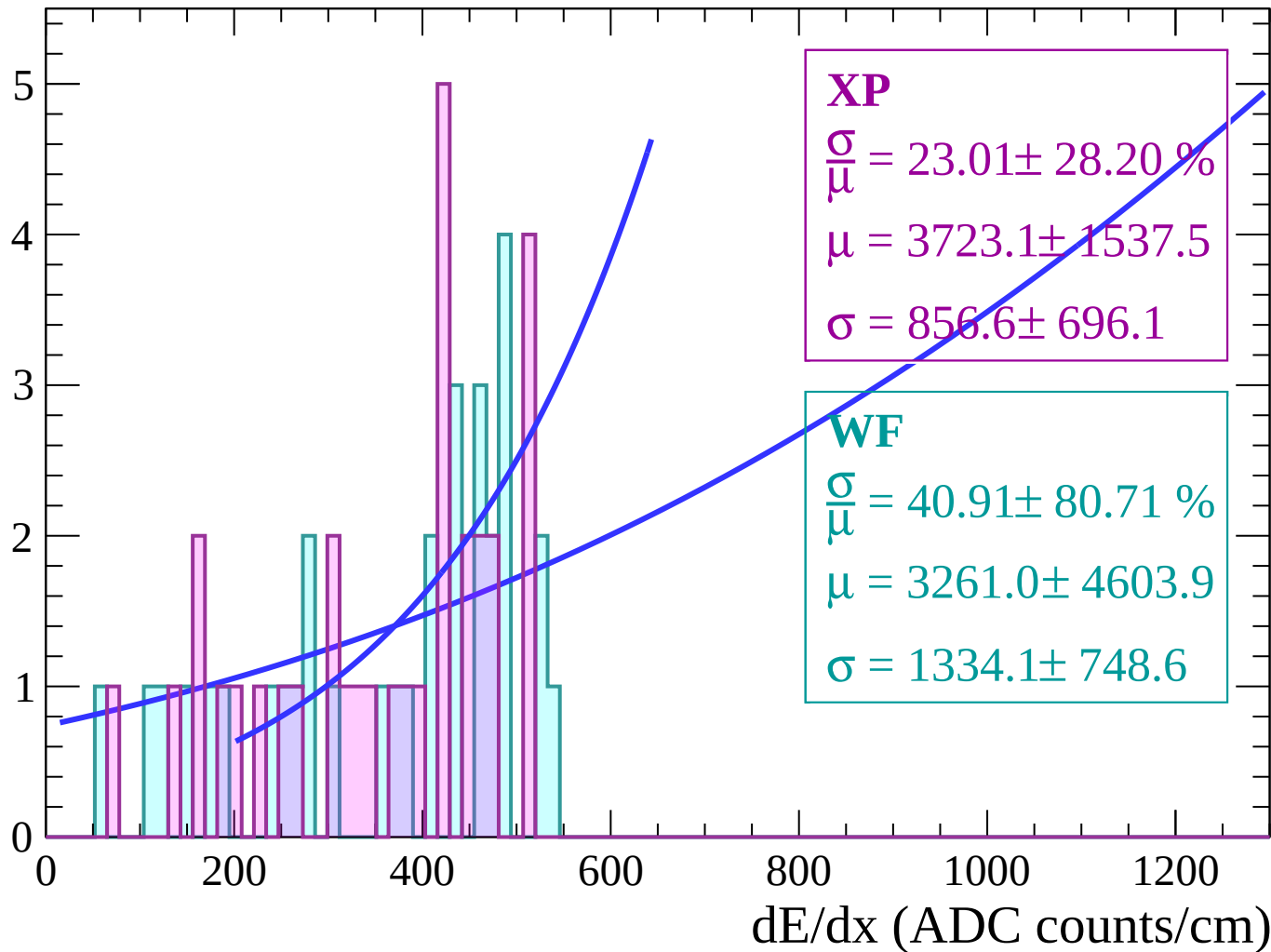
Energy loss | $-74 < \theta < -72$

Count



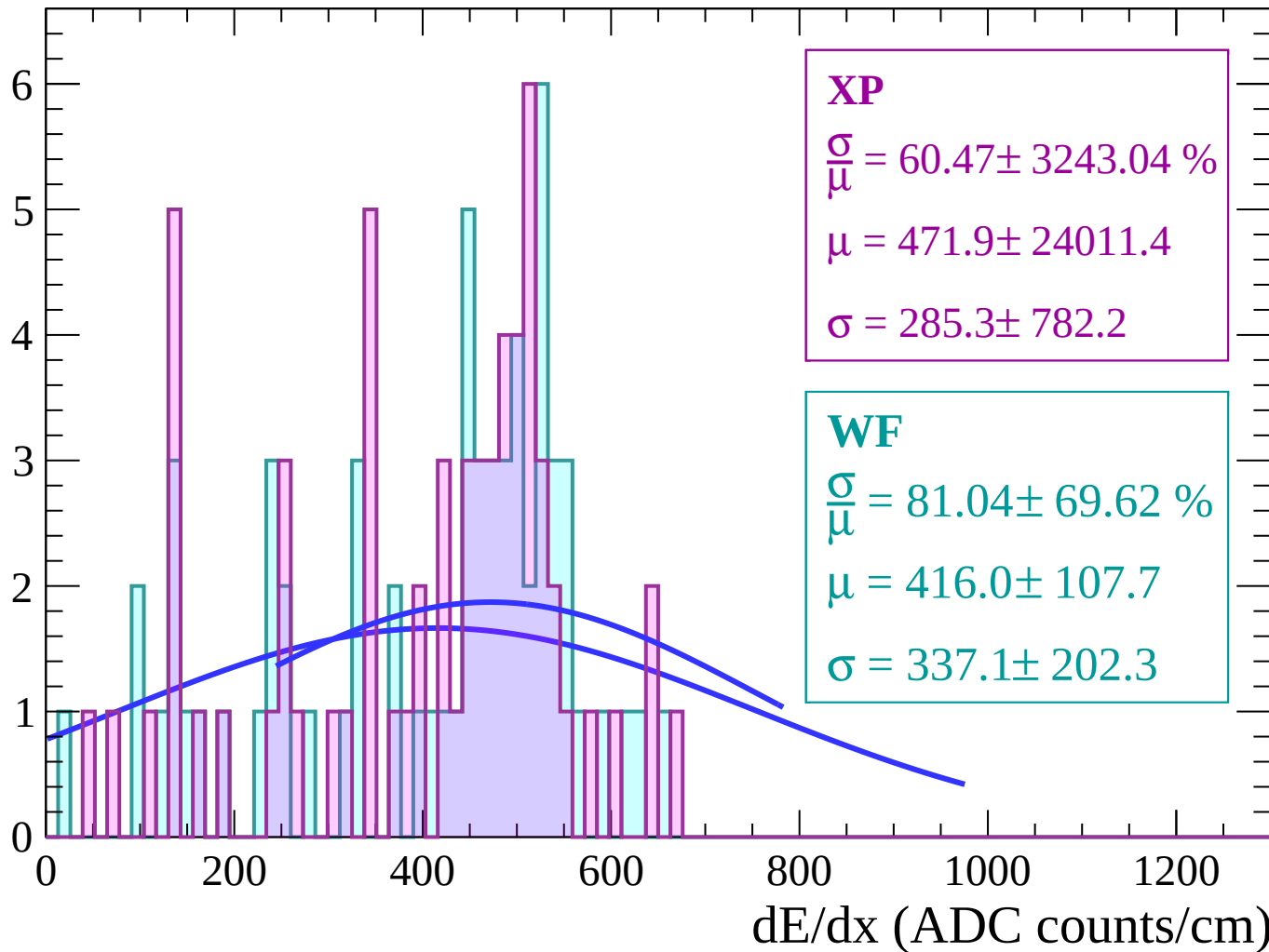
Energy loss | $-72 < \theta < -70$

Count



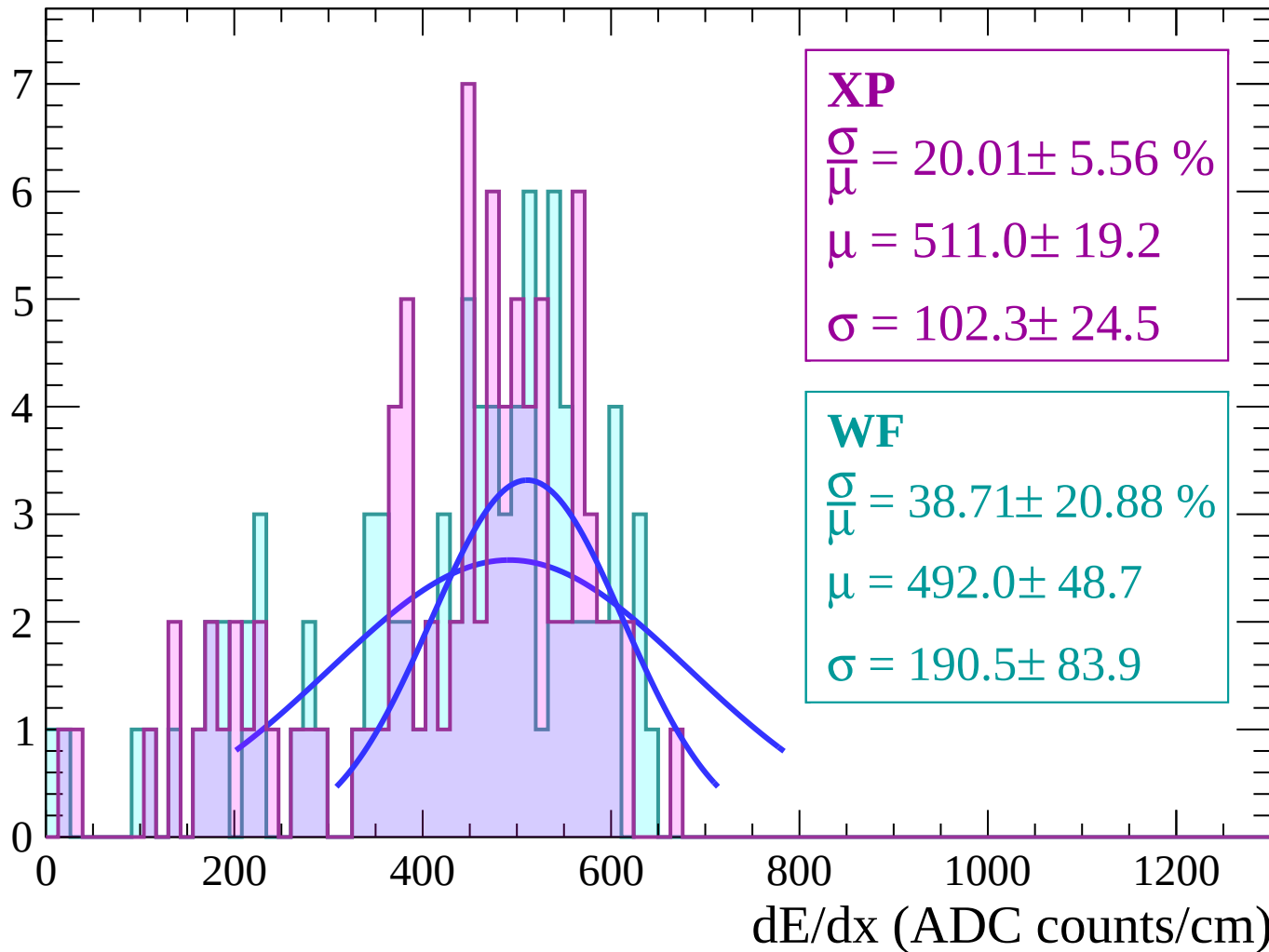
Energy loss | $-70 < \theta < -68$

Count



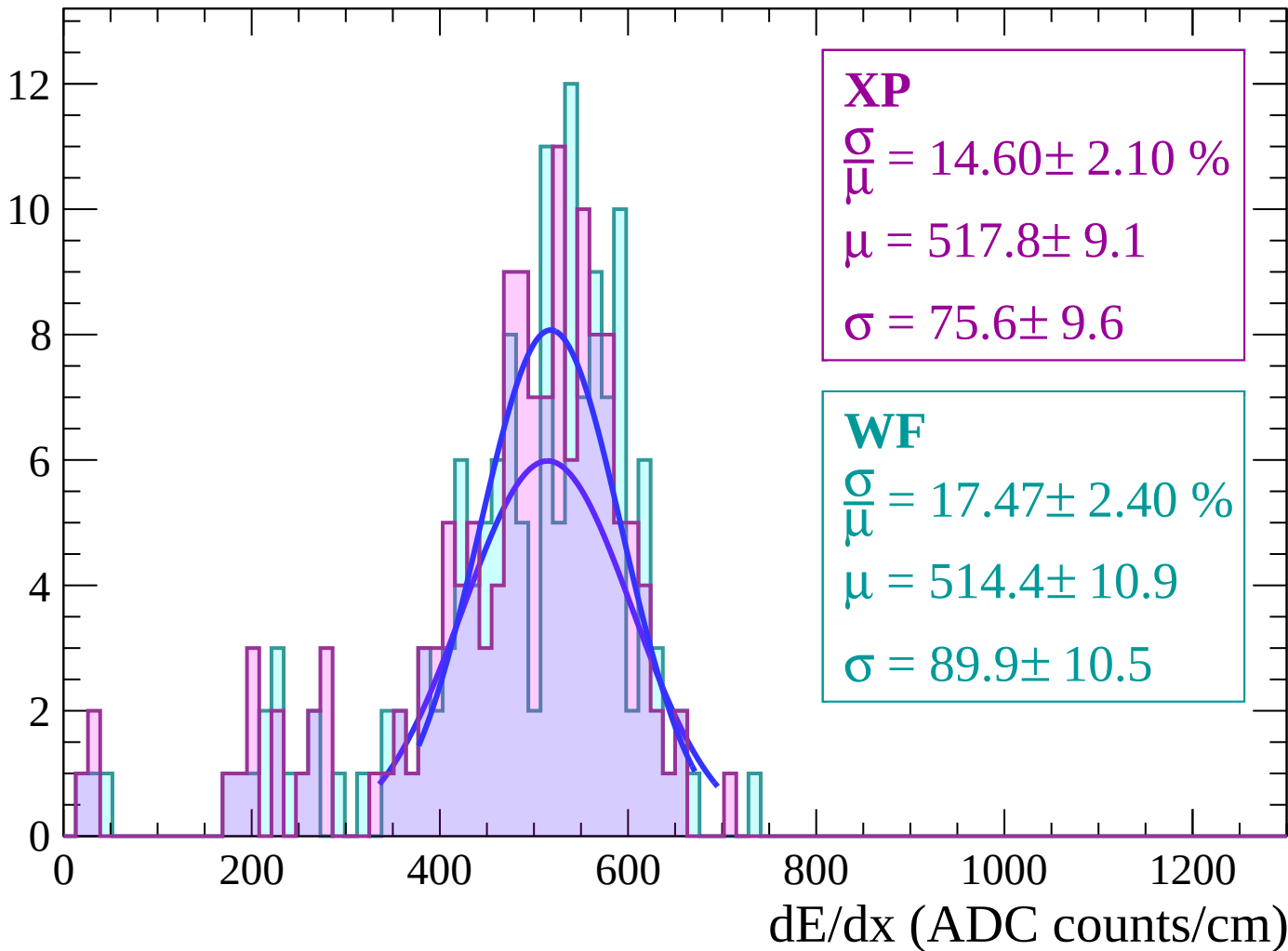
Energy loss | $-68 < \theta < -66$

Count



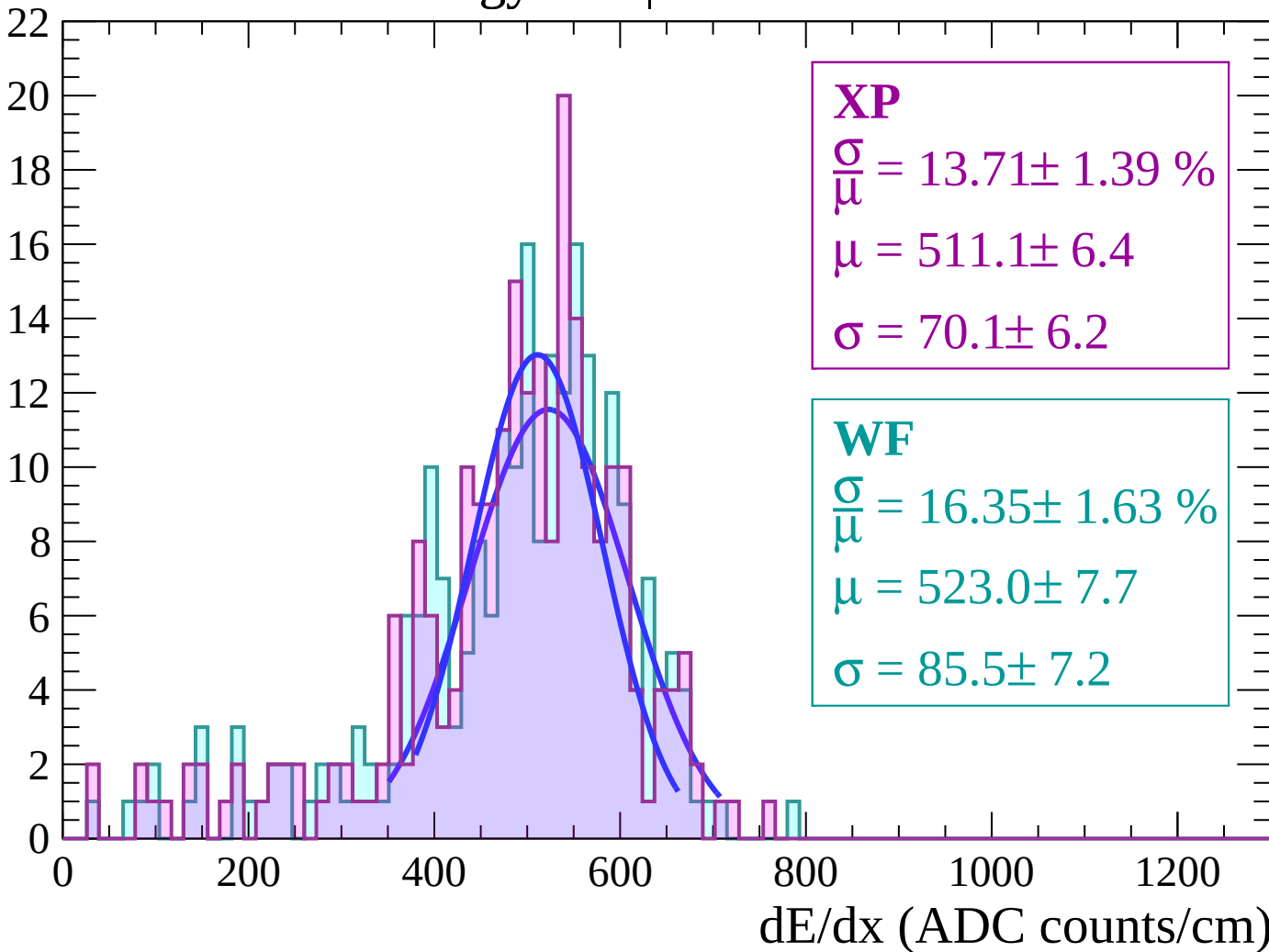
Energy loss | $-66 < \theta < -64$

Count



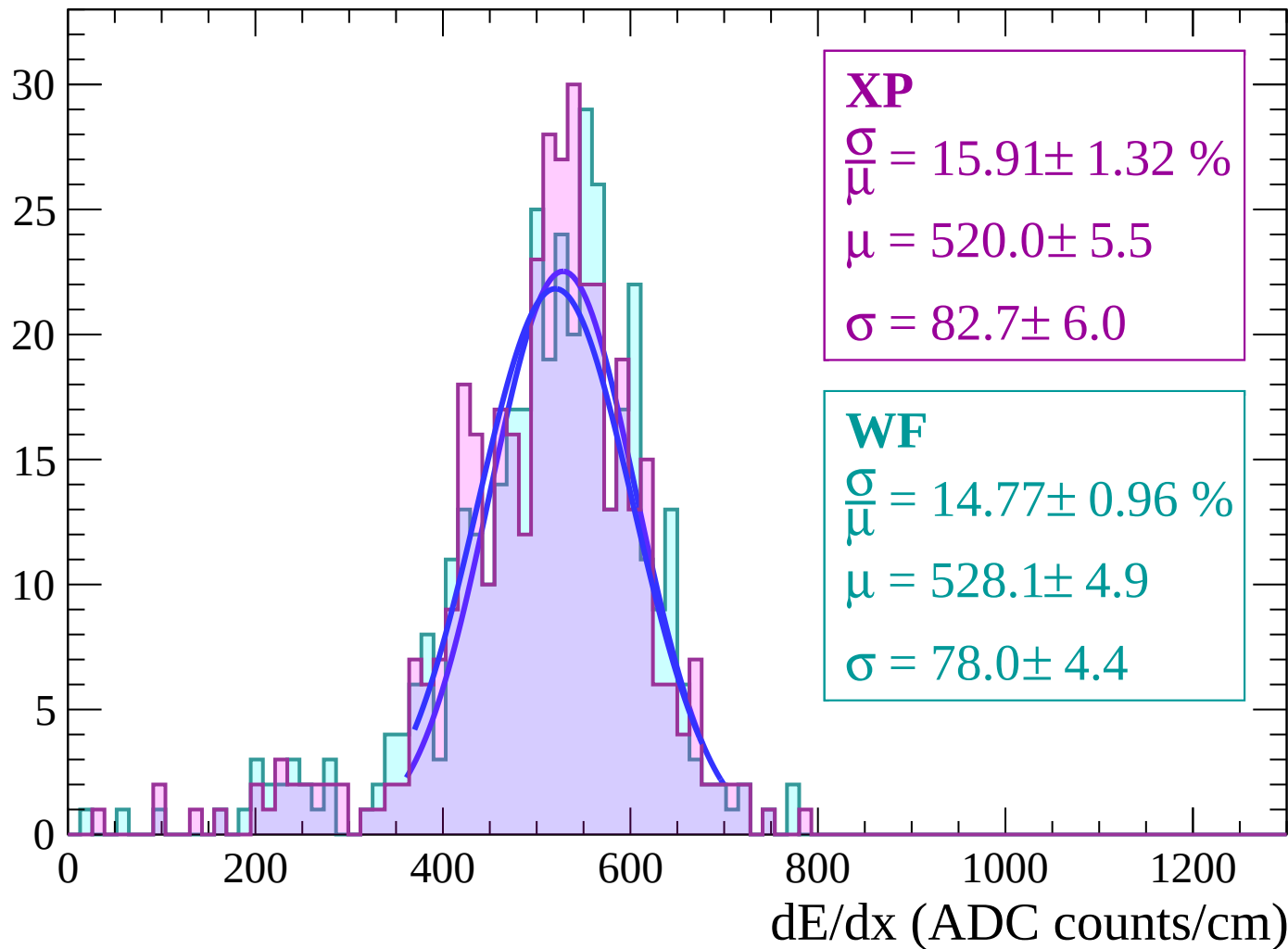
Energy loss | $-64 < \theta < -62$

Count



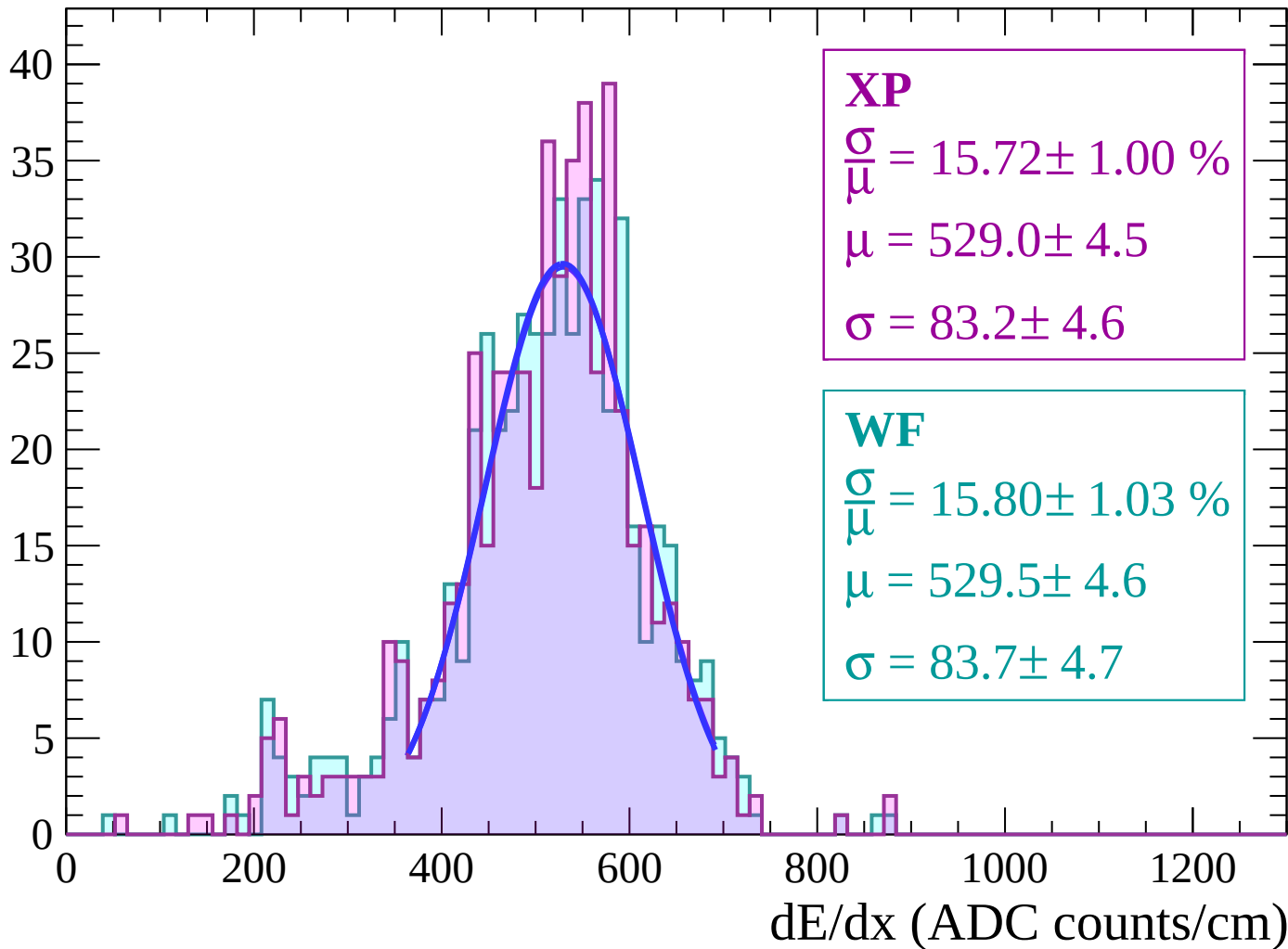
Energy loss | $-62 < \theta < -60$

Count



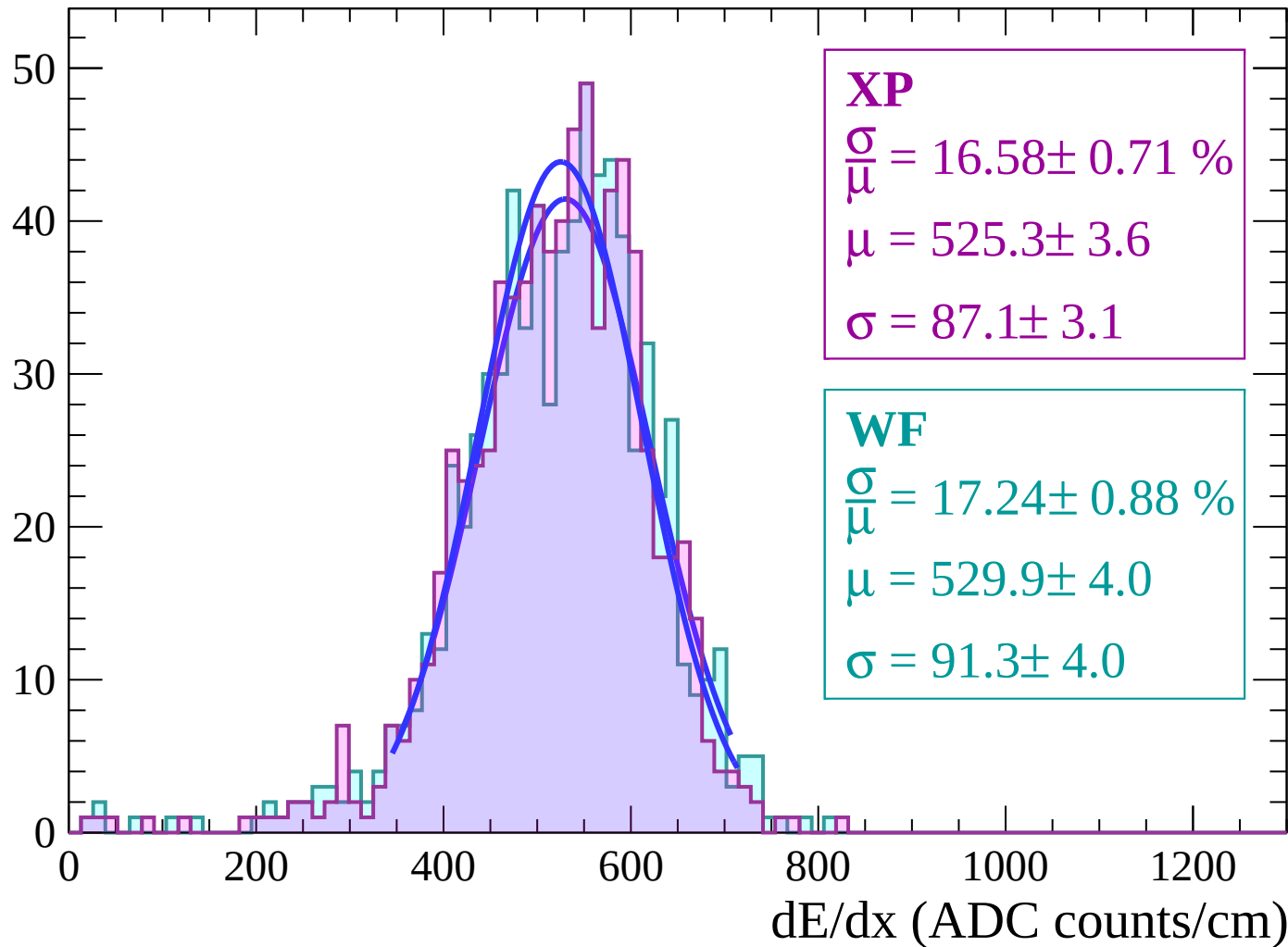
Energy loss | $-60 < \theta < -58$

Count



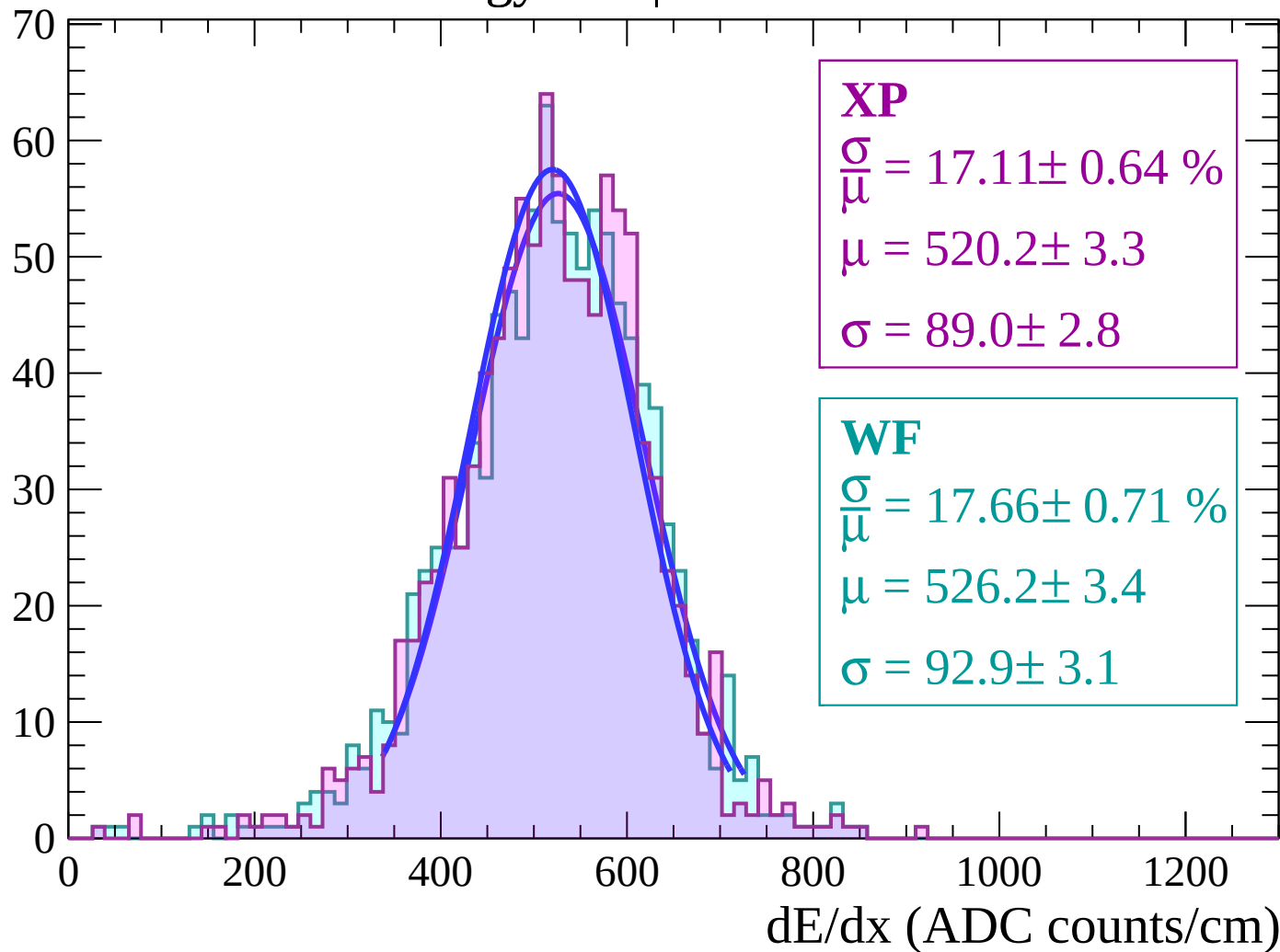
Energy loss | $-58 < \theta < -56$

Count



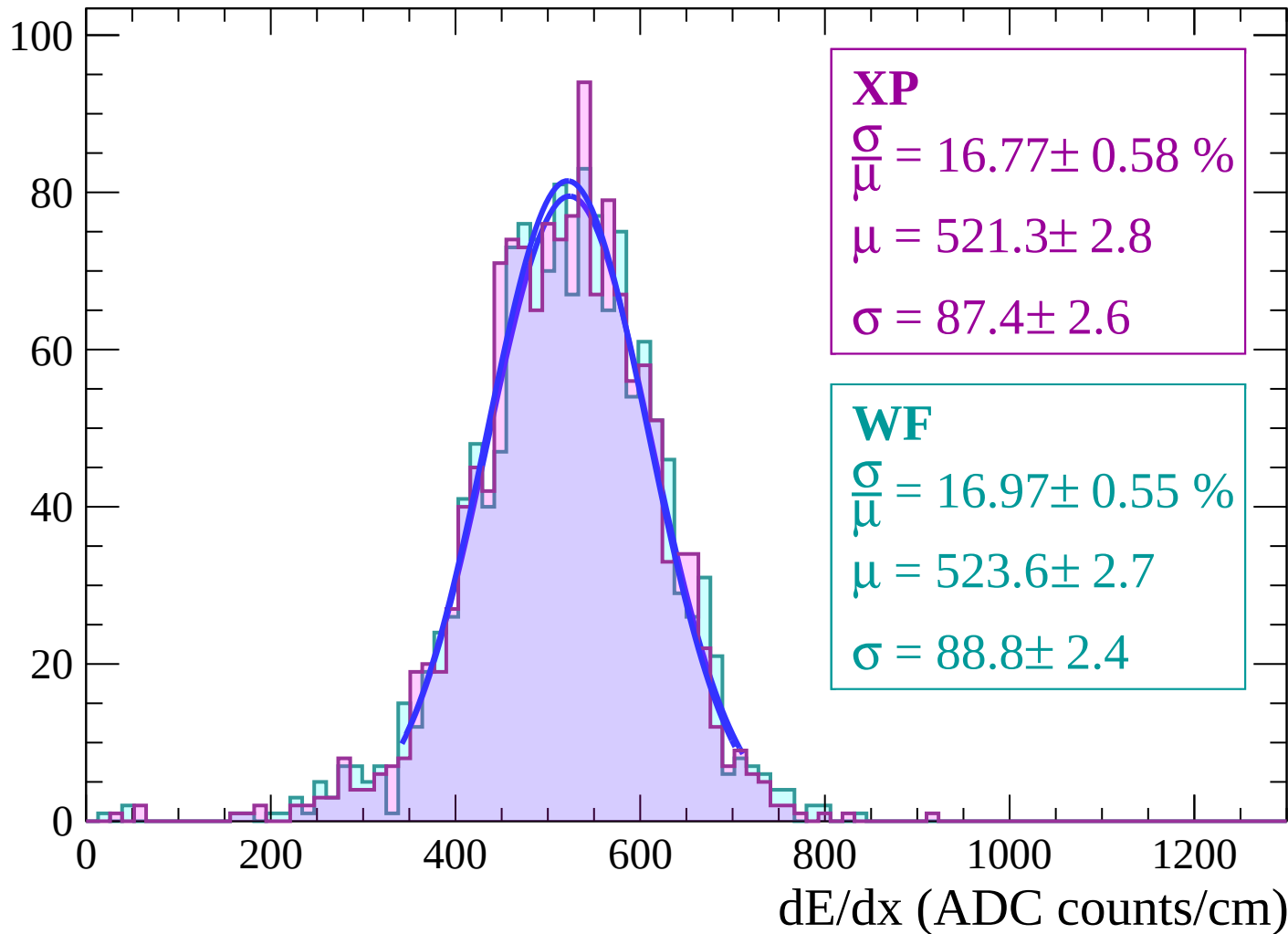
Energy loss | $-56 < \theta < -54$

Count



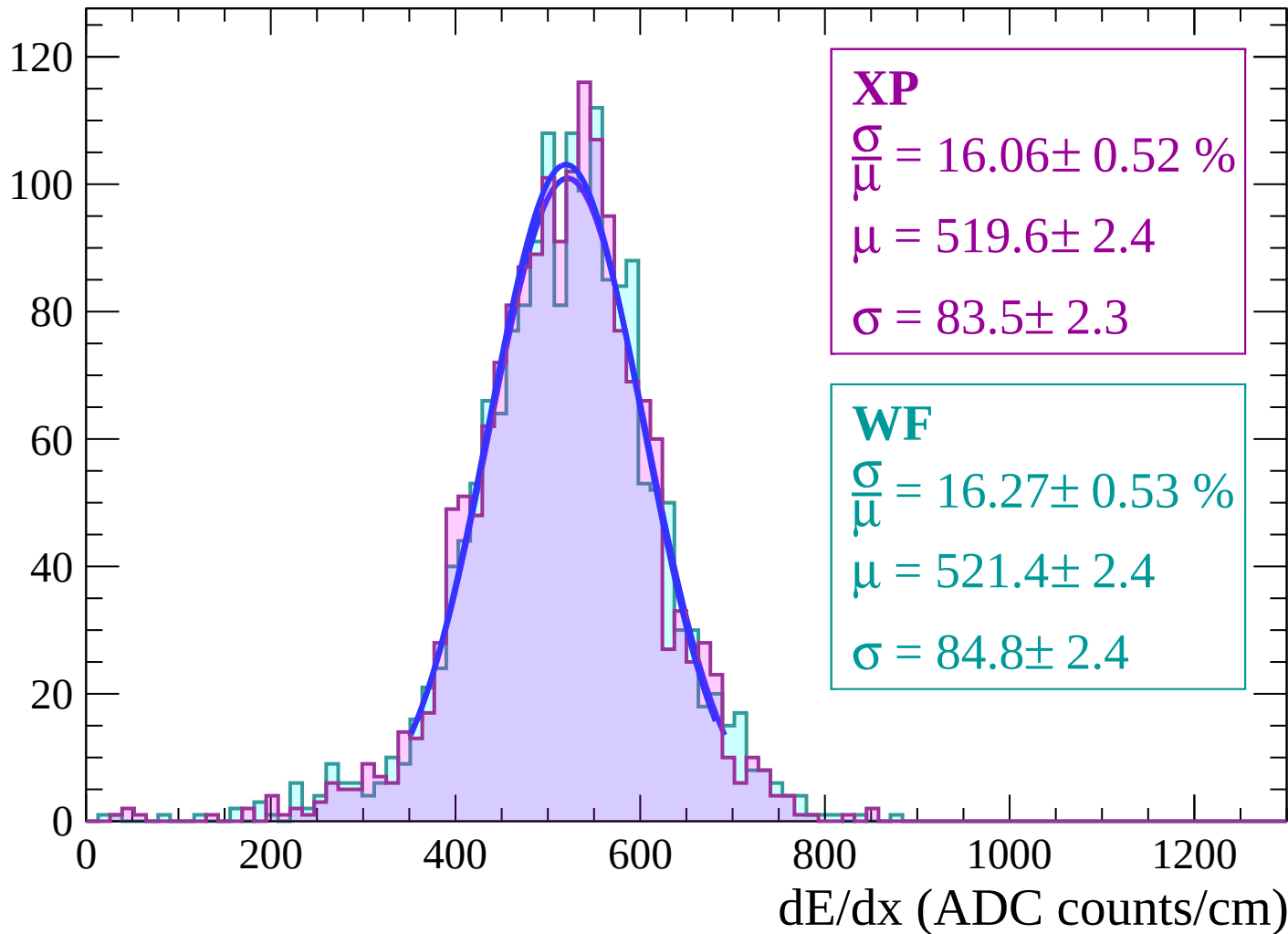
Energy loss | $-54 < \theta < -52$

Count



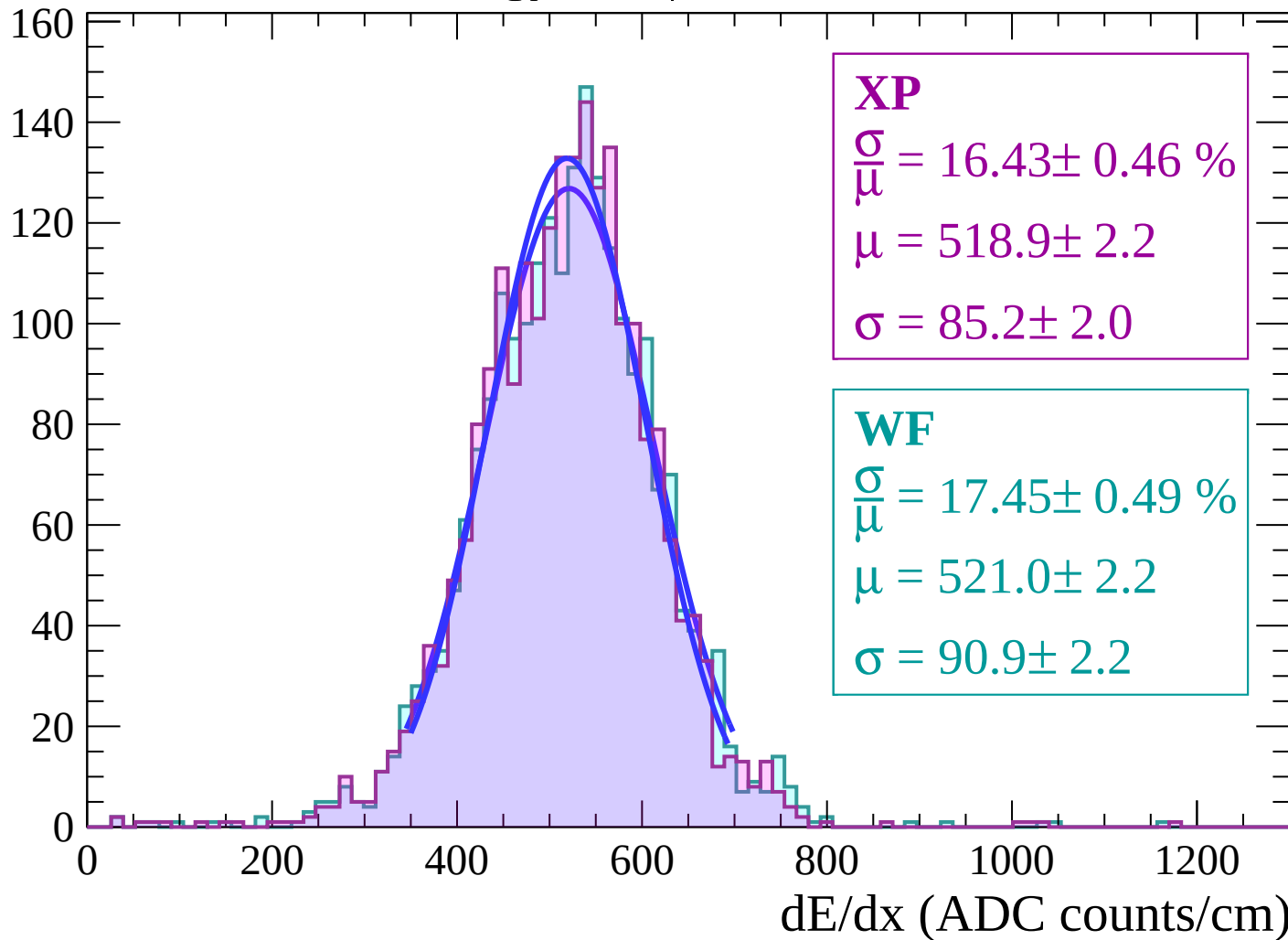
Energy loss | $-52 < \theta < -50$

Count



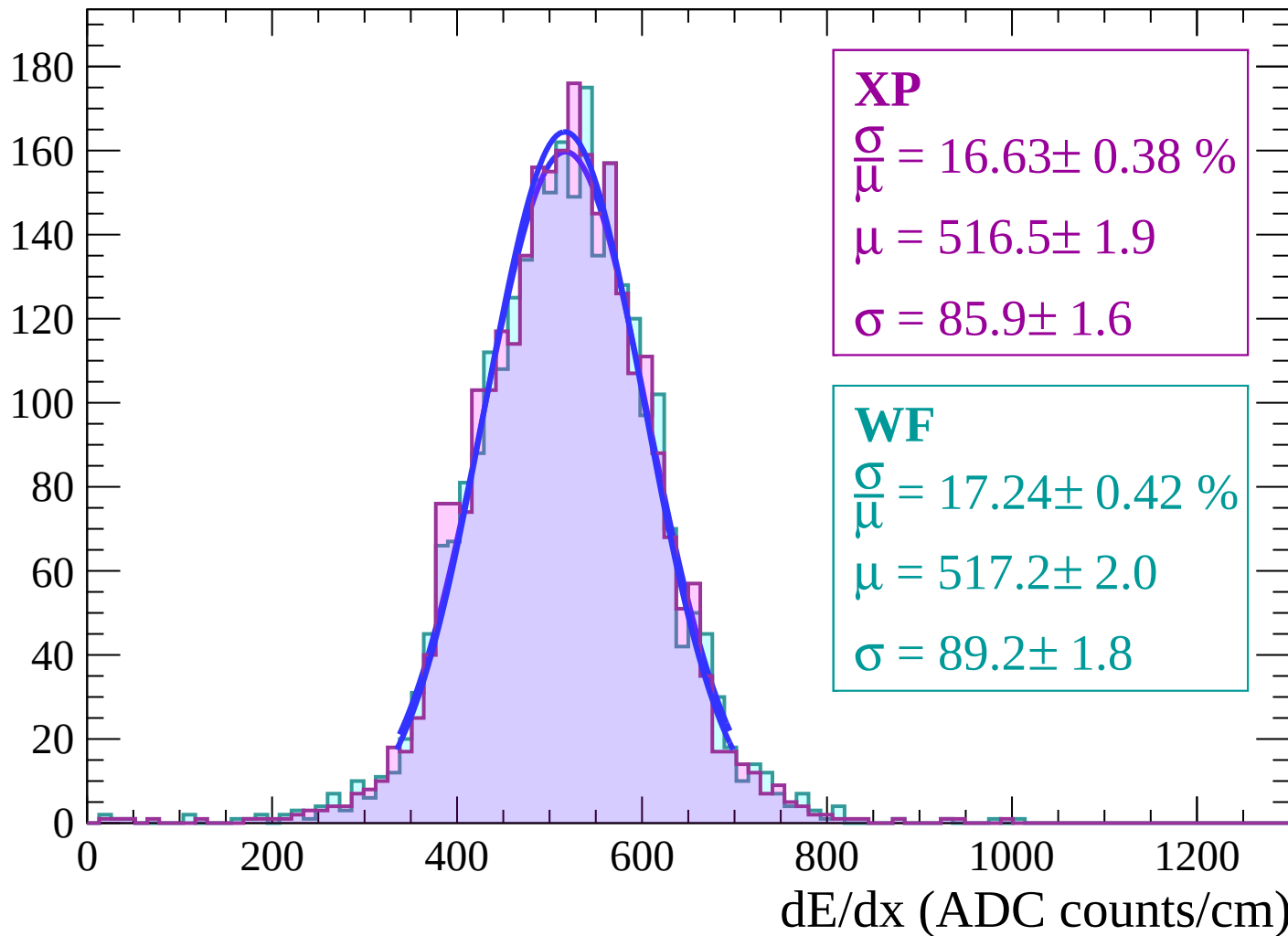
Energy loss | $-50 < \theta < -48$

Count



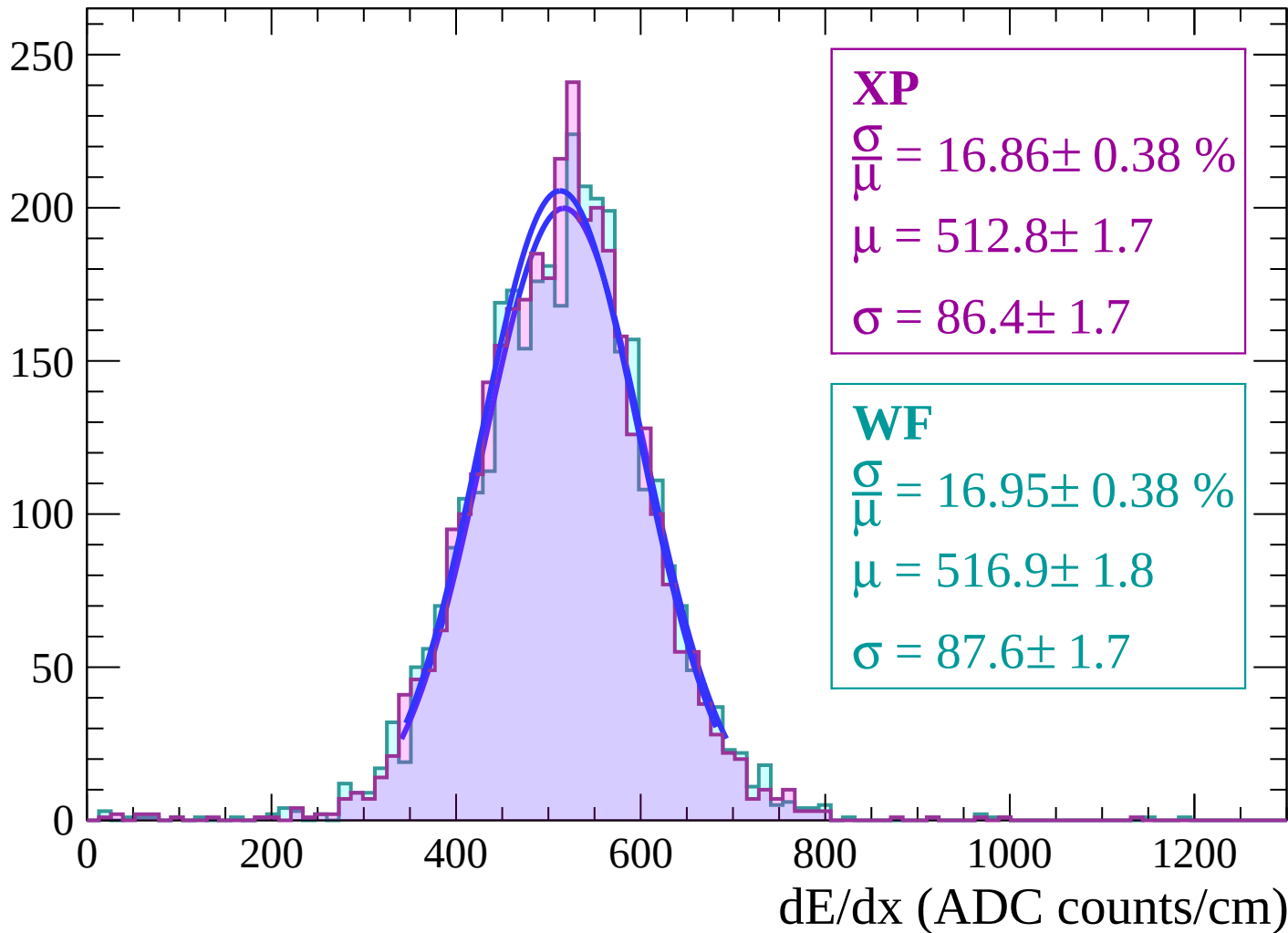
Energy loss | $-48 < \theta < -46$

Count



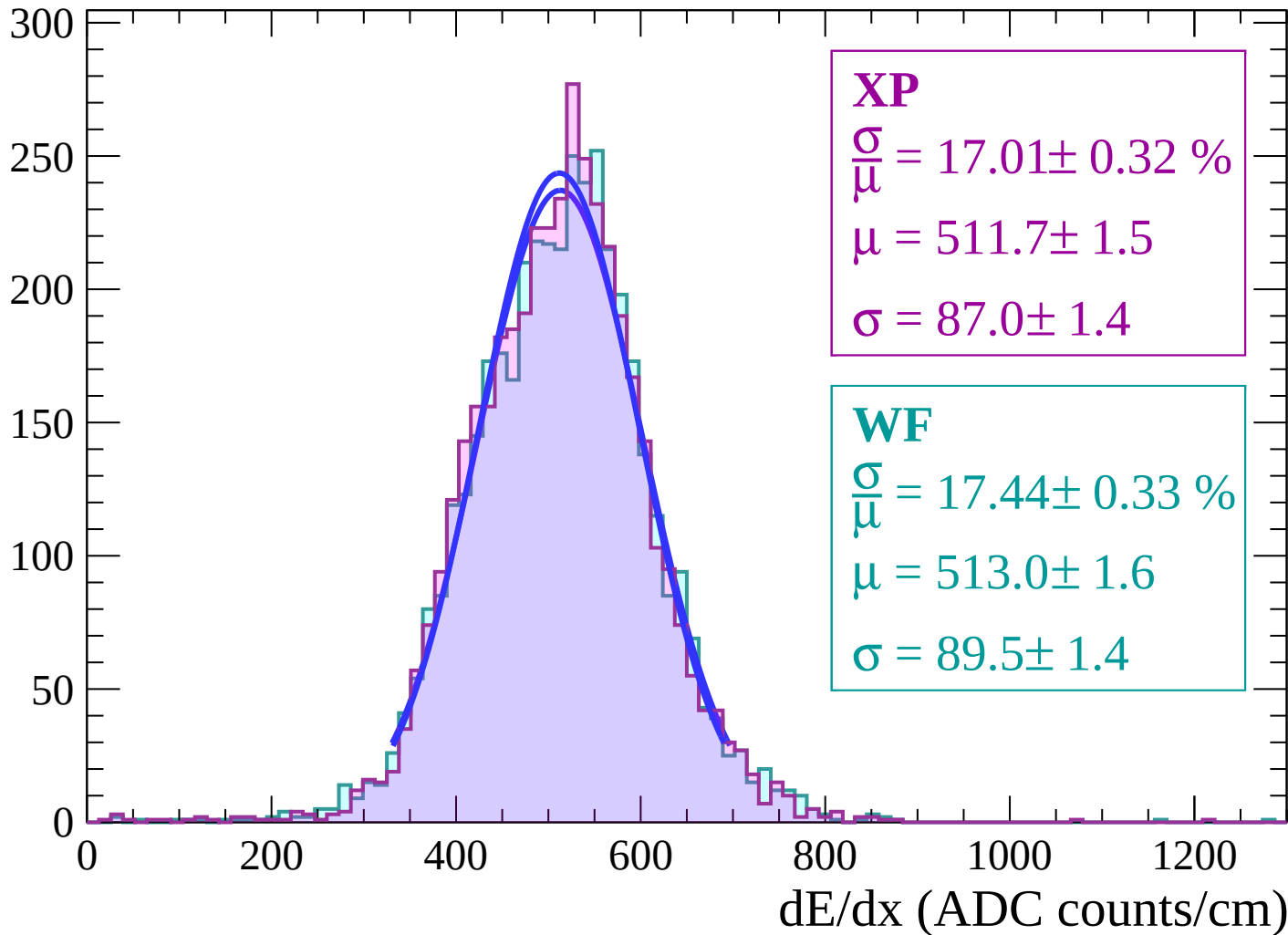
Energy loss | $-46 < \theta < -44$

Count



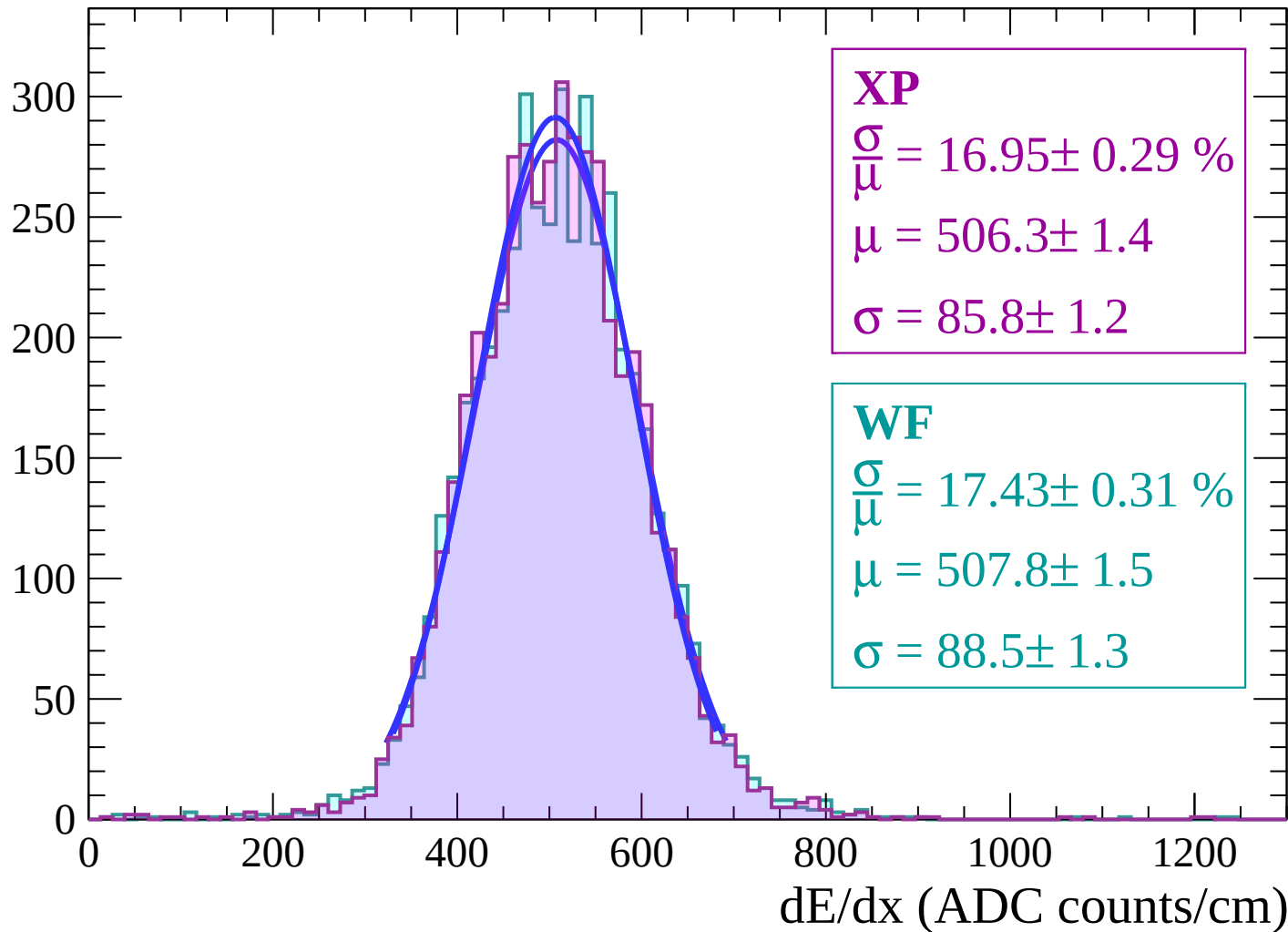
Energy loss | $-44 < \theta < -42$

Count



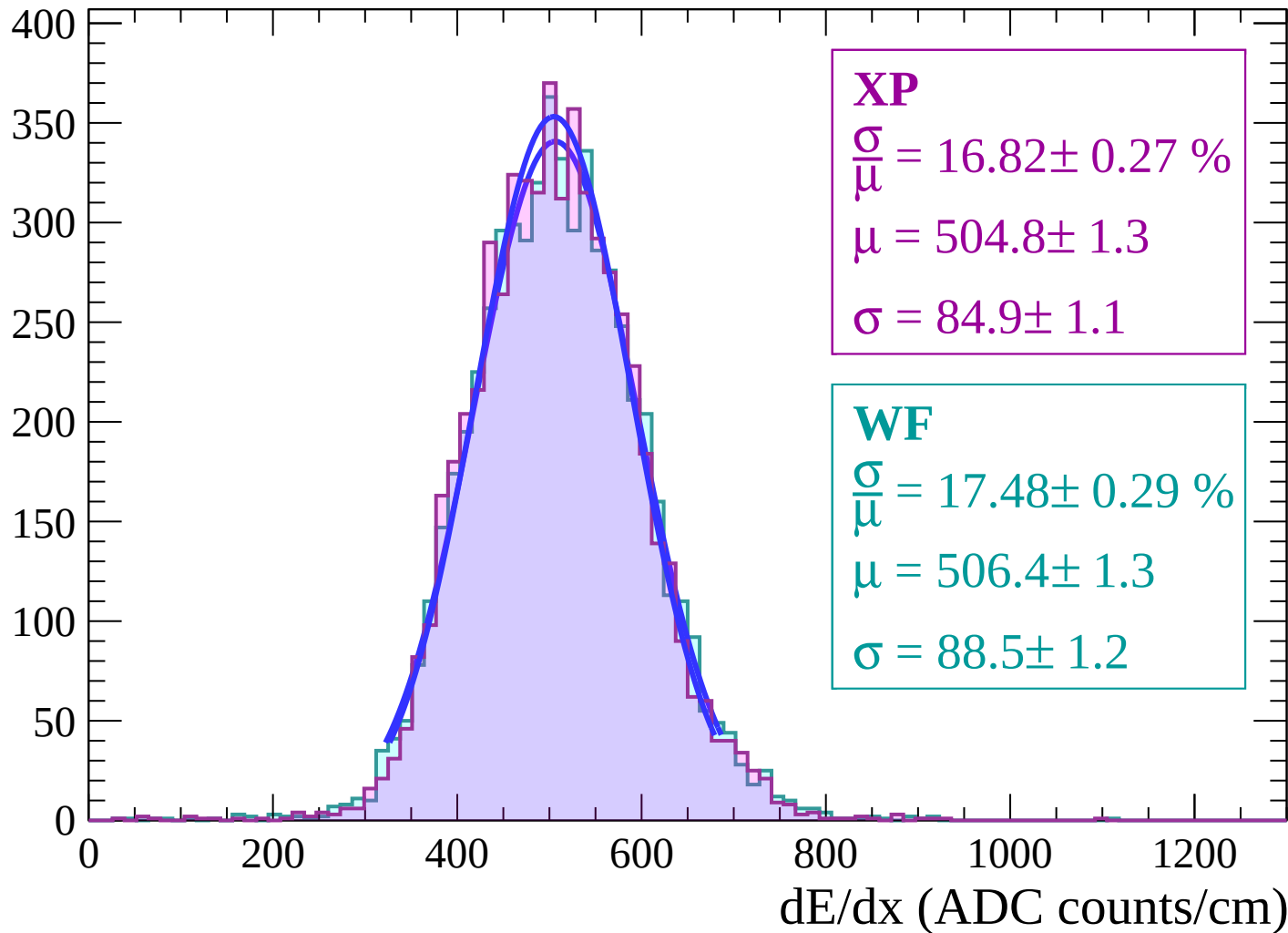
Energy loss | $-42 < \theta < -40$

Count



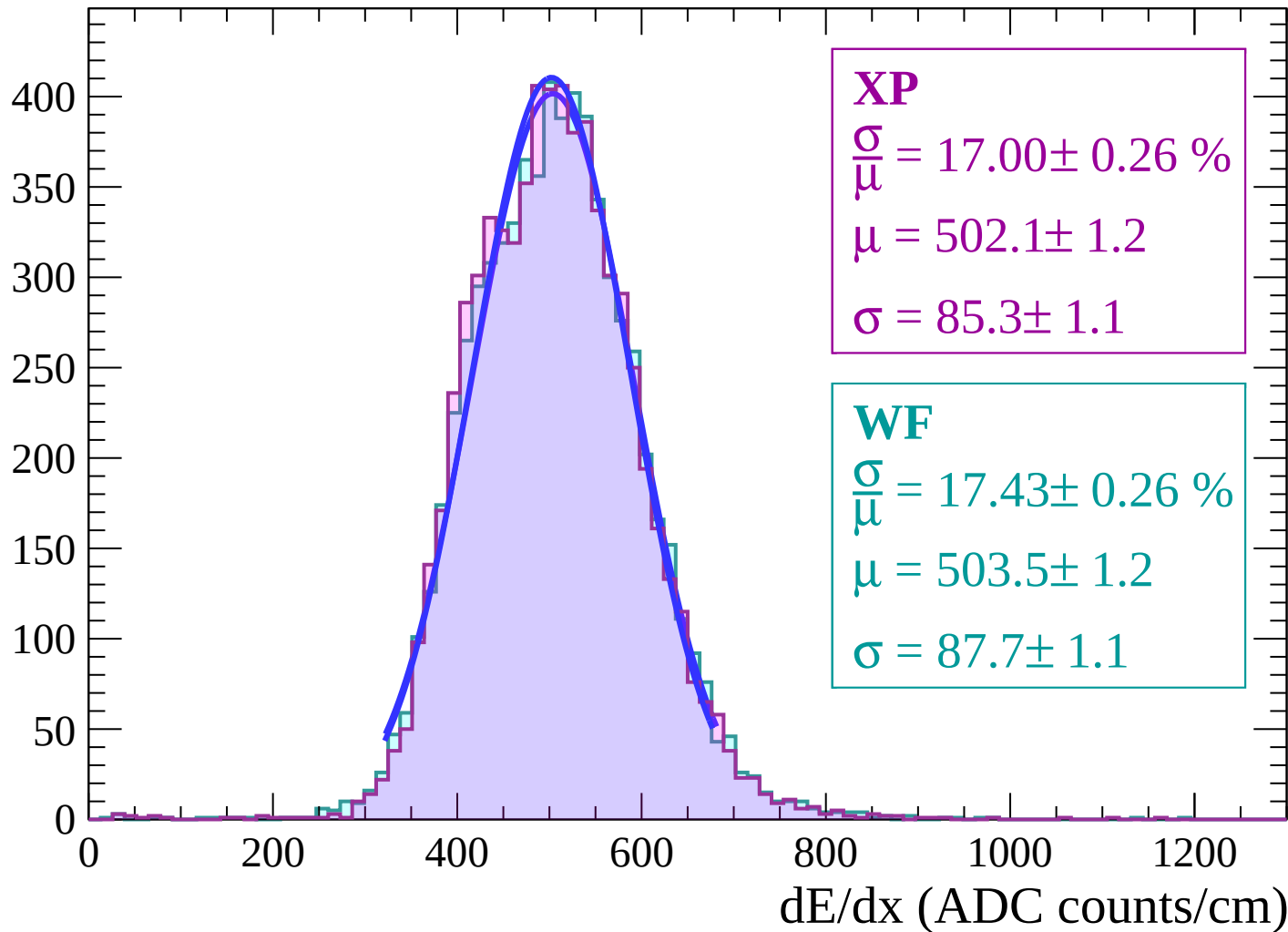
Energy loss | $-40 < \theta < -38$

Count



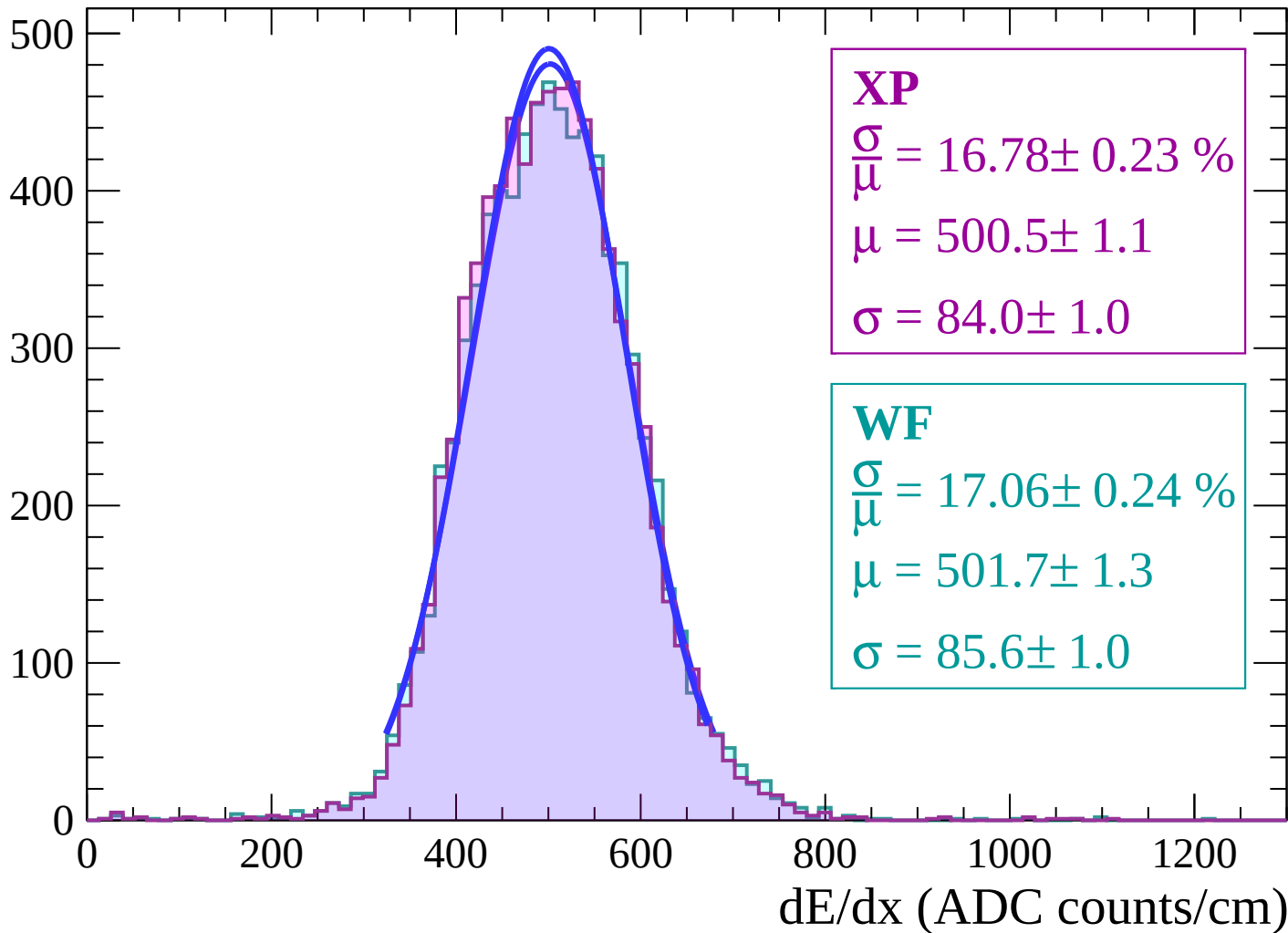
Energy loss | $-38 < \theta < -36$

Count



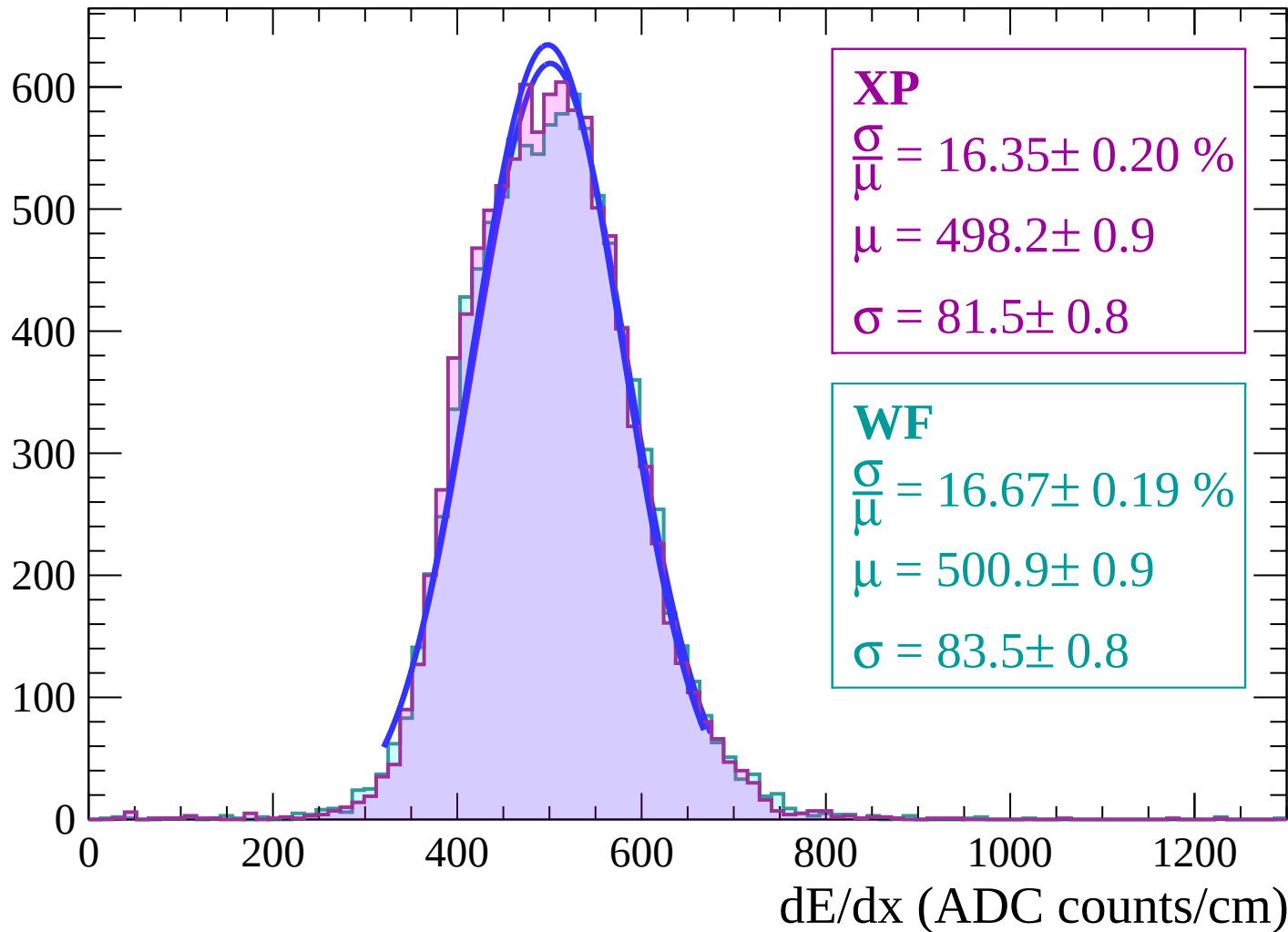
Energy loss | $-36 < \theta < -34$

Count



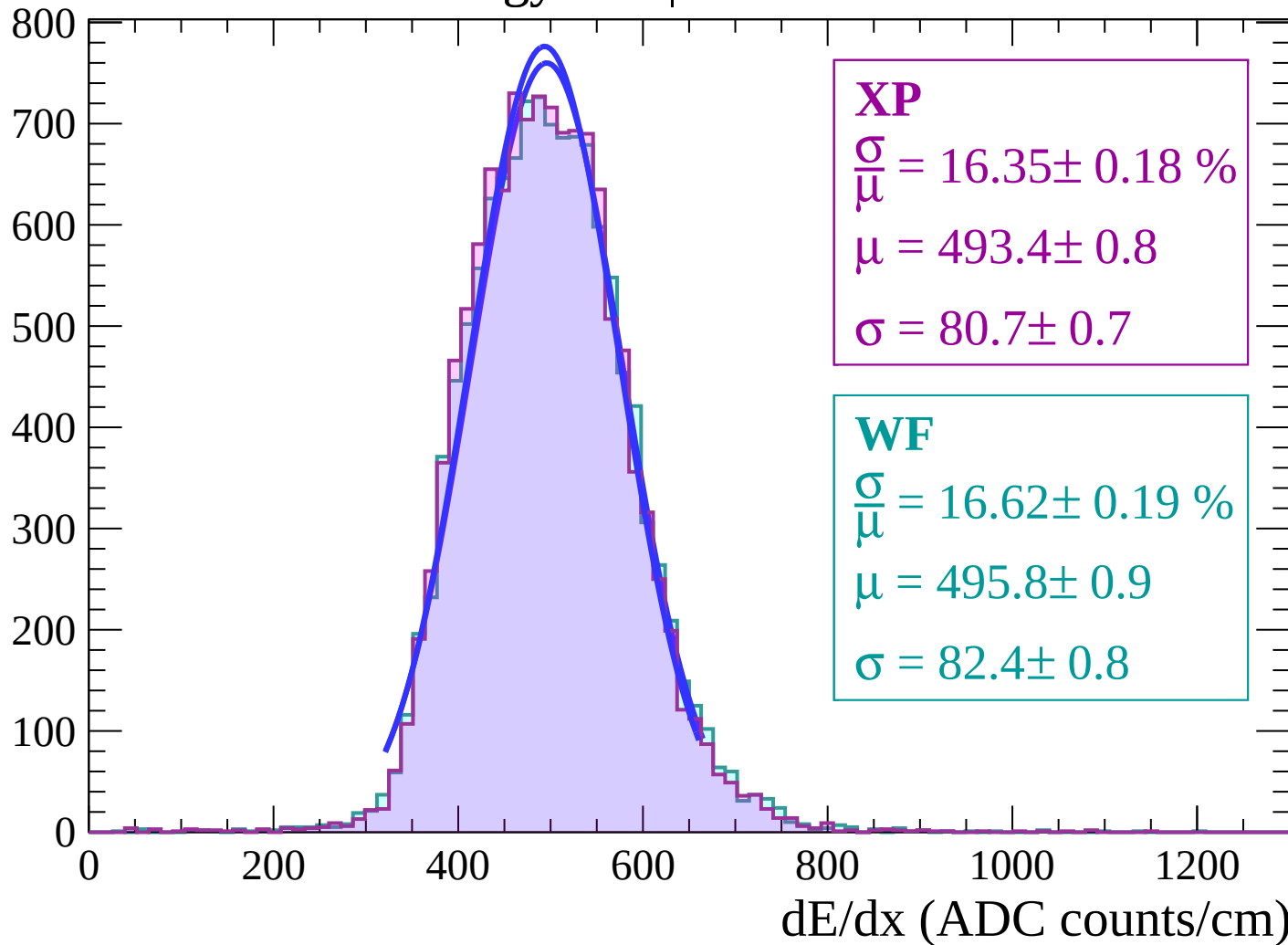
Energy loss | $-34 < \theta < -32$

Count



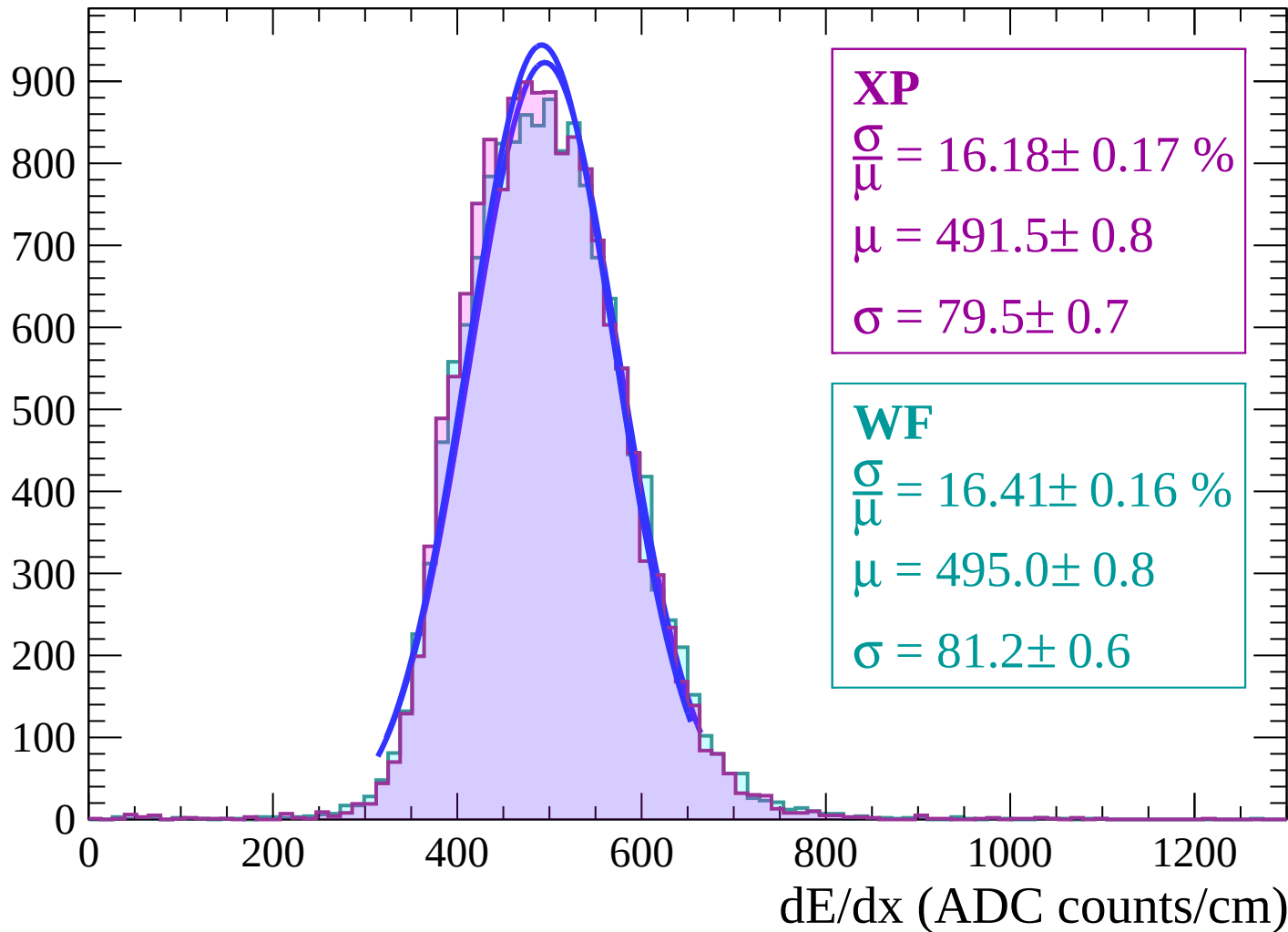
Energy loss | $-32 < \theta < -30$

Count



Energy loss | $-30 < \theta < -28$

Count



Energy loss | $-28 < \theta < -26$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 16.01 \pm 0.15 \%$$

$$\mu = 488.2 \pm 0.7$$

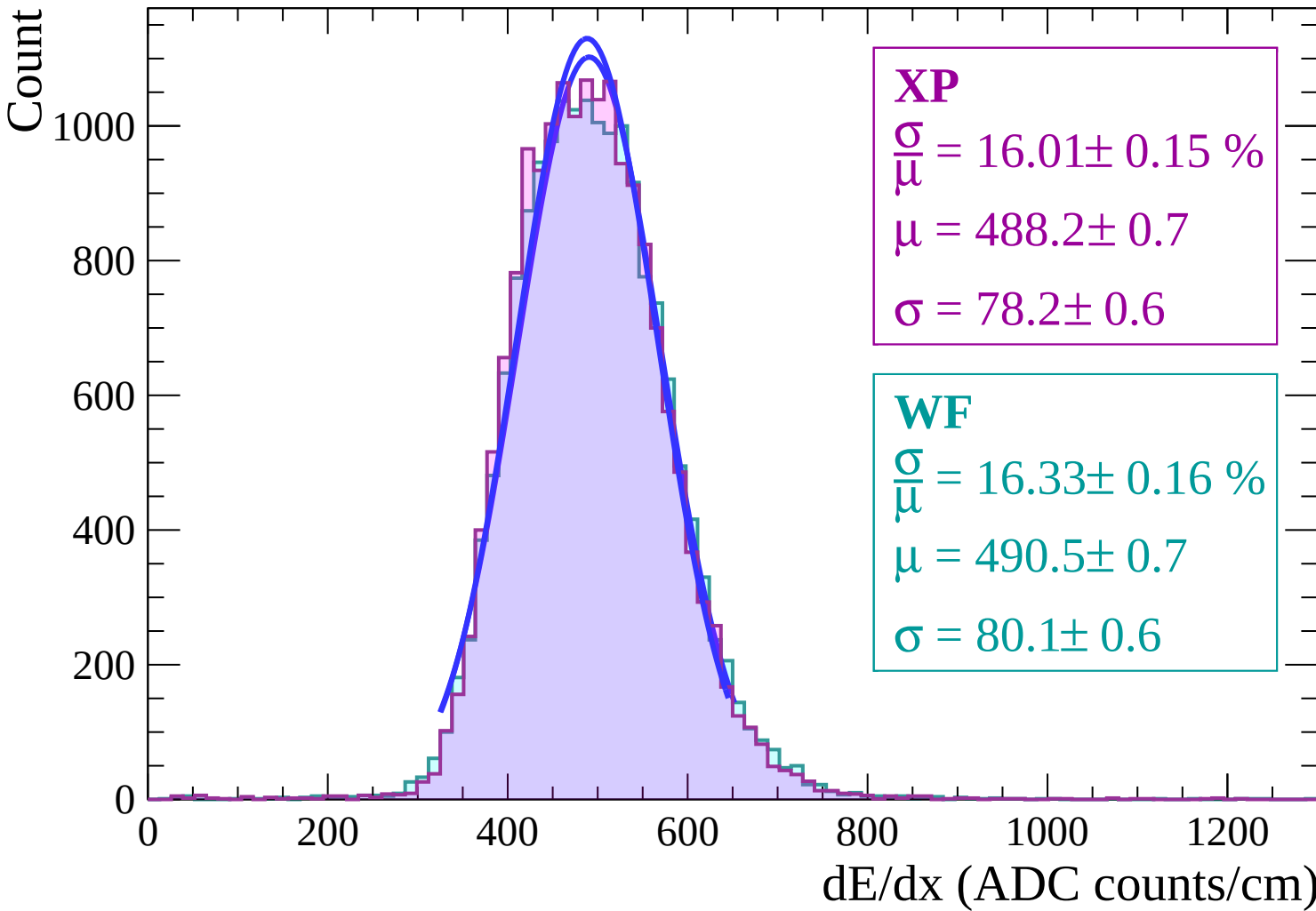
$$\sigma = 78.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 16.33 \pm 0.16 \%$$

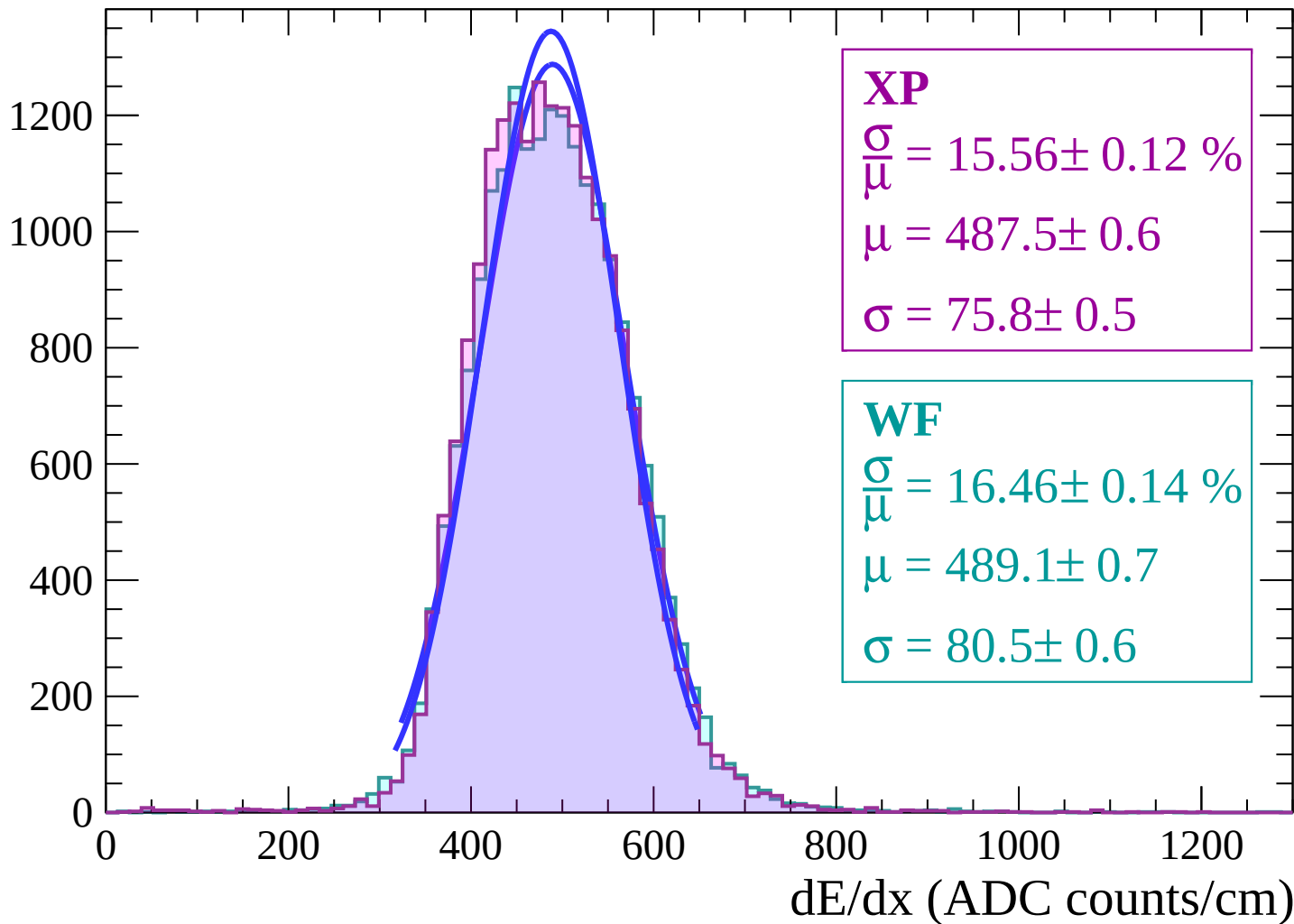
$$\mu = 490.5 \pm 0.7$$

$$\sigma = 80.1 \pm 0.6$$

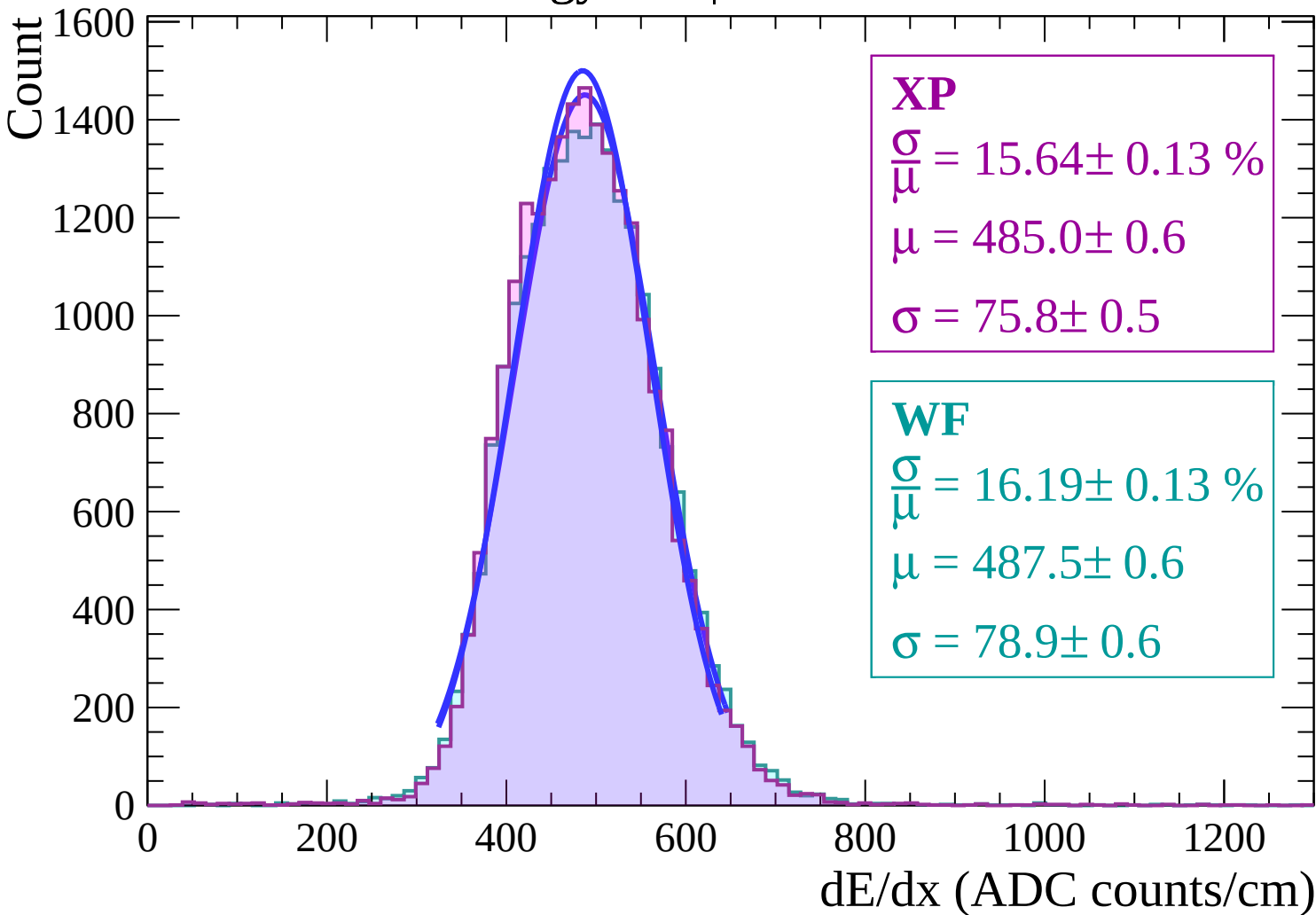


Energy loss | $-26 < \theta < -24$

Count

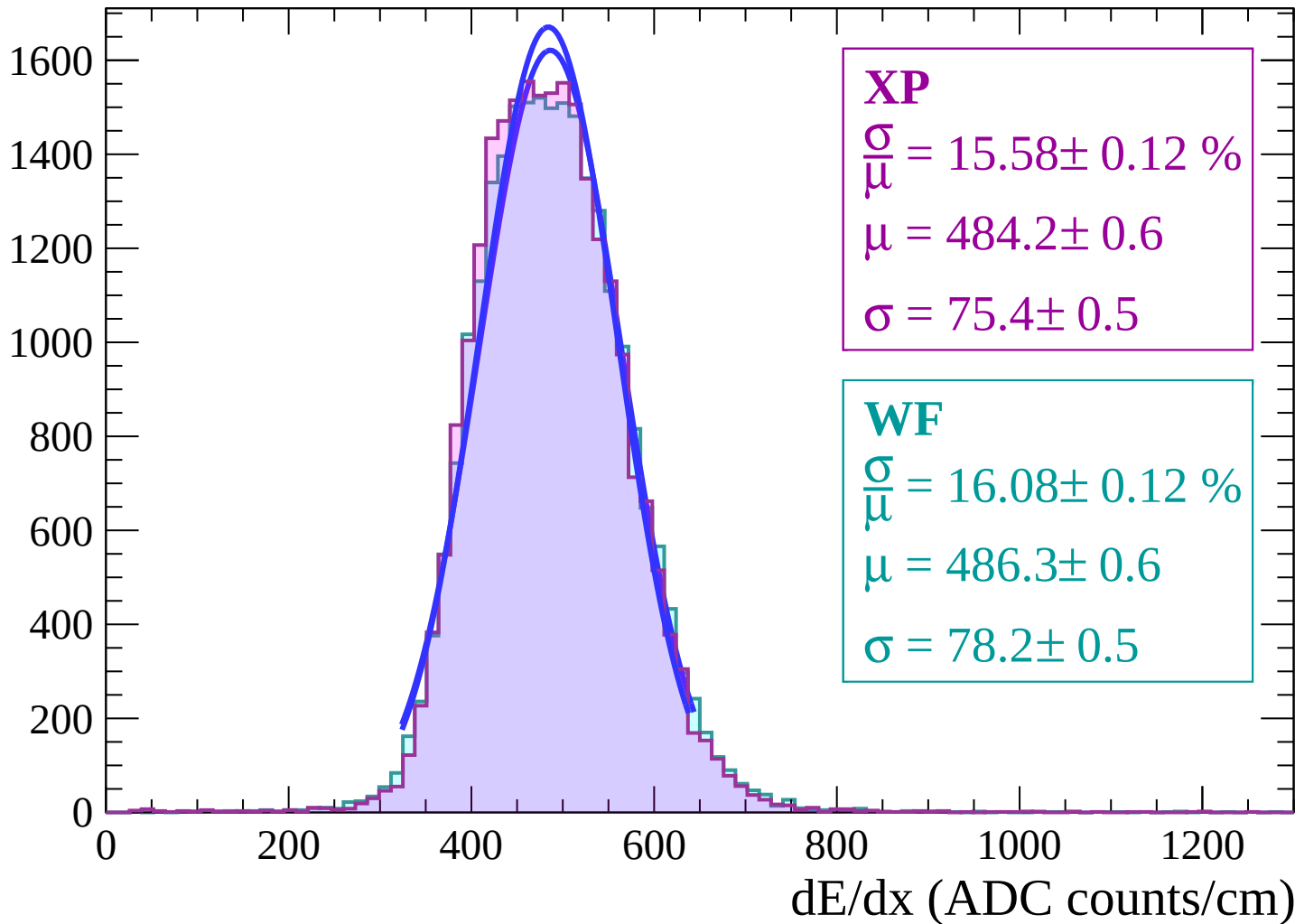


Energy loss | $-24 < \theta < -22$

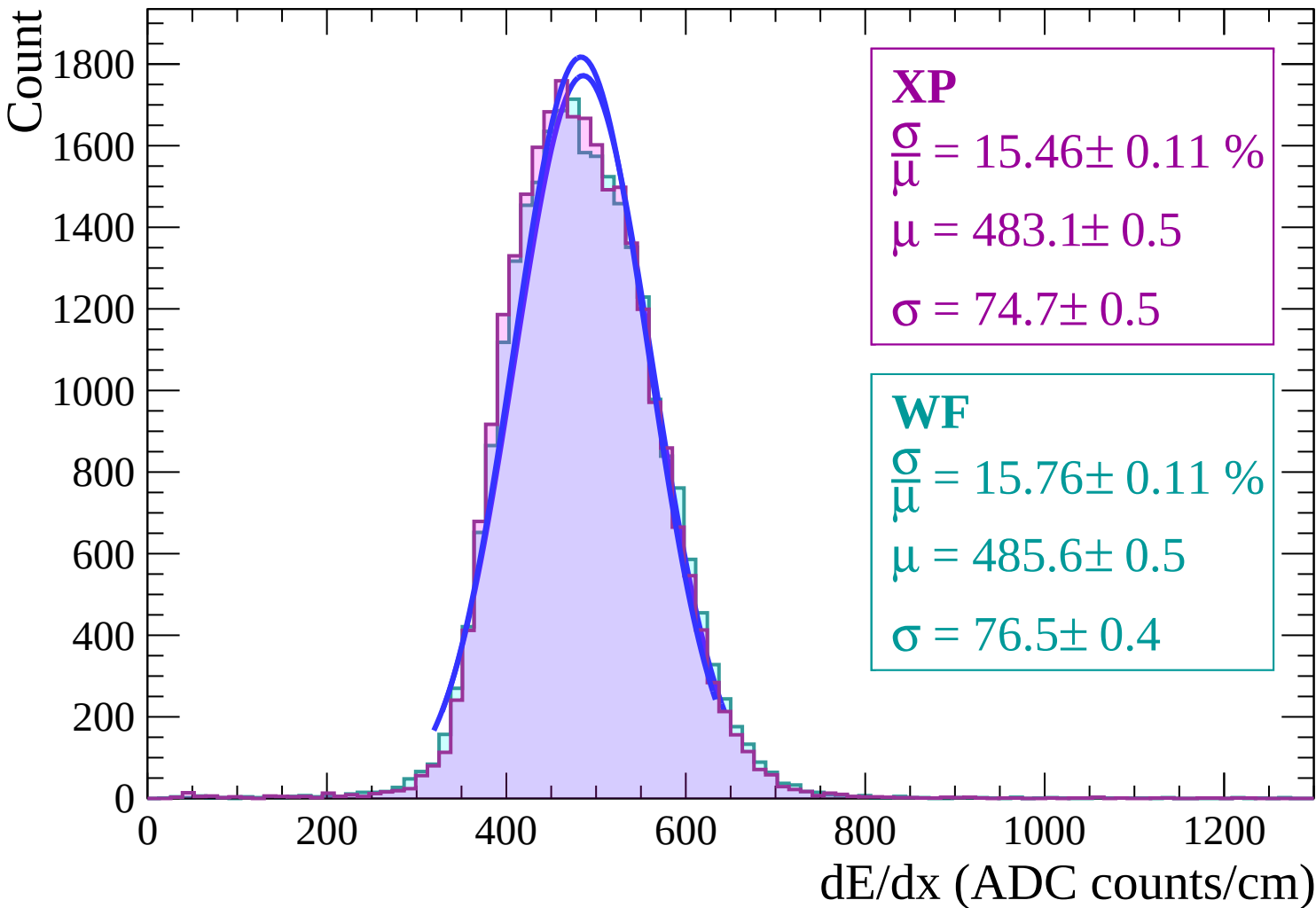


Energy loss | $-22 < \theta < -20$

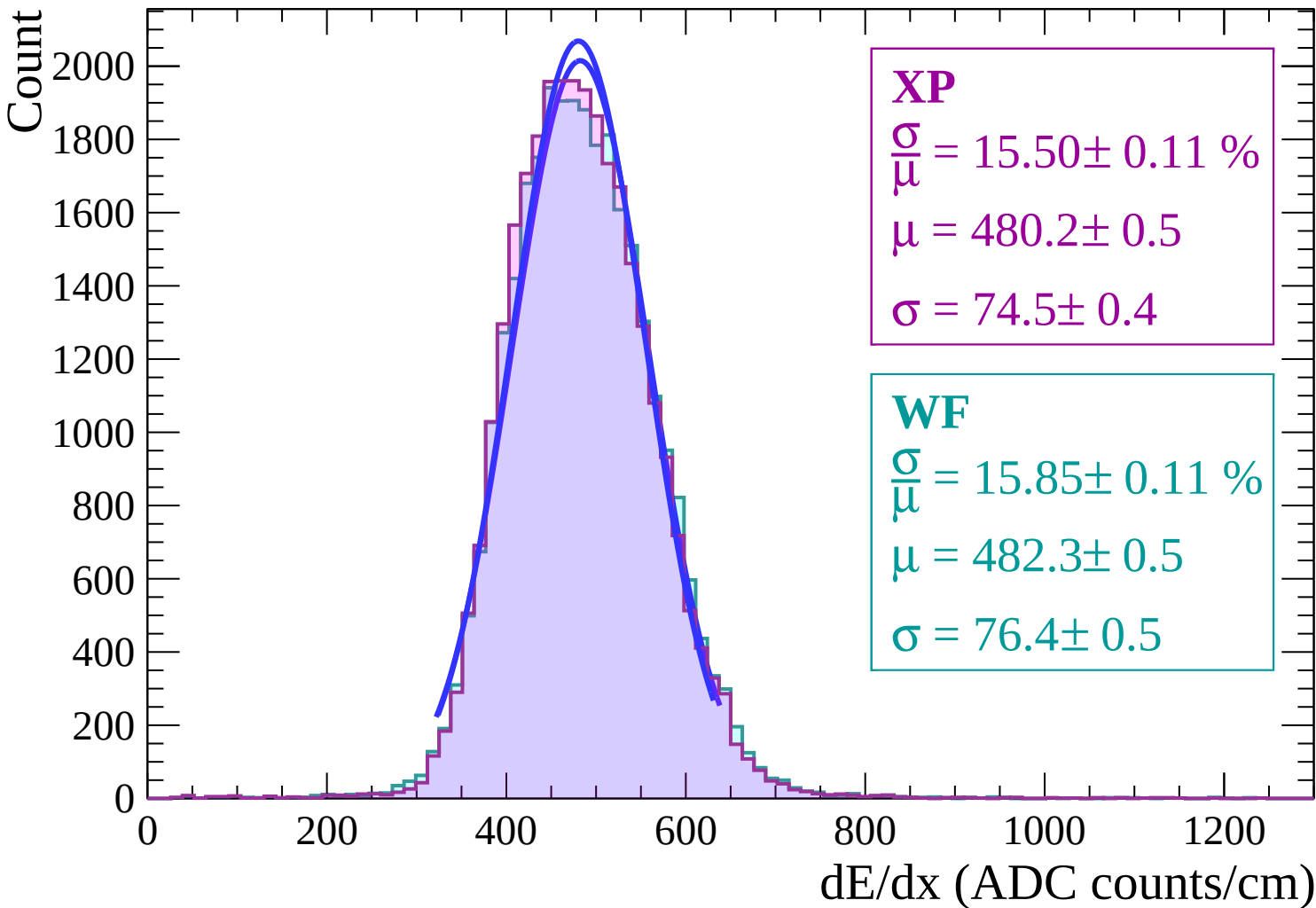
Count



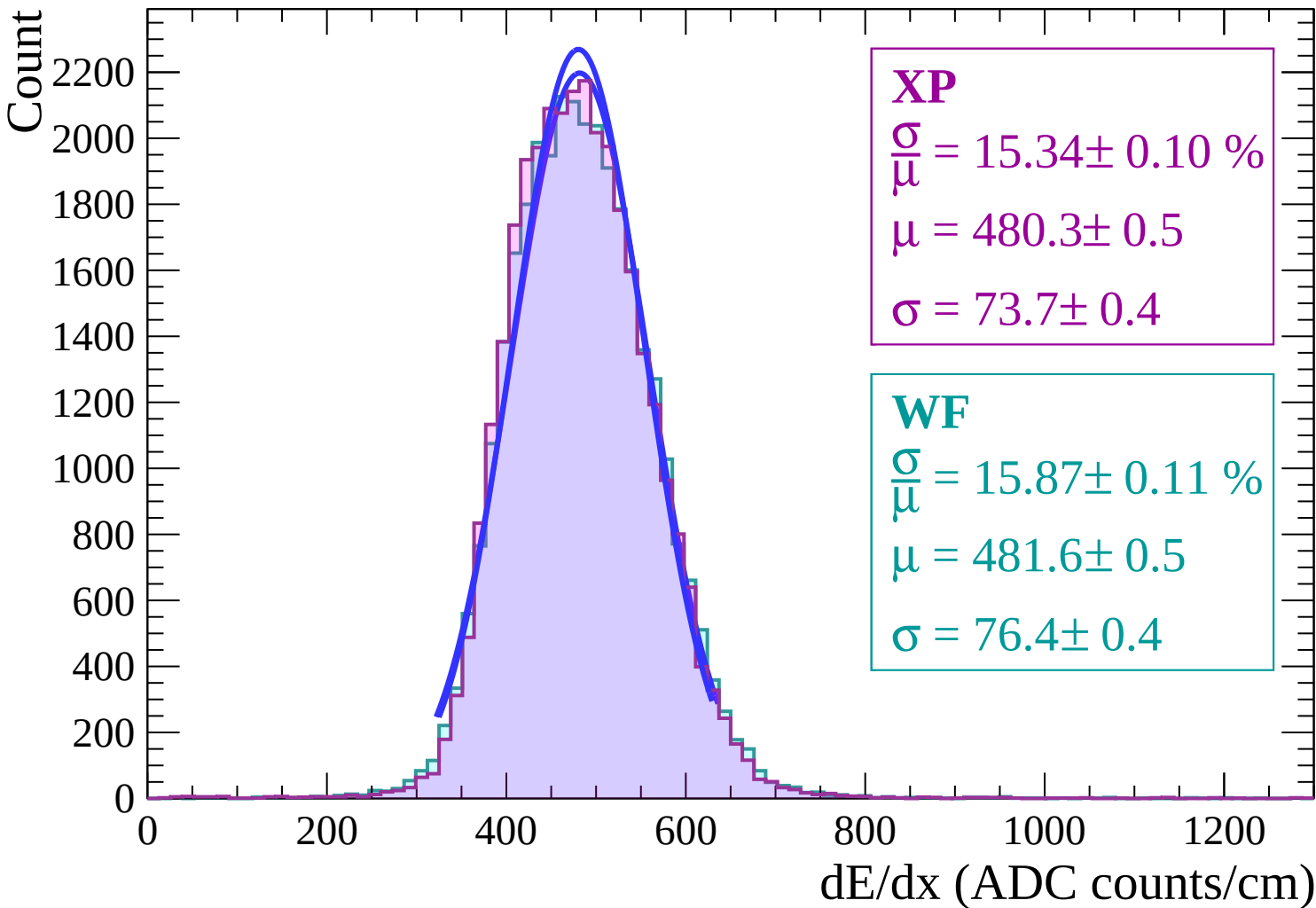
Energy loss | $-20 < \theta < -18$



Energy loss | $-18 < \theta < -16$

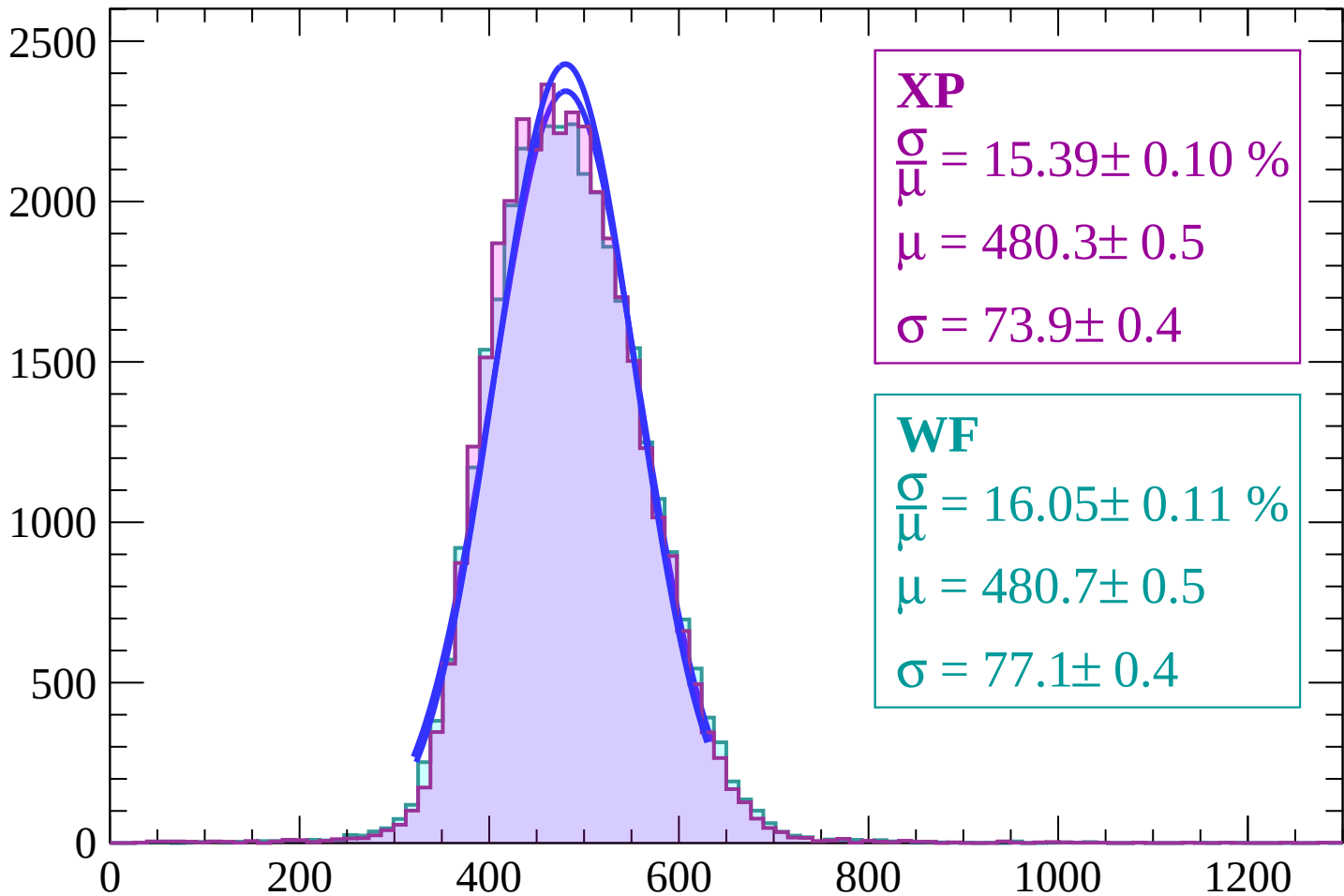


Energy loss | $-16 < \theta < -14$



Energy loss | $-14 < \theta < -12$

Count



XP

$$\frac{\sigma}{\mu} = 15.39 \pm 0.10 \%$$

$$\mu = 480.3 \pm 0.5$$

$$\sigma = 73.9 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 16.05 \pm 0.11 \%$$

$$\mu = 480.7 \pm 0.5$$

$$\sigma = 77.1 \pm 0.4$$

dE/dx (ADC counts/cm)

Energy loss | $-12 < \theta < -10$

Count

2500

2000

1500

1000

500

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 15.21 \pm 0.09 \%$$

$$\mu = 480.5 \pm 0.4$$

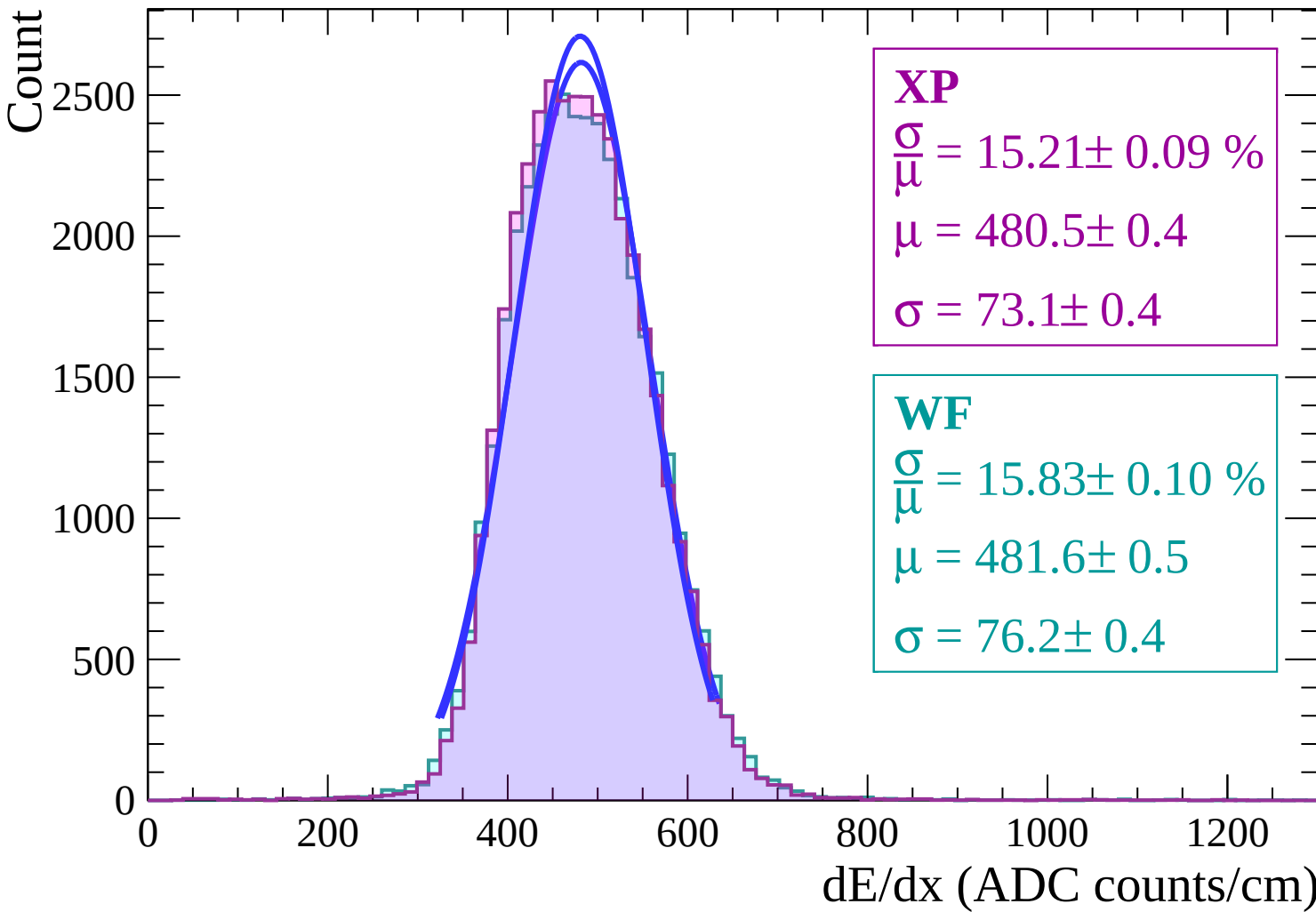
$$\sigma = 73.1 \pm 0.4$$

WF

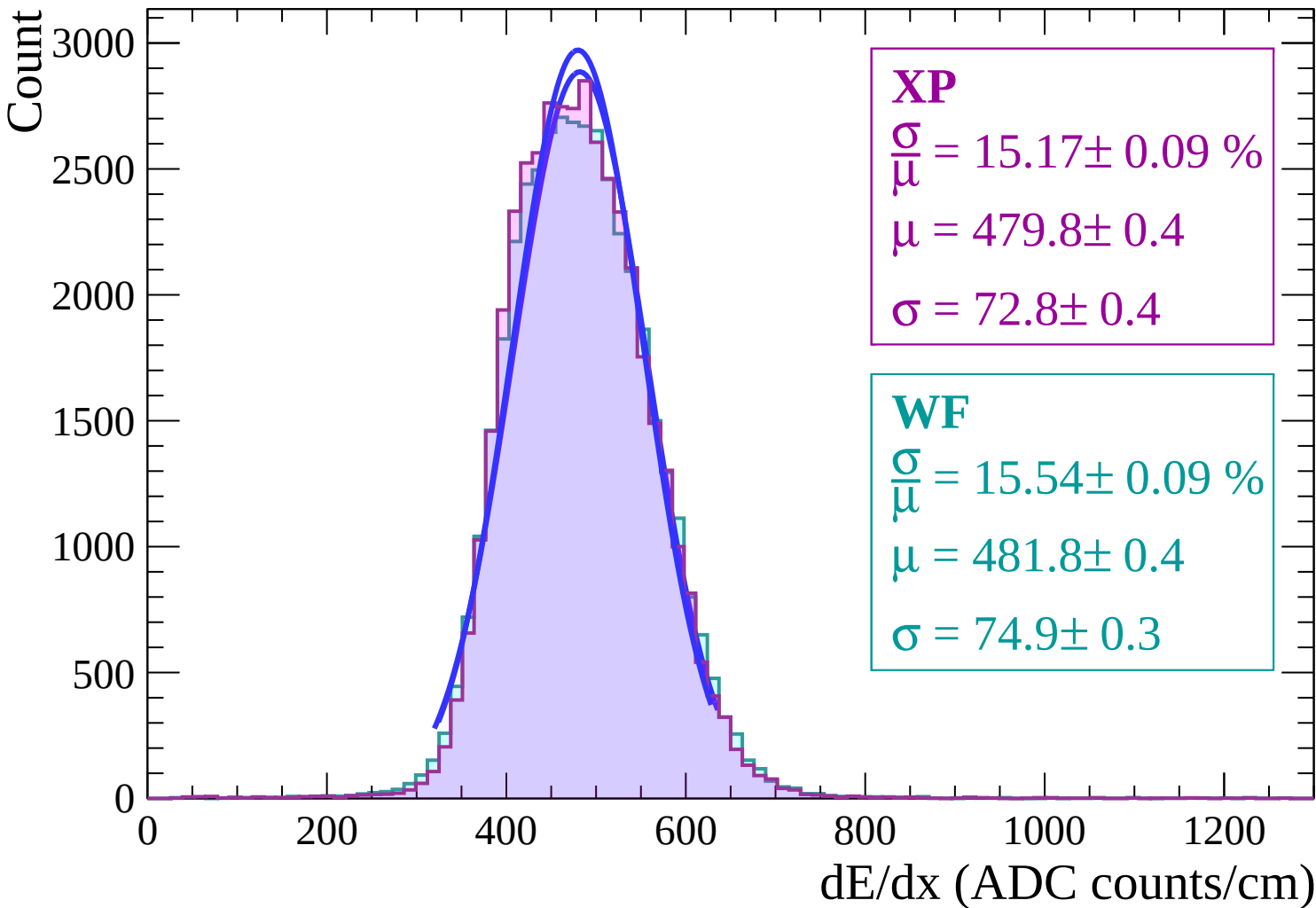
$$\frac{\sigma}{\mu} = 15.83 \pm 0.10 \%$$

$$\mu = 481.6 \pm 0.5$$

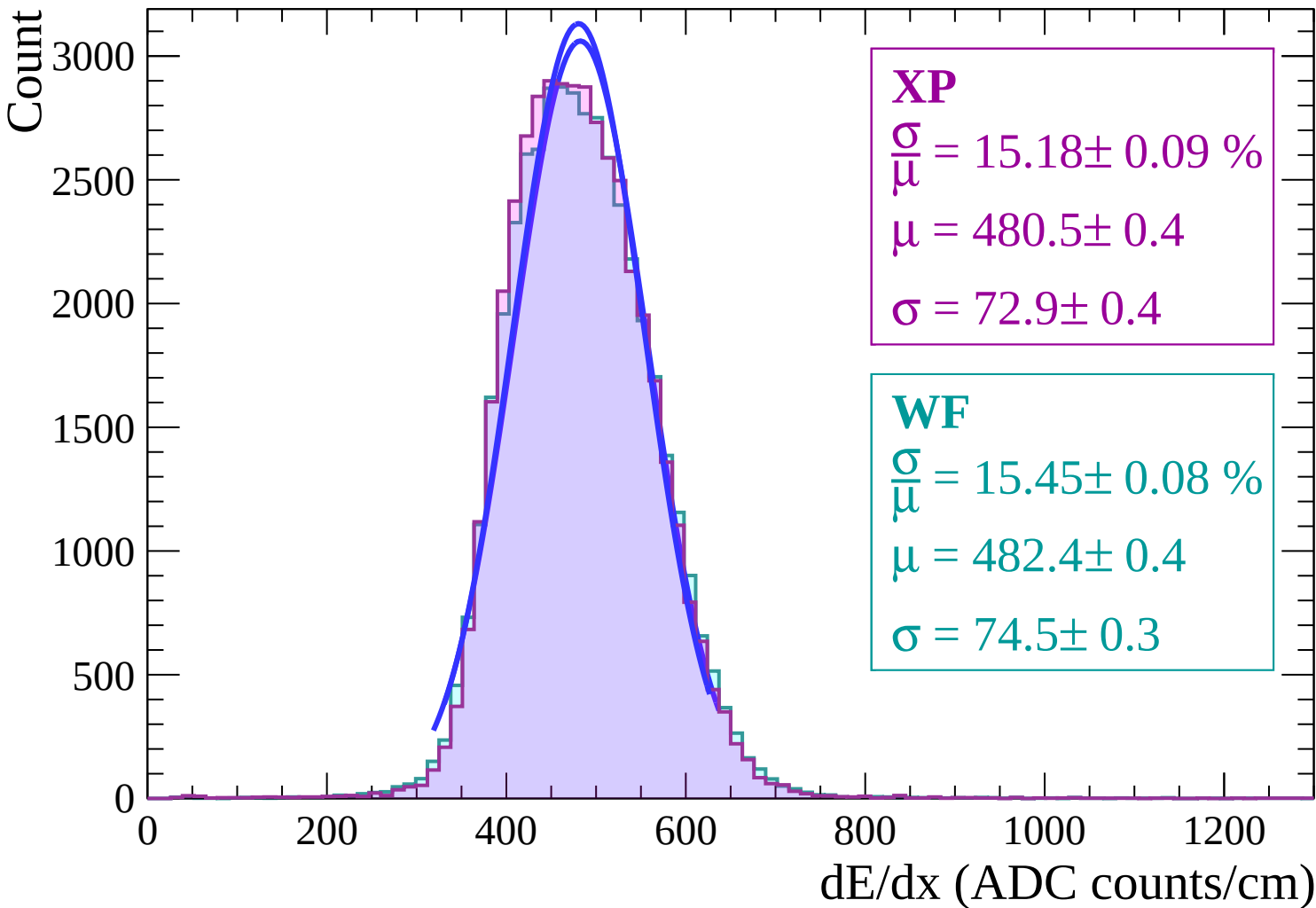
$$\sigma = 76.2 \pm 0.4$$



Energy loss | $-10 < \theta < -8$



Energy loss | $-8 < \theta < -6$



Energy loss | $-6 < \theta < -4$

Count

3500

3000

2500

2000

1500

1000

500

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 14.98 \pm 0.08 \%$$

$$\mu = 480.0 \pm 0.4$$

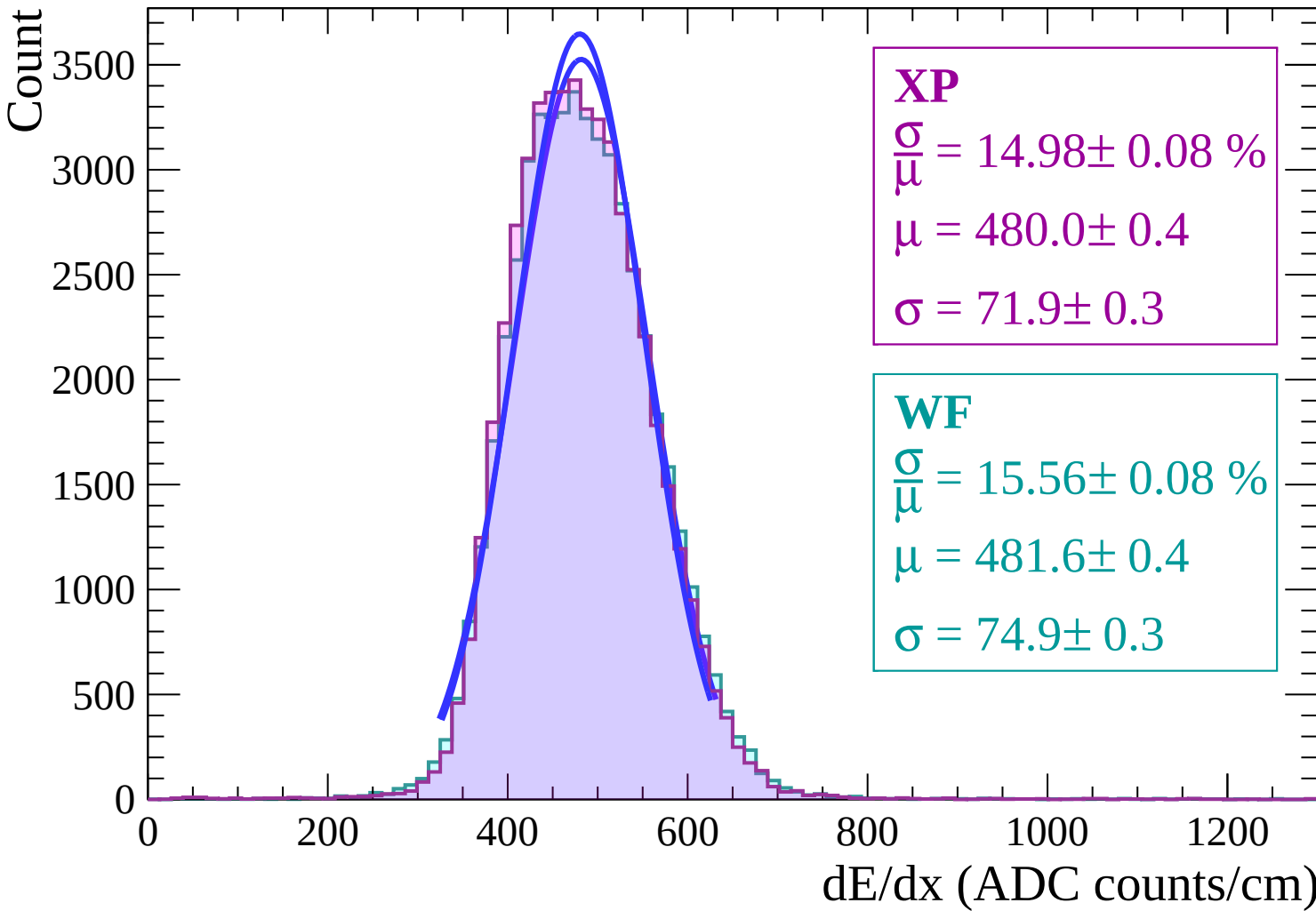
$$\sigma = 71.9 \pm 0.3$$

WF

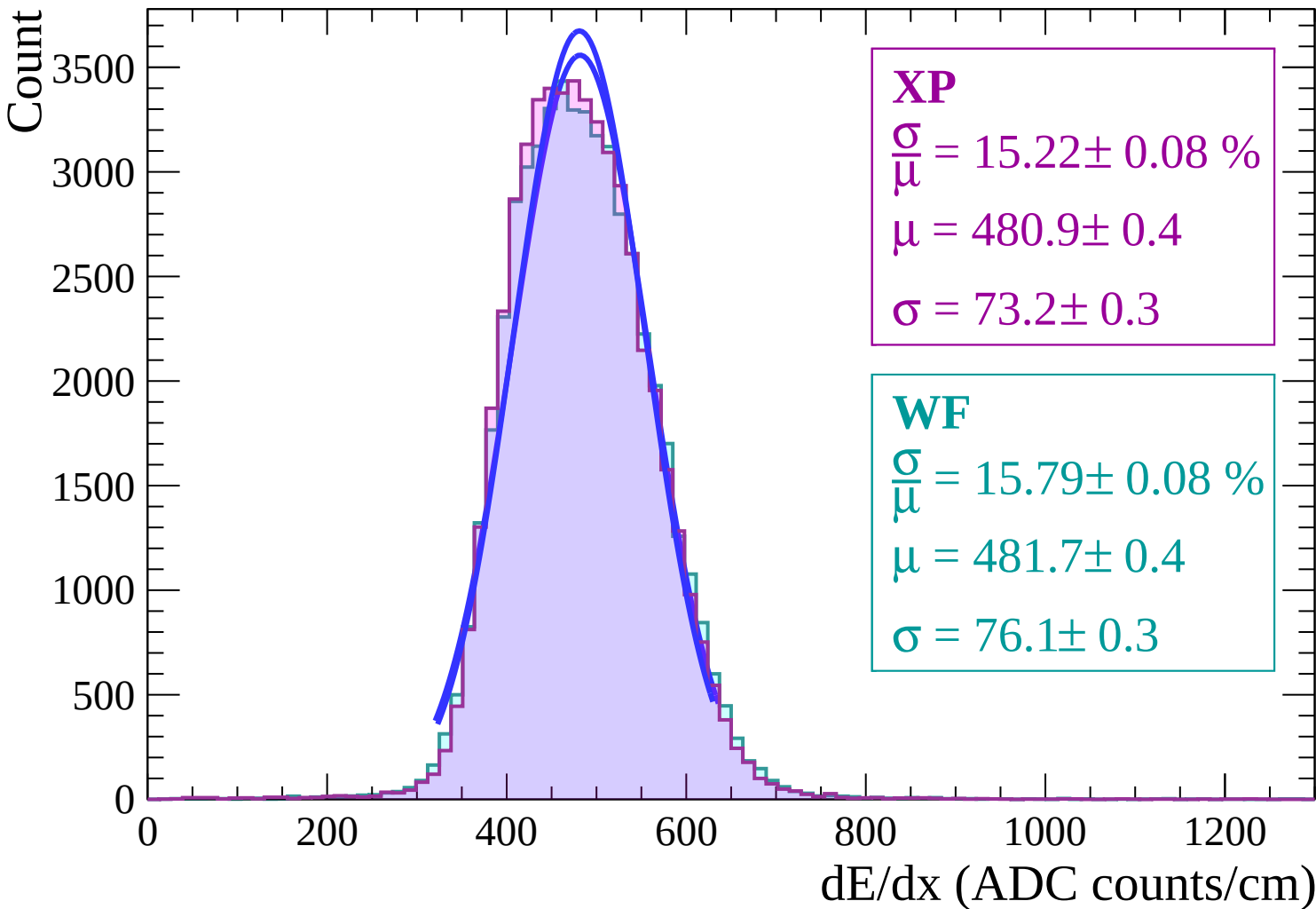
$$\frac{\sigma}{\mu} = 15.56 \pm 0.08 \%$$

$$\mu = 481.6 \pm 0.4$$

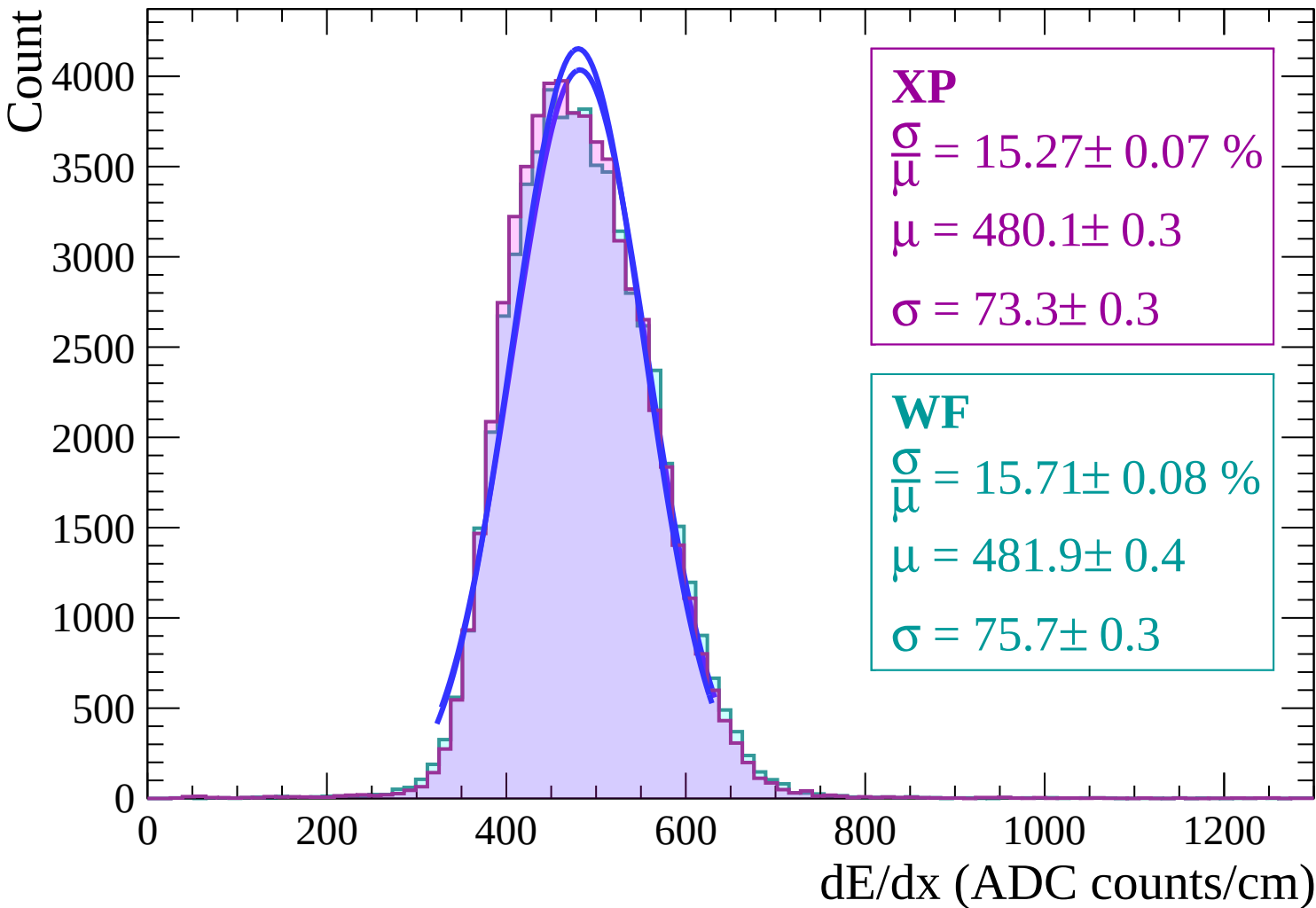
$$\sigma = 74.9 \pm 0.3$$



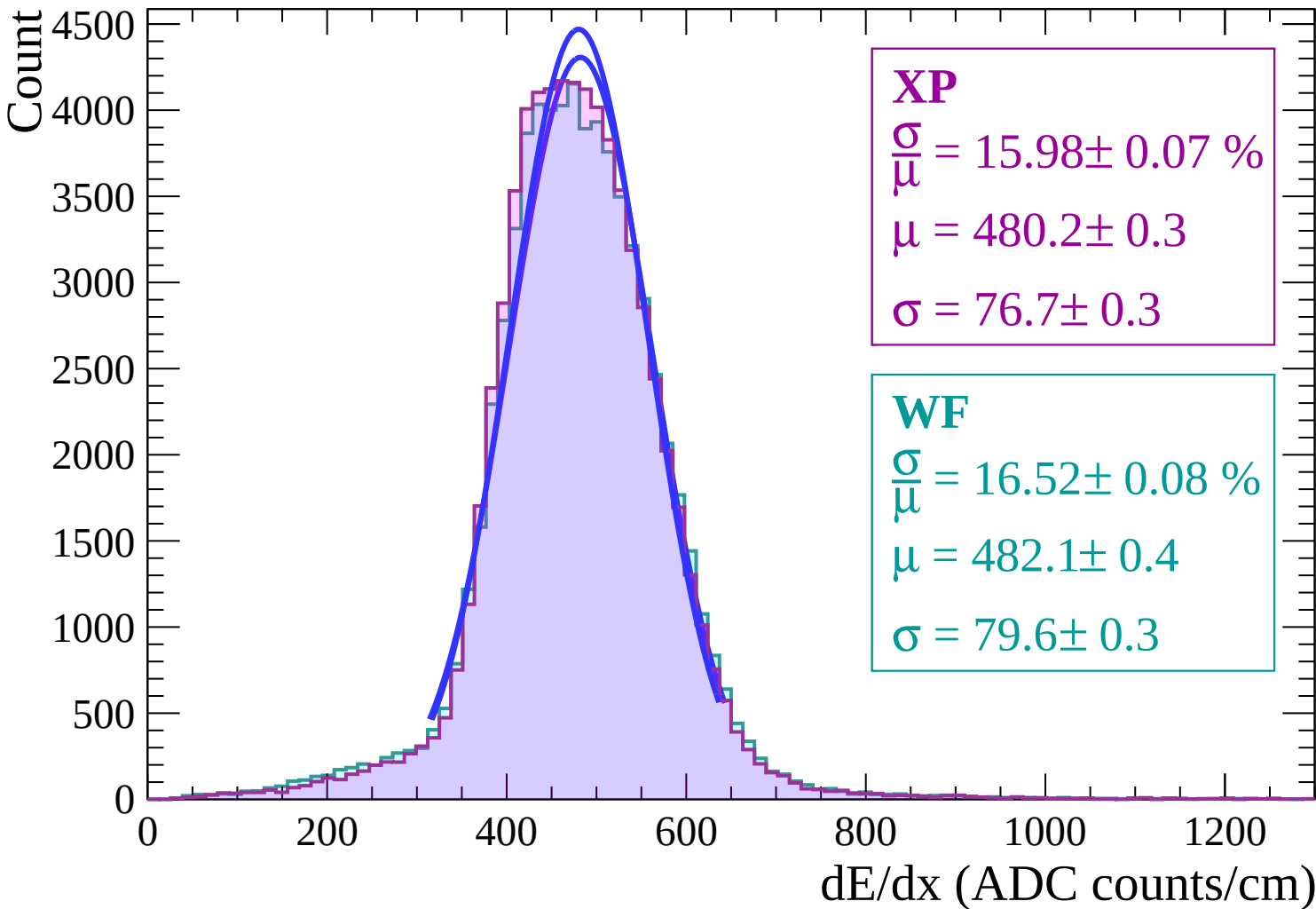
Energy loss | $-4 < \theta < -2$



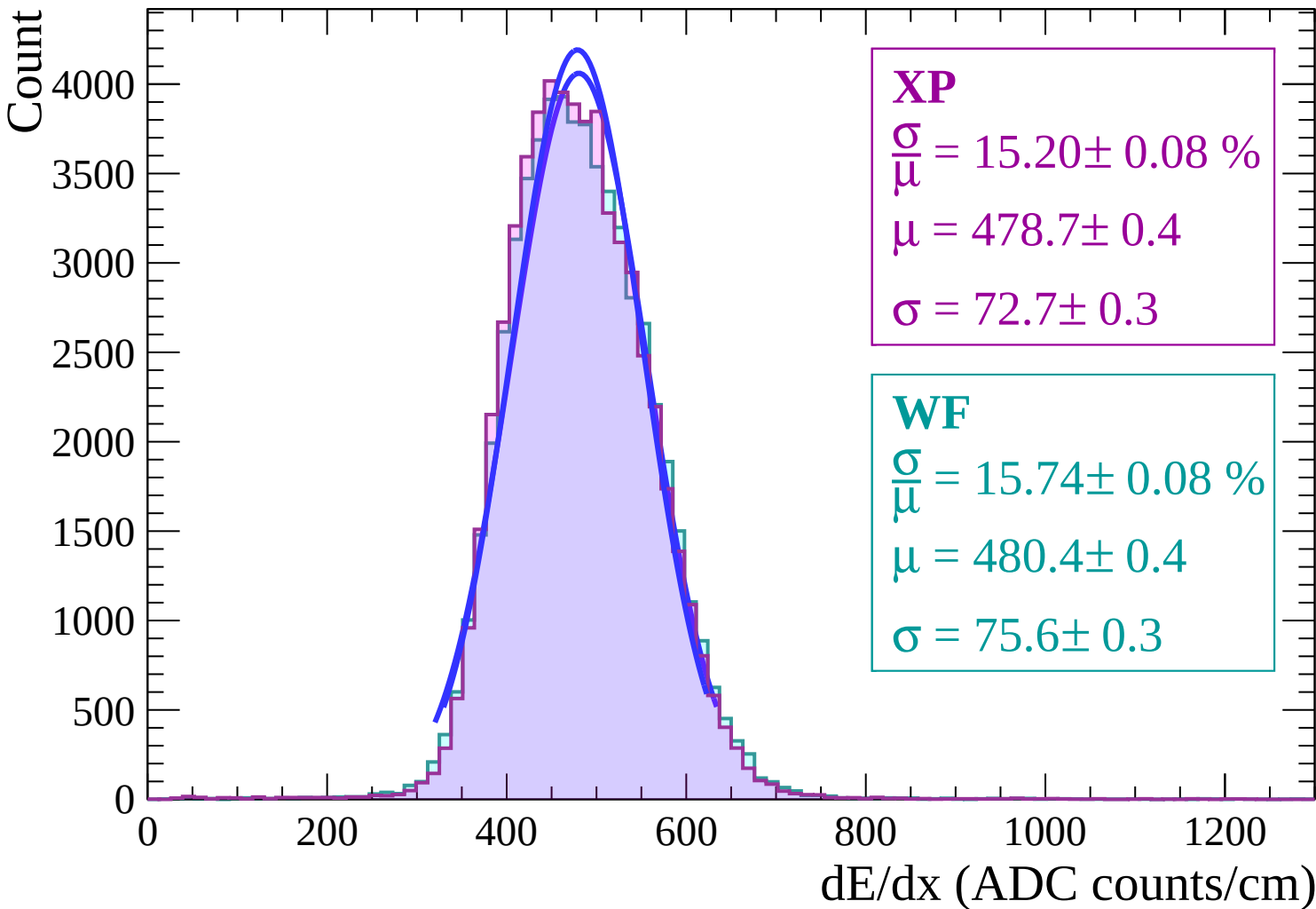
Energy loss | $-2 < \theta < 0$



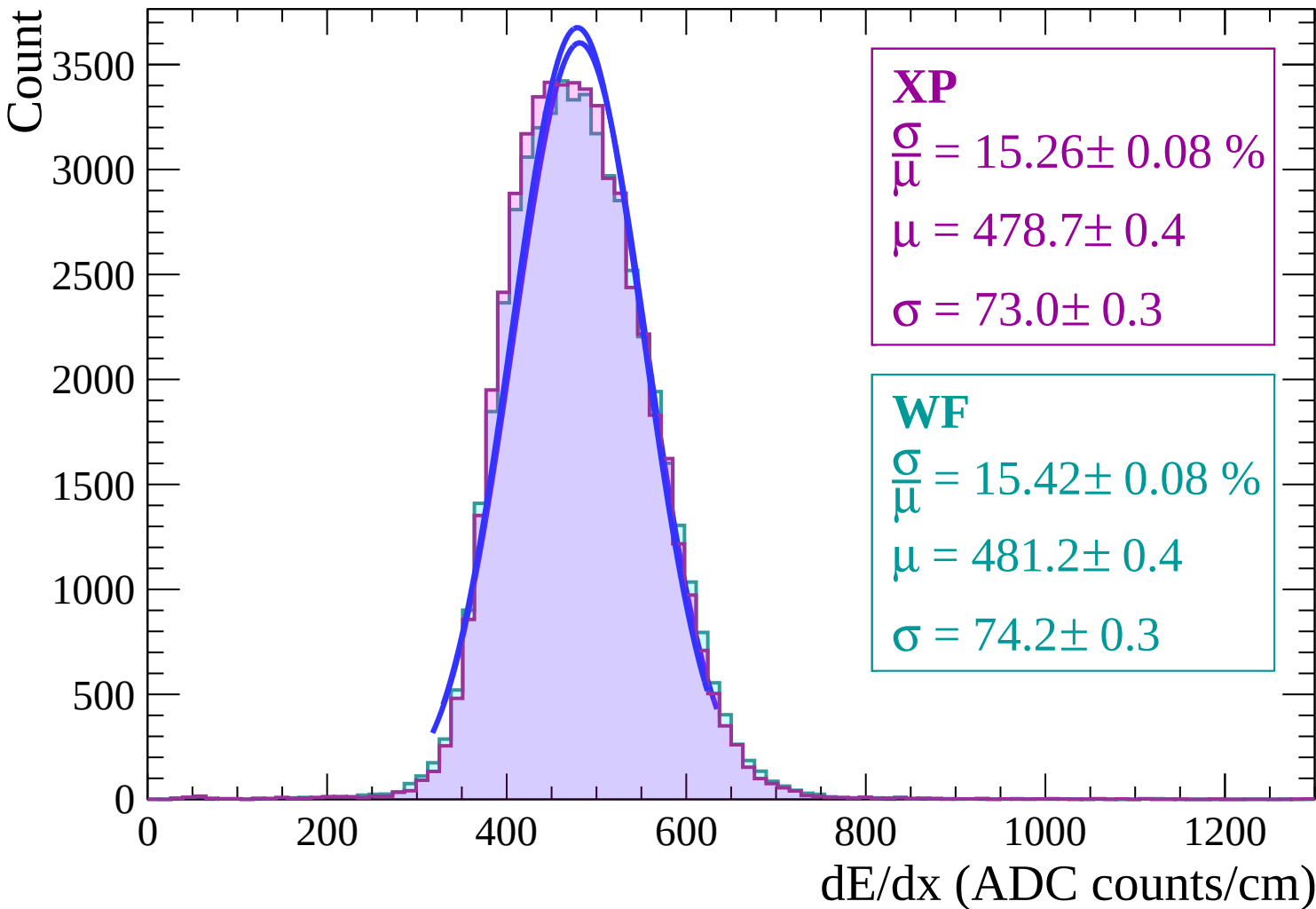
Energy loss | $0 < \theta < 2$



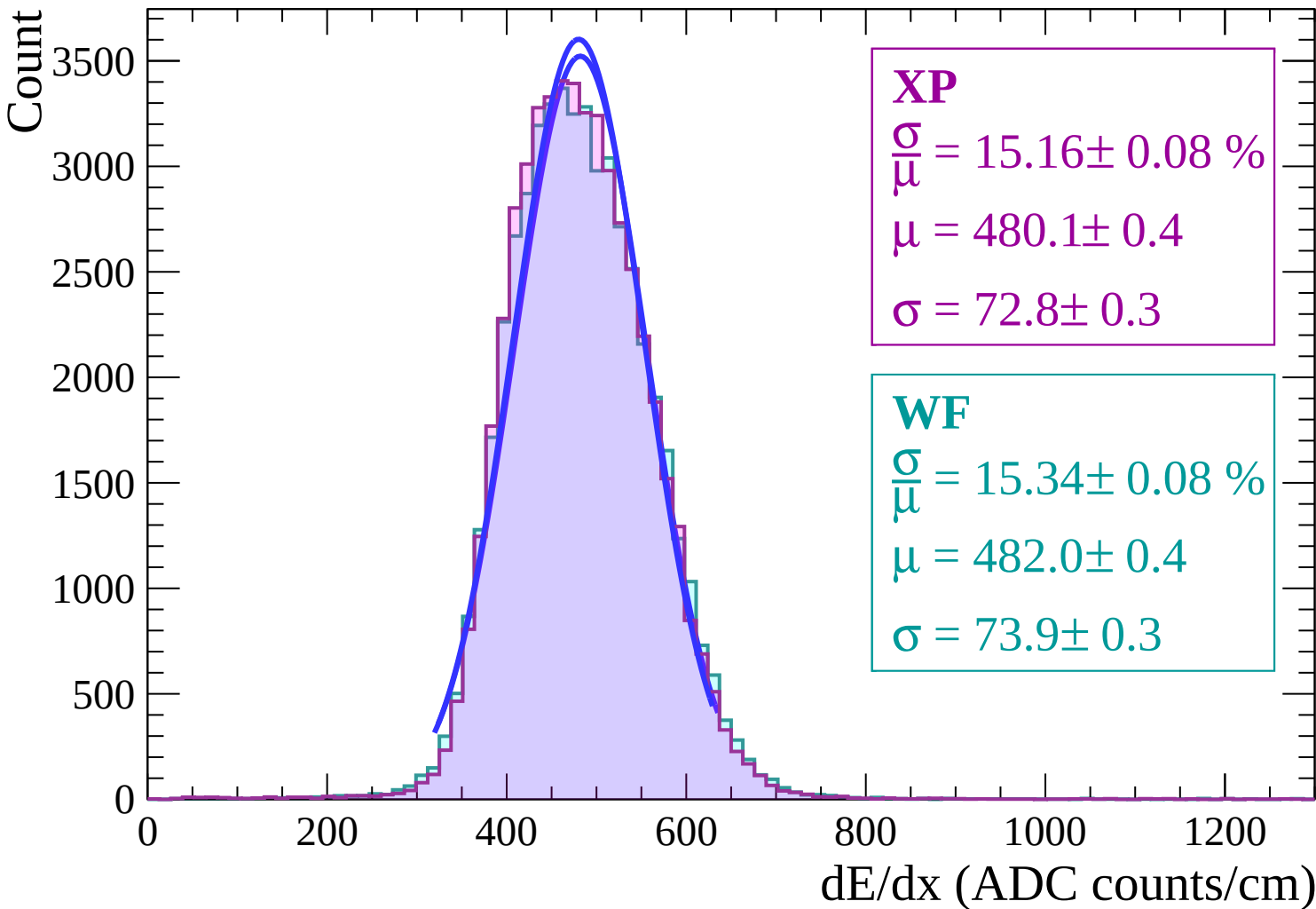
Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$



Energy loss | $6 < \theta < 8$



Energy loss | $8 < \theta < 10$

Count

3000
2500
2000
1500
1000
500
0

XP

$$\frac{\sigma}{\mu} = 15.20 \pm 0.09 \%$$

$$\mu = 479.3 \pm 0.4$$

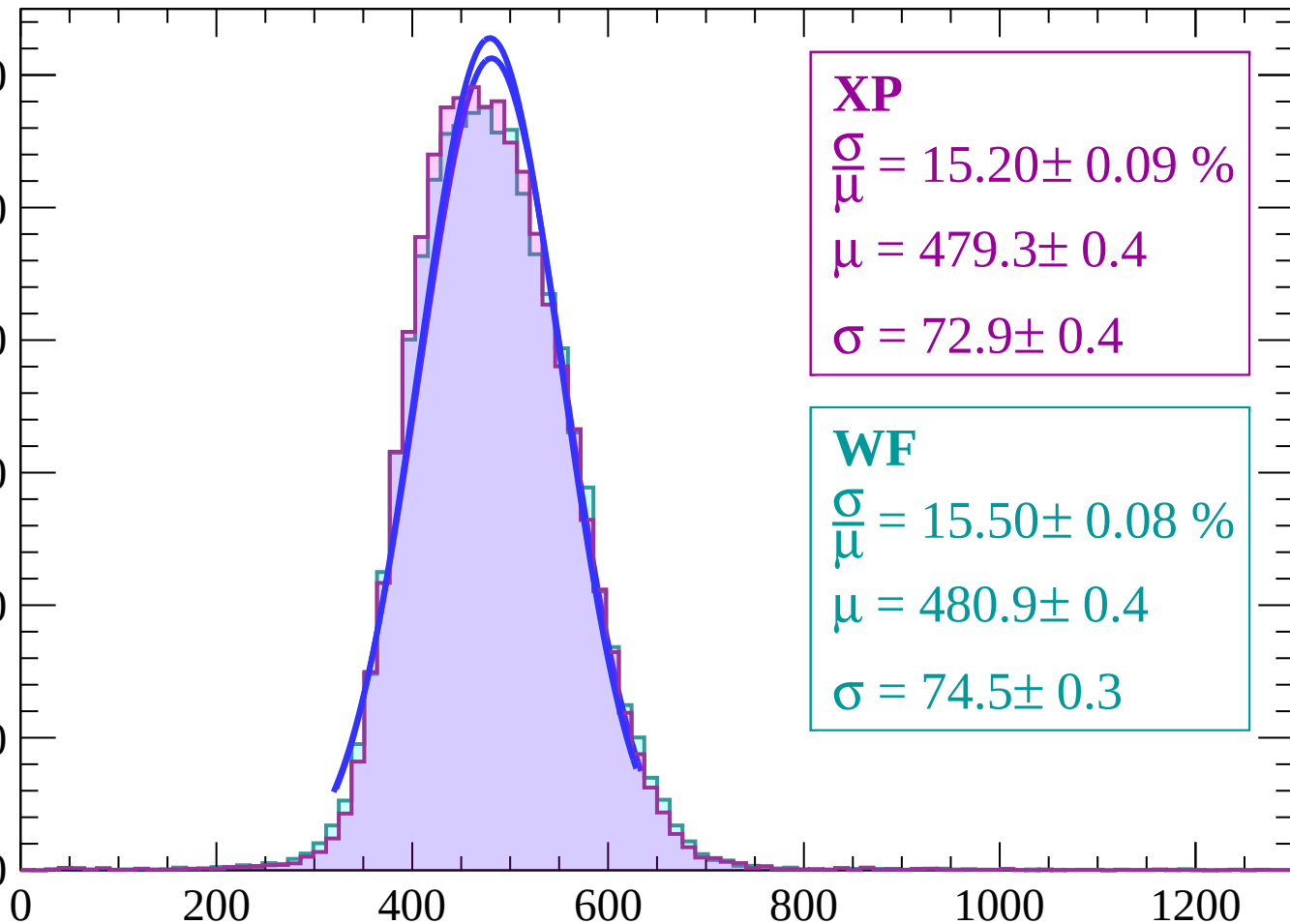
$$\sigma = 72.9 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 15.50 \pm 0.08 \%$$

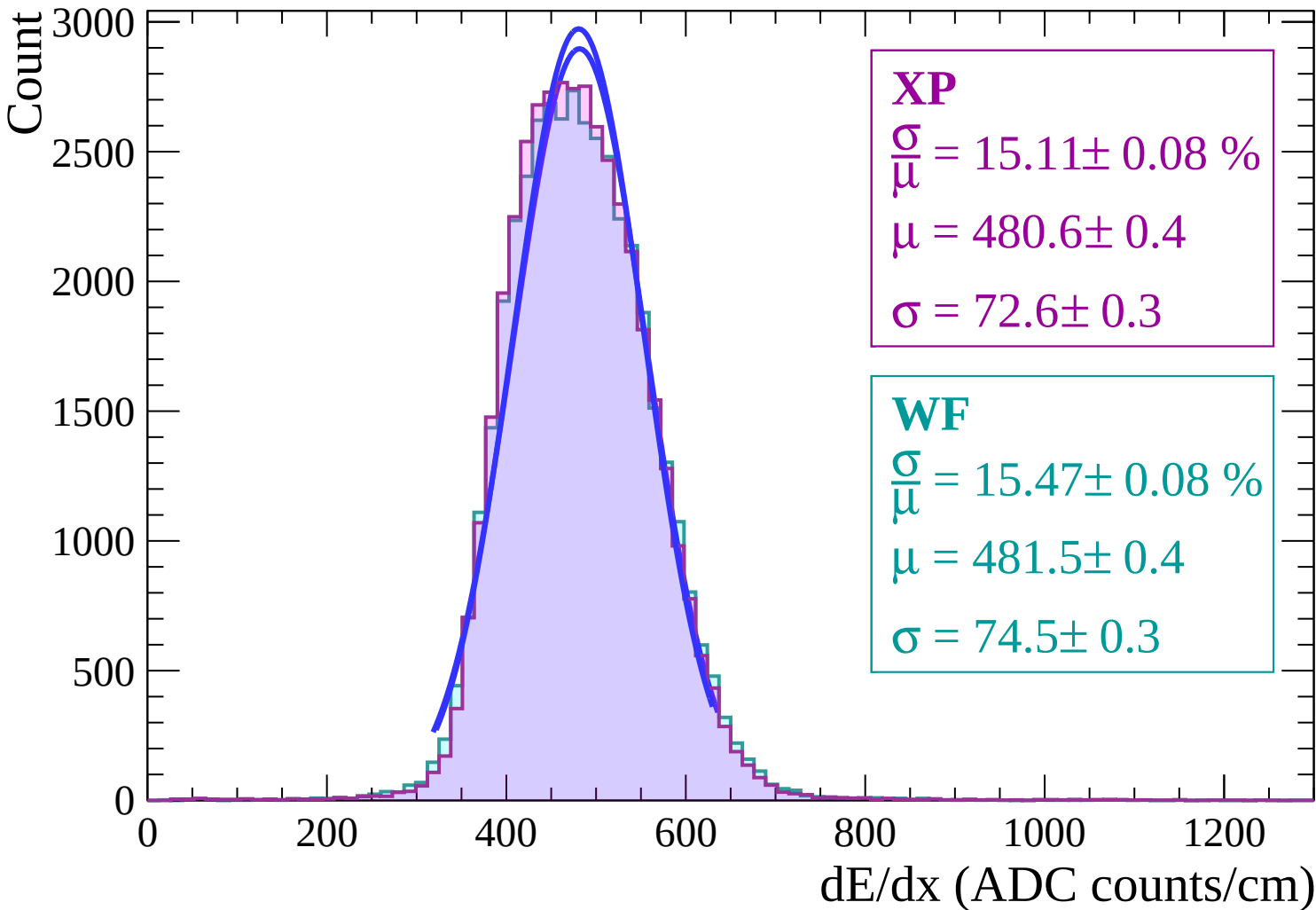
$$\mu = 480.9 \pm 0.4$$

$$\sigma = 74.5 \pm 0.3$$



dE/dx (ADC counts/cm)

Energy loss | $10 < \theta < 12$



Energy loss | $12 < \theta < 14$

Count

2500

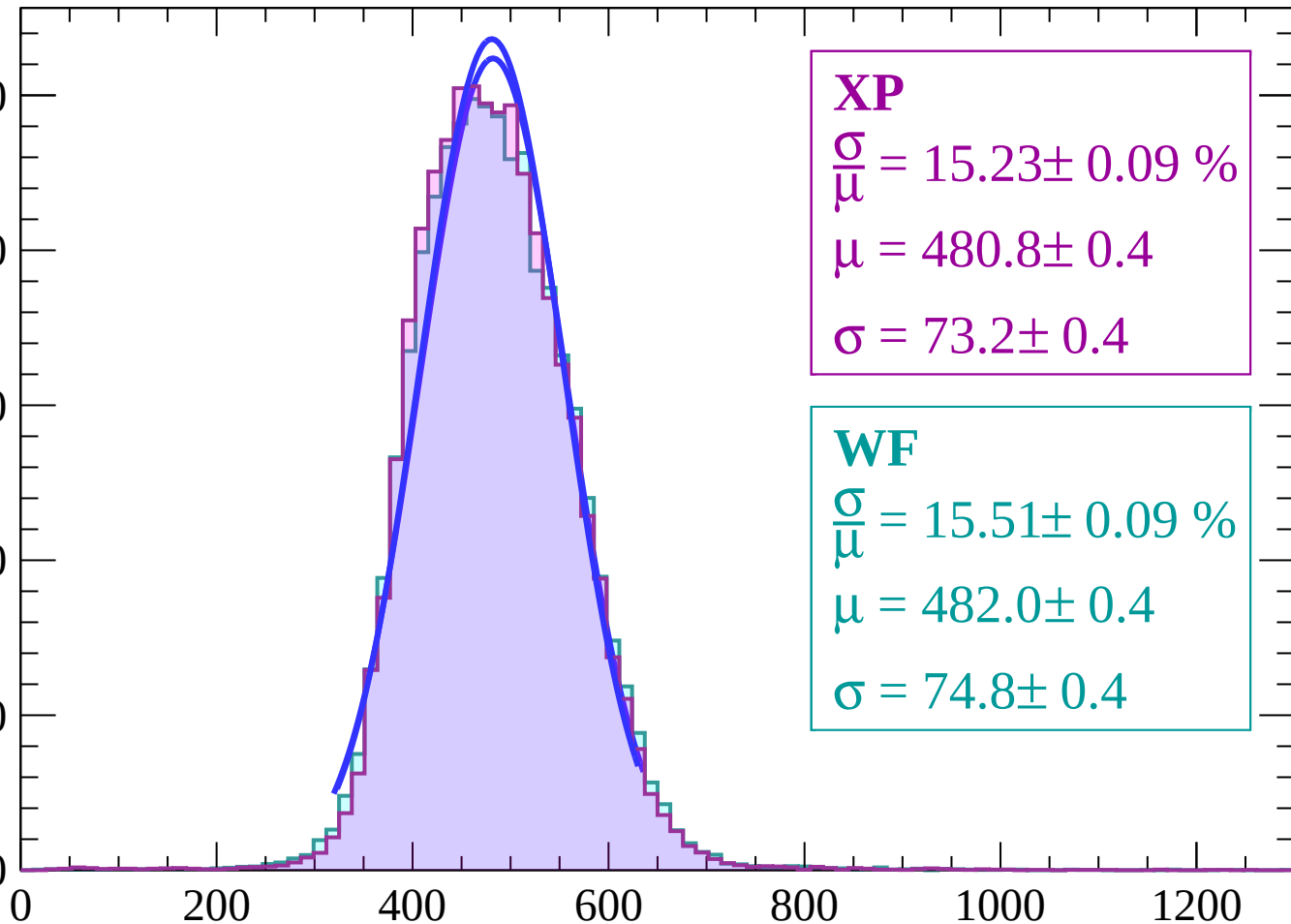
2000

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 15.23 \pm 0.09 \%$$

$$\mu = 480.8 \pm 0.4$$

$$\sigma = 73.2 \pm 0.4$$

WF

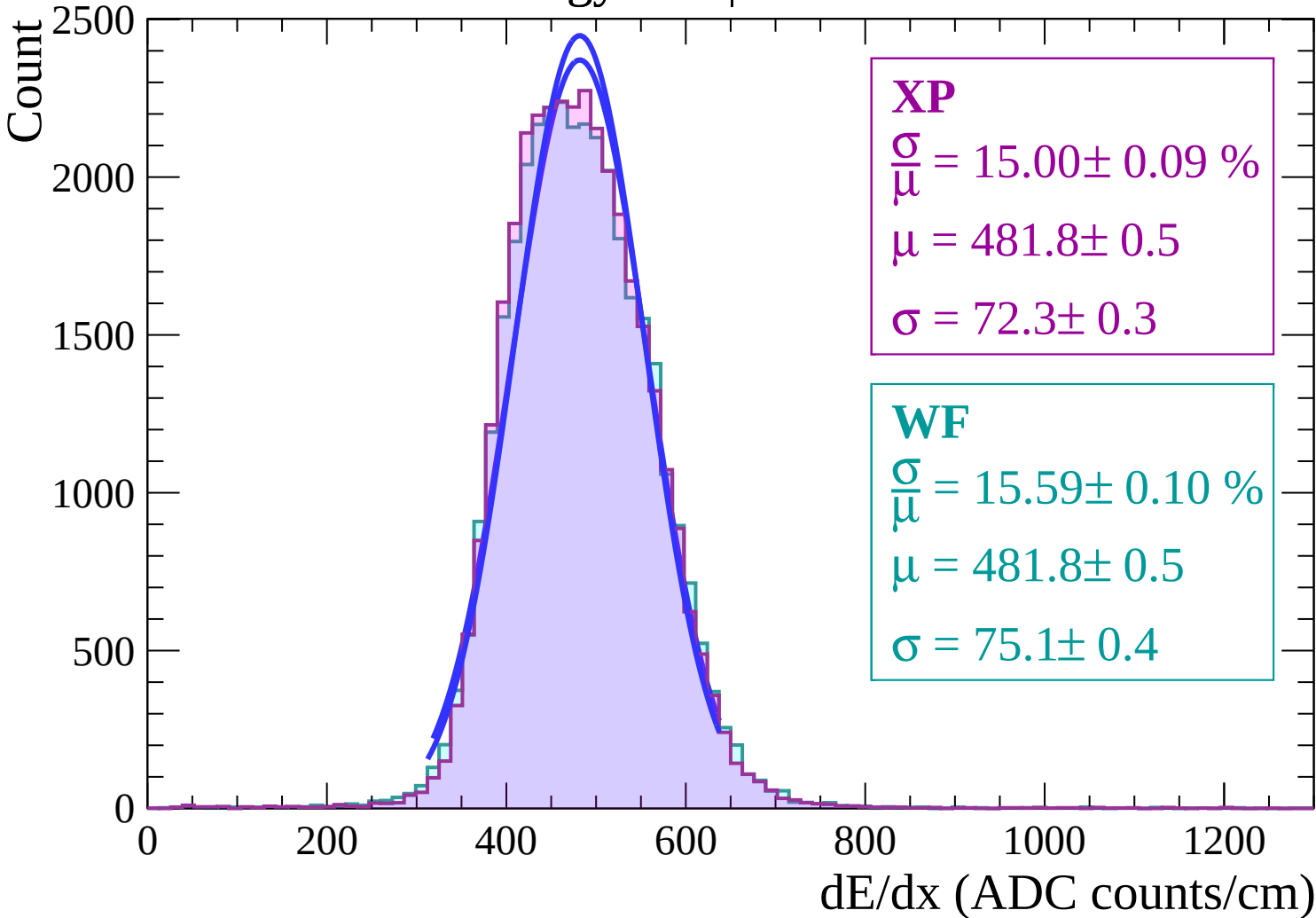
$$\frac{\sigma}{\mu} = 15.51 \pm 0.09 \%$$

$$\mu = 482.0 \pm 0.4$$

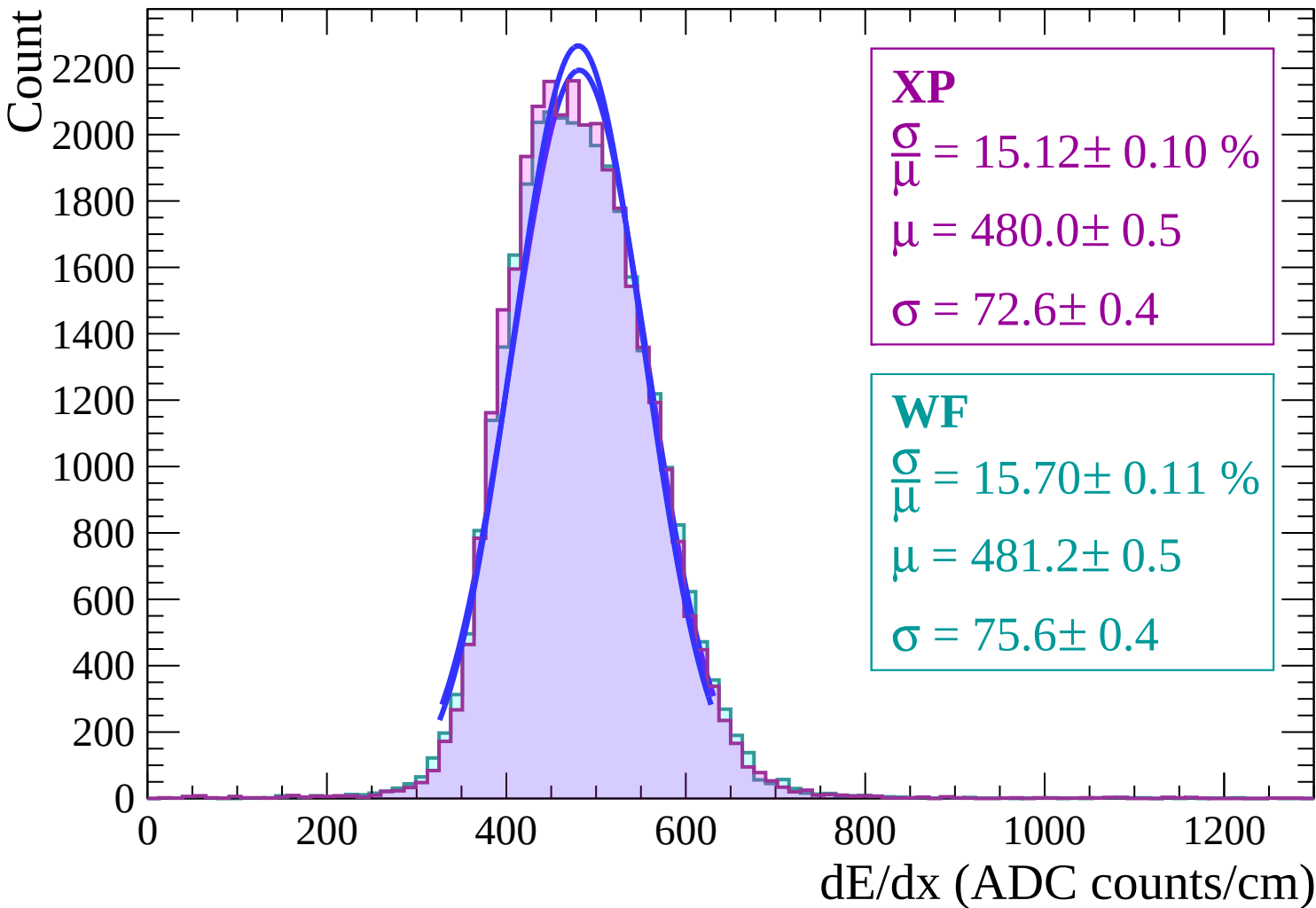
$$\sigma = 74.8 \pm 0.4$$

dE/dx (ADC counts/cm)

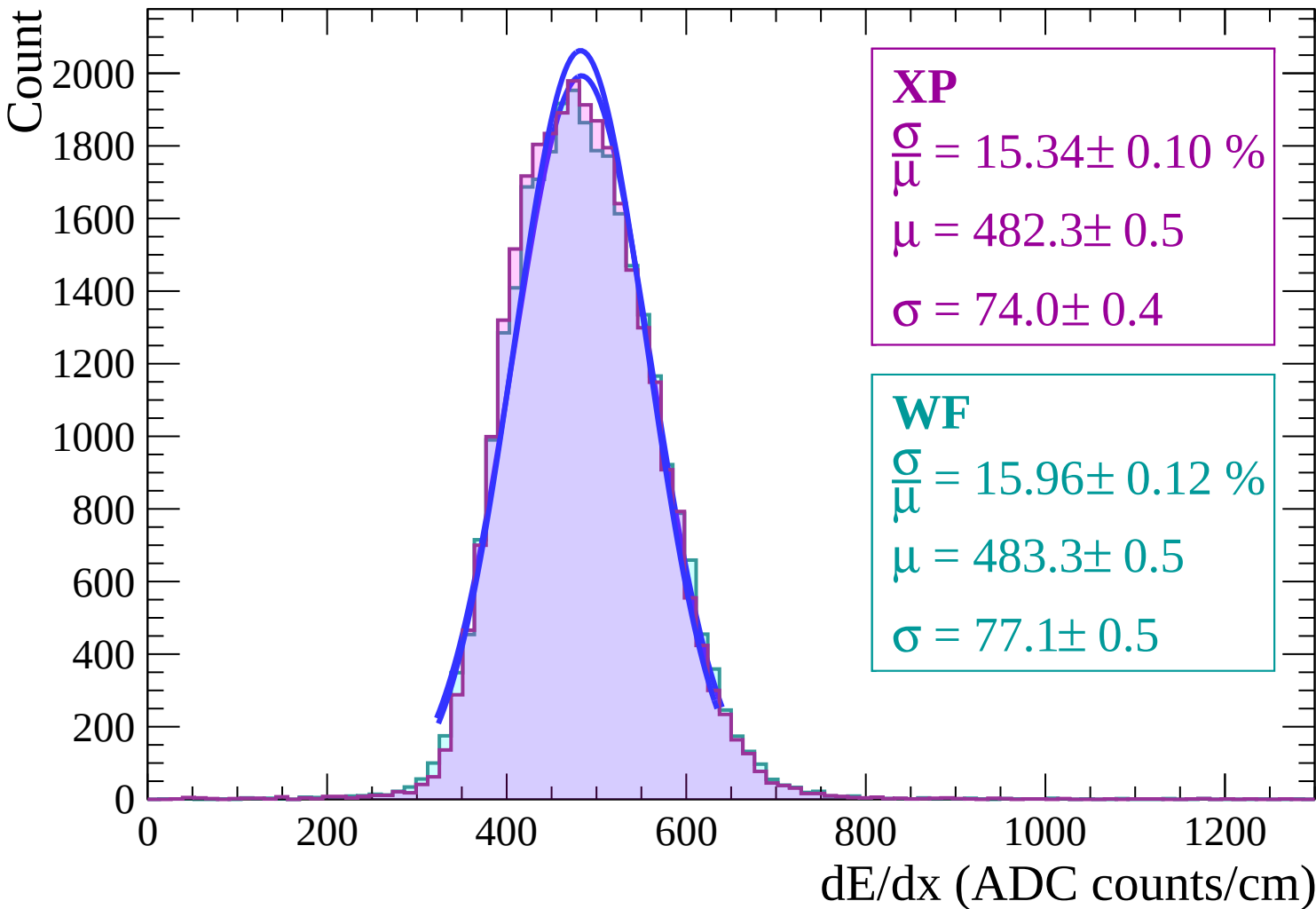
Energy loss | $14 < \theta < 16$



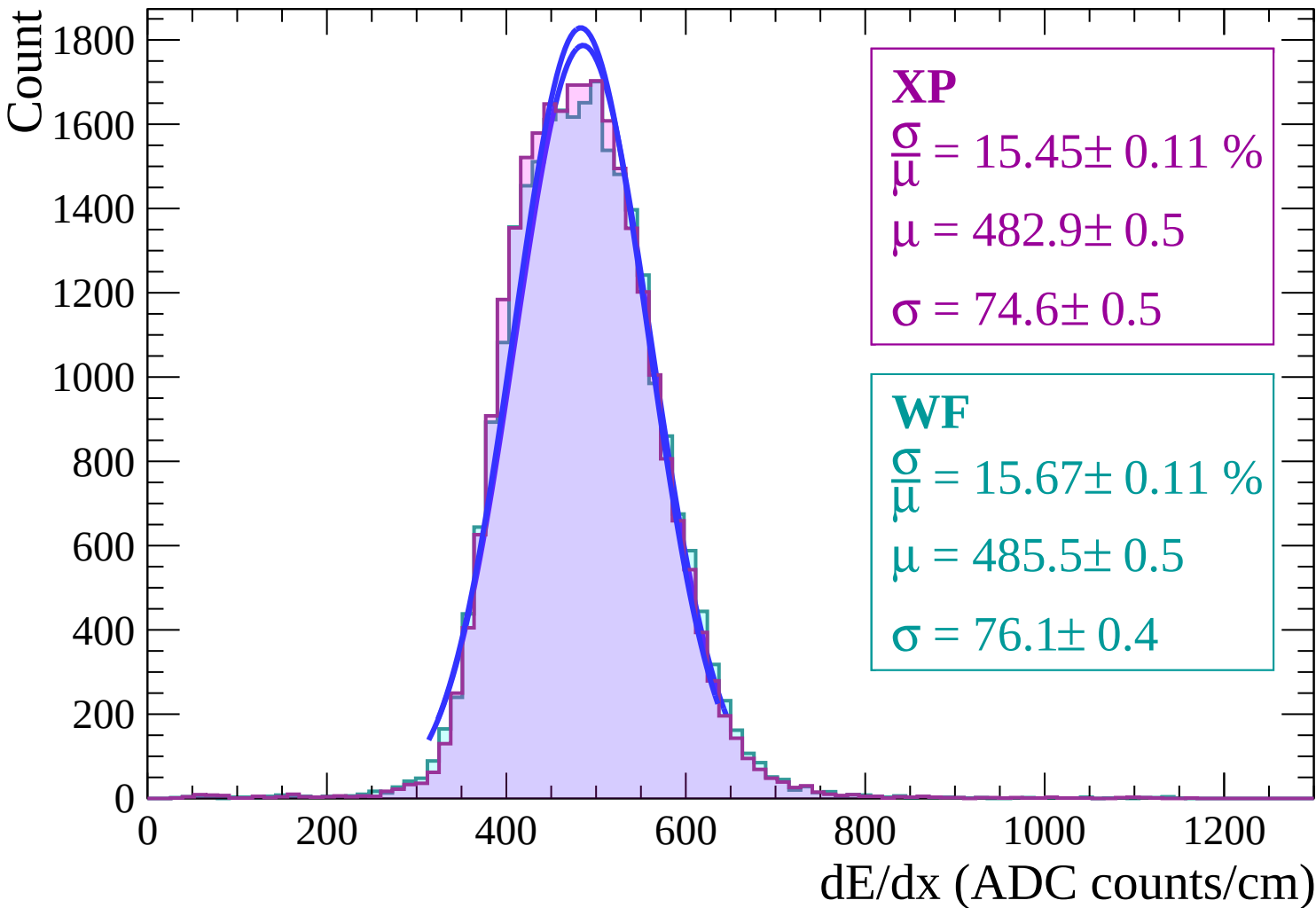
Energy loss | $16 < \theta < 18$



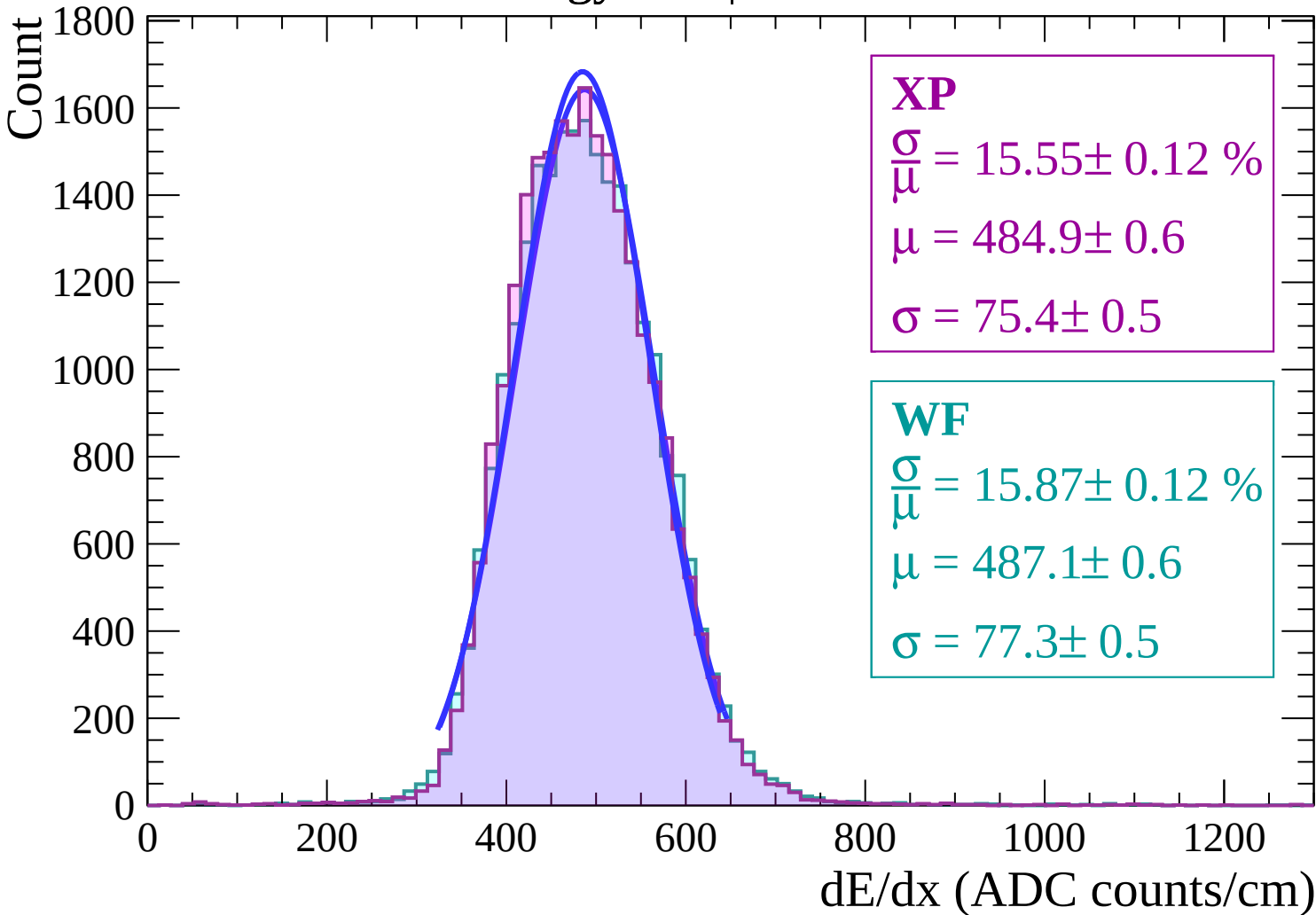
Energy loss | $18 < \theta < 20$



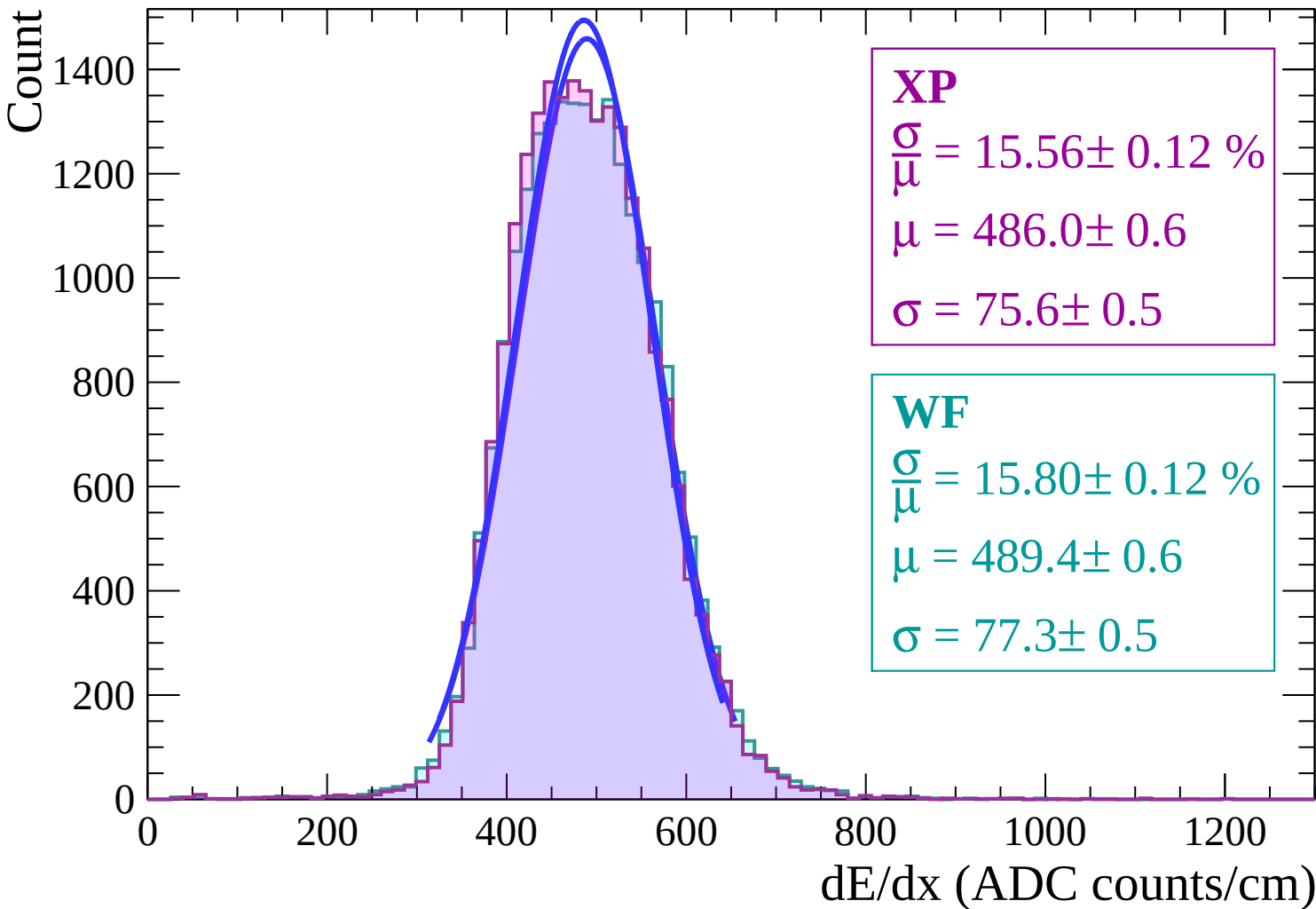
Energy loss | $20 < \theta < 22$



Energy loss | $22 < \theta < 24$

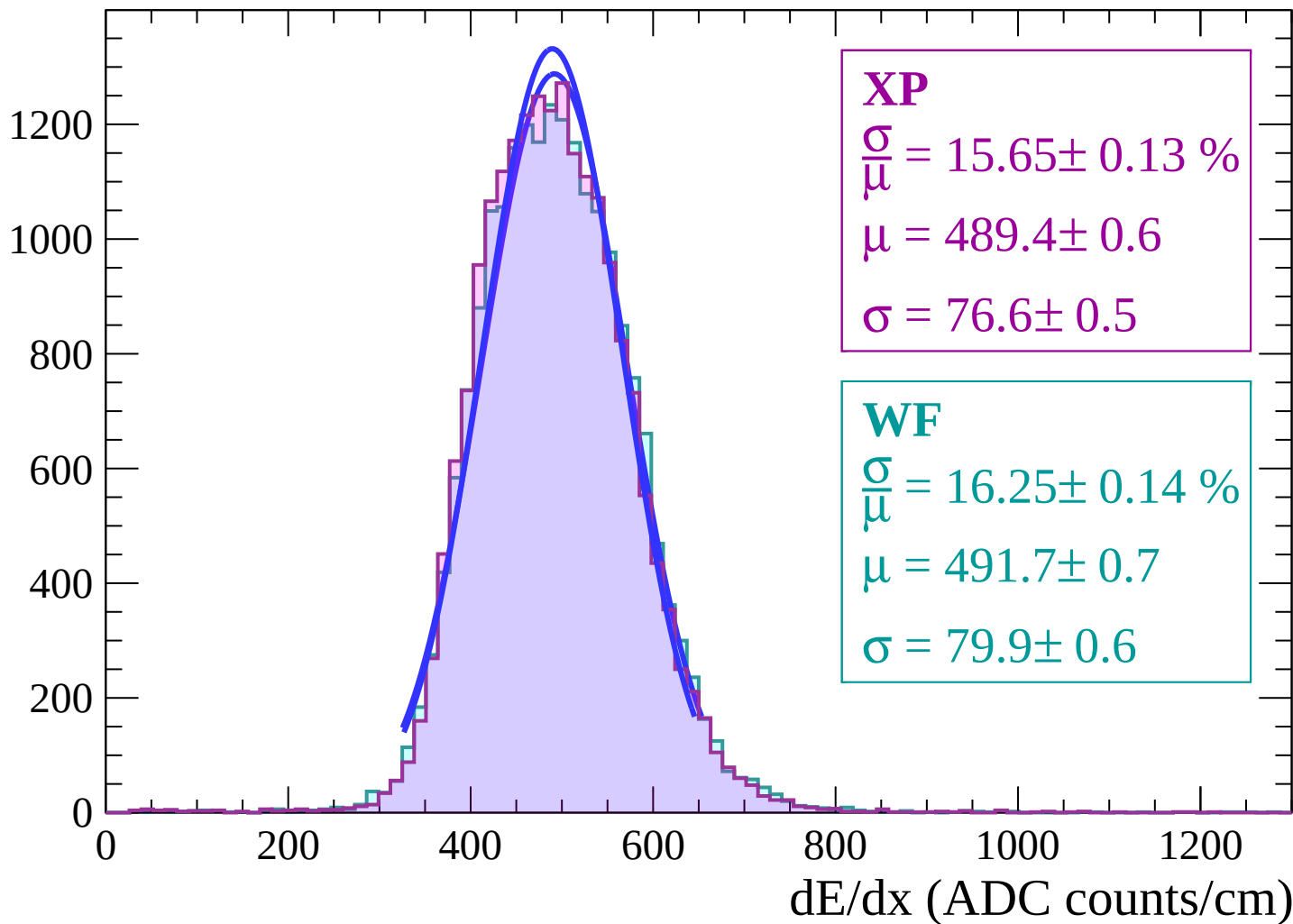


Energy loss | $24 < \theta < 26$



Energy loss | $26 < \theta < 28$

Count



Energy loss | $28 < \theta < 30$

Count

1000

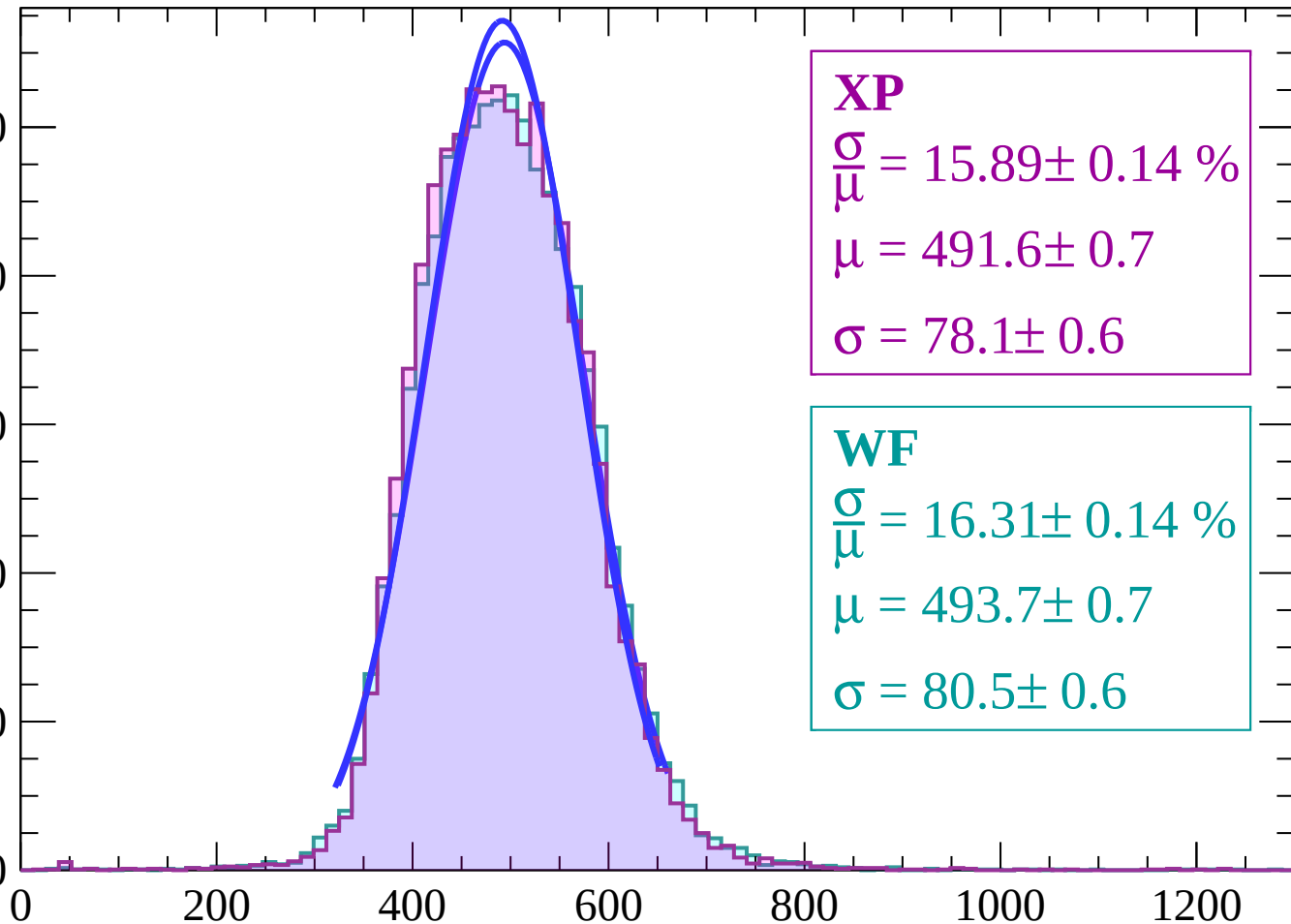
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 15.89 \pm 0.14 \%$$

$$\mu = 491.6 \pm 0.7$$

$$\sigma = 78.1 \pm 0.6$$

WF

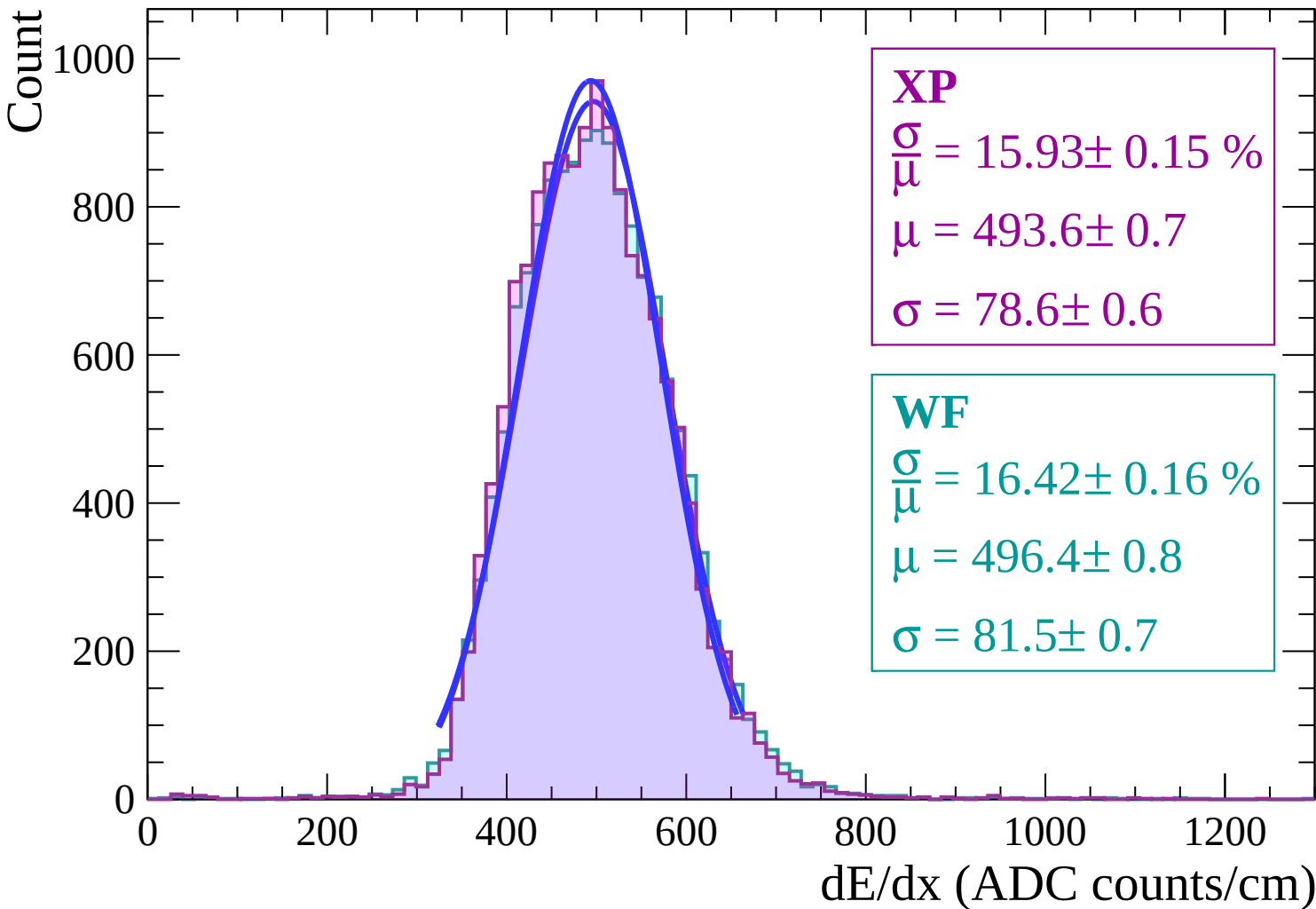
$$\frac{\sigma}{\mu} = 16.31 \pm 0.14 \%$$

$$\mu = 493.7 \pm 0.7$$

$$\sigma = 80.5 \pm 0.6$$

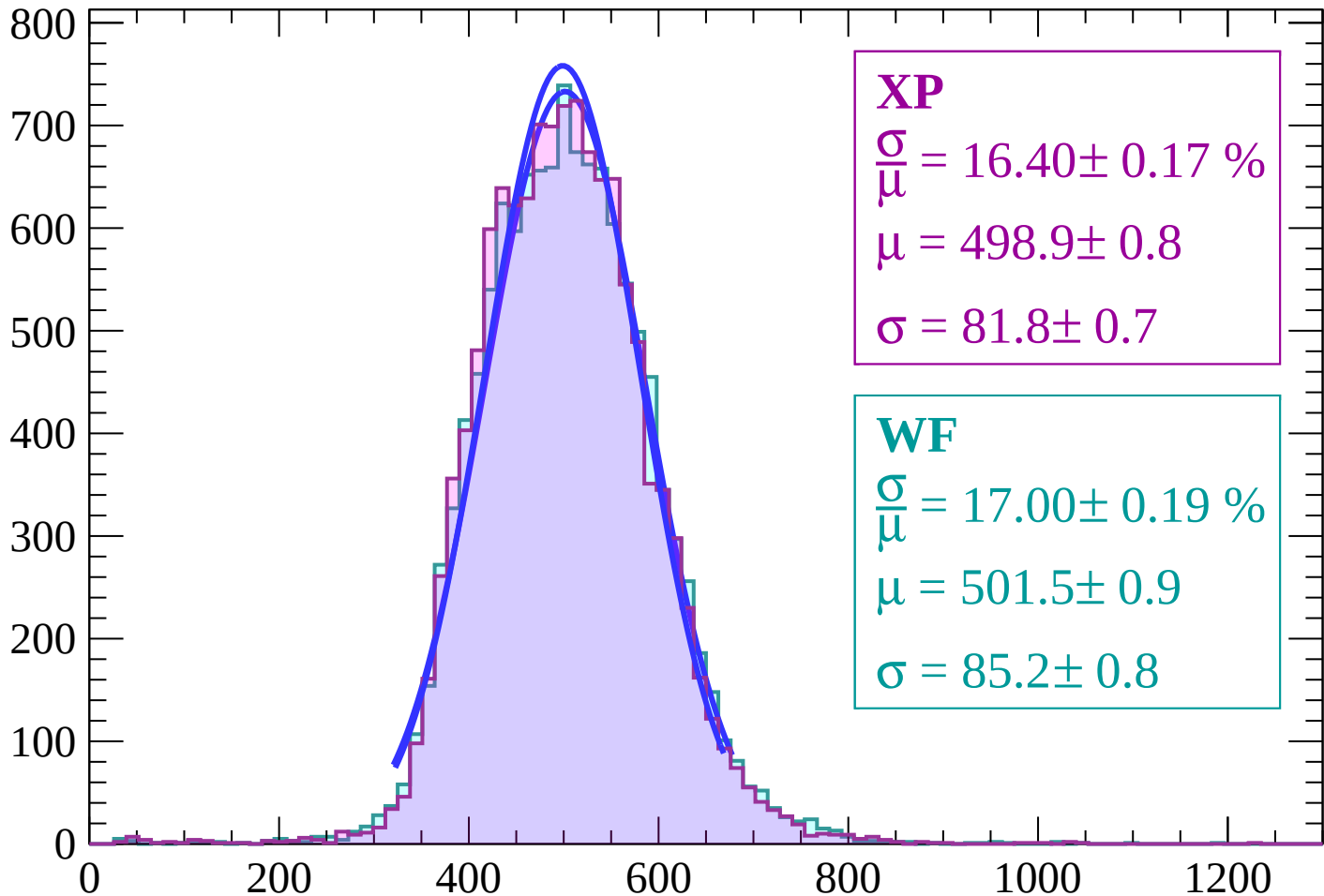
dE/dx (ADC counts/cm)

Energy loss | $30 < \theta < 32$



Energy loss | $32 < \theta < 34$

Count



XP

$$\frac{\sigma}{\mu} = 16.40 \pm 0.17 \%$$

$$\mu = 498.9 \pm 0.8$$

$$\sigma = 81.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 17.00 \pm 0.19 \%$$

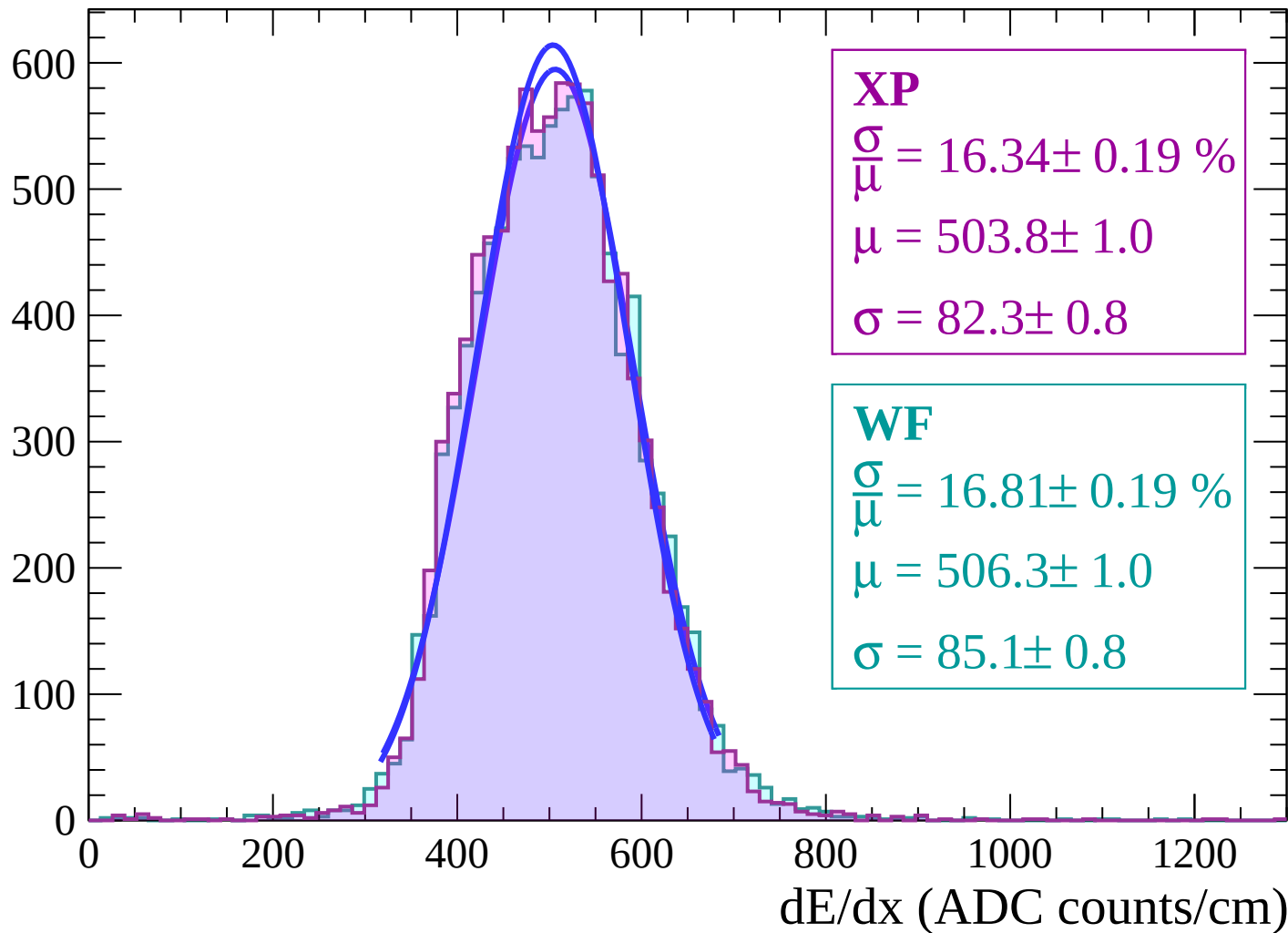
$$\mu = 501.5 \pm 0.9$$

$$\sigma = 85.2 \pm 0.8$$

dE/dx (ADC counts/cm)

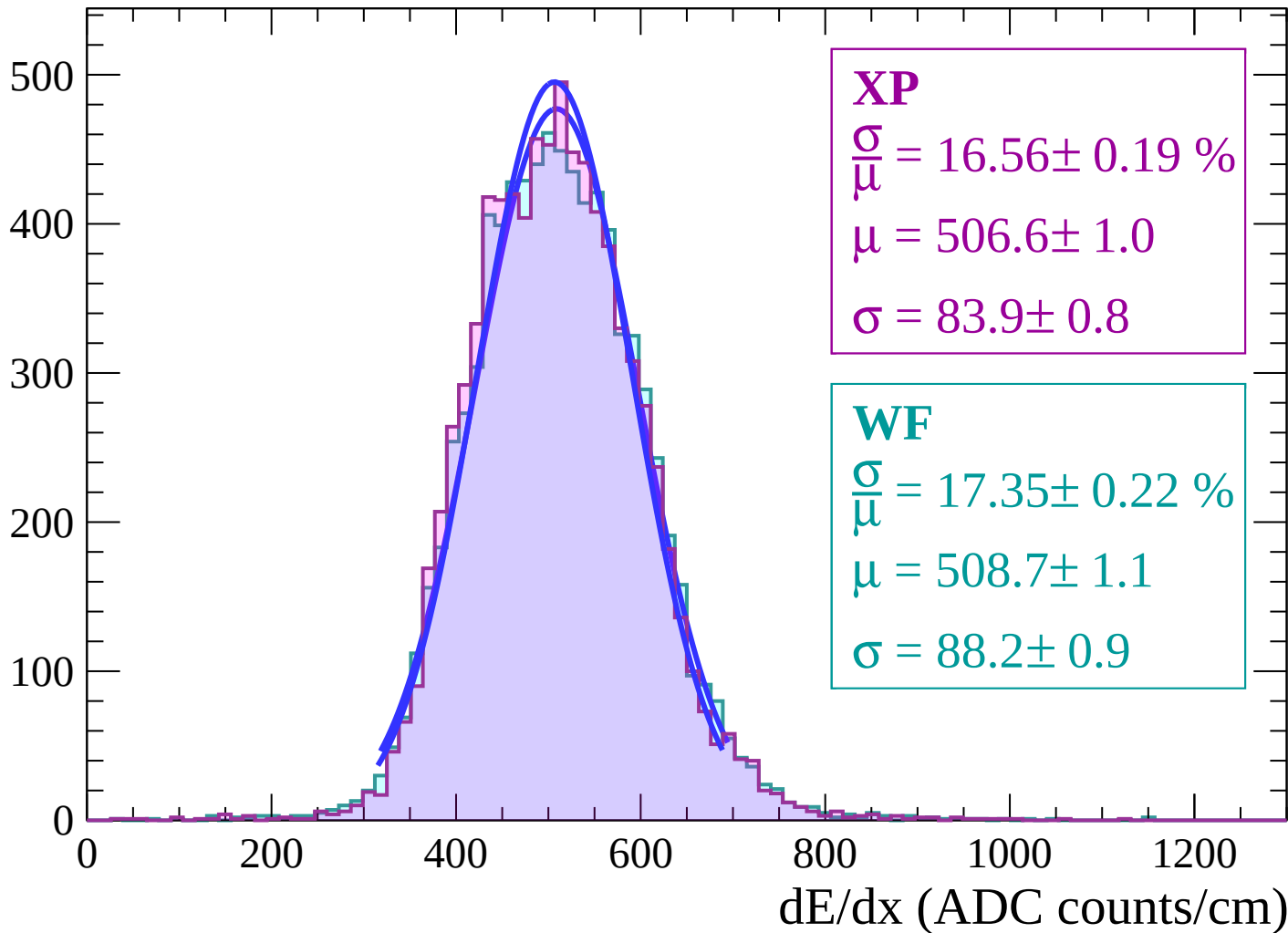
Energy loss | $34 < \theta < 36$

Count



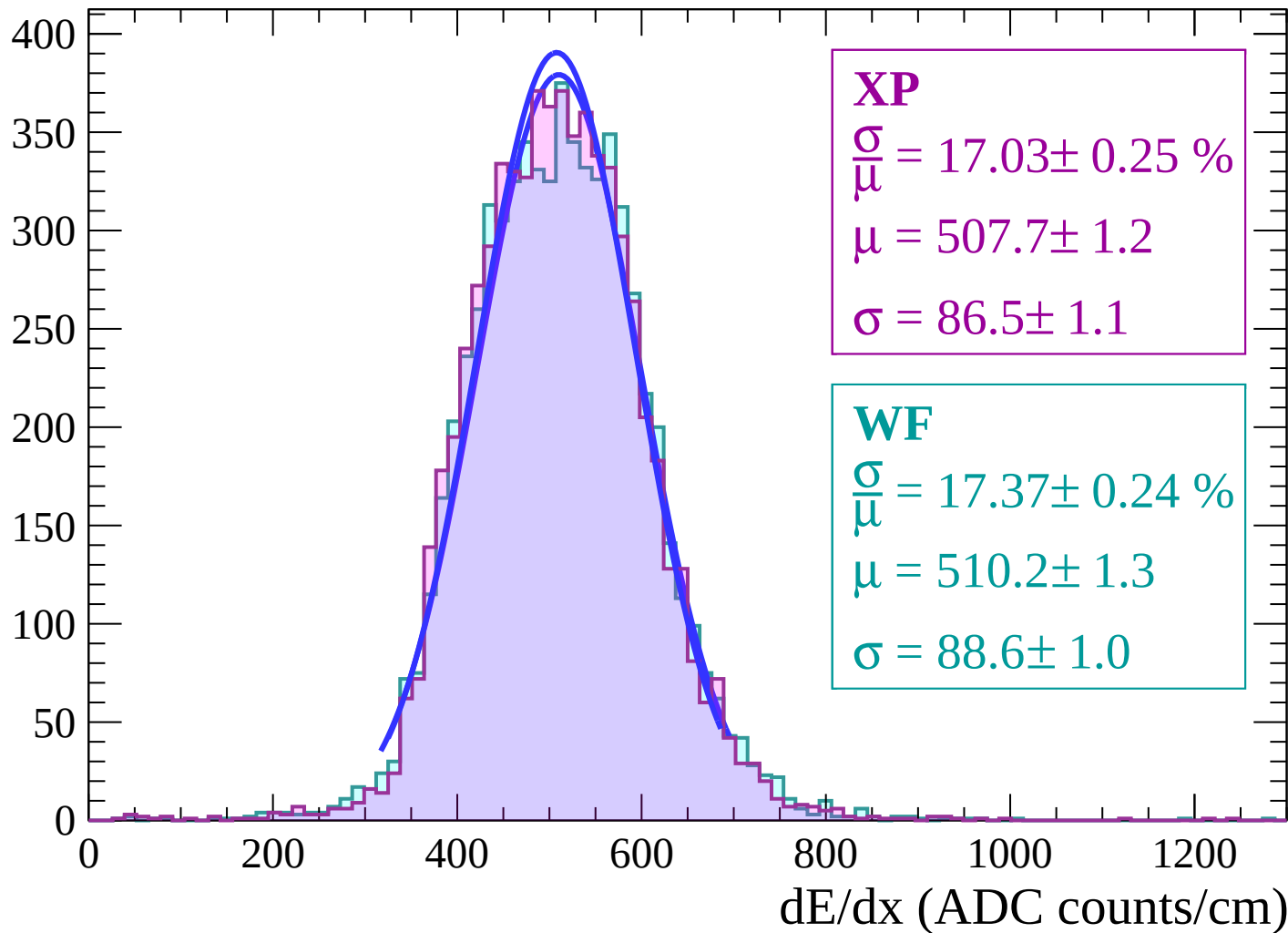
Energy loss | $36 < \theta < 38$

Count



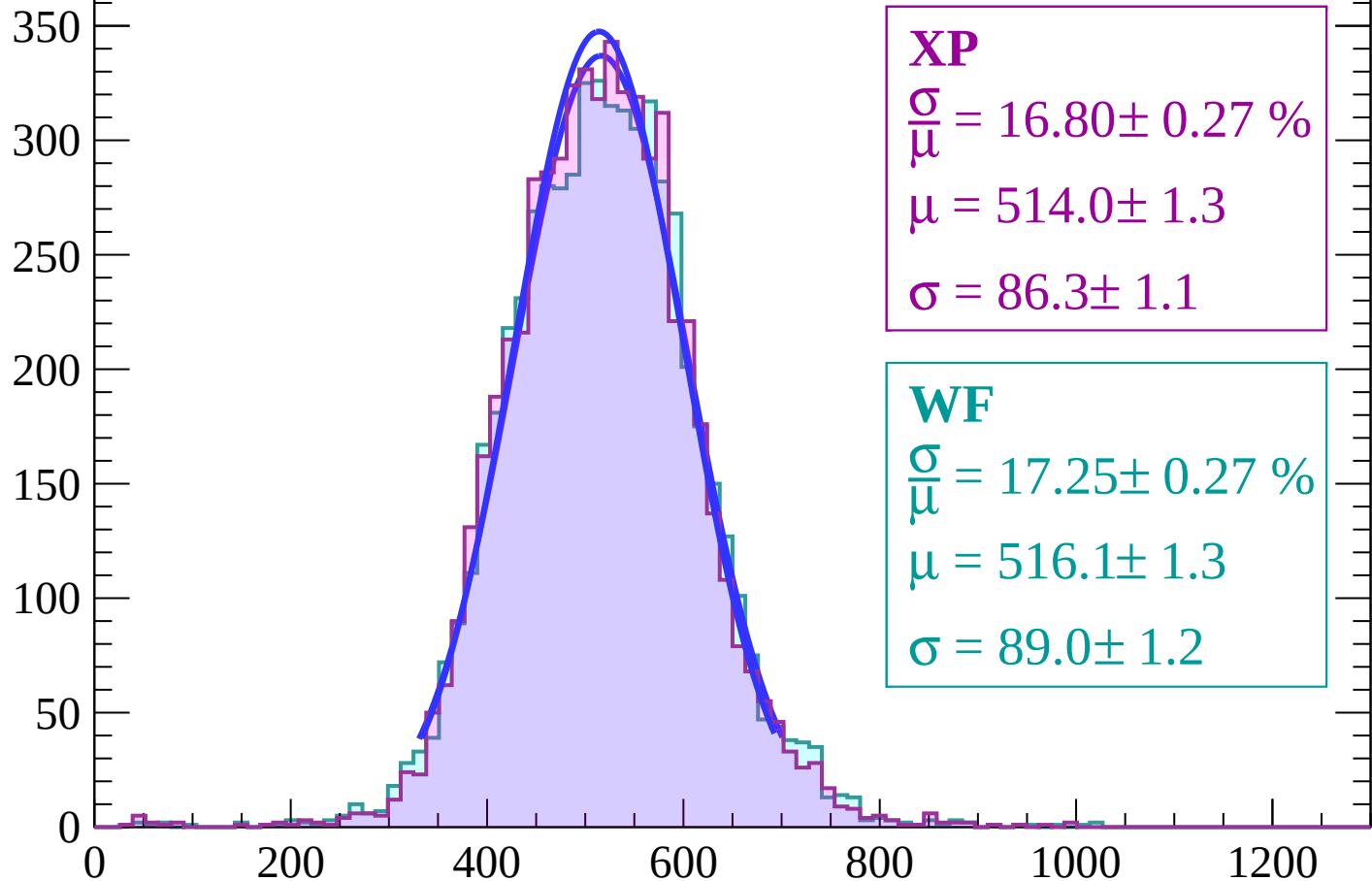
Energy loss | $38 < \theta < 40$

Count



Energy loss | $40 < \theta < 42$

Count



XP

$$\frac{\sigma}{\mu} = 16.80 \pm 0.27 \%$$

$$\mu = 514.0 \pm 1.3$$

$$\sigma = 86.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 17.25 \pm 0.27 \%$$

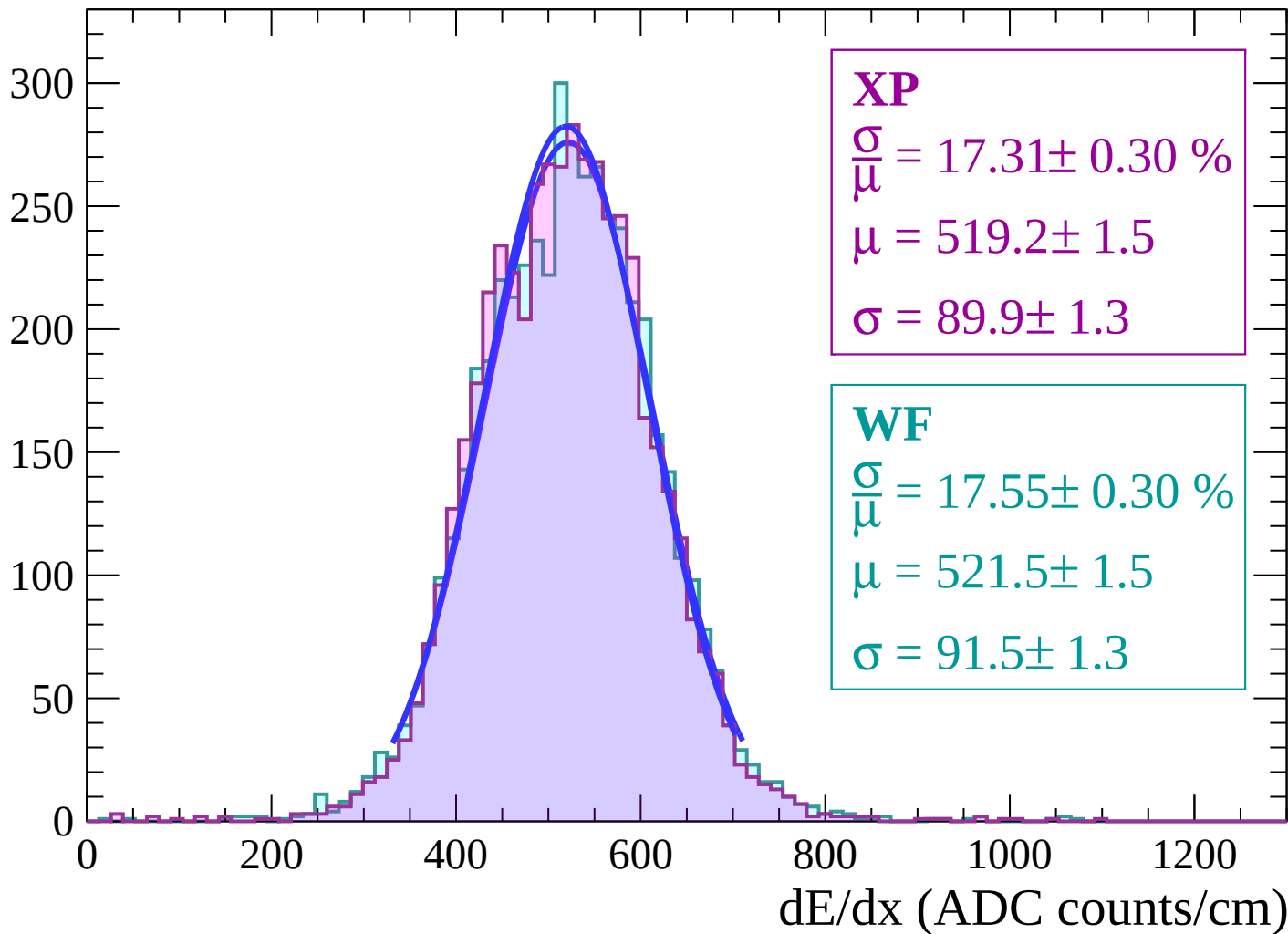
$$\mu = 516.1 \pm 1.3$$

$$\sigma = 89.0 \pm 1.2$$

dE/dx (ADC counts/cm)

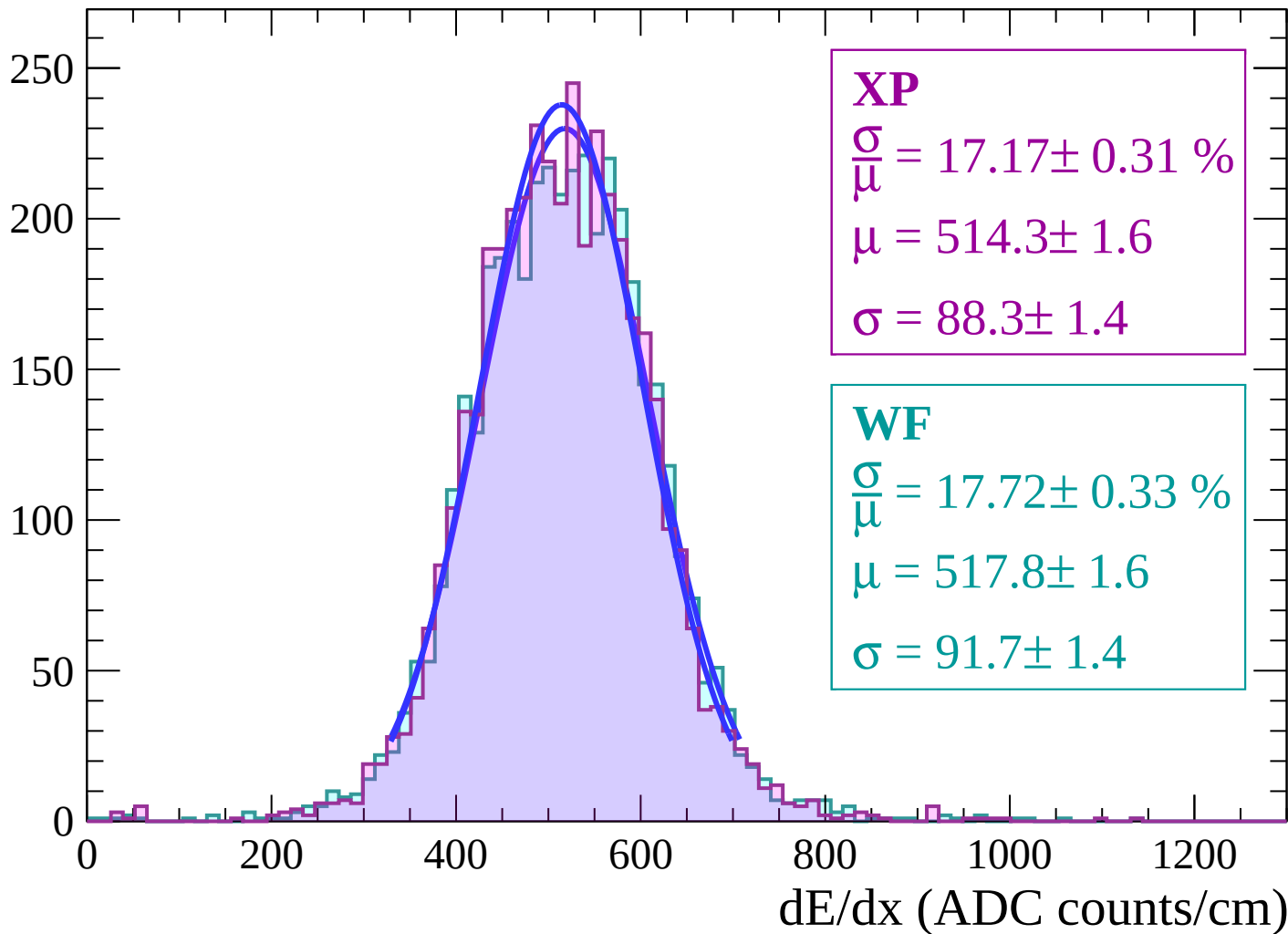
Energy loss | $42 < \theta < 44$

Count



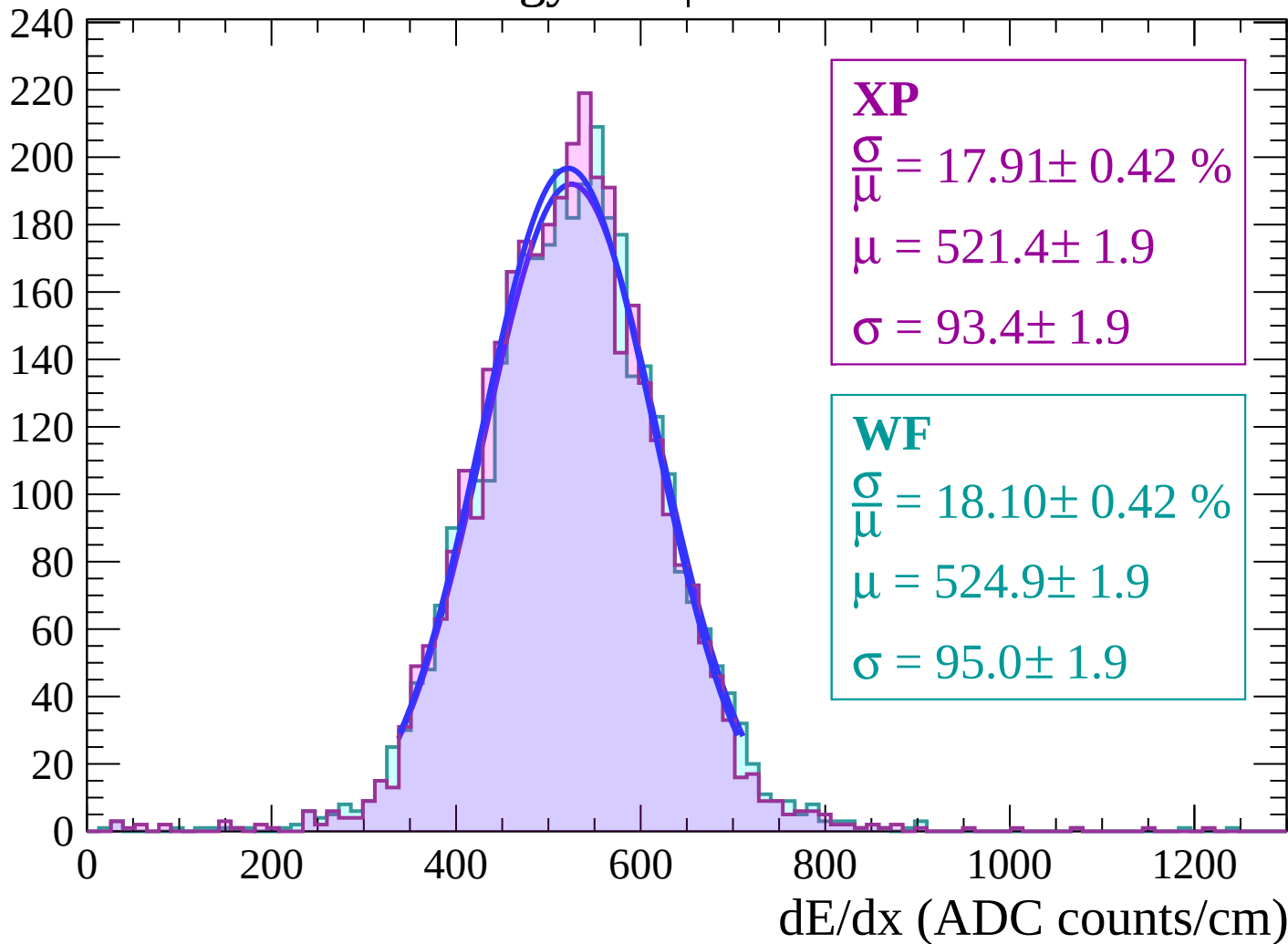
Energy loss | $44 < \theta < 46$

Count



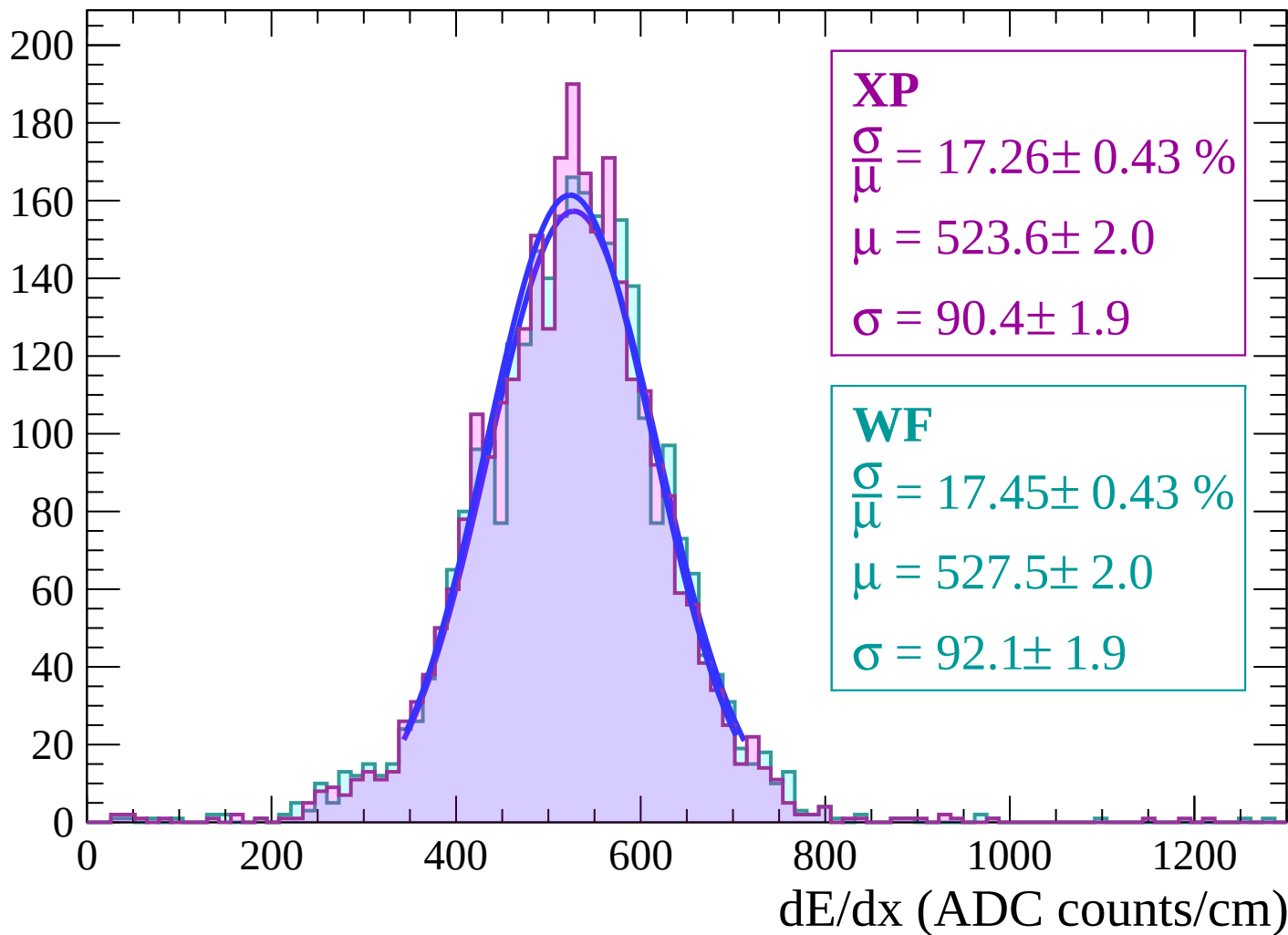
Energy loss | $46 < \theta < 48$

Count



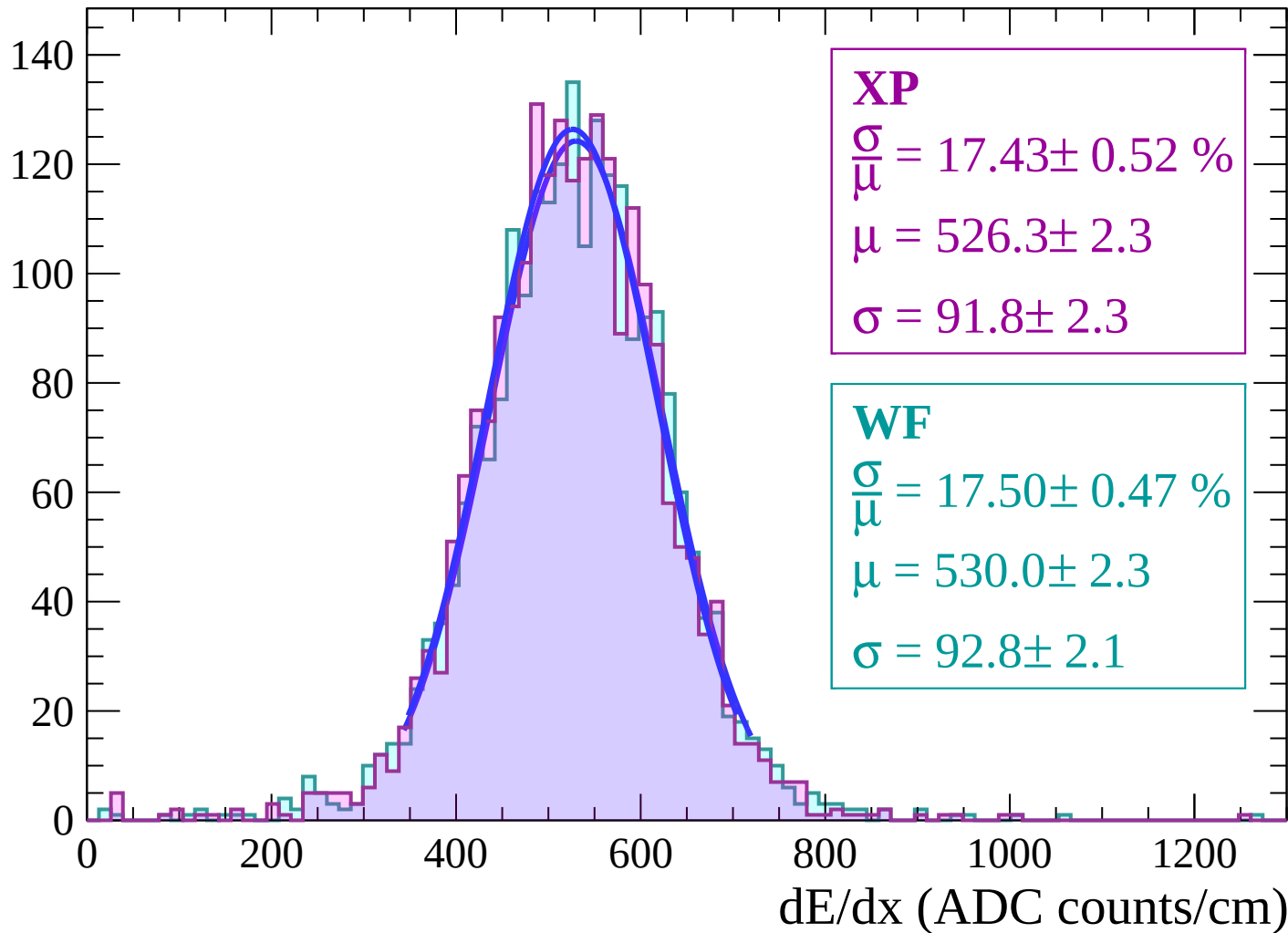
Energy loss | $48 < \theta < 50$

Count



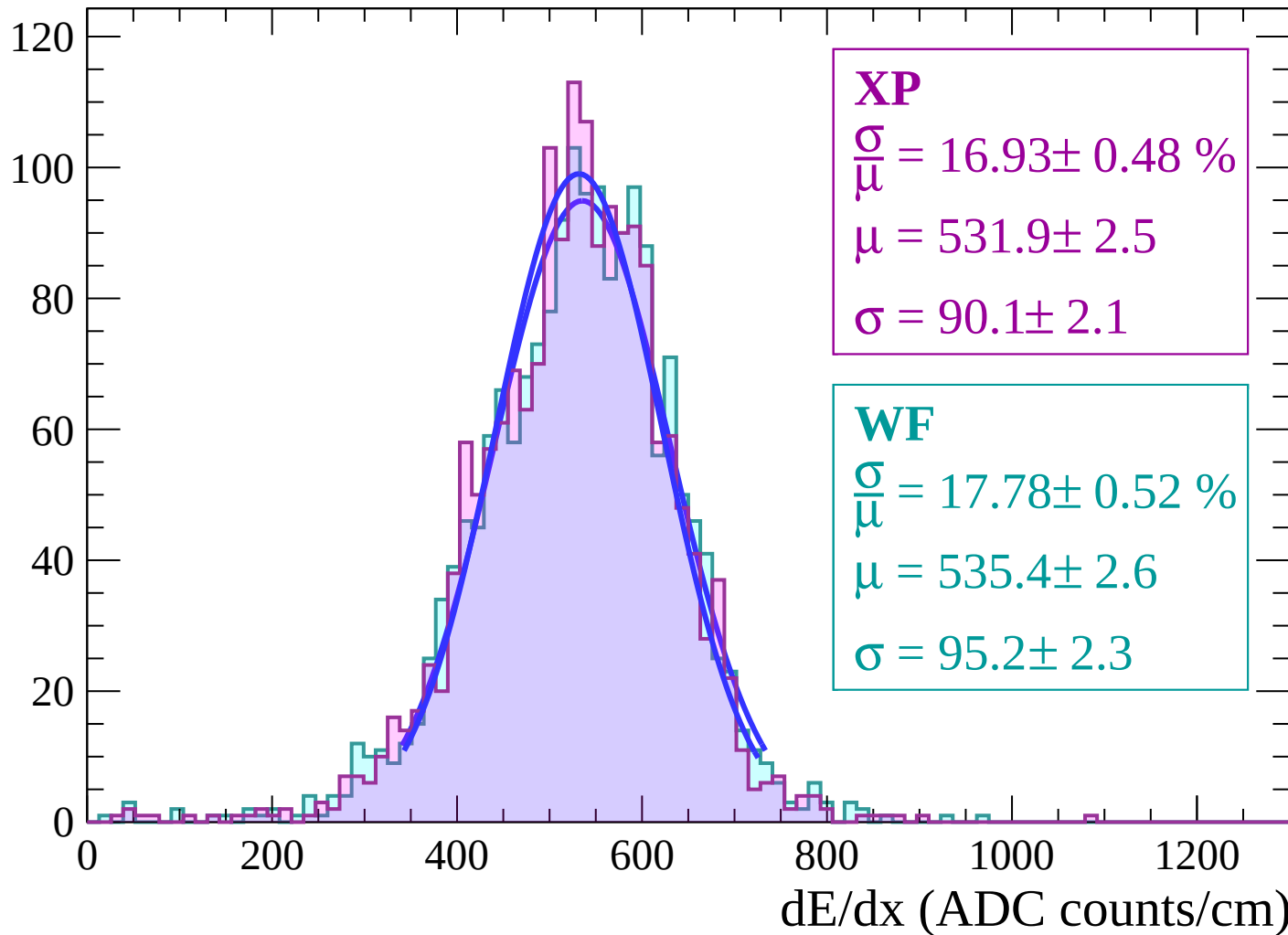
Energy loss | $50 < \theta < 52$

Count



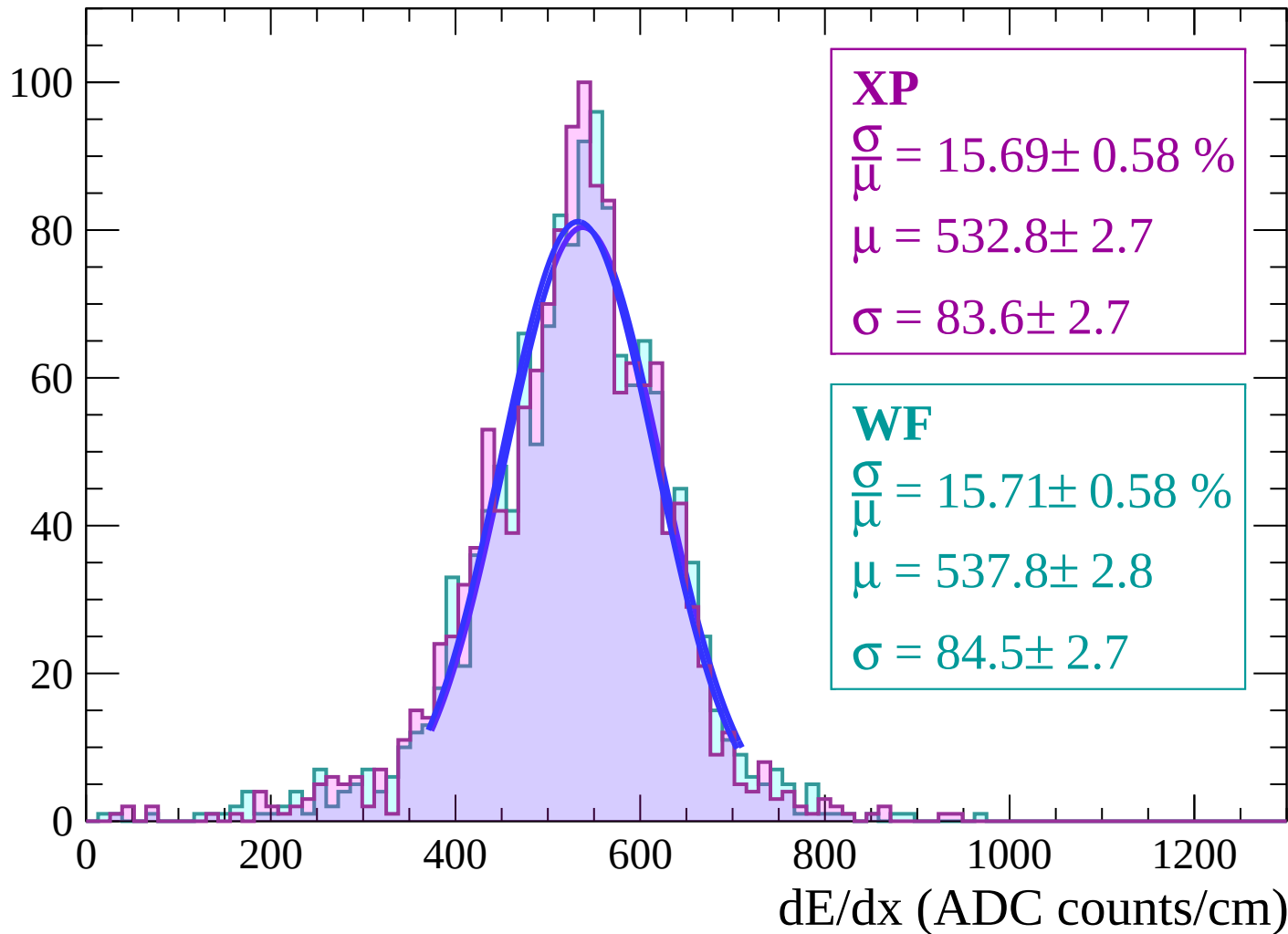
Energy loss | $52 < \theta < 54$

Count



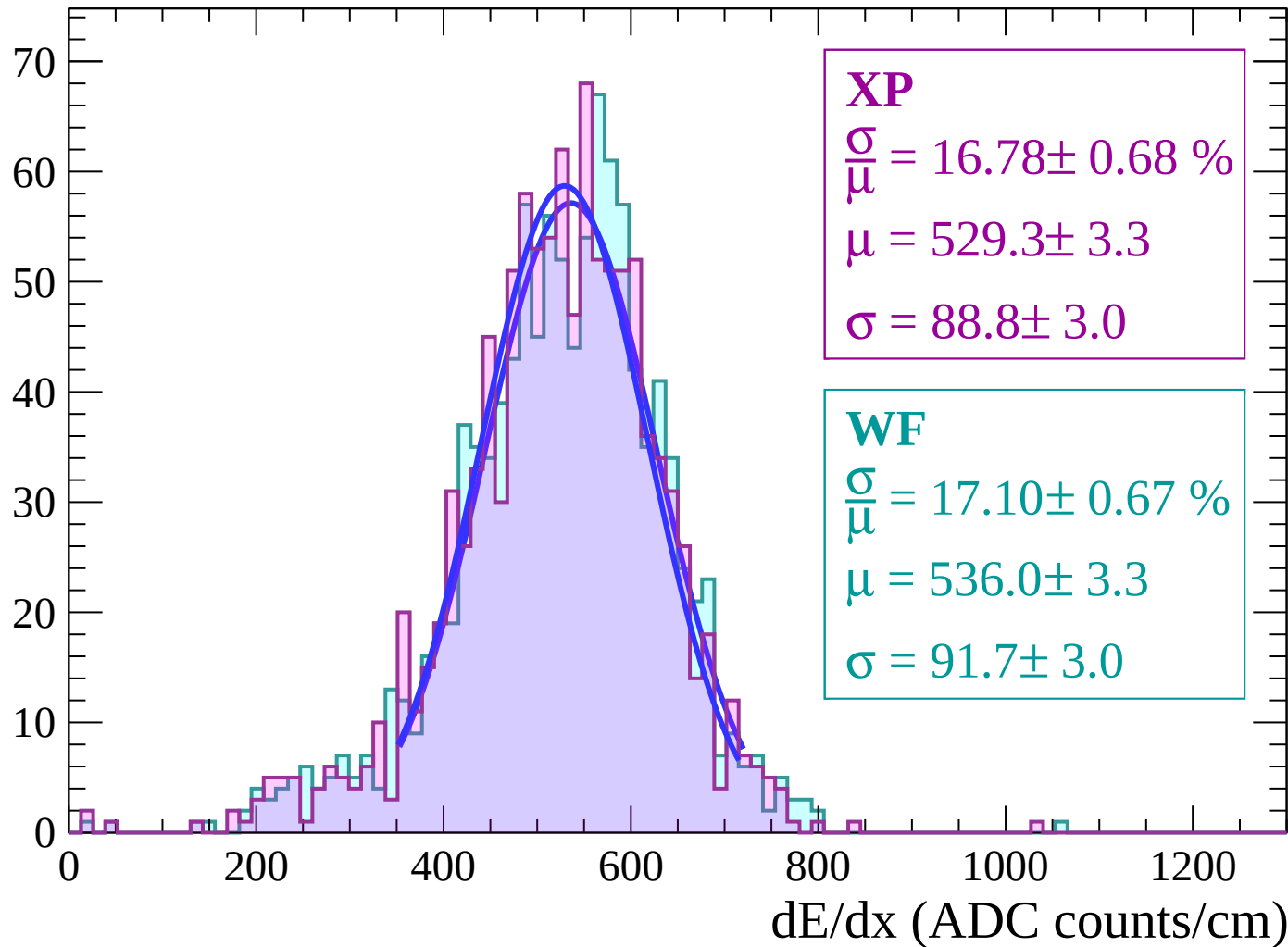
Energy loss | $54 < \theta < 56$

Count



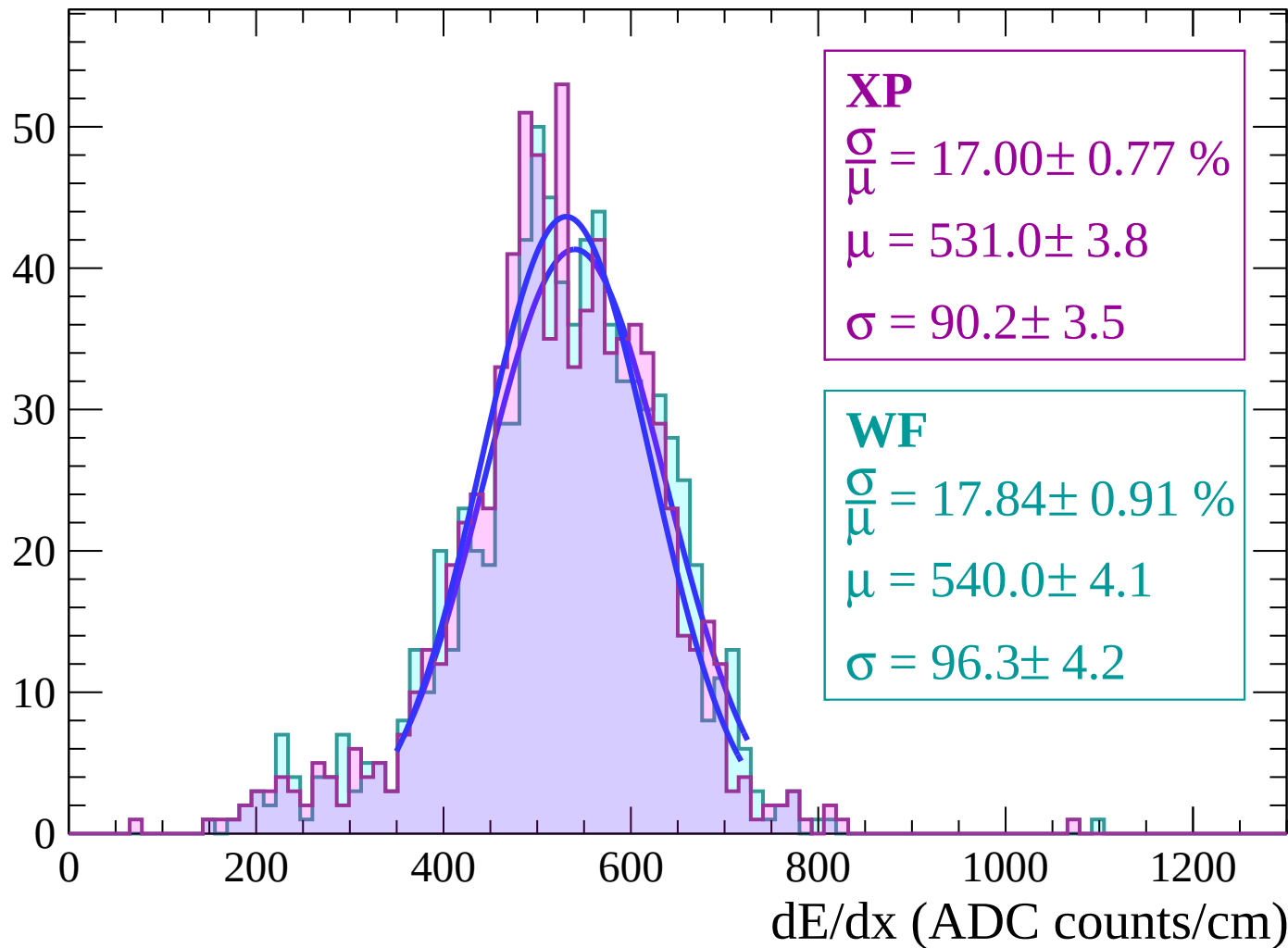
Energy loss | $56 < \theta < 58$

Count



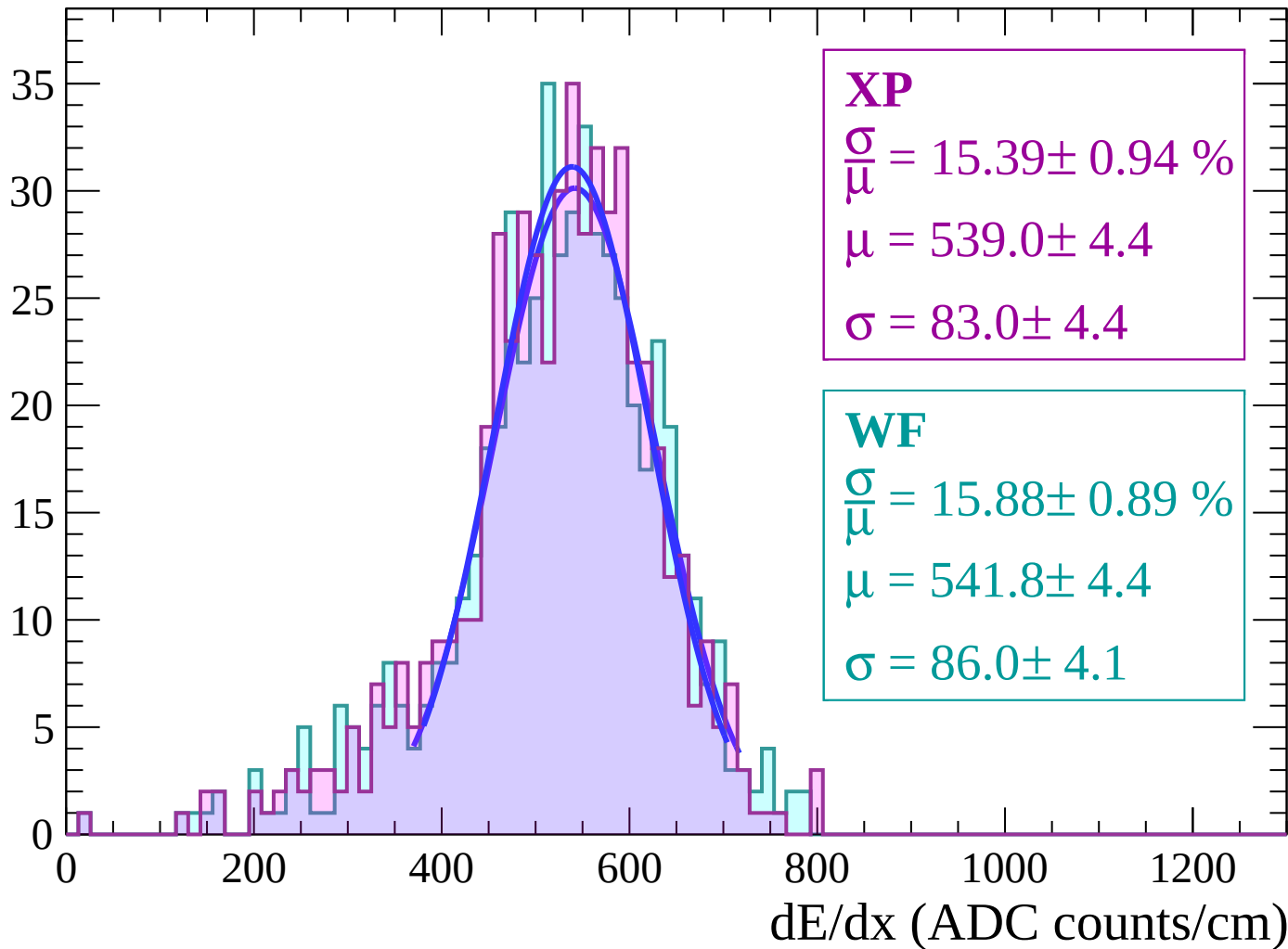
Energy loss | $58 < \theta < 60$

Count



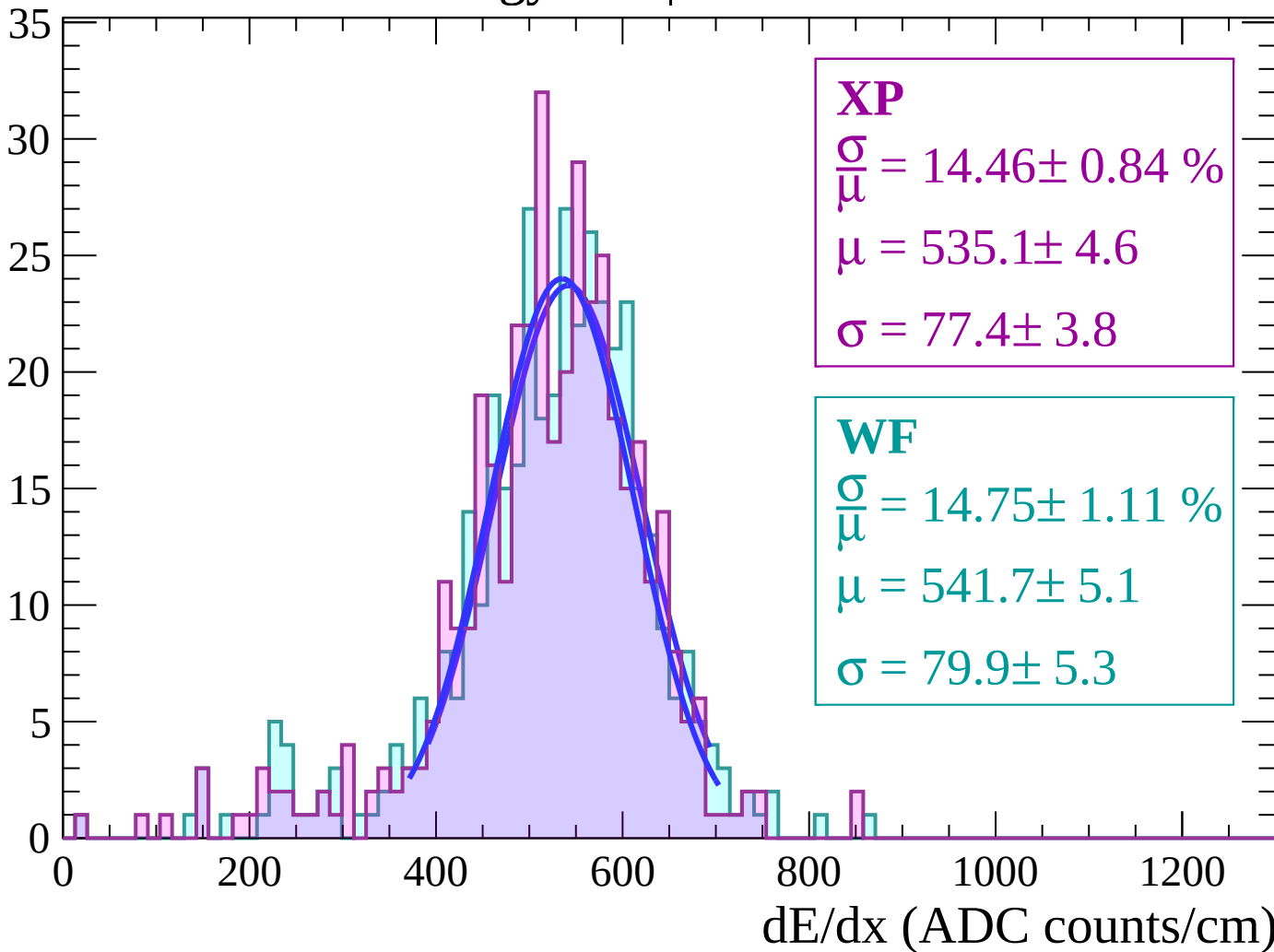
Energy loss | $60 < \theta < 62$

Count



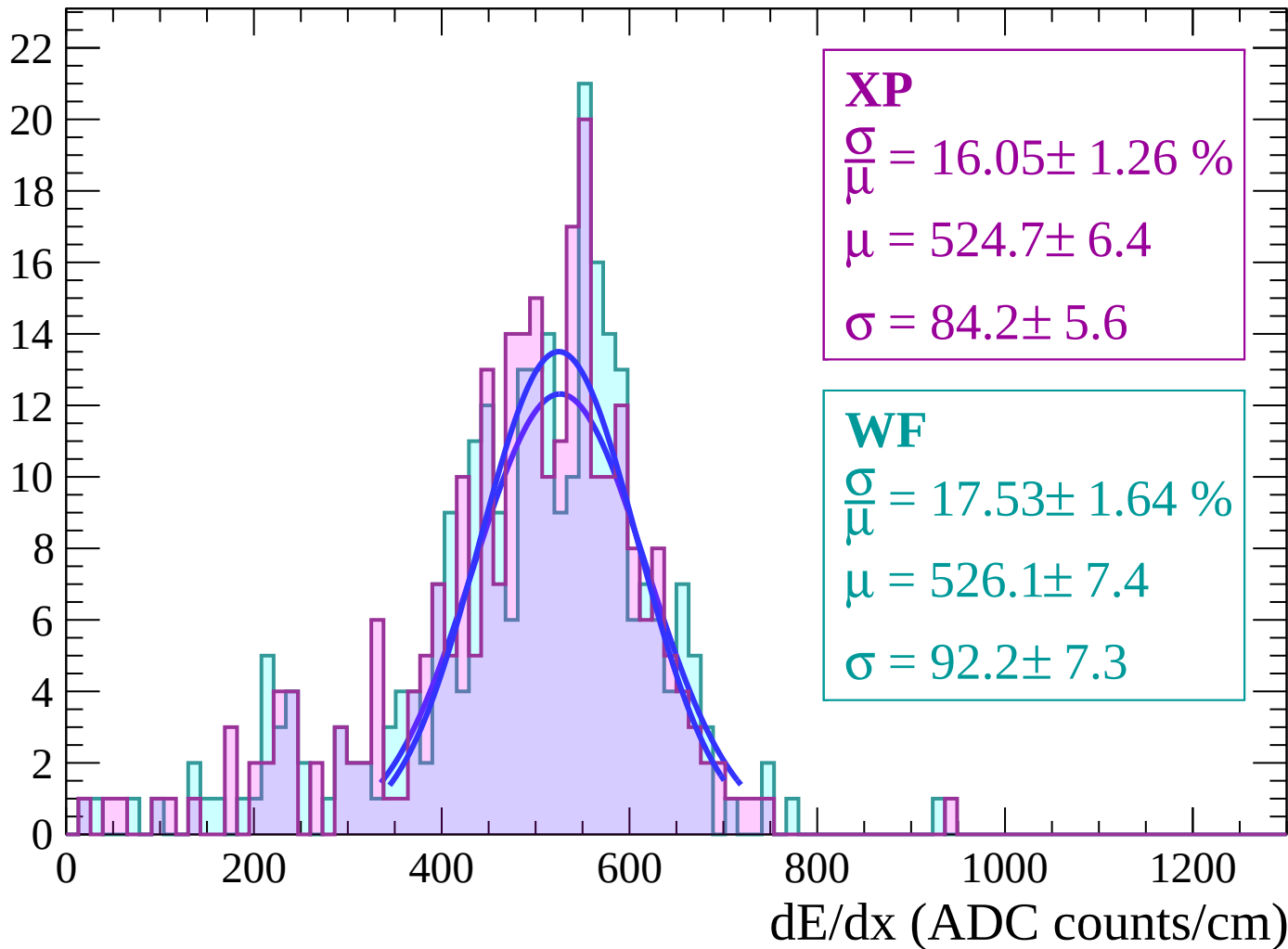
Energy loss | $62 < \theta < 64$

Count



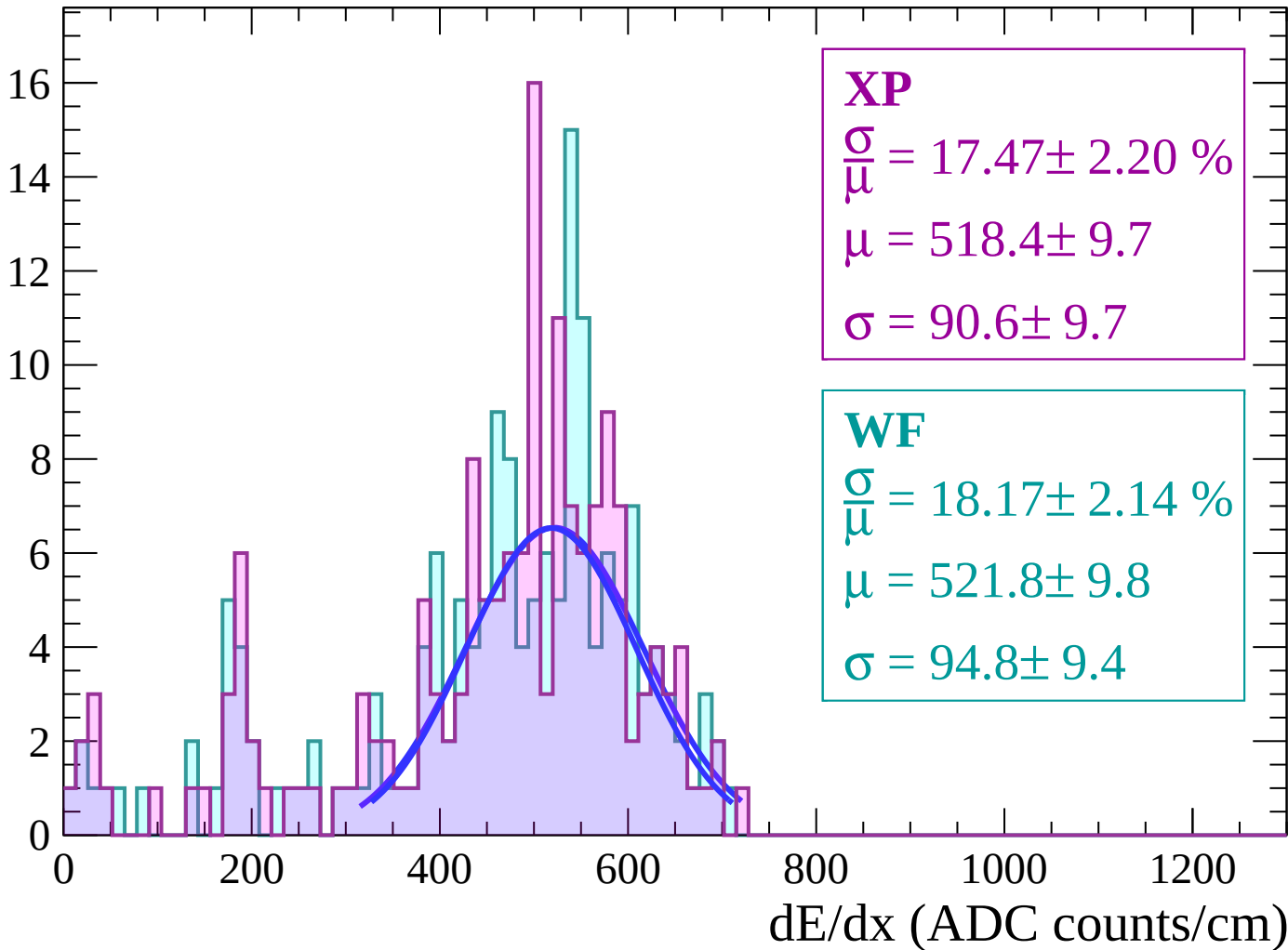
Energy loss | $64 < \theta < 66$

Count



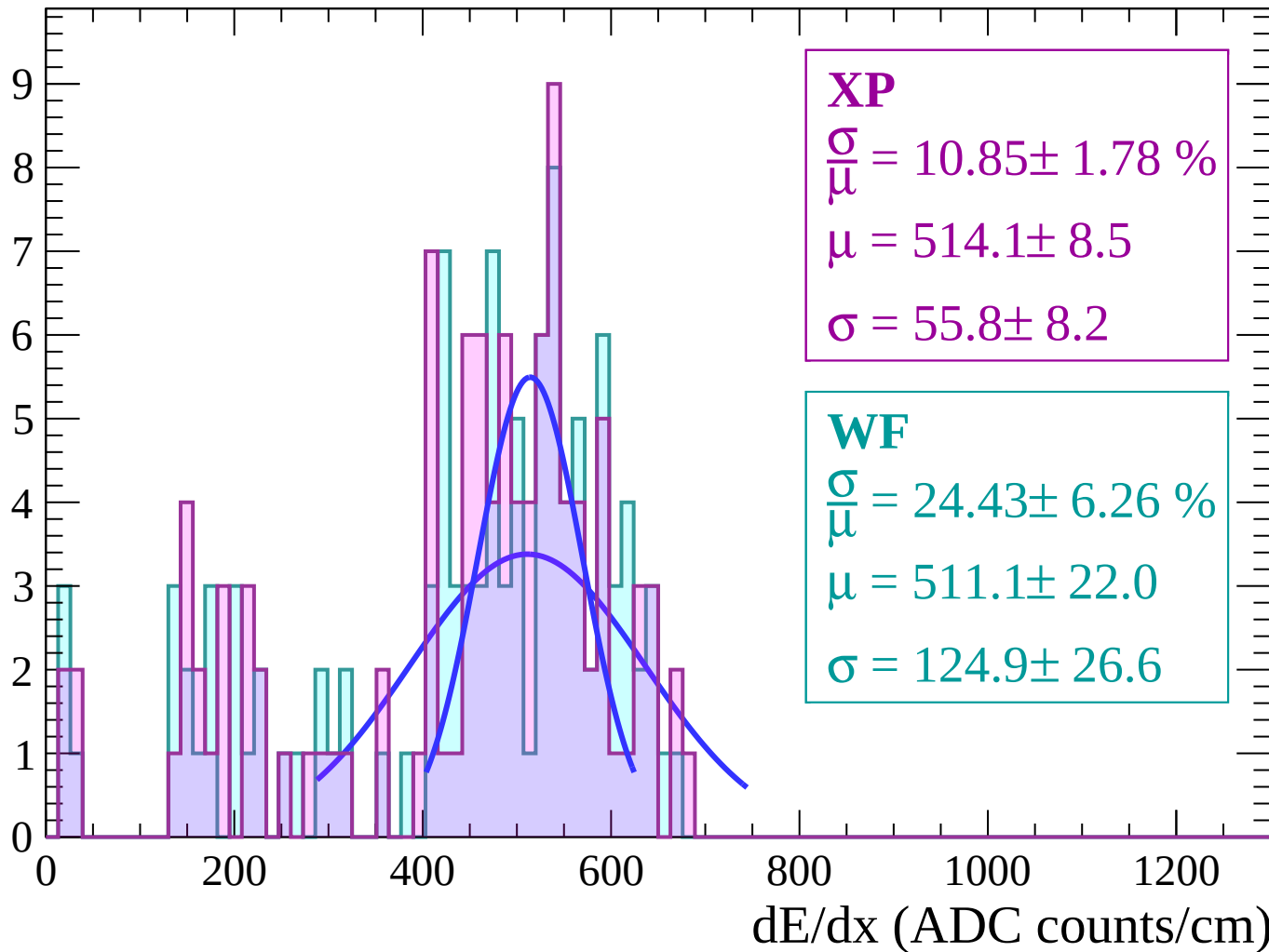
Energy loss | $66 < \theta < 68$

Count



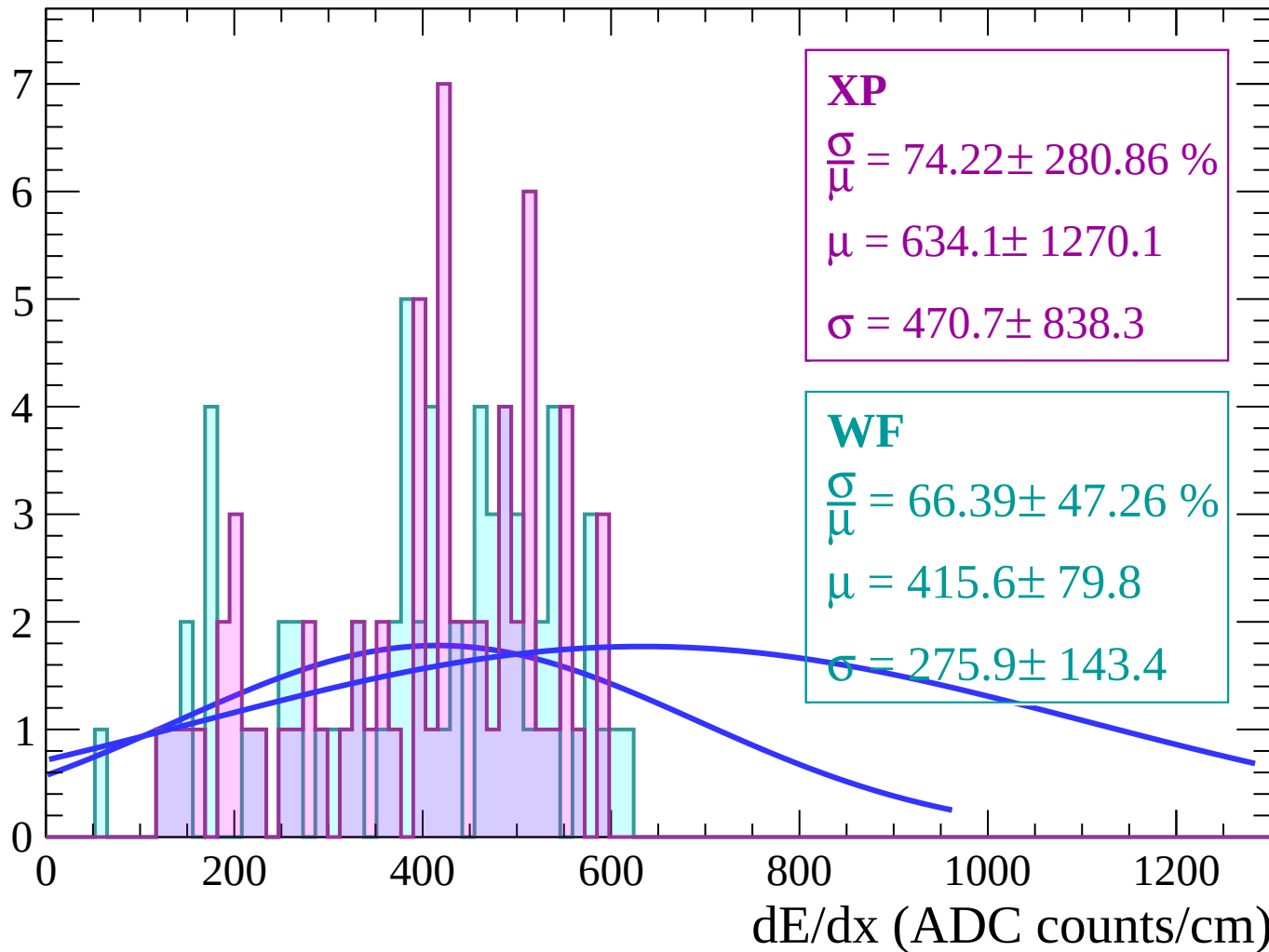
Energy loss | $68 < \theta < 70$

Count



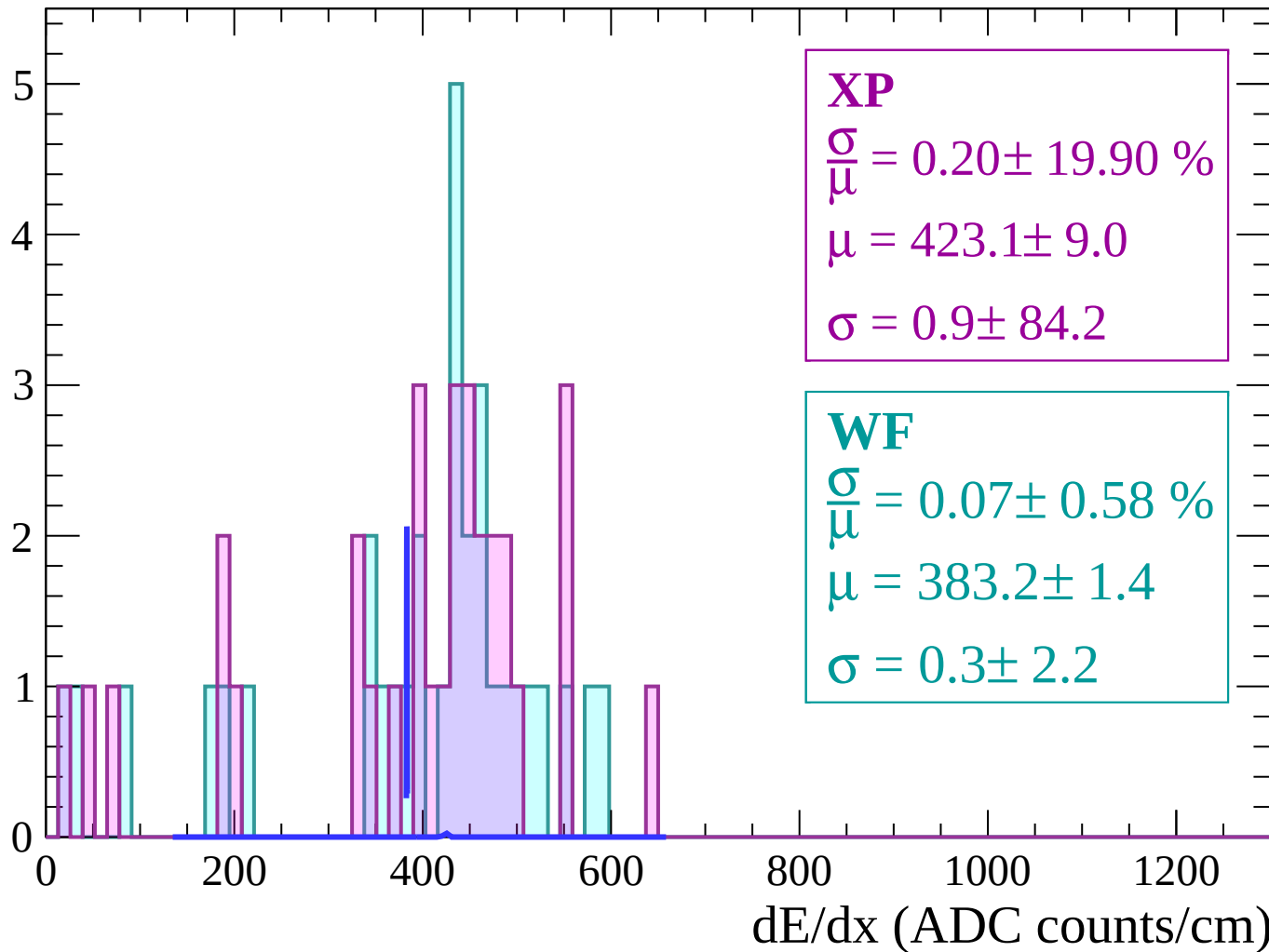
Energy loss | $70 < \theta < 72$

Count



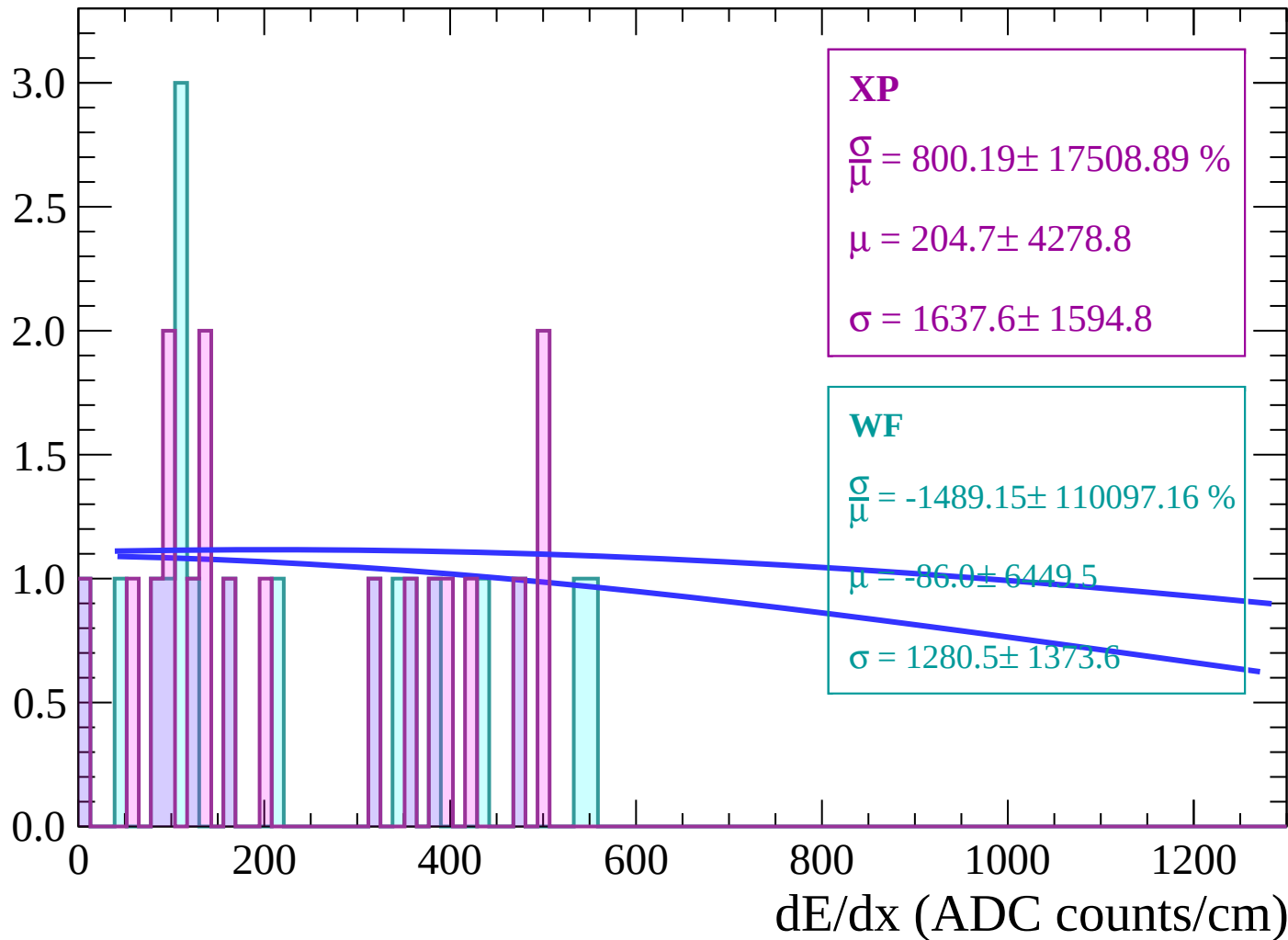
Energy loss | $72 < \theta < 74$

Count



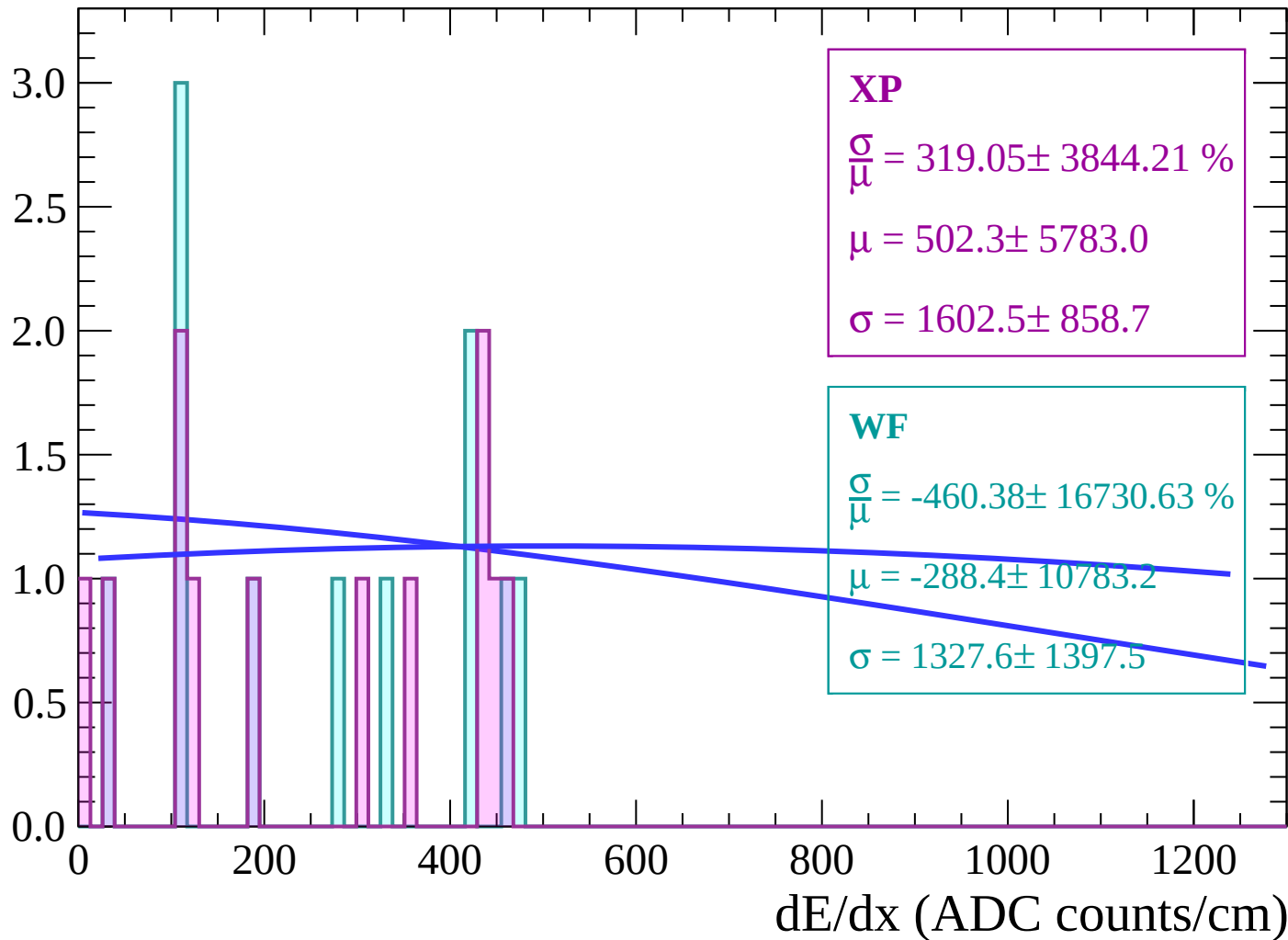
Energy loss | $74 < \theta < 76$

Count



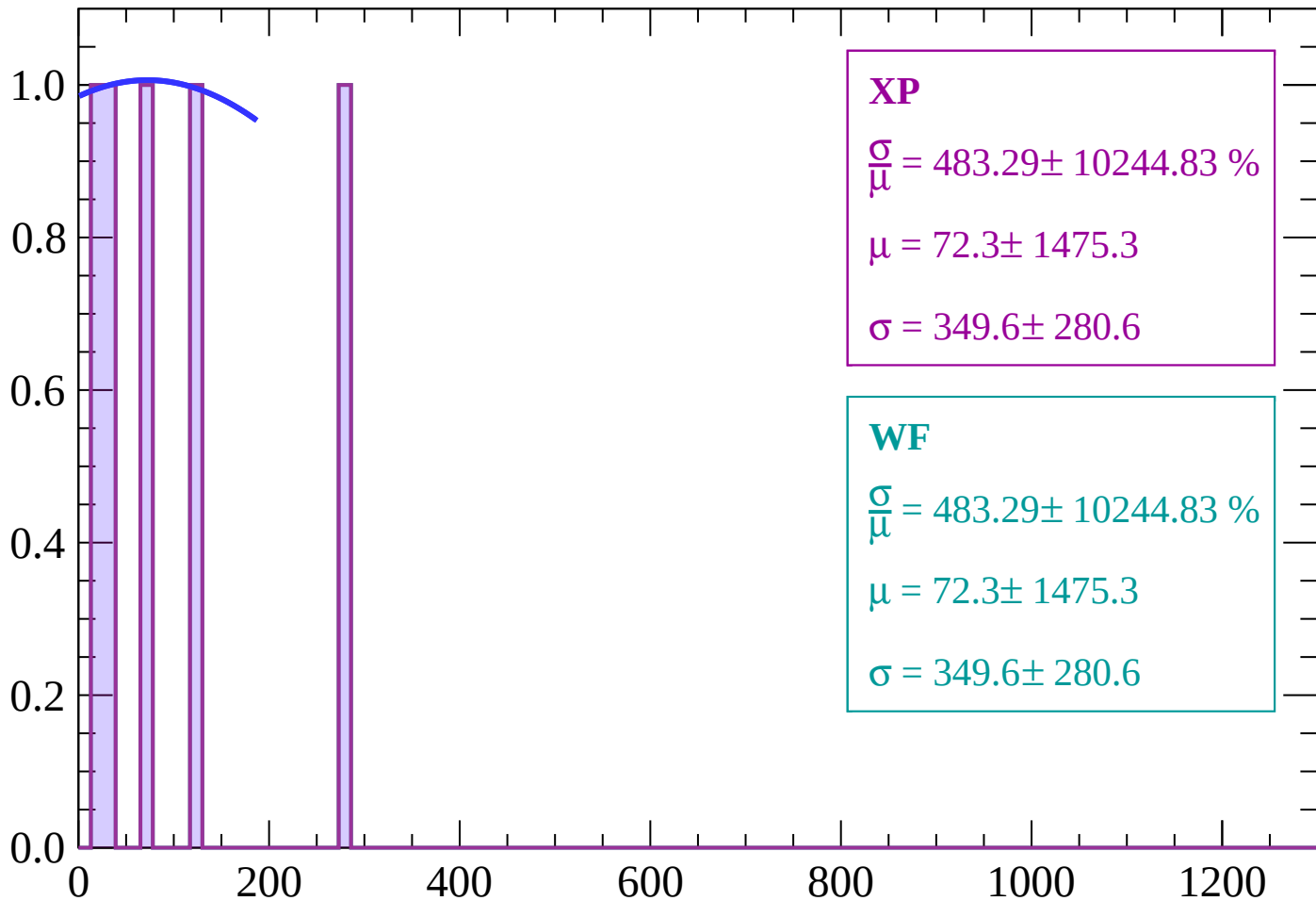
Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

Count



XP

$$\frac{\sigma}{\mu} = 483.29 \pm 10244.83 \%$$

$$\mu = 72.3 \pm 1475.3$$

$$\sigma = 349.6 \pm 280.6$$

WF

$$\frac{\sigma}{\mu} = 483.29 \pm 10244.83 \%$$

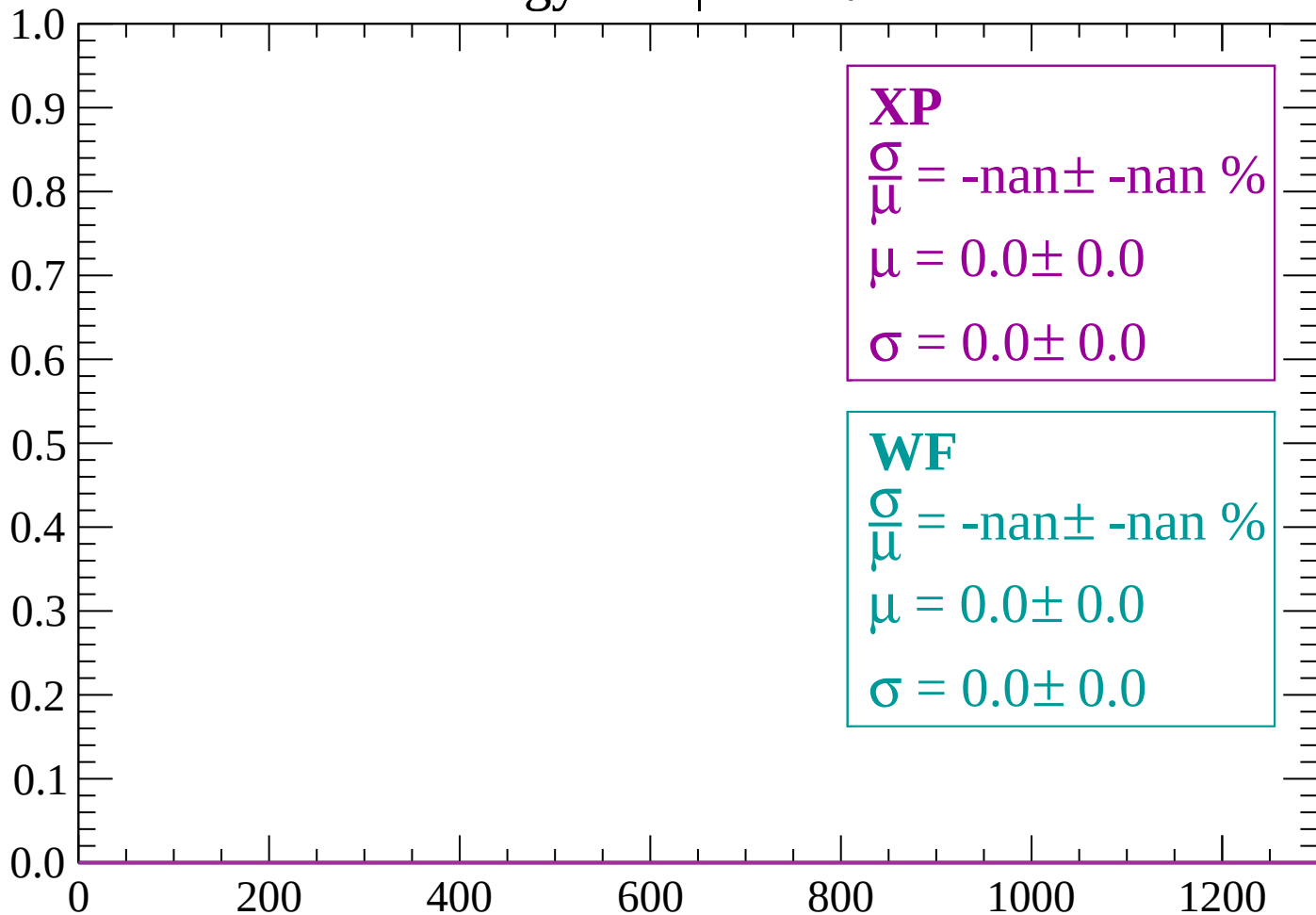
$$\mu = 72.3 \pm 1475.3$$

$$\sigma = 349.6 \pm 280.6$$

dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

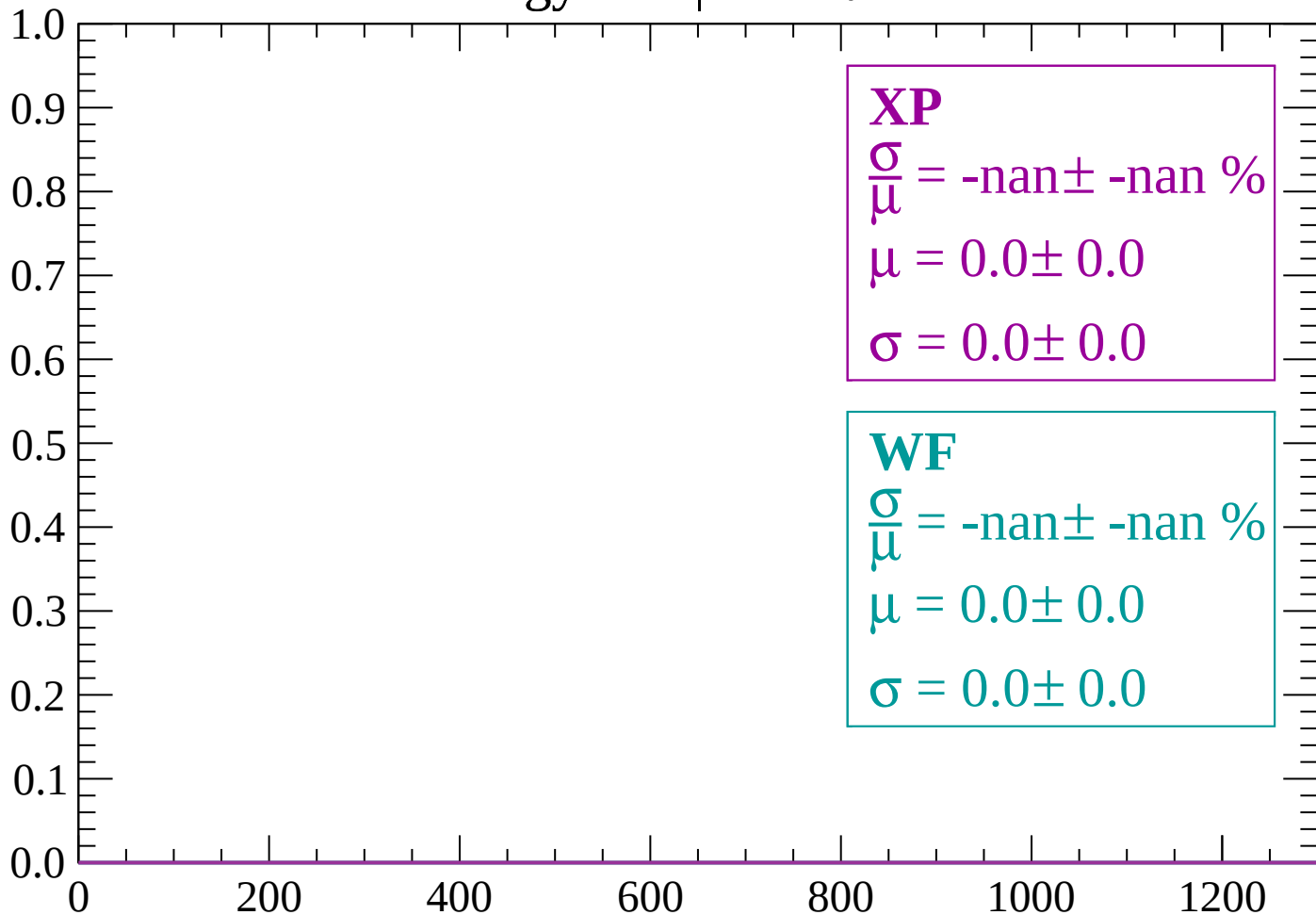
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

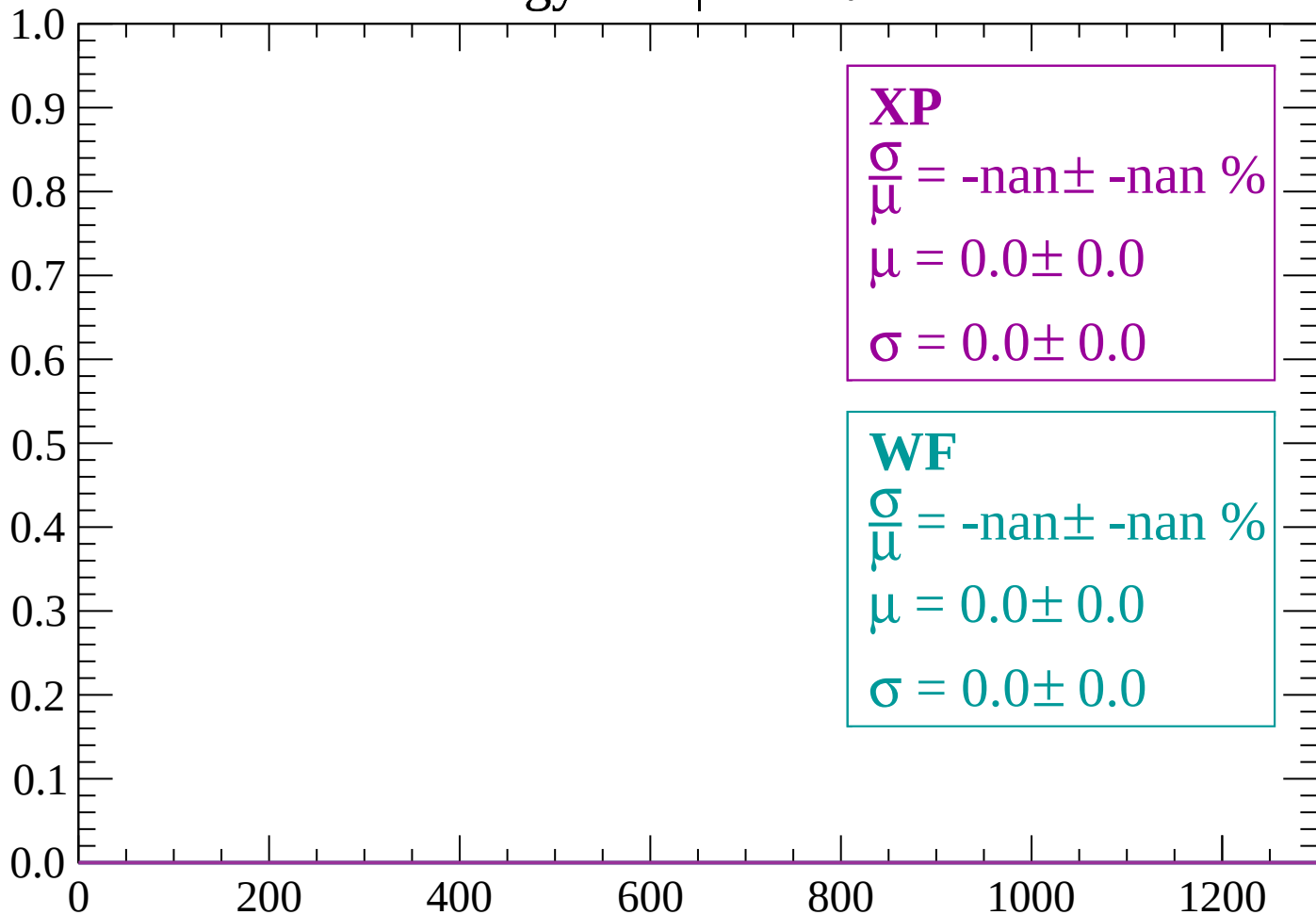
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

WF

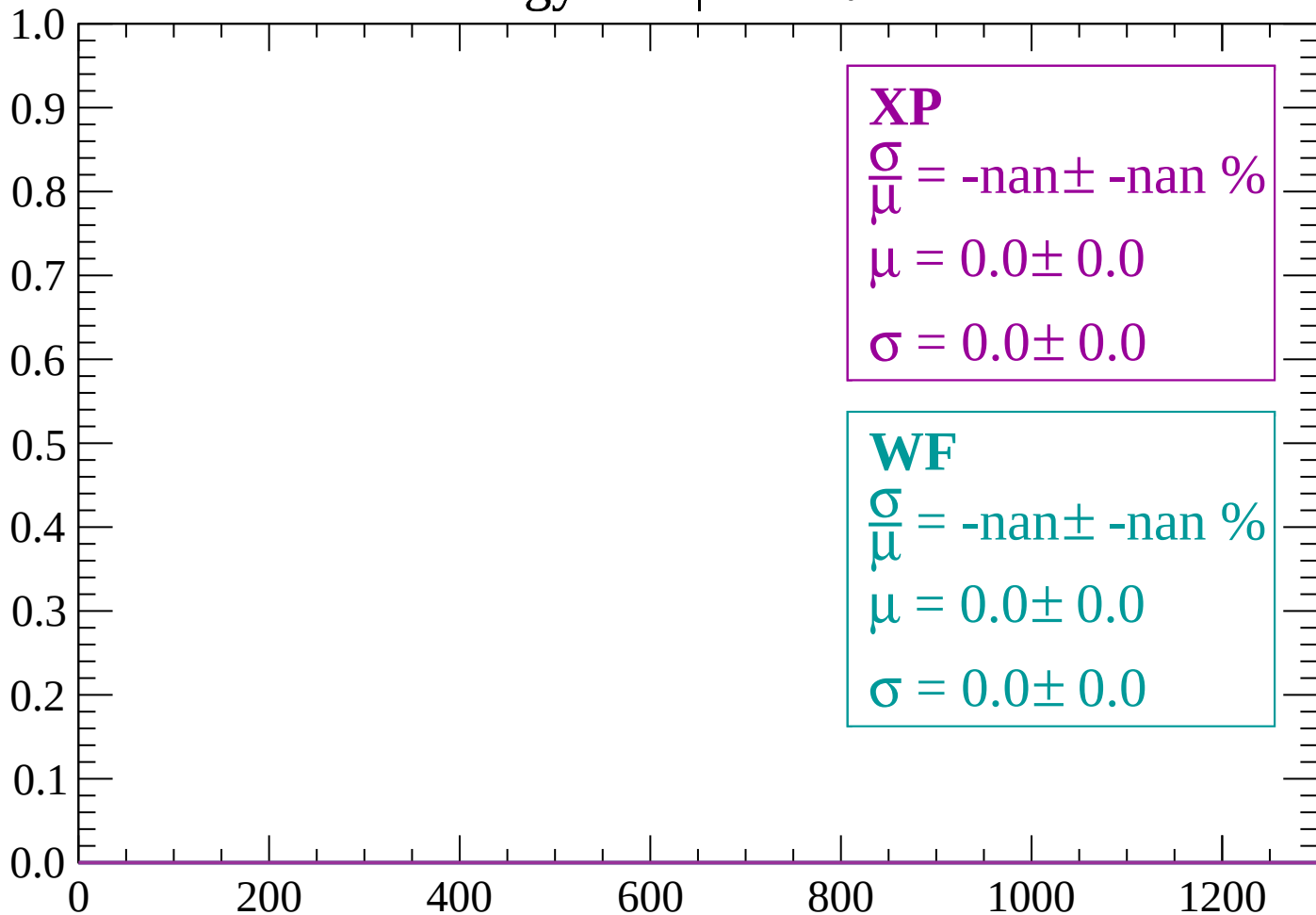
$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

Energy loss | $88 < \theta < 90$

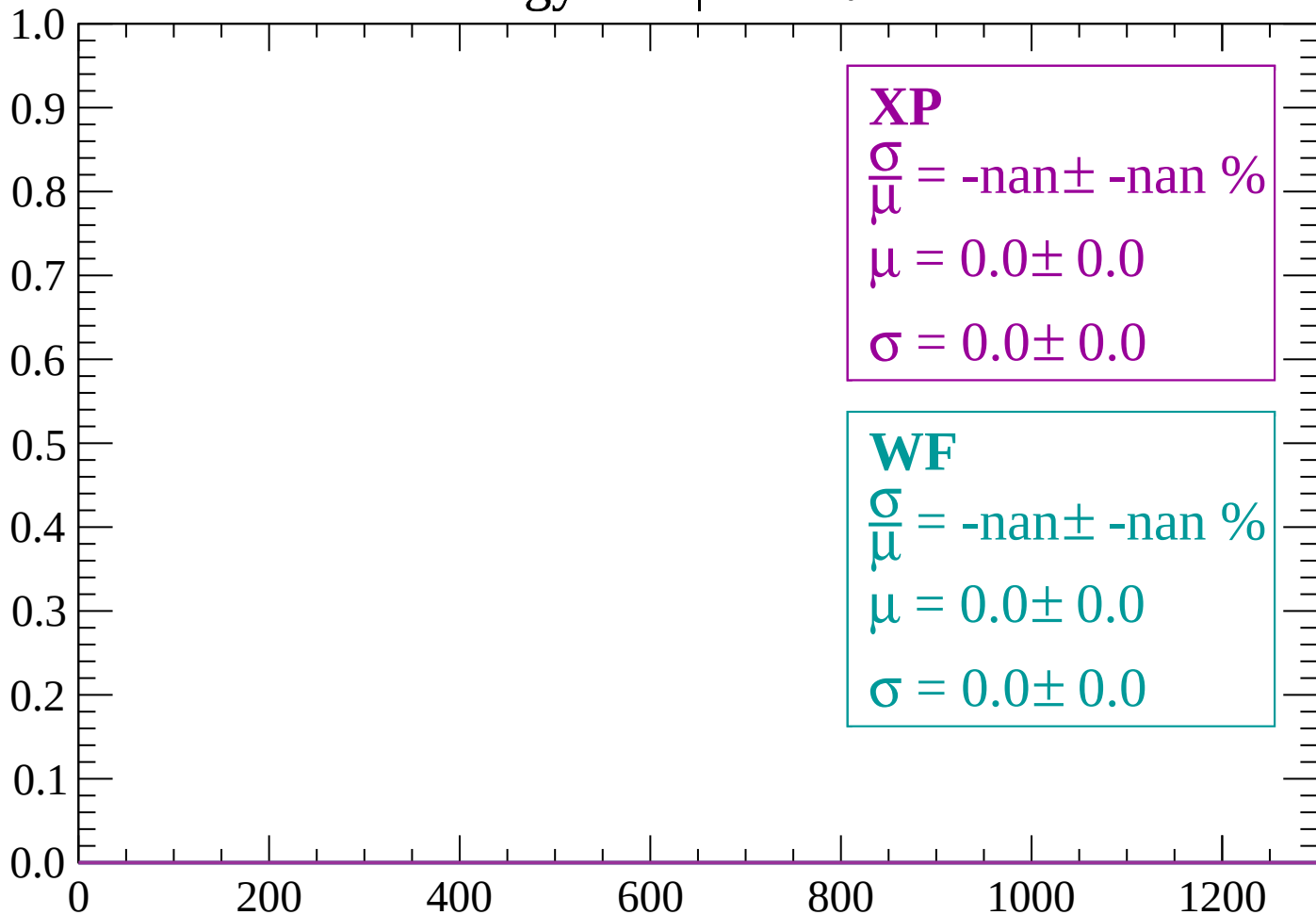
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

